

IEEE P802.3.2™/D3.3

Draft Standard for Ethernet YANG Data Model Definition

Prepared by the

LAN/MAN Standards Committee
of the
IEEE Computer Society

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Abstract: YANG models for IEEE Std 802.3 are defined in this standard. This standard also publishes these models in a machine-readable format.

Keywords: 802.3, 802.3.2, Ethernet, YANG

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Introduction

This introduction is not part of IEEE Std 802.3.2-202x, IEEE Draft Standard for Ethernet YANG Data Model Definitions.

The YANG modules included in this standard provide YANG versions of attributes defined in IEEE Std 802.3™-2022, Clause 30, as well as derivative attributes defined in other management information bases (e.g., SNMP attributes included in IEEE Std 802.3.1, YANG versions of IETF Etherlike MIB attributes, etc.). The YANG modules defined in this standard accommodate IEEE Std 802.3-2022, excluding any currently published or future amendments.

IEEE Std 802.3 will continue to evolve. New Ethernet capabilities are anticipated to be added within the next few years as amendments to this standard.

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IEEE Standard for Ethernet YANG Data Model Definitions

1. Overview

This standard defines YANG modules for various Ethernet functions specified in IEEE Std 802.3.

1.1 Scope

This standard defines YANG data models for IEEE Std 802.3 Ethernet.

1.2 Purpose

The purpose of the standard is to define YANG modules for IEEE Std 802.3 and publish these modules in a machine-readable format.

1.3 Word usage

The word shall indicates mandatory requirements strictly to be followed in order to conform to the standard and from which no deviation is permitted (shall equals is required to).^{1,2}

The word should indicates that among several possibilities one is recommended as particularly suitable, without mentioning or excluding others; or that a certain course of action is preferred but not necessarily required (should equals is recommended that).

The word may is used to indicate a course of action permissible within the limits of the standard (may equals is permitted to).

The word can is used for statements of possibility and capability, whether material, physical, or causal (can equals is able to).

The use of the word must is deprecated and cannot be used when stating mandatory requirements, must is used only to describe unavoidable situations.

The use of *will* is deprecated and cannot be used when stating mandatory requirements, *will* is only used in statements of fact

1.4 Machine-readable YANG modules

Editor's Note (to be removed prior to publication):

Yang files contained in <https://github.com/YangModels/yang/tree/main/standard/ieee/published/802.3> are IEEE 802.3.2-2019 version and will be updated at the publication time.

The authoritative machine-readable YANG files are attached to this standard. A copy is also available for download at the following URL: <https://github.com/YangModels/yang/tree/master/standard/ieee/published/802.3> as text files with a *.yang* extension, e.g., *ieee802-ethernet-interface.yang*. The use of specialized tools to view YANG modules may be useful to create tree, UML image, and HTML outputs from the YANG modules.

Like other languages, YANG (see IETF RFC 7950) has an accepted style for machine-readable files, which was followed during the development of this standard. This formatting may not be preserved when importing the machine-readable YANG modules into the PDF. In case of any discrepancies, the machine-readable files attached to this standard take precedence.

1.5 Summary of YANG-based management framework

The structure of YANG-based management framework closely resembles the structure of the Internet-Standard Management Framework, described in detail in section 7 of IETF RFC 3410.

Managed objects defined using YANG modeling language are hosted on the managed device and accessed through NETCONF (see IETF RFC 7803) or RESTCONF (see IETF RFC 8040). This standard specifies YANG modules that are compliant to YANG 1.1 (see IETF RFC 7950).

1.6 Security considerations

The YANG modules defined in this standard are designed to be accessed via network management protocols, including NETCONF (see IETF RFC 7803) or RESTCONF (see IETF RFC 8040). The lowest NETCONF layer is the secure transport layer, and the mandatory-to-implement secure transport is Secure Shell (SSH) (see IETF RFC 6242) or TLS (see IETF RFC 8446). The lowest RESTCONF layer is HTTPS, and the mandatory-to-implement secure transport is TLS (see IETF RFC 8446).

The NETCONF access control model (see IETF RFC 8341) provides the means to restrict access for particular NETCONF or RESTCONF users to a pre-configured subset of all available NETCONF or RESTCONF protocol operations and content.

There are a number of data nodes defined in these YANG modules that are writable/creatable/deletable, i.e., have the *config* property set to *true*, which is the default setting. These data nodes can be considered sensitive or vulnerable in some network environments. Write operations (e.g., *edit-config*) to these data nodes without proper protection can have a negative effect on network operations.

Some of the readable data nodes in these YANG modules can be considered sensitive or vulnerable in some network environments. It is thus important to control read access (e.g., via *get*, *get-config*, or *notification*) to these data nodes.

Some of the RPC operations in these YANG modules can be considered sensitive or vulnerable in some network environments. Therefore, it is important to control access to these operations.

1.7 YANG module syntax validation

All YANG modules included in this standard are YANG 1.1 (see IETF RFC 7950) compliant and pass automated checks using tools available at the time of publication.

The following open source and/or free versions of YANG validation tools may be used: Pyang (see <https://github.com/mbj4668/pyang>), ConfD (see <http://www.tail-f.com/confd-basic>), as well as other YANG model validation tools listed at <http://www.yangvalidator.com>.

2. Bibliography

2.1 Normative references

The following referenced documents are indispensable for the application of this document (i.e., they must be understood and used, so each referenced document is cited in text and its relationship to this document is explained). For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments or corrigenda) applies.

IEEE Std 802.3™-2022, IEEE Standard for Ethernet.

Editor's Note (to be removed prior to publication):

IEEE Std 802.3.1 to be updated to correct date once project is complete

IEEE Std 802.3.1™-2013, IEEE Standard for Management Information Base (MIB) Definitions for Ethernet.

IETF RFC 3410, *Introduction and Applicability Statements for Internet Standard Management Framework*, J. Case, R. Mundy, D. Partain, B. Stewart, December 2002.

IETF RFC 6242, *Using the NETCONF Protocol over Secure Shell (SSH)*, Wasserman M, June 2011.

IETF RFC 7803, *Changing the Registration Policy for the NETCONF Capability URNs Registry*, B. Leiba February 2016.

IETF RFC 7950, *The YANG 1.1 Data Modeling Language*, Bjorklund M., August 2016.

IETF RFC 8040, *RESTCONF Protocol*, Bierman A., Bjorklund M., and Watsen K., January 2017.

IETF RFC 8341, *Network Configuration Access Control Model*, A. Bierman and M. Bjorklund, March 2018.

IETF RFC 8407, *Guidelines for Authors and Reviewers of YANG Data Model Documents*, Bierman A., October 2018.

IETF RFC 8446, *The Transport Layer Security (TLS) Protocol Version 1.3*, E. Rescorla, August 2018.

3. Definitions

For the purposes of this document, the following terms and definitions apply. Some terms used in this document are defined in IEEE Std 802.3, and where alternative definitions occur in the IEEE Standards Dictionary, the IEEE Std 802.3 definition should be used. The *IEEE Standards Dictionary Online* should be consulted for terms not defined in this clause.^f

3.1 data model. model that describes how data are represented and accessed

3.2 YANG module. module that defines a hierarchy of nodes that can be used for NETCONF-based (see IETF RFC 7803) and RESTCONF-based (see IETF RFC 8040) operations

NOTE—With its definitions and the definitions it imports or includes from elsewhere, a module is self-contained and can be compiled.

^f*IEEE Standards Dictionary Online* is available at: <http://dictionary.ieee.org/>.

4. Abbreviations

This standard contains the following abbreviations:

CO	Central Office
CPE	Customer Premise Equipment
CSMA/CD	carrier sense multiple access with collision detection
DTE	data terminal equipment
EPON	Ethernet passive optical networks
IEEE	Institute of Electrical and Electronics Engineers
IETF	Internet Engineering Task Force
LLID	Link Local Identifier
MPCP	Multi-Point Control Protocol
NETCONF	Network Configuration Protocol
OAM	Operations, Administration, and Maintenance
ONU	Optical Network Unit
OLT	Optical Line terminal
PoE	Power over Ethernet
PoDL	Power over Data Line
RESTCONF	RESTful Configuration Protocol
TDM	Time Division Multiplexing
TDMA	Time Division Multiple Access
WDM	Wavelength Division Multiplexing
YANG	Yet Another Next Generation

5. Ethernet YANG Module

5.1 YANG module structure

Four modules defined in this clause are focused on the configuration and monitoring of IEEE Std 802.3 Ethernet interfaces.

ieee802-ethernet-interface YANG module contains definitions of current attributes used widely in the industry in current products,

ieee802-ethernet-interface-half-duplex YANG module contains definitions of half-duplex attributes.

ieee802-ethernet-lldp YANG module contains definitions for configuring LLDP for IEEE Std 802.3 compliant interfaces.

ieee802-ethernet-mac-merge modules contain definition for configuration of MAC Merge for IEEE Std 802.1Qcw frame preemption.

ieee802-ethernet-phy YANG module contain definition for Ethernet Phy, derived from IANA-MAU-MIB REVISION 202309070000Z, the YANG and definitions correct as per release of 2024-11-09.

ieee802-ethernet-phy-type YANG module contain definition for Ethernet Phy Types, derived IANA-MAU-MIB REVISION 202309070000Z using smidump, the YANG and definitions are correct as per 2024-11-09.

NOTE—*ieee802-ethernet-phy-type* this module may in future be maintained by an external organization (IANA).

This standard does not have a normative requirement for data nodes of the base ietf-interfaces YANG module, but the following data nodes are supported: name, description, type, enabled, admin-status, oper-status, if-index, and phys-address.

5.2 Mapping of IEEE Std 802.3, Clause 30 managed objects

This subclause contains the mapping between YANG data nodes included in *ieee802-ethernet-interface* (see Table 5–1), *ieee802-ethernet-interface-half-duplex* (see Table 5–4), *ieee802-ethernet-mac-merge* (see Table 5–5), and *ieee802-ethernet-lldp* (see Table 5–6) YANG modules, managed objects, and attributes defined in IEEE Std 802.3, Clause 30.

Table 5–1—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-ethernet-interface* YANG data nodes

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-interface</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
oAutoNegotiation	acAutoNegAdminControl	30.6.1.2.2	interfaces/interface/ethernet/	auto-negotiation/enable	R/W
	aAutoNegAutoConfig	30.6.1.1.4		negotiation-status	R
N/A	N/A			flow-control/pause/direction	R/W
oMACControlFunctionEntity	aPAUSEMACCtrlFramesReceived	30.3.4.3		flow-control/pause/statistics/in-frames-pause	R
	aPAUSEMACCtrlFramesTransmitted	30.3.4.2		flow-control/pause/statistics/out-frames-pause	R
N/A	dot3HCOutPFCFrames				
N/A	N/A			flow-control/force-flow-control	R/W
N/A	N/A			speed	R/W
oMACEntity	aDuplexStatus	30.3.1.1.32		duplex	R/W
	aMaxFrameLength	30.3.1.1.37		max-frame-length	R
	aSlowProtocolFrameLimit	30.3.1.1.38		frame-limit-slow-protocol	R
oEXTENSION	aEXTENSIONMACCtrlStatus	30.3.8.3		mac-control-extension-control	R
N/A	N/A			capabilities/auto-negotiation	R

Table 5–1—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-ethernet-interface* YANG data nodes (continued)

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-interface</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
oMACEntity	aFramesReceivedOK	30.3.1.1.5	interfaces/interface/ethernet/statistics/frame	in-frames	R
	aMulticastFramesReceivedOK	30.3.1.1.21		in-multicast-frames	R
	aBroadcastFramesReceivedOK	30.3.1.1.22		in-broadcast-frames	R
	aFrameCheckSequenceErrors + aAlignmentErrors	30.4.3.1.6, 30.4.3.1.7		in-error-fcs-frames	R
oMACEntity	aFrameTooLongErrors	30.3.1.1.25		in-error-oversize-frames	R
	aFramesLostDueToIntMACRcvError	30.3.1.1.15		in-error-mac-internal-frames	R
	aFramesTransmittedOK	30.3.1.1.2		out-frames	R
	aMulticastFramesXmittedOK	30.3.1.1.18		out-multicast-frames	R
	aBroadcastFramesXmittedOK	30.3.1.1.19		out-broadcast-frames	R
	aFramesLostDueToIntMACXmitError	30.3.1.1.12		out-error-mac-internal-frames	R
oPHYEntity	aSymbolErrorDuringCarrier	30.3.2.1.5	interfaces/interface/ethernet/statistics/phy	in-error-symbol	R
	aReceiveLPITransitions	30.3.2.1.11	interfaces/interface/ethernet/statistics/phy/lpi	in-lpi-transitions	R
	aReceiveLPIMicroseconds	30.3.2.1.9		in-lpi-time	R
	aTransmitLPITransitions	30.3.2.1.10		out-lpi-transitions	R
	aTransmitLPIMicroseconds	30.3.2.1.8		out-lpi-time	R

Table 5–1—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-ethernet-interface* YANG data nodes (continued)

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-interface</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
oMACControlEntity	aUnsupportedOpcodesReceived	30.3.3.5	interfaces/interface/ethernet/statistics/mac-control	in-frames-mac-control-unknown	R
oEXTENSION	aEXTENSIONMACCtrlFramesReceived	30.3.8.2		in-frames-mac-control-extension	R
	aEXTENSIONMACCtrlFramesTransmitted	30.3.8.1		out-frames-mac-control-extension	R

Table 5–2—Mapping between IETF RFC 2819 managed objects and *ieee802-ethernet-interface* YANG data nodes

IETF RFC 2819 Attribute(s)	Corresponding <i>ieee802-ethernet-interface</i> YANG data nodes		
	Container(s)	Data node(s)	R/W
no direct object ^a	interfaces/interface/ethernet/statistics/frame	in-total-frames	R
etherStatsOctets		in-total-octets	R
etherStatsUndersizePkts + etherStatsFragments		in-error-undersize-frames	R

^a Can be calculated as: aFramesReceivedOK + aFrameCheckSequenceErrors + aAlignmentErrors + aFrameTooLongErrors + aFramesLostDueToIntMACRcvError.

Table 5–3—Mapping between IETF RFC 3635 managed objects and *ieee802-ethernet-interface* YANG data nodes

ETHERLIKE MIB Attribute(s)	Corresponding <i>ieee802-ethernet-interface</i> YANG data nodes		
	Container(s)	Data node(s)	R/W
dot3HCInPFCFrames	interfaces/interface/ethernet/	flow-control/pfc{ethernet-pfc} / statistics/in-frames-pfc	R
dot3HCOutPFCFrames		flow-control/pfc{ethernet-pfc} / statistics/out-frames-pfc	R

Table 5–4—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-ethernet-interface-half-duplex* YANG data nodes

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-interface-half-duplex</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
oMACEntity	aRateControlAbility	30.3.1.1.33	interfaces/interface/ethernet	dynamic-rate-control	R/W
			interfaces/interface/ethernet/capability	dynamic-rate-control-supported	R
oPHYEntity	aSQETestErrors	30.3.2.1.4	interfaces/interface/ethernet/statistics/frame/csmacd{csma-cd}	in-errors-sqe-test	R
oMACEntity	aSingleCollisionFrames	30.3.1.1.3		out-frames-collision-single	R
	aMultipleCollisionFrames	30.3.1.1.4		out-frames-collision-multiple	R
	aFramesWithDeferredXmissions	30.3.1.1.9		out-frames-deferred	R
	aFramesAbortedDueToXSColls	30.3.1.1.11		out-frames-collisions-excessive	R
	aLateCollisions	30.3.1.1.10		out-collisions-late	R
	aCarrierSenseErrors	30.3.1.1.13		out-errors-carrier-sense	R
	aCollisionFrames	30.3.1.1.30		collision-histogram/collision-count	R
collision-histogram/collision-count-frames				R	

Table 5–5—Mapping between IEEE Std 802.3, 30.14 managed objects and *ieee802-ethernet-mac-merge* YANG data nodes

IEEE Std 802.3, 30.14		Reference	Corresponding <i>ieee802-ethernet-mac-merge</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
oMacMergeEntity	aMACMergeSupport	30.14.1.1	Interfaces/interface/ethernet/mac-merge/admin-status	merge-support	R
	aMACMergeStatusVerify	30.14.1.2		verify-status	R
	aMACMergeStatusTx	30.14.1.5		status-tx	R
	aMACMergeEnableTx	30.14.1.3	Interfaces/interface/ethernet/mac-merge/admin-control	merge-enable-tx	R/W
	aMACMergeVerifyDisableTx	30.14.1.4		verify-disable-tx	R/W
	aMACMergeVerifyTime	30.14.1.6		verify-time	R/W
	aMACMergeAddFragSize	30.14.1.7		frag-size	R/W
	aMACMergeFrameAssErrorCount	30.14.1.8	Interfaces/interface/ethernet/mac-merge/statistics	assembly-error-count	R
	aMACMergeFrameSmdErrorCount	30.14.1.9		smd-error-count	R
	aMACMergeFrameAssOkCount	30.14.1.10		assembly-ok-count	R
	aMACMergeFragCountRx	30.14.1.11		fragment-count-rx	R
	aMACMergeFragCountTx	30.14.1.12		fragment-count-tx	R
	aMACMergeHoldCount	30.14.1.13		hold-count	R

Table 5–6—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-lldp* YANG data nodes

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-llDP</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
oLldpXdot3Config	aLldpXdot3PortConfigTLVsTxEnable	30.12.1.1.1	lldp/port	tlvs-port-config-enable	R/W
oLldpXdot3LocSystemsGroup	aLldpXdot3LocPortAutoNegSupported	30.12.2.1.1		auto-negotiation-supported	R
	aLldpXdot3LocPortAutoNegEnabled	30.12.2.1.2		auto-negotiation-enabled	R
	aLldpXdot3LocPortAutoNegAdvertisedCap	30.12.2.1.3		auto-negotiation-cap	R
	aLldpXdot3LocPortOperMauType	30.12.2.1.4		operational-mau-type	R
	aLldpXdot3LocPowerPortClass	30.12.2.1.5		power-port-class	R
	aLldpXdot3LocPowerMDISupported	30.12.2.1.6		mdi-power-supported	R
	aLldpXdot3LocPowerMDIEnabled	30.12.2.1.7		mdi-power-enabled	R
	aLldpXdot3LocPowerPairControllable	30.12.2.1.8		power-pair-controlable	R

Table 5–6—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-lldp* YANG data nodes

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-llDP</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
	aLldpXdot3LocPowerPairs	30.12.2.1.9		power-pairs	R
	aLldpXdot3LocPowerClass	30.12.2.1.10		local-power-class	R
	aLldpXdot3LocLinkAggStatus	30.12.2.1.11		link-aggregation-status	R
	aLldpXdot3LocLinkAggPortId	30.12.2.1.12		aggregation-port-id	R
	aLldpXdot3LocMaxFrameSize	30.12.2.1.13		local-max-frame-size	R
	aLldpXdot3LocPowerType	30.12.2.1.14		power-type	R
	aLldpXdot3LocPowerSource	30.12.2.1.15		power-source	R
	aLldpXdot3LocPowerPriority	30.12.2.1.16		local-power-priority	R/W
	aLldpXdot3LocPDRequestedPowerValue	30.12.2.1.17		pd-requested-power-value	R
	aLldpXdot3LocPDRequestedPowerValueA	30.12.2.1.18		pd-requested-power-value-a	R
	aLldpXdot3LocPDRequestedPowerValueB	30.12.2.1.19		pd-requested-power-value-b	R
	aLldpXdot3LocPSEAllocatedPowerValue	30.12.2.1.20		pse-allocated-power-value	R
	aLldpXdot3LocPSEAllocatedPowerValueA	30.12.2.1.21		pse-allocated-power-value-a	R
	aLldpXdot3LocPSEAllocatedPowerValueB	30.12.2.1.22		pse-allocated-power-value-b	R

Table 5–6—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-lldp* YANG data nodes

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-lldp</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
	aLldpXdot3LocPSEPoweringStatus	30.12.2.1.23		pse-powering-status	R
	aLldpXdot3LocPDPoweredStatus	30.12.2.1.24		pd-powered-status	R
	aLldpXdot3LocPowerPairsExt	30.12.2.1.25		power-pairs-ext	R
	aLldpXdot3LocPowerClassExtA	30.12.2.1.26		power-class-ext-A	R
	aLldpXdot3LocPowerClassExtB	30.12.2.1.27		power-class-ext-B	R
	aLldpXdot3LocPowerClassExt	30.12.2.1.28		power-class-ext	R
	aLldpXdot3LocPowerTypeExt	30.12.2.1.29		power-type-ext	R
	aLldpXdot3LocPDLload	30.12.2.1.30		pd-load	R
	aLldpXdot3LocPD4PID	30.12.2.1.31		pd-4pid	R
	aLldpXdot3LocPSEMaxAvailPower	30.12.2.1.32		pse-max-avail-power	R
	aLldpXdot3LocPSEAutoclassSupport	30.12.2.1.33		pse-autoclass-support	R
	aLldpXdot3LocAutoclassCompleted	30.12.2.1.34		autoclass-completed	R
	aLldpXdot3LocAutoclassRequest	30.12.2.1.35		autoclass-request	R
	aLldpXdot3LocPowerDownRequest	30.12.2.1.36		power-down-request	R
	aLldpXdot3LocPowerDownTime	30.12.2.1.37		power-down-time	R

Table 5–6—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-lldp* YANG data nodes

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-llp</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
	aLldpXdot3LocMeasVoltageSupport	30.12.2.1.38		meas-voltage-support	R
	aLldpXdot3LocMeasCurrentSupport	30.12.2.1.39		meas-current-support	R
	aLldpXdot3LocMeasPowerSupport	30.12.2.1.40		meas-power-support	R
	aLldpXdot3LocMeasEnergySupport	30.12.2.1.41		meas-energy-support	R
	aLldpXdot3LocMeasurementSource	30.12.2.1.42		measurement-source	R
	aLldpXdot3LocMeasVoltageRequest	30.12.2.1.43		meas-voltage-request	R
	aLldpXdot3LocMeasCurrentRequest	30.12.2.1.44		meas-current-request	R
	aLldpXdot3LocMeasCurrentRequest	30.12.2.1.45		meas-power-request	R
	aLldpXdot3LocMeasEnergyRequest	30.12.2.1.46		meas-energy-request	R
	aLldpXdot3LocMeasVoltageValid	30.12.2.1.47		meas-voltage-valid	R
	aLldpXdot3LocMeasCurrentValid	30.12.2.1.48		meas-current-valid	R
	aLldpXdot3LocMeasPowerValid	30.12.2.1.49		meas-power-valid	R
	aLldpXdot3LocMeasEnergyValid	30.12.2.1.50		meas-energy-valid	R

Table 5–6—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-lldp* YANG data nodes

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-llDP</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
	aLldpXdot3LocMeasVoltageUncertainty	30.12.2.1.51		meas-voltage-uncertainty	R
	aLldpXdot3LocMeasCurrentUncertainty	30.12.2.1.52		meas-current-uncertainty	R
	aLldpXdot3LocMeasPowerUncertainty	30.12.2.1.53		meas-power-uncertainty	R
	aLldpXdot3LocMeasEnergyUncertainty	30.12.2.1.54		meas-energy-uncertainty	R
	aLldpXdot3LocVoltageMeasurement	30.12.2.1.55		voltage-measurement	R
	aLldpXdot3LocCurrentMeasurement	30.12.2.1.56		current-measurement	R
	aLldpXdot3LocPowerMeasurement	30.12.2.1.57		power-measurement	R
	aLldpXdot3LocEnergyMeasurement	30.12.2.1.58		energy-measurement	R
	aLldpXdot3LocPSEPowerPriceIndex	30.12.2.1.59		pse-power-price-index	R
	aLldpXdot3LocResponseTime	30.12.2.1.60		local-response	R
	aLldpXdot3LocReady	30.12.2.1.61		local-system-ready	R
	aLldpXdot3LocTxTwSys	30.12.2.1.62		tx-system-value	R

Table 5–6—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-lldp* YANG data nodes

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-llDP</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
	aLldpXdot3LocTxTwSysEcho	30.12.2.1.63		tx-system-value-echo	R
	aLldpXdot3LocRxTwSys	30.12.2.1.64		rx-system-value	R
	aLldpXdot3LocRxTwSysEcho	30.12.2.1.65		rx-system-value-echo	R
	aLldpXdot3LocFbTwSys	30.12.2.1.66		fallback-system-value	R
	aLldpXdot3TxDllReady	30.12.2.1.67		tx-dll-ready	R
	aLldpXdot3RxDllReady	30.12.2.1.68		rx-dll-ready	R
	aLldpXdot3LocDllEnabled	30.12.2.1.69		dll-ready	R
	aLldpXdot3LocTxFw	30.12.2.1.70		tx-system-fw	R
	aLldpXdot3LocTxFwEcho	30.12.2.1.71		tx-system-fw-echo	R
	aLldpXdot3LocRxFw	30.12.2.1.72		rx-system-fw	R
	aLldpXdot3LocRxFwEcho	30.12.2.1.73		rx-system-fw-echo	R
	aLldpXdot3LocPreemptSupported	30.12.2.1.74		preemption-supported	R
	aLldpXdot3LocPreemptEnabled	30.12.2.1.75		preemption-enabled	R
	aLldpXdot3LocPreemptActive	30.12.2.1.76		preemption-active	R
	aLldpXdot3LocAddFragSize	30.12.2.1.77		additional-fragment-size	R

Table 5–6—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-lldp* YANG data nodes

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-lldp</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
oLldpXdot3RemSystemsGroup	aLldpXdot3RemPortAutoNegSupported	30.12.3.1.1	lldp/port/remote-systems-data	auto-negotiation-supported	R
	aLldpXdot3RemPortAutoNegEnabled	30.12.3.1.2		auto-negotiation-enabled	R
	aLldpXdot3RemPortAutoNegAdvertisedCap	30.12.3.1.3		auto-negotiation-cap	R
	aLldpXdot3RemPortOperMauType	30.12.3.1.4		operational-mau-type	R
	aLldpXdot3RemPowerPortClass	30.12.3.1.5		power-port-class	R
	aLldpXdot3RemPowerMDISupported	30.12.3.1.6		mdi-power-supported	R

Table 5–6—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-lldp* YANG data nodes

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-llDP</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
	aLldpXdot3RemPowerMDIEnabled	30.12.3.1.7		mdi-power-enabled	R
	aLldpXdot3RemPowerPairControllable	30.12.3.1.8		power-pair-controllable	R
	aLldpXdot3RemPowerPairs	30.12.3.1.9		power-pairs	R
	aLldpXdot3RemPowerClass	30.12.3.1.10		power-class	R
	aLldpXdot3RemLinkAggStatus	30.12.3.1.11		link-aggregation-status	R
	aLldpXdot3RemLinkAggPortId	30.12.3.1.12		aggregation-port-id	R
	aLldpXdot3RemMaxFrameSize	30.12.3.1.13		local-max-frame-size	R
	aLldpXdot3RemPowerType	30.12.3.1.14		power-type	R
	aLldpXdot3RemPowerSource	30.12.3.1.15		power-source	R
	aLldpXdot3RemPowerPriority	30.12.3.1.16		power-priority	RW
	aLldpXdot3RemPDRequestedPowerValue	30.12.3.1.17		pd-requested-power-value	R
	aLldpXdot3RemPDRequestedPowerValueA	30.12.3.1.18		pd-requested-power-value-a	R
	aLldpXdot3RemPDRequestedPowerValueB	30.12.3.1.19		pd-requested-power-value-b	R
	aLldpXdot3RemPSEAllocatedPowerValue	30.12.3.1.20		pse-allocated-power-value	R
	aLldpXdot3RemPSEAllocatedPowerValueA	30.12.3.1.21		pse-allocated-power-value-a	R
	aLldpXdot3RemPSEAllocatedPowerValueB	30.12.3.1.22		pse-allocated-power-value-b	R

Table 5–6—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-lldp* YANG data nodes

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-llp</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
	aLldpXdot3RemPSEPoweringStatus	30.12.3.1.23		pse-powering-status	R
	aLldpXdot3RemPDPoweredStatus	30.12.3.1.24		pd-powered-status	R
	aLldpXdot3RemPowerPairsExt	30.12.3.1.25		power-pairs-ext	R
	aLldpXdot3RemPowerClassExtA	30.12.3.1.26		power-class-ext-A	R
	aLldpXdot3RemPowerClassExtB	30.12.3.1.27		power-class-ext-B	R
	aLldpXdot3RemPowerClassExt	30.12.3.1.28		power-class-ext	R
	aLldpXdot3RemPowerTypeExt	30.12.3.1.29		power-type-ext	R
	aLldpXdot3RemPDLload	30.12.3.1.30		pd-load	R
	aLldpXdot3RemPD4PID	30.12.3.1.31		pd-4pid	R
	aLldpXdot3RemPSEMaxAvailPower	30.12.3.1.32		pse-max-avail-power	R
	aLldpXdot3RemPSEAutoclassSupport	30.12.3.1.33		pse-autoclass-support	R
	aLldpXdot3RemAutoclassCompleted	30.12.3.1.34		autoclass-completed	R
	aLldpXdot3RemAutoclassRequest	30.12.3.1.35		autoclass-request	R
	aLldpXdot3RemPowerDownRequest	30.12.3.1.36		power-down-request	R
	aLldpXdot3RemPowerDownTime	30.12.3.1.37		power-down-time	R

Table 5–6—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-lldp* YANG data nodes

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-llDP</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
	aLldpXdot3RemMeasVoltageSupport	30.12.3.1.38		meas-voltage-support	R
	aLldpXdot3RemMeasCurrentSupport	30.12.3.1.39		meas-current-support	R
	aLldpXdot3RemMeasPowerSupport	30.12.3.1.40		meas-power-support	R
	aLldpXdot3RemMeasEnergySupport	30.12.3.1.41		meas-energy-support	R
	aLldpXdot3RemMeasurementSource	30.12.3.1.42		measurement-source	R
	aLldpXdot3RemMeasVoltageRequest	30.12.3.1.43		meas-voltage-request	R
	aLldpXdot3RemMeasCurrentRequest	30.12.3.1.44		meas-current-request	R
	aLldpXdot3RemMeasCurrentRequest	30.12.3.1.45		meas-power-request	R
	aLldpXdot3RemMeasEnergyRequest	30.12.3.1.46		meas-energy-request	R
	aLldpXdot3RemMeasVoltageValid	30.12.3.1.47		meas-voltage-valid	R
	aLldpXdot3RemMeasCurrentValid	30.12.3.1.48		meas-current-valid	R
	aLldpXdot3RemMeasPowerValid	30.12.3.1.49		meas-power-valid	R
	aLldpXdot3RemMeasEnergyValid	30.12.3.1.50		meas-energy-valid	R

Table 5–6—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-lldp* YANG data nodes

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-lldp</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
	aLldpXdot3RemMeasVoltageUncertainty	30.12.3.1.51		meas-voltage-uncertainty	R
	aLldpXdot3RemMeasCurrentUncertainty	30.12.3.1.52		meas-current-uncertainty	R
	aLldpXdot3RemMeasPowerUncertainty	30.12.3.1.53		meas-power-uncertainty	R
	aLldpXdot3RemMeasEnergyUncertainty	30.12.3.1.54		meas-energy-uncertainty	R
	aLldpXdot3RemVoltageMeasurement	30.12.3.1.55		voltage-measurement	R
	aLldpXdot3RemCurrentMeasurement	30.12.3.1.56		current-measurement	R
	aLldpXdot3RemPowerMeasurement	30.12.3.1.57		power-measurement	R
	aLldpXdot3RemEnergyMeasurement	30.12.3.1.58		energy-measurement	R
	aLldpXdot3RemPSEPowerPriceIndex	30.12.3.1.59		pse-power-price-index	R
	aLldpXdot3RemTxTwSys	30.12.3.1.60		tx-system-value	R
	aLldpXdot3RemTxTwSysEcho	30.12.3.1.61		tx-system-value-echo	R
	aLldpXdot3RemRxTwSys	30.12.3.1.62		rx-system-value	R

Table 5–6—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-lldp* YANG data nodes

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-llDP</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
	aLldpXdot3RemRxTwSysEcho	30.12.3.1.63		rx-system-value-echo	R
	aLldpXdot3RemFbTwSys	30.12.3.1.64		fallback-system-value	R
	aLldpXdot3RemTxFw	30.12.3.1.65		tx-system-fw	R
	aLldpXdot3RemTxFwEcho	30.12.3.1.66		tx-system-fw-echo	R
	aLldpXdot3RemRxFw	30.12.3.1.67		rx-system-fw	R
	aLldpXdot3RemRxFwEcho	30.12.3.1.68		rx-system-fw-echo	R
	aLldpXdot3RemPreemptSupported	30.12.3.1.69		preemption-supported	R
	aLldpXdot3RemPreemptEnabled	30.12.3.1.70		preemption-enabled	R
	aLldpXdot3RemPreemptActive	30.12.3.1.71		preemption-active	R
	aLldpXdot3RemAddFragSize	30.12.3.1.72		additonal-fragment-size	R

5.3 YANG module definition⁹

The YANG module tree hierarchy uses terms defined in IETF RFC 8407.

5.3.1 Tree hierarchy

5.3.1.1 ieee802-ethernet-interface

```

module: ieee802-ethernet-interface
  augment /if:interfaces/if:interface:
    +--rw ethernet
      +--rw auto-negotiation!
        | +--rw enable?          boolean
        | +--ro negotiation-status? enumeration
      +--ro phy-type?            identityref
      +--ro pmd-type?            identityref
      +--rw duplex?              duplex-type
      o--rw speed?               eth-if-speed-type
      x--rw flow-control
        | +--rw pause {ethernet-pause}?
        | | +--rw direction?    pause-fc-direction-type
        | | +--ro statistics
        | |   +--ro in-frames-pause? yang:counter64
        | |   +--ro out-frames-pause? yang:counter64
        | +--rw pfc {ethernet-pfc}?
        | | +--rw enable?        boolean
        | | x--ro statistics
        | |   +--ro in-frames-pfc? yang:counter64
        | |   +--ro out-frames-pfc? yang:counter64
        | +--rw force-flow-control? boolean
      +--ro max-frame-length?    uint16
      +--ro mac-control-extension-control? boolean
      +--ro frame-limit-slow-protocol? uint64
      +--ro capabilities
        | +--ro auto-negotiation? boolean

```

⁹Copyright release for YANG modules: Users of this standard may freely reproduce the YANG module contained in this subclause so that it can be used for its intended purpose.

```

1  +---rw ethernet-pause {ethernet-pause}?
2  |   +---rw control-and-status
3  |   |   +---rw pause-admin-control?      pause-fc-direction-type
4  |   |   +---ro pause-oper-status?        pause-fc-direction-type
5  |   |   +---rw link-delay-allowance?      uint32
6  |   |   +---ro pfc-enable-status?         boolean
7  |   +---ro statistics
8  |       +---ro in-frames-pause?           yang:counter64
9  |       +---ro out-frames-pause?          yang:counter64
10 +---ro statistics
11     +---ro frame
12         |   +---ro in-total-frames?        yang:counter64
13         |   +---ro in-total-octets?        yang:counter64
14         |   +---ro in-frames?              yang:counter64
15         |   +---ro in-multicast-frames?    yang:counter64
16         |   +---ro in-broadcast-frames?    yang:counter64
17         |   +---ro in-error-fcs-frames?    yang:counter64
18         |   +---ro in-error-undersize-frames? yang:counter64
19         |   +---ro in-error-oversize-frames? yang:counter64
20         |   +---ro in-error-mac-internal-frames? yang:counter64
21         |   +---ro out-frames?             yang:counter64
22         |   +---ro out-multicast-frames?   yang:counter64
23         |   +---ro out-broadcast-frames?   yang:counter64
24         |   +---ro out-error-mac-internal-frames? yang:counter64
25     +---ro phy
26         |   +---ro in-error-symbol?        yang:counter64
27         |   +---ro lpi
28         |       +---ro in-lpi-transitions?  yang:counter64
29         |       +---ro in-lpi-time?         decimal64
30         |       +---ro out-lpi-transitions? yang:counter64
31         |       +---ro out-lpi-time?        decimal64
32     +---ro mac-control
33         +---ro in-frames-mac-control-unknown? yang:counter64
34         +---ro in-frames-mac-control-extension? yang:counter64
35         +---ro out-frames-mac-control-extension? yang:counter64
36
37
38
39
40
41
42
43

```

5.3.1.2 ieee802-ethernet-interface-half-duplex

```

module: ieee802-ethernet-interface-half-duplex

augment /if:interfaces/if:interface/ieee802-eth-if:ethernet:
  o--rw dynamic-rate-control?    dynamic-rate-control-type {dynamic-rate-control}?
augment /if:interfaces/if:interface/ieee802-eth-if:ethernet/ieee802-eth-if:capabilities:
  o--ro dynamic-rate-control-supported?    boolean {dynamic-rate-control}?
augment /if:interfaces/if:interface/ieee802-eth-if:ethernet/ieee802-eth-if:statistics/ieee802-eth-if:frame:
  +--ro csma-cd {csma-cd}?
    +--ro in-errors-sqe-test?                yang:counter64
    +--ro out-frames-collision-single?        yang:counter64
    +--ro out-frames-collision-multiple?      yang:counter64
    +--ro out-frames-deferred?                yang:counter64
    +--ro out-frames-collisions-excessive?    yang:counter64
    +--ro out-collisions-late?                yang:counter64
    +--ro out-errors-carrier-sense?           yang:counter64
    +--ro collision-histogram* [collision-count]
      +--ro collision-count                  yang:counter64
      +--ro collision-count-frames?          yang:counter64

```

5.3.1.3 ieee802-ethernet-mac-merge

```

module: ieee802-ethernet-mac-merge

augment /if:interfaces/if:interface/ieee802-eth-if:ethernet:
  +--rw mac-merge {mac-merge}?
    +--rw admin-control
      | +--rw merge-enable-tx?      enumeration
      | +--rw verify-disable-tx?    enumeration
      | +--rw verify-time?          uint16
      | +--rw frag-size?            uint8
    +--ro admin-status
      | +--ro merge-support?        enumeration
      | +--ro verify-status?        enumeration
      | +--ro status-tx?            enumeration
    +--ro statistics
      +--ro assembly-error-count?    yang:counter64

```

```

1      +---ro smd-error-count?          yang:counter64
2      +---ro assembly-ok-count?        yang:counter64
3      +---ro fragment-count-rx?        yang:counter64
4      +---ro fragment-count-tx?        yang:counter64
5      +---ro hold-count?                yang:counter64
6
7
8
9  5.3.1.4 ieee802-ethernet-lldp
10
11  module: ieee802-ethernet-lldp
12
13    augment /lldp:lldp/lldp:port:
14      +---ro auto-negotiation-supported?  boolean
15      +---ro auto-negotiation-enabled?    boolean
16      x--ro auto-negotiation-cap?         binary
17      x--ro operational-mau-type?         int32
18      +---rw auto-negotiation-cap-bits?   ieee802-phy:auto-neg-ability-bits
19      +---rw operational-pmd-type?        identityref
20      +---rw power-port-class?            port-class-type
21      +---rw mdi-power-supported?         boolean
22      +---rw mdi-power-enabled?          boolean
23      +---rw power-pair-controlable?      boolean
24      +---rw power-pairs?                 pse-pinout-type
25      +---rw local-power-class?           pse-power-class-type
26      o--ro link-aggregation-status?      bits
27      o--ro aggregation-port-id?          int32
28      o--ro local-max-frame-size?         int32
29      +---ro power-type?                  bits
30      +---rw power-support?               bits
31      +---ro power-source?                power-source-type
32      +---ro pd-requested-power-value?    int32
33      +---ro pd-requested-power-value-a?  int32
34      +---ro pd-requested-power-value-b?  int32
35      +---ro pse-allocated-power-value?   int32
36      +---ro pse-allocated-power-value-a? int32
37      +---ro pse-allocated-power-value-b? int32
38      +---ro pse-powering-status?         powering-status-type
39      +---ro pd-powered-status?           powered-status-type
40      +---ro power-pairs-ext?             power-pairs-type

```

1	+--ro power-class-ext-A?	power-class-ext-AB-type
2	+--ro power-class-ext-B?	power-class-ext-AB-type
3	+--ro power-class-ext?	power-class-ext-type
4	+--ro power-type-ext?	power-type
5	+--ro pd-load?	boolean
6	+--ro pd-4pid?	boolean
7	+--ro pse-max-avail-power?	int32
8	+--ro pse-autoclass-support?	boolean
9	+--ro autoclass-completed?	boolean
10	+--ro autoclass-request?	boolean
11	+--rw power-down-request?	int32
12	+--rw power-down-time?	int32
13	+--rw meas-voltage-support?	boolean
14	+--rw meas-current-support?	boolean
15	+--rw meas-power-support?	boolean
16	+--rw meas-energy-support?	boolean
17	o--rw measurement-source?	bits
18	+--ro meas-voltage-request?	boolean
19	+--ro meas-current-request?	boolean
20	+--ro meas-power-request?	boolean
21	+--ro meas-energy-request?	boolean
22	+--ro meas-voltage-valid?	boolean
23	+--ro meas-current-valid?	boolean
24	+--ro meas-power-valid?	boolean
25	+--ro meas-energy-valid?	boolean
26	+--ro meas-voltage-uncertainty?	int32
27	+--ro meas-current-uncertainty?	int32
28	+--ro meas-power-uncertainty?	int32
29	+--ro meas-energy-uncertainty?	int32
30	+--ro voltage-measurement?	int32
31	+--ro current-measurement?	int32
32	+--ro power-measurement?	int32
33	+--ro energy-measurement?	int32
34	+--ro pse-power-price-index?	int32
35	+--ro local-response?	int32
36	+--ro local-system-ready?	boolean
37	+--ro tx-system-value?	int32
38	+--ro tx-system-value-echo?	int32
39	+--ro rx-system-value?	int32
40		
41		
42		
43		

```

1  +--ro rx-system-value-echo?          int32
2  +--ro fallback-system-value?         int32
3  +--ro tx-dll-ready?                  boolean
4  +--ro rx-dll-ready?                  boolean
5  +--ro dll-enabled?                   boolean
6  +--ro tx-system-fw?                  boolean
7  +--ro tx-system-fw-echo?             boolean
8  +--ro rx-system-fw?                  boolean
9  +--ro rx-system-fw-echo?             boolean
10 +--ro preemption-supported?           boolean
11 +--ro preemption-enabled?            boolean
12 +--ro preemption-active?             boolean
13 +--ro additional-fragment-size?       int32
14
15 augment /lldp:lldp/lldp:port:
16   +--rw tlvs-port-config-enable?      bits
17   +--rw local-power-priority?          power-priority-type
18   +--rw local-measurement-source?      measurement-source-type
19
20 augment /lldp:lldp/lldp:port/lldp:remote-systems-data:
21   +--ro auto-negotiation-supported?    boolean
22   +--ro auto-negotiation-enabled?      boolean
23   o--ro auto-negotiation-cap?          binary
24   o--ro operational-mau-type?          int32
25   +--ro auto-negotiation-cap-bits?     ieee802-phy:auto-neg-ability-bits
26   +--ro operational-pmd-type?          identityref
27   +--ro power-port-class?              port-class-type
28   +--ro mdi-power-supported?           boolean
29   +--ro mdi-power-enabled?             boolean
30   +--ro power-pair-controlable?        boolean
31   +--ro power-pairs?                   pse-pinout-type
32   +--ro power-class?                   pse-power-class-type
33   o--ro link-aggregation-status?       bits
34   o--ro aggregation-port-id?           int32
35   o--ro local-max-frame-size?          int32
36   o--ro power-type?                    bits
37   +--ro power-source?                  power-source-type
38   +--ro power-priority?                power-priority-type
39   +--ro pd-requested-power-value?      int32
40   +--ro pd-requested-power-value-a?    int32
41   +--ro pd-requested-power-value-b?    int32

```

```

1  +---ro pse-allocated-power-value?      int32
2  +---ro pse-allocated-power-value-a?    int32
3  +---ro pse-allocated-power-value-b?    int32
4  +---ro pse-powering-status?            powering-status-type
5  +---ro pd-powered-status?              powered-status-type
6  +---ro power-pairs-ext?                power-pairs-type
7  +---ro power-class-ext-A?              power-class-ext-AB-type
8  +---ro power-class-ext-B?              power-class-ext-AB-type
9  +---ro power-class-ext?                power-class-ext-type
10 +---ro power-type-ext?                  power-type
11 +---ro pd-load?                         boolean
12 +---ro pd-4pid?                         boolean
13 +---ro pse-max-avail-power?             int32
14 +---ro pse-autoclass-support?           boolean
15 +---ro autoclass-completed?            boolean
16 +---ro autoclass-request?              boolean
17 +---ro power-down-request?             int32
18 +---ro power-down-time?                int32
19 +---ro meas-voltage-support?            boolean
20 +---ro meas-current-support?            boolean
21 +---ro meas-power-support?              boolean
22 +---ro meas-energy-support?              boolean
23 +---ro meas-energy-support?              boolean
24 o---ro measurement-source?              bits
25 +---ro rem-measurement-source?          measurement-source-type
26 +---ro meas-voltage-request?            boolean
27 +---ro meas-current-request?            boolean
28 +---ro meas-power-request?              boolean
29 +---ro meas-energy-request?              boolean
30 +---ro meas-voltage-valid?              boolean
31 +---ro meas-current-valid?              boolean
32 +---ro meas-power-valid?                boolean
33 +---ro meas-energy-valid?                boolean
34 +---ro meas-voltage-uncertainty?        int32
35 +---ro meas-current-uncertainty?        int32
36 +---ro meas-power-uncertainty?          int32
37 +---ro meas-energy-uncertainty?          int32
38 +---ro voltage-measurement?             int32
39 +---ro current-measurement?             int32
40 +---ro power-measurement?               int32

```

```

1  +---ro energy-measurement?          int32
2  +---ro pse-power-price-index?       int32
3  +---ro tx-system-value?             int32
4  +---ro tx-system-value-echo?        int32
5  +---ro rx-system-value?             int32
6  +---ro rx-system-value-echo?        int32
7  +---ro fallback-system-value?       int32
8  +---ro tx-system-fw?                boolean
9  +---ro tx-system-fw-echo?           boolean
10 +---ro rx-system-fw?                boolean
11 +---ro rx-system-fw-echo?           boolean
12 +---ro preemption-supported?        boolean
13 +---ro preemption-enabled?          boolean
14 +---ro preemption-active?           boolean
15 +---ro additional-fragment-size?    int32
16
17
18
19

```

59.3.1.5 ieee802-ethernet-phy

```

20 module: ieee802-ethernet-phy
21
22
23
24
25
26   augment /if:interfaces/if:interface:
27     +---rw phy-table
28     |   +---rw control
29     |   |   +---rw phy-reset?          boolean
30     |   |   +---rw phy-admin-control?  enumeration
31     |   |   +---rw fec-control! {fec-support}?
32     |   |   |   +---rw fec-mode-control?  enumeration
33     |   |   |   +---rw rs-fec-bypass-control?  enabled-state
34     |   +---ro status
35     |   |   +---ro phy-type?            identityref
36     |   |   +---ro pmd-type?            identityref
37     |   |   +---ro pmd-type-list?       ieee802-phy:pmd-type-list
38     |   |   +---ro phy-status?          enumeration
39     |   |   +---ro phy-media-available?  ieee802-phy:media-available-type
40     |   |   +---ro jabber-state?        enumeration
41     |   |   +---ro jabbering-state-enters?  yang:counter64
42     |   |   +---ro fec-status! {fec-support}?
43     |

```

```

1 | | | +--ro fec-ability? bits
2 | | | +--ro fec-mode-status? enumeration
3 | | | +--ro rs-fec-bypass-enabled-status? enabled-state
4 | | +--ro baset-snr! {baset-snr-support}?
5 | | | +--ro snr-op-margin-chnl-a? snr-value
6 | | | +--ro snr-op-margin-chnl-b? snr-value
7 | | | +--ro snr-op-margin-chnl-c? snr-value
8 | | | +--ro snr-op-margin-chnl-d? snr-value
9 | | +--ro time-sync! {time-sync-support}?
10 | | +--ro time-sync-ability? bits
11 | | +--ro time-sync-delay-tx-max? uint32
12 | | +--ro time-sync-delay-tx-min? uint32
13 | | +--ro time-sync-delay-rx-max? int32
14 | | +--ro time-sync-delay-rx-min? int32
15 | +--ro statistics
16 | | +--ro phy-media-losses? yang:counter64
17 | | +--ro jabbering-state-enters? yang:counter64
18 | | +--ro false-carriers? yang:counter64
19 | | +--ro pcs-coding-violations? yang:counter64
20 | | +--ro fast-retrain-count-local? counter-1000-second
21 | | +--ro fast-retrain-count-peer? counter-1000-second
22 | +--ro pcs-lane-statistics {pcs-lane-statistics-support}?
23 | | +--ro pcs-lane* [pcs-lane-index]
24 | | | +--ro pcs-lane-index uint32
25 | | | +--ro fec-corrected-blocks? counter-pcs-lane
26 | | | +--ro fec-uncorrected-blocks? counter-pcs-lane
27 +--rw auto-neg-table {auto-negotiation-support}?
28 | +--rw control
29 | | +--rw auto-neg-control? boolean
30 | | +--rw auto-neg-restart? boolean
31 | +--ro status
32 | | +--ro auto-neg-enabled? boolean
33 | | +--ro auto-neg-remote-signaling? boolean
34 | | +--ro auto-neg-status? enumeration
35 | | +--ro auto-neg-capability-bits? ieee802-phy:auto-neg-ability-bits
36 | | +--ro auto-neg-cap-advertised-bits? ieee802-phy:auto-neg-ability-bits
37 | | +--ro auto-neg-cap-received-bits? ieee802-phy:auto-neg-ability-bits

```

5.3.2 YANG modules

In the YANG module definition, should any discrepancy between the text of the description for individual YANG nodes in the YANG modules of this clause and the corresponding definition in 5.2 through 5.3 of this clause occur, the definitions in the YANG modules attached to this standard shall take precedence.

An ASCII text version of the Ethernet YANG module can be found at the following URL:^h

Editor's Note (to be removed prior to publication):

Yang files contained in <https://github.com/YangModels/yang/tree/main/standard/ieee/published/802.3> are IEEE 802.3.2-2019 version and will be updated at the publication time.

<https://github.com/YangModels/yang/tree/master/standard/ieee/published/802.3>

5.3.2.1 Ethernet interface module

Editor's Note (to be removed prior to publication):

Pretty printing of `ieee802-ethernet-interface.yang` file may change the appearance by adding whitespace and reformatting lines

Editor's Note (to be removed prior to publication):

IEEE Std 802.3.1 and IEEE Std 802.3.2 to be updated at the publication time

```
module ieee802-ethernet-interface {
  yang-version 1.1;
  namespace "urn:ieee:std:802.3:yang:ieee802-ethernet-interface";
  prefix ieee802-eth-if;

  import ietf-yang-types {
    prefix yang;
    reference
      "IETF RFC 6991";
  }
  import iana-if-type {
    prefix ianaift;
    reference
      "http://www.iana.org/assignments/yang-parameters/
      iana-if-type@2023-01-26.yang";
  }
  import ietf-interfaces {
    prefix if;
    reference
      "IETF RFC 8343";
  }
  import ieee802-ethernet-phy-type {
    prefix ieee802-phy;
    reference
      "IEEE Std 802.3-2022";
  }
}
```

^hCopyright release for YANG modules: Users of this standard may freely reproduce the YANG module contained in this subclause so that it can be used for its intended purpose.

```
1 organization
2   "IEEE Std 802.3 Ethernet Working Group
3   Web URL: http://www.ieee802.org/3/";
4 contact
5   "Web URL: http://www.ieee802.org/3/";
6 description
7   "This module contains YANG definitions for configuring IEEE Std
8   802.3 Ethernet Interfaces.
9   In this YANG module, 'Ethernet interface' can be interpreted
10  as referring to 'IEEE Std 802.3 compliant Ethernet
11  interfaces'.";
12
13
14 revision 2025-04-17 {
15   description
16     "Updates under IEEE Std 802.3.2-202x, Draft D3.3";
17   reference
18     "IEEE Std 802.3-2022 and IEEE Std 802.3.1-202X, unless dated
19     explicitly";
20 }
21
22
23 feature ethernet-pfc {
24   description
25     "This device supports Ethernet priority flow-control.";
26 }
27
28
29 feature ethernet-pause {
30   description
31     "This device supports Ethernet PAUSE.";
32 }
33
34 typedef eth-if-speed-type {
35   type decimal64 {
36     fraction-digits 3;
37   }
38   units "Gb/s";
39   status obsolete;
40   description
41     "Used to represent the configured, negotiated, or actual
42     speed of an Ethernet interface in Gigabits per second
43     (Gb/s), accurate to 3 decimal places (i.e., accurate to 1
44     Mb/s).";
45 }
46
47
48 typedef duplex-type {
49   type enumeration {
50     enum full {
51       description
52         "Full duplex.";
53     }
54     enum half {
55       description
56         "Half duplex.";
57     }
58     enum unknown {
59       description
60         "Link is currently disconnected or initializing.";
61     }
62   }
63 }
64
65 default "full";
```

```
1      description
2          "Used to represent the configured, negotiated, or actual
3          duplex mode of an Ethernet interface.";
4      reference
5          "IEEE Std 802.3, 30.3.1.1.32, aDuplexStatus";
6  }
7
8
9  typedef pause-fc-direction-type {
10     type enumeration {
11         enum disabled {
12             description
13                 "Flow-control disabled in both ingress and egress
14                 directions.";
15         }
16         enum ingress-only {
17             description
18                 "PAUSE frame based flow control is enabled in the ingress
19                 direction only.";
20         }
21         enum egress-only {
22             description
23                 "PAUSE frame based flow control is enabled in the egress
24                 direction only.";
25         }
26         enum bi-directional {
27             description
28                 "PAUSE frame based flow control is enabled in both
29                 ingress and egress directions.";
30         }
31         enum undefined {
32             description
33                 "Link is currently disconnected or initializing.";
34         }
35     }
36 }
37
38 description
39     "Used to represent the configured, negotiated, or actual
40     PAUSE frame-based flow control setting.";
41 reference
42     "IEEE Std 802.3.1, dot3PauseAdminMode and dot3PauseOperMode";
43
44 }
45
46 augment "/if:interfaces/if:interface" {
47     when "derived-from-or-self(if:type, 'ianaift:ethernetCsmacd')" {
48         description
49             "Applies to all Ethernet interfaces.";
50     }
51 }
52 description
53     "Augment interface model with Ethernet interface
54     specific configuration nodes.";
55 container ethernet {
56     description
57         "Contains all Ethernet interface related configuration.";
58     container auto-negotiation {
59         presence "The presence of this container indicates that
60             auto-negotiation is supported on this Ethernet
61             interface.";
62         description
63             "Contains auto-negotiation transmission parameters
64
65
```

```
1         This container contains a data node that allows the
2         advertised duplex value in the negotiation to be
3         restricted.
4
5         If not specified then the default behavior for the
6         duplex data node is to negotiate all available values for
7         the particular type of Ethernet PHY associated with the
8         interface.";
9
10        reference
11        "IEEE Std 802.3, Clause 28 and Annexes 28A-D";
12    leaf enable {
13        type boolean;
14        default "true";
15        description
16        "Controls whether auto-negotiation is enabled or
17        disabled.
18        For interface types that support auto-negotiation then
19        it defaults to being enabled.
20
21        For interface types that do not support
22        auto-negotiation, the related configuration data is
23        ignored.";
24    }
25    leaf negotiation-status {
26        when "../enable = 'true'";
27        type enumeration {
28            enum in-progress {
29                description
30                "The auto-negotiation protocol is running and
31                negotiation is currently in-progress.";
32            }
33            enum complete {
34                description
35                "The auto-negotiation protocol has completed
36                successfully.";
37            }
38            enum failed {
39                description
40                "The auto-negotiation protocol has failed.";
41            }
42            enum unknown {
43                description
44                "The auto-negotiation status is not currently known,
45                this could be because it is still negotiating or the
46                protocol cannot run
47                (e.g., if no medium is present).";
48            }
49            enum no-negotiation {
50                description
51                "No auto-negotiation is executed.
52                The auto-negotiation function is either not
53                supported on this interface or has not been
54                enabled.";
55            }
56        }
57    }
58    config false;
59    description
60    "The status of the auto-negotiation protocol.";
61    reference
```

```
1         "IEEE 802.3, 30.6.1.1.4, aAutoNegAutoConfig";
2     }
3 }
4 leaf phy-type {
5     type identityref {
6         base ieee802-phy:phy-type;
7     }
8     config false;
9     description
10        "A value that uniquely identifies the IEEE 802.3 PHY type
11        of the interface.";
12    reference
13        "IEEE Std 802.3, 30.3.2.1.2 aPhyType";
14 }
15 leaf pmd-type {
16     type identityref {
17         base ieee802-phy:pmd-type;
18     }
19     config false;
20     description
21        "A value that uniquely identifies the IEEE 802.3 PMD type
22        of the interface.";
23    reference
24        "IEEE Std 802.3, 30.5.1.1.2 aMAUType";
25 }
26 leaf duplex {
27     type duplex-type;
28     description
29        "Operational duplex mode of the Ethernet interface.";
30    reference
31        "IEEE Std 802.3, 30.3.1.1.32 aDuplexStatus";
32 }
33 leaf speed {
34     type eth-if-speed-type;
35     units "Gb/s";
36     status obsolete;
37     description
38        "Operational speed (data rate) of the Ethernet interface.
39        The default value is implementation-dependent.
40        Supersceded by speed in /if:interfaces/if:interface";
41 }
42 /* deprecated flow-control container */
43 container flow-control {
44     status deprecated;
45     description
46        "Holds the different types of Ethernet PAUSE frame based
47        flow control that can be enabled.
48        Supersceded by new container - pause.";
49     container pause {
50         if-feature "ethernet-pause";
51         description
52            "IEEE Std 802.3 PAUSE frame based PAUSE frame based
53            flow control.";
54         reference
55            "IEEE Std 802.3, Annex 31B";
56         leaf direction {
57             type pause-fc-direction-type;
58             description
59                "Indicates which direction PAUSE frame based flow
```

```
1         control is enabled in, or whether it is disabled.
2         The default flow-control settings are vendor
3         specific. If auto-negotiation is enabled, then PAUSE
4         based flow-control is negotiated by default.
5         The default value is implementation-dependent.";
6     }
7 container statistics {
8     config false;
9     description
10        "Contains the number of PAUSE frames received or
11        transmitted.
12
13        Discontinuities in the values of counters in
14        this container can occur at re-initialization of
15        the management system, and at other times as
16        indicated by the value of the 'discontinuity-time'
17        leaf defined in the ietf-interfaces YANG module
18        (IETF RFC 8343).";
19    leaf in-frames-pause {
20        type yang:counter64;
21        units "frames";
22        description
23            "A count of PAUSE MAC Control frames transmitted on
24            this Ethernet interface.";
25        reference
26            "IEEE Std 802.3, 30.3.4.3
27            aPAUSEMACCtrlFramesReceived";
28    }
29    leaf out-frames-pause {
30        type yang:counter64;
31        units "frames";
32        description
33            "A count of PAUSE MAC Control frames transmitted on
34            this Ethernet interface.";
35        reference
36            "IEEE Std 802.3, 30.3.4.2
37            aPAUSEMACCtrlFramesTransmitted";
38    }
39 }
40 }
41 }
42 container pfc {
43     if-feature "ethernet-pfc";
44     description
45         "IEEE Std 802.3 Priority-based flow control.";
46     reference
47         "IEEE Std 802.3, Annex 31D";
48     leaf enable {
49         type boolean;
50         description
51             "True indicates that IEEE Std 802.3 priority-based
52             flow control is enabled, false indicates that
53             IEEE Std 802.3 priority-based flow control is
54             disabled. For interfaces that have auto-negotiation,
55             the priority-based flow control is enabled by
56             default.";
57     }
58 }
59 container statistics {
60     config false;
61     status deprecated;
```

```
1      description
2          "This container collects all statistics for
3          Ethernet interfaces.
4
5          Discontinuities in the values of counters in
6          this container can occur at re-initialization of
7          the management system, and at other times as
8          indicated by the value of the 'discontinuity-time'
9          leaf defined in the ietf-interfaces YANG module
10         (IETF RFC 8343).";
11     leaf in-frames-pfc {
12         type yang:counter64;
13         units "frames";
14         description
15             "Deprecated in-frames-pfc as not defined in base
16             standard. A count of PFC MAC Control frames
17             received on this Ethernet interface.";
18         reference
19             "IEEE Std 802.3.1, dot3HCInPFCFrames";
20     }
21     leaf out-frames-pfc {
22         type yang:counter64;
23         units "frames";
24         description
25             "Deprecated out-frames-pfc as not defined in base
26             standard. A count of PFC MAC Control frames
27             transmitted on this interface.";
28         reference
29             "IEEE Std 802.3.1, dot3HCInPFCFrames";
30     }
31 }
32
33 leaf force-flow-control {
34     type boolean;
35     default "false";
36     description
37         "Explicitly forces the local PAUSE frame based flow
38         control settings regardless of what has been
39         negotiated.
40
41         Since the auto-negotiation of flow-control settings
42         does not allow all same combinations to be negotiated
43         (e.g., consider a device that is only capable of
44         sending PAUSE frames connected to a peer device that
45         is only capable of receiving and acting on PAUSE
46         frames) and failing to agree on the flow-control
47         settings does not cause the auto-negotiation to fail
48         completely, then it is sometimes useful to be able to
49         explicitly enable particular PAUSE frame based flow
50         control settings on the local device regardless of
51         what is being advertised or negotiated.";
52     reference
53         "IEEE Std 802.3, Table 28B-3";
54 }
55
56 leaf max-frame-length {
57     type uint16;
58     units "octets";
59     config false;
```

```
1      description
2          "This indicates the MAC frame length (including FCS bytes)
3          at which frames are dropped for being too long.";
4      reference
5          "IEEE Std 802.3, 30.3.1.1.37 aMaxFrameLength";
6  }
7  leaf mac-control-extension-control {
8      type boolean;
9      config false;
10     description
11         "A value that identifies the current EXTENSION
12         MAC Control function, as specified in
13         IEEE Std 802.3, Annex 31C.";
14     reference
15         "IEEE Std 802.3, 30.3.8.3 aEXTENSIONMACCtrlStatus
16         IEEE Std 802.3.1, dot3ExtensionMacCtrlStatus ";
17 }
18 leaf frame-limit-slow-protocol {
19     type uint64;
20     units "f/s";
21     default "10";
22     config false;
23     description
24         "The maximum number of Slow Protocol frames of a given
25         subtype that can be transmitted in a one second
26         interval.
27         The default value is 10.";
28     reference
29         "IEEE Std 802.3, 30.3.1.1.38 aSlowProtocolFrameLimit";
30 }
31 container capabilities {
32     config false;
33     description
34         "Container all Ethernet interface specific capabilities.";
35     leaf auto-negotiation {
36         type boolean;
37         description
38             "Indicates whether auto-negotiation may be configured on
39             this interface.";
40     }
41 }
42 container ethernet-pause {
43     if-feature "ethernet-pause";
44     description
45         "IEEE Std 802.3 PAUSE flow control.
46
47         Discontinuities in the values of counters in
48         this container can occur at re-initialization of
49         the management system, and at other times as
50         indicated by the value of the 'discontinuity-time'
51         leaf defined in the ietf-interfaces YANG module
52         (IETF RFC 8343).";
53     reference
54         "IEEE Std 802.3, 30.3.4 PAUSE entity managed object class and
55         Clause 31, Annex 31B";
56     container control-and-status {
57         description
58             "PAUSE function control and status objects.";
59         leaf pause-admin-control {
```

```

1         type pause-fc-direction-type;
2         default "disabled";
3         description
4             "What PAUSE functionality will run on this interface when
5             Auto-Negotiation is not implemented or disabled.";
6         reference
7             "IEEE Std 802.3, ????" ;
8     }
9
10    leaf pause-oper-status {
11        type pause-fc-direction-type;
12        config false;
13        description
14            "What PAUSE functionality is running on this
15            interface, as a result of Auto-Negotiation or
16            setting pause-admin-control.";
17        reference
18            "IEEE Std 802.3, ????" ;
19    }
20
21    leaf link-delay-allowance {
22        type uint32;
23        description
24            "The value in bit-times of the allowance made by the
25            MAC Control PAUSE entity for round-trip propagation
26            delay.
27            The default value is implementation-dependent.";
28    }
29
30    leaf pfc-enable-status {
31        type boolean;
32        config false;
33        description
34            "Is IEEE 802.1 Priority-based Flow Control (PFC)
35            enabled on the interface. If PFC is enabled, then
36            802.3 PAUSE flow control is disabled.";
37        reference
38            "IEEE Std 802.3, 30.3.3.6 aPFCEnableStatus";
39    }
40
41    }
42    container statistics {
43        config false;
44        description
45            "PAUSE frame counters.";
46        leaf in-frames-pause {
47            type yang:counter64;
48            units "frames";
49            description
50                "A count of MAC Control PAUSE frames transmitted on
51                this Ethernet interface.";
52            reference
53                "IEEE Std 802.3, 30.3.4.3
54                aPAUSEMACCtrlFramesReceived";
55        }
56
57        leaf out-frames-pause {
58            type yang:counter64;
59            units "frames";
60            description
61                "A count of MAC Control PAUSE frames transmitted on
62                this Ethernet interface.";
63            reference
64                "IEEE Std 802.3, 30.3.4.2

```

```

1          aPAUSEMACCtrlFramesTransmitted";
2      }
3  }
4  }
5  container statistics {
6      config false;
7      description
8          "Contains statistics specific to Ethernet interfaces.
9
10         Discontinuities in the values of counters in the
11         container can occur at re-initialization of the
12         management system, and at other times as indicated by
13         the value of the 'discontinuity-time' leaf defined in
14         the ietf-interfaces YANG module (IETF RFC 8343).";
15     container frame {
16         description
17             "Contains frame statistics specific to Ethernet
18             interfaces.
19
20             All octet frame lengths include the 4 byte FCS.
21
22             Error counters are only reported once. The count
23             represented by an instance of this object is
24             incremented when the frameCheckError status is
25             returned by the MAC service to the MAC Client.
26             Received frames for which multiple error conditions
27             pertain are, according to the conventions of
28             IEEE Std 802.3 Layer Management, counted exclusively
29             according to the error status presented to the MAC
30             Client.
31
32             A frame that is counted by an instance of this object
33             is also counted by the corresponding instance of
34             'in-errors' leaf defined in the ietf-interfaces YANG
35             module (IETF RFC 8343).";
36     leaf in-total-frames {
37         type yang:counter64;
38         units "frames";
39         description
40             "The total number of frames (including bad frames)
41             received on the Ethernet interface.
42
43             This counter is calculated by summing the following
44             IEEE Std 802.3, Clause 30 counters:
45             aFramesReceivedOK +
46             aFrameCheckSequenceErrors +
47             aAlignmentErrors +
48             aFrameTooLongErrors +
49             aFramesLostDueToIntMACRcvError";
50         reference
51             "IEEE Std 802.3, Clause 30 counters, as specified
52             in the description above.";
53     }
54     leaf in-total-octets {
55         type yang:counter64;
56         units "octets";
57         description
58             "The total number of octets of data (including those
59             in bad frames) received on the Ethernet interface.

```

```
1
2         Includes the 4-octet FCS.";
3     reference
4         "IETF RFC 2819, etherStatsOctets";
5 }
6 leaf in-frames {
7     type yang:counter64;
8     units "frames";
9     description
10        "A count of frames (including unicast, multicast and
11         broadcast) that have been successfully received on
12         the Ethernet interface.
13
14         This count does not include frames received with
15         frame-too-long, FCS, length or alignment errors, or
16         frames lost due to internal MAC sublayer error.";
17     reference
18         "IEEE Std 802.3, 30.3.1.1.5 aFramesReceivedOK";
19 }
20 leaf in-multicast-frames {
21     type yang:counter64;
22     units "frames";
23     description
24        "A count of multicast frames that have been
25         successfully received on the Ethernet interface.
26
27         This counter represents a subset of the frames
28         counted by in-frames.
29
30         This count does not include frames received with
31         frame-too-long, FCS, length or alignment errors, or
32         frames lost due to internal MAC sublayer error.";
33     reference
34         "IEEE Std 802.3, 30.3.1.1.21
35         aMulticastFramesReceivedOK";
36 }
37 leaf in-broadcast-frames {
38     type yang:counter64;
39     units "frames";
40     description
41        "A count of broadcast frames that have been
42         successfully received on the Ethernet interface.
43
44         This counter represents a subset of the frames
45         counted by in-frames.
46
47         This count does not include frames received with
48         frame-too-long, FCS, length or alignment errors, or
49         frames lost due to internal MAC sublayer error.";
50     reference
51         "IEEE Std 802.3, 30.3.1.1.22
52         aBroadcastFramesReceivedOK";
53 }
54 leaf in-error-fcs-frames {
55     type yang:counter64;
56     units "frames";
57     description
58        "A count of receive frames that are of valid length,
59         but do not pass the FCS check, regardless of whether
```

```
1           or not the frames are an integral number of octets
2           in length.
3
4           This counter is calculated by summing the following
5           counters:
6               aFrameCheckSequenceErrors +
7               aAlignmentErrors";
8
9           reference
10              "IEEE Std 802.3, 30.3.1.1.6 aFrameCheckSequenceErrors;
11              IEEE Std 802.3, 30.3.1.1.7 aAlignmentErrors";
12      }
13      leaf in-error-undersize-frames {
14          type yang:counter64;
15          units "frames";
16          status deprecated;
17          description
18              "Deprecated in-error-undersize-frames as not defined
19              in base standard. A count of frames received on a
20              particular Ethernet interface that are less than
21              64 bytes in length, and are discarded.
22
23              This counter is incremented regardless of whether
24              the frame passes the FCS check.";
25          reference
26              "IETF RFC 2819, etherStatsUndersizePkts and
27              etherStatsFragments";
28      }
29
30      leaf in-error-oversize-frames {
31          type yang:counter64;
32          units "frames";
33          description
34              "A count of frames received on a particular Ethernet
35              interface that exceed the maximum permitted frame
36              size, that is specified in max-frame-length, and are
37              discarded.
38
39              This counter is incremented regardless of whether
40              the frame passes the FCS check.";
41          reference
42              "IEEE Std 802.3, 30.3.1.1.25
43              aFrameTooLongErrors";
44      }
45
46      leaf in-error-mac-internal-frames {
47          type yang:counter64;
48          units "frames";
49          description
50              "A count of frames for which reception on a
51              particular Ethernet interface fails due to an
52              internal MAC sublayer receive error.
53
54              A frame is only counted by an instance of this
55              object if it is not counted by the corresponding
56              instance of either the in-error-fcs-frames,
57              in-error-undersize-frames, or
58              in-error-oversize-frames. The precise meaning of
59              the count represented by an instance of this object
60              is implementation-specific.
61
62              In particular, an instance of this object may
```

```

1         represent a count of receive errors on a particular
2         Ethernet interface that are not otherwise counted.";
3     reference
4         "IEEE Std 802.3, 30.3.1.1.15
5         aFramesLostDueToIntMACRcvError";
6     }
7     leaf out-frames {
8         type yang:counter64;
9         units "frames";
10        description
11            "A count of frames (including unicast, multicast and
12            broadcast) that have been successfully transmitted
13            on the Ethernet interface.";
14        reference
15            "IEEE Std 802.3, 30.3.1.1.2 aFramesTransmittedOK";
16    }
17    leaf out-multicast-frames {
18        type yang:counter64;
19        units "frames";
20        description
21            "A count of multicast frames that have been
22            successfully transmitted on the Ethernet interface.
23
24            This counter represents a subset of the frames
25            counted by out-frames.";
26        reference
27            "IEEE Std 802.3, 30.3.1.1.18
28            aMulticastFramesXmittedOK";
29    }
30    leaf out-broadcast-frames {
31        type yang:counter64;
32        units "frames";
33        description
34            "A count of broadcast frames that have been
35            successfully transmitted on the Ethernet interface.
36
37            This counter represents a subset of the frames
38            counted by out-frames.";
39        reference
40            "IEEE Std 802.3, 30.3.1.1.19
41            aBroadcastFramesXmittedOK";
42    }
43    leaf out-error-mac-internal-frames {
44        type yang:counter64;
45        units "frames";
46        description
47            "A count of frames for which transmission on a
48            particular Ethernet interface fails due to an
49            internal MAC sublayer transmit error.
50
51            The precise meaning of the count represented by an
52            instance of this object is implementation-specific.
53            In particular, an instance of this object may
54            represent a count of transmission errors on a
55            particular Ethernet interface that are not otherwise
56            counted.";
57        reference
58            "IEEE Std 802.3, 30.3.1.1.12
59            aFramesLostDueToIntMACXmitError";
60    }

```

```
1      }
2    }
3    container phy {
4      description
5        "Ethernet statistics related to the PHY layer.";
6      leaf in-error-symbol {
7        type yang:counter64;
8        units "errors";
9        description
10         "A count of the number of symbol errors that have
11          occurred.
12
13          For the precise definition of when the symbol error
14          counter is incremented, please see the 'description'
15          text associated with aSymbolErrorDuringCarrier,
16          specified in IEEE Std 802.3, 30.3.2.1.5.";
17      reference
18        "IEEE Std 802.3, 30.3.2.1.5
19        aSymbolErrorDuringCarrier";
20    }
21    container lpi {
22      description
23        "Physical Ethernet statistics for the energy
24         efficiency related low power idle indications.";
25      leaf in-lpi-transitions {
26        type yang:counter64;
27        units "transitions";
28        description
29          "The number of times the link partner transitioned to
30           Low Power Idle.
31           This counter has a maximum per second increment rate
32           of:
33             50 thousand for 100 Mb/s;
34             90 thousand for 1000 Mb/s
35             230 thousand for 10 Gb/s.";
36      reference
37        "IEEE Std 802.3, 30.3.2.1.11
38        aReceiveLPITransitions";
39    }
40    leaf in-lpi-time {
41      type decimal64 {
42        fraction-digits 6;
43      }
44      units "seconds";
45      description
46        "The amount of time the link partner has
47         been in Low Power Idle.";
48      reference
49        "IEEE Std 802.3, 30.3.2.1.9
50        aReceiveLPIMicroseconds";
51    }
52    leaf out-lpi-transitions {
53      type yang:counter64;
54      units "transitions";
55      description
56        "The number of times this port transitioned to
57         Low Power Idle.
58         This counter has a maximum per second increment rate
59         of:
60
```

```

1           50 thousand for 100 Mb/s;
2           90 thousand for 1000 Mb/s
3           230 thousand for 10 Gb/s.";
4       reference
5           "IEEE Std 802.3, 30.3.2.1.10
6           aTransmitLPITransitions";
7       }
8   leaf out-lpi-time {
9       type decimal64 {
10          fraction-digits 6;
11      }
12      units "seconds";
13      description
14          "The amount of time in this port has
15          been in Low Power Idle.";
16      reference
17          "IEEE Std 802.3, 30.3.2.1.8
18          aTransmitLPIMicroseconds";
19  }
20  }
21  }
22  }
23  }
24  container mac-control {
25      description
26          "A group of statistics specific to MAC Control
27          operation of selected Ethernet interfaces.";
28      reference
29          "IEEE Std 802.3.1, dot3ExtensionTable";
30      leaf in-frames-mac-control-unknown {
31          type yang:counter64;
32          units "frames";
33          description
34              "A count of MAC Control frames with an unsupported
35              opcode received on this Ethernet interface.
36
37              Frames counted against this counter are also counted
38              against in-discards defined in the ietf-interfaces
39              YANG module (IETF RFC 8343).";
40          reference
41              "IEEE Std 802.3, 30.3.3.5
42              aUnsupportedOpCodesReceived";
43      }
44      leaf in-frames-mac-control-extension {
45          type yang:counter64;
46          units "frames";
47          description
48              "The count of Extension MAC Control frames received
49              on this Ethernet interface.";
50          reference
51              "IEEE Std 802.3, 30.3.8.2
52              aEXTENSIONMACCtrlFramesReceived";
53      }
54      leaf out-frames-mac-control-extension {
55          type yang:counter64;
56          units "frames";
57          description
58              "The count of Extension MAC Control frames
59              transmitted on this Ethernet interface.";
60          reference
61              "IEEE Std 802.3, 30.3.8.1

```

```
1         aEXTENSIONMACCtrlFramesTransmitted";
2     }
3 }
4 }
5 }
6 }
7 }
8 }
9 }
```

5.3.2.2 Ethernet interface module (half-duplex)

Editor's Note (to be removed prior to publication):

Pretty printing of ieee802-ethernet-interface-half-duplex.yang file may change the appearance by adding whitespace and reformatting lines

Editor's Note (to be removed prior to publication):

IEEE Std 802.3.1 and IEEE Std 802.3.2 to be updated at the publication time

```
24 module ieee802-ethernet-interface-half-duplex {
25     yang-version 1.1;
26     namespace
27         "urn:ieee:std:802.3:yang:ieee802-ethernet-interface-half-duplex";
28     prefix ieee802-eth-half-duplex;
29
30     import ietf-yang-types {
31         prefix yang;
32         reference
33             "IETF RFC 6991";
34     }
35     import ietf-interfaces {
36         prefix if;
37         reference
38             "IETF RFC 8343";
39     }
40     import iana-if-type {
41         prefix ianaift;
42         reference
43             "http://www.iana.org/assignments/yang-parameters/
44             iana-if-type@2023-01-26.yang";
45     }
46     import ieee802-ethernet-interface {
47         prefix ieee802-eth-if;
48     }
49
50     organization
51         "IEEE Std 802.3 Ethernet Working Group
52         Web URL: http://www.ieee802.org/3/";
53     contact
54         "Web URL: http://www.ieee802.org/3/";
55     description
56         "This module contains YANG definitions for half duplex
57         Ethernet interfaces, e.g., 10BASE-T1S.";
58
59     revision 2025-04-17 {
60         description
61             "Updates under IEEE Std 802.3.2-202x, Draft D3.3";
62     }
```

```
1     reference
2         "IEEE Std 802.3-2022, unless dated explicitly";
3     }
4
5     feature dynamic-rate-control {
6         status obsolete;
7         description
8             "This feature indicates that the device supports Ethernet
9             interfaces lowering the average data rate of the MAC
10             sublayer, with frame granularity, by using Rate Control to
11             dynamically increase the inter-packet gap for some types of
12             Ethernet interface.
13             Only valid for Ethernet interfaces operating at speeds
14             (data rates) above 1000 Mb/s.";
15         reference
16             "IEEE Std 802.3, 30.3.1.1.33 aRateControlAbility";
17     }
18
19
20     feature csma-cd {
21         description
22             "This feature indicates that the device supports Ethernet
23             interfaces running at half-duplex using CSMA/CD.";
24     }
25
26
27     typedef dynamic-rate-control-type {
28         type enumeration {
29             enum disabled {
30                 description
31                     "Dynamic rate control is disabled";
32             }
33             enum sonet-oc192 {
34                 value 2;
35                 description
36                     "Dynamic rate control is enabled for a 10 Gb/s Ethernet
37                     interface to SONET/SDH OC192/STM64.";
38             }
39         }
40     }
41
42     default "disabled";
43     status obsolete;
44     description
45         "Allowed values for dynamic-rate-control.";
46     reference
47         "IEEE Std 802.3, 4.4.2 ipgStretchRatio and 30.3.1.1.34
48         aRateControlStatus";
49 }
50
51
52 augment "/if:interfaces/if:interface/ieee802-eth-if:ethernet" {
53     when "derived-from-or-self(..if:type, 'ianaift:ethernetCsmacd')
54         and ieee802-eth-if:duplex = 'half'" {
55         description
56             "Applies to half-duplex Ethernet interfaces.";
57     }
58     status obsolete;
59     description
60         "Augment with Ethernet interface configuration parameters
61         for half-duplex operation.";
62     leaf dynamic-rate-control {
63         if-feature "dynamic-rate-control";
64         type dynamic-rate-control-type;
65     }
```

```
1      status obsolete;
2      description
3          "Enables dynamic rate control and specifies what speed
4            (data rate) the dynamic rate control is operating at.
5            The value of this attribute is constrained by the MAC
6            data rate and hardware support.
7            The default value is implementation-dependent.";
8      reference
9          "IEEE Std 802.3, 30.3.1.1.34 aRateControlStatus";
10  }
11  }
12  }
13
14  augment "/if:interfaces/if:interface/ieee802-eth-if:ethernet/"
15      + "ieee802-eth-if:capabilities" {
16      when "derived-from-or-self(..../if:type,
17          'ianaift:ethernetCsmacd')
18          and ../ieee802-eth-if:duplex = 'half'" {
19          description
20              "Applies to half-duplex Ethernet interfaces";
21      }
22      status obsolete;
23      description
24          "Augment with configuration capabilities for half-duplex
25            Ethernet interface.";
26      leaf dynamic-rate-control-supported {
27          if-feature "dynamic-rate-control";
28          type boolean;
29          default "false";
30          status obsolete;
31          description
32              "Indicates whether the Ethernet interface supports lowering
33                the average data rate of the MAC sublayer, with frame
34                granularity, by using Rate Control to dynamically increase
35                the inter-packet gap.
36                Only valid for Ethernet interfaces operating at speeds
37                (data rates) above 1000 Mb/s.";
38          reference
39              "IEEE Std 802.3, 30.3.1.1.33 aRateControlAbility";
40      }
41  }
42  }
43  }
44  }
45
46  augment "/if:interfaces/if:interface/ieee802-eth-if:ethernet/"
47      + "ieee802-eth-if:statistics/ieee802-eth-if:frame" {
48      when "derived-from-or-self(..../if:type,
49          'ianaift:ethernetCsmacd') and
50          ../ieee802-eth-if:duplex = 'half'" {
51          description
52              "Applies to half-duplex Ethernet interfaces.";
53      }
54      description
55          "Augment with statistics for half-duplex Ethernet interface.";
56      container csma-cd {
57          if-feature "csma-cd";
58          description
59              "Holds counters that are specific to CDMA/CD half-duplex
60                operation of Ethernet interfaces.
61                This counter does not increment on Ethernet interfaces
62                operating at speeds (data rates) greater than 10 Mb/s, or
63                on Ethernet interfaces operating in full-duplex mode.
64                on Ethernet interfaces operating in full-duplex mode.
65                on Ethernet interfaces operating in full-duplex mode.
```

```
1      Discontinuities in the value of this counter can occur at
2      re-initialization of the management system, and at other
3      times as indicated by the value of the
4      'discontinuity-time' leaf defined in the ietf-interfaces
5      YANG module (IETF RFC 8343).";
6
7      leaf in-errors-sqe-test {
8          type yang:counter64;
9          units "errors";
10         description
11             "A count of times that the SQE TEST ERROR is received on
12             a particular interface. The SQE TEST ERROR is set in
13             accordance with the rules for verification of the SQE
14             detection mechanism in the PLS Carrier Sense Function as
15             described in IEEE Std 802.3, 7.2.4.6.";
16         reference
17             "IEEE Std 802.3, 7.2.4.6, and 30.3.2.1.4 aSQETestErrors";
18     }
19
20     leaf out-frames-collision-single {
21         type yang:counter64;
22         units "frames";
23         description
24             "A count of frames that are involved in a single
25             collision, and are subsequently transmitted
26             successfully. A frame that is counted by an instance of
27             this object is also counted by the corresponding
28             instance of either 'out-unicast-frames',
29             'out-broadcast-frames', or 'out-multicast-frames',
30             and is not counted by the corresponding instance of the
31             'out-frames-collision-multiple'.";
32         reference
33             "IEEE Std 802.3, 30.3.1.1.3 aSingleCollisionFrames";
34     }
35
36     leaf out-frames-collision-multiple {
37         type yang:counter64;
38         units "frames";
39         description
40             "A count of frames that are involved in multiple
41             collisions, and are subsequently transmitted
42             successfully. A frame that is counted by an instance of
43             this object is also counted by the corresponding
44             instance of either 'out-unicast-frames',
45             'out-broadcast-frames', or 'out-multicast-frames', and
46             is not counted by the corresponding instance of the
47             'out-frames-collision-single'.";
48         reference
49             "IEEE Std 802.3, 30.3.1.1.4 aMultipleCollisionFrames";
50     }
51
52
53     leaf out-frames-deferred {
54         type yang:counter64;
55         units "frames";
56         description
57             "A count of frames for which the first transmission
58             attempt on a particular Ethernet interface is delayed
59             because the medium is busy.
60             A deferred frame that is not subject to any number of
61             collisions is not counted by an instance of
62             'out-frames-collision-single' or
63             'out-frames-collision-multiple' objects.";
64         reference
```

```
1         "IEEE Std 802.3, 30.3.1.1.9
2         aFramesWithDeferredXmissions";
3     }
4     leaf out-frames-collisions-excessive {
5         type yang:counter64;
6         units "frames";
7         description
8             "A count of frames for which transmission on a particular
9             Ethernet interface fails due to excessive collisions.";
10        reference
11            "IEEE Std 802.3, 30.3.1.1.11 aFramesAbortedDueToXSColls";
12    }
13    leaf out-collisions-late {
14        type yang:counter64;
15        units "collisions";
16        description
17            "The number of times that a collision is detected on a
18            particular Ethernet interface later than one slotTime
19            into the transmission of a packet.
20            A (late) collision included in a count represented by an
21            instance of this object is also considered as a
22            (generic) collision for purposes of other
23            collision-related statistics.";
24        reference
25            "IEEE Std 802.3, 30.3.1.1.10 aLateCollisions";
26    }
27    leaf out-errors-carrier-sense {
28        type yang:counter64;
29        units "errors";
30        description
31            "The number of times that the carrier sense condition was
32            lost or never asserted when attempting to transmit a
33            frame on a particular Ethernet interface.
34            The count represented by an instance of this object is
35            incremented at most once per transmission attempt, even
36            if the carrier sense condition fluctuates during a
37            transmission attempt.";
38        reference
39            "IEEE Std 802.3, 30.3.1.1.13 aCarrierSenseErrors";
40    }
41    list collision-histogram {
42        key "collision-count";
43        description
44            "A collection of collision histograms for a particular
45            interface.";
46        reference
47            "IEEE Std 802.3, 30.3.1.1.30 aCollisionFrames";
48        leaf collision-count {
49            type yang:counter64;
50            units "collisions";
51            description
52                "The number of per-frame media collisions for which a
53                particular collision histogram cell represents the
54                frequency on a particular interface.";
55        }
56        leaf collision-count-frames {
57            type yang:counter64;
58            units "frames";
59            description
```

```
1      "A count of individual MAC frames for which the
2      transmission (successful or otherwise) on a particular
3      interface occurs after the frame has experienced
4      exactly the number of collisions in the associated
5      dot3CollCount object.
6      For example, a frame which is transmitted on an
7      interface after experiencing exactly 4 collisions
8      would be indicated by incrementing only
9      collision-count-frames object associated with the
10     collision-count value of 4. No other instance of
11     collision-count-frames would be incremented in this
12     example.";
13
14     }
15   }
16 }
17 }
18 }
19 }
```

5.3.2.3 Ethernet MAC merge module

Editor's Note (to be removed prior to publication):

Pretty printing of ieee802-mac-merge.yang file may change the appearance by adding whitespace and reformatting lines

Editor's Note (to be removed prior to publication):

IEEE Std 802.3.1 and IEEE Std 802.3.2 to be updated at the publication time

```
37 module ieee802-ethernet-mac-merge {
38   yang-version 1.1;
39   namespace "urn:ieee:std:802.3:yang:ieee802-ethernet-mac-merge";
40   prefix mac-merge;
41
42   import ietf-yang-types {
43     prefix yang;
44     reference
45       "IETF RFC 6991";
46   }
47   import ietf-interfaces {
48     prefix if;
49     reference
50       "IETF RFC 8343";
51   }
52   import ieee802-ethernet-interface {
53     prefix ieee802-eth-if;
54     reference
55       "IEEE Std 802.3.2-2019";
56   }
57
58   organization
59     "IEEE Std 802.3 Ethernet Working Group
60     Web URL: http://www.ieee802.org/3/";
61   contact
62     "Web URL: http://www.ieee802.org/3/";
63 }
```

```
1  description
2    "The Yang model for managing devices that support the MAC Merge
3    (aka preemption) sublayer as defined in Clause 99. Unless
4    otherwise indicated, the references in this model module are to
5    IEEE Std 802.3-2022.";
6
7  revision 2025-04-17 {
8    description
9      "Updates under IEEE Std 802.3.2-202x, Draft D3.3";
10   reference
11     "IEEE Std 802.3-2022, unless dated explicitly";
12  }
13
14
15  feature mac-merge {
16    description
17      "MAC Merge is supported.";
18    reference
19      "IEEE Std 802.3-2022";
20  }
21
22
23  augment "/if:interfaces/if:interface/ieee802-eth-if:ethernet" {
24    if-feature "mac-merge";
25    description
26      "Interfaces supporting MAC Merge.";
27    container mac-merge {
28      description
29        "Container for MAC Merge objects.";
30      container admin-control {
31        description
32          "Container for MAC Merge control objects.";
33        leaf merge-enable-tx {
34          type enumeration {
35            enum Disabled {
36              description
37                "Transmit preemption is disabled";
38            }
39            enum Enabled {
40              description
41                "Transmit preemption is enabled";
42            }
43          }
44          default "Disabled";
45          description
46            "Is MAC Merge enabled when transmitting?
47            It may be modified using an edit operation.";
48          reference
49            "IEEE Std 802.3, 30.14.1.3";
50        }
51      }
52      leaf verify-disable-tx {
53        type enumeration {
54          enum Disabled {
55            description
56              "Verify is disabled";
57          }
58          enum Enabled {
59            description
60              "Verify is enabled";
61          }
62        }
63      }
64    }
65  }
```

```
1      default "Disabled";
2      description
3          "Does MAC Merge check to make sure the link partner
4          supports MAC Merge?
5          The check is performed by sending a verify request and
6          getting a verify response.
7          It may be modified using an edit operation.>";
8      reference
9          "IEEE Std 802.3, 30.14.1.4";
10     }
11     leaf verify-time {
12         type uint16 {
13             range "1..128";
14         }
15         units "milliseconds";
16         default "10";
17         description
18             "The retry timer for verification attempts.
19             The valid range is 1 to 128 inclusive.
20             The default value is 10.";
21         reference
22             "IEEE Std 802.3, 30.14.1.6";
23     }
24     leaf frag-size {
25         type uint8 {
26             range "0..3";
27         }
28         default "0";
29         description
30             "This controls minimum frame size for packet fragments.
31             It's specified in units of 64 bytes above the minimum
32             of 64 bytes. The equation is:
33             minimum-size = 64 + (frag-size*64)-4";
34         reference
35             "IEEE Std 802.3, 30.14.1.7";
36     }
37 }
38
39 container admin-status {
40     config false;
41     description
42         "Container for MAC Merge status objects.";
43     leaf merge-support {
44         type enumeration {
45             enum Supported {
46                 description
47                     "MAC Merge is supported on the device";
48             }
49             enum NotSupported {
50                 description
51                     "MAC Merge is not supported on the device";
52             }
53         }
54         description
55             "Is MAC Merge is supported?";
56         reference
57             "IEEE Std 802.3, 30.14.1.1";
58     }
59     leaf verify-status {
60         type enumeration {
```

```
1         enum unknown {
2             description
3                 "unknown";
4         }
5         enum initial {
6             description
7                 "initial";
8         }
9
10        enum verifying {
11            description
12                "verifying";
13        }
14        enum succeeded {
15            description
16                "succeeded";
17        }
18        enum failed {
19            description
20                "failed";
21        }
22
23        enum disabled {
24            description
25                "disabled";
26        }
27    }
28    description
29        "The status of MAC Merge Verification.";
30    reference
31        "IEEE Std 802.3, 30.14.1.2";
32    }
33    leaf status-tx {
34        type enumeration {
35            enum unknown {
36                description
37                    "unknown";
38            }
39            enum inactive {
40                description
41                    "inactive";
42            }
43            enum active {
44                description
45                    "active";
46            }
47        }
48    }
49    description
50        "The status of MAC Merge in the transmit direction.";
51    reference
52        "IEEE Std 802.3, 30.14.1.5";
53    }
54    }
55    container statistics {
56        config false;
57        description
58            "Container for MAC Merge counters.
59            Discontinuities in the values of counters in
60            this container can occur at re-initialization of
61            the management system, and at other times as
62            indicated by the value of the 'discontinuity-time'
63            indicated by the value of the 'discontinuity-time'
64            indicated by the value of the 'discontinuity-time'
65            indicated by the value of the 'discontinuity-time'";
```

```
1         leaf defined in the ietf-interfaces YANG module
2         (IETF RFC 8343).";
3     leaf assembly-error-count {
4         type yang:counter64;
5         description
6             "A count of MAC frames with reassembly errors.";
7         reference
8             "IEEE Std 802.3, 30.14.1.8";
9     }
10
11     leaf smd-error-count {
12         type yang:counter64;
13         description
14             "A count of received MAC frames / MAC frame fragments
15             rejected with an unexpected preemption header.";
16         reference
17             "IEEE Std 802.3, 30.14.1.9";
18     }
19
20     leaf assembly-ok-count {
21         type yang:counter64;
22         description
23             "A count of preemptable MAC frames that were successfully
24             reassembled and delivered to MAC.";
25         reference
26             "IEEE Std 802.3, 30.14.1.10";
27     }
28
29     leaf fragment-count-rx {
30         type yang:counter64;
31         description
32             "The number of preemptable frame fragments recieved.";
33         reference
34             "IEEE Std 802.3, 30.14.1.11";
35     }
36
37     leaf fragment-count-tx {
38         type yang:counter64;
39         description
40             "The number of preemptable frame fragments transmitted.";
41         reference
42             "IEEE Std 802.3, 30.14.1.12";
43     }
44
45     leaf hold-count {
46         type yang:counter64;
47         description
48             "The number of MAC Merge hold requests (i.e., to stop
49             transmissions of preemptable traffic) received from
50             the MAC";
51         reference
52             "IEEE Std 802.3, 30.14.1.13";
53     }
54 }
55 }
56 }
57 }
58 }
59 }
60 }
61 }
62 }
63 }
64 }
65 }
```

5.3.2.4 Ethernet LLDP module

Editor's Note (to be removed prior to publication):

Pretty printing of ieee802-ethernet-lldp.yang file may change the appearance by adding whitespace and reformatting lines

Editor's Note (to be removed prior to publication):

IEEE Std 802.3.1 and IEEE Std 802.3.2 to be updated at the publication time

```
module ieee802-ethernet-lldp {
  yang-version 1.1;
  namespace "urn:ieee:std:802.3:yang:ieee802-ethernet-lldp";
  prefix ieee802-eth-lldp;

  import ieee802-dot1ab-lldp {
    prefix lldp;
    reference
      "IEEE Std 802.1ABcu-2021";
  }
  import ieee802-ethernet-phy-type {
    prefix ieee802-phy;
    reference
      "IEEE Std 802.3-2022";
  }

  organization
    "IEEE Std 802.3 Ethernet Working Group
     Web URL: http://www.ieee802.org/3/";
  contact
    "Web URL: http://www.ieee802.org/3/";
  description
    "This module contains YANG definitions for configuring LLDP for
     802.3 Ethernet Interfaces.
     In this YANG module, 'Ethernet interface' can be interpreted
     as referring to 'IEEE Std 802.3 compliant Ethernet
     interfaces'.";

  revision 2025-04-17 {
    description
      "Updates under IEEE Std 802.3.2-202x, Draft D3.3";
    reference
      "IEEE Std 802.3-2022, unless dated explicitly";
  }

  typedef port-class-type {
    type enumeration {
      enum p-class-pse {
        value 0;
        description
          "Power Sourcing Equipment";
      }
      enum p-class-pd {
        value 1;
        description
          "Powered Device";
      }
    }
  }
```

```
1      }
2      description
3          "Enumeration for the power port class";
4      reference
5          "IEEE Std 802.3, 30.12.2.1.5";
6  }
7
8
9  typedef pse-pinout-type {
10     type enumeration {
11         enum signal {
12             value 0;
13             description
14                 "PSE Pinout Alternative A";
15         }
16         enum spare {
17             value 1;
18             description
19                 "PSE Pinout Alternative B";
20         }
21     }
22 }
23 description
24     "Enumeration for the pinout alternatives used for PD
25     detection and power ";
26 reference
27     "IEEE Std 802.3, 30.12.2.1.9";
28 }
29
30 typedef pse-power-class-type {
31     type enumeration {
32         enum class0 {
33             value 0;
34             description
35                 "Class 0 PD";
36         }
37         enum class1 {
38             value 1;
39             description
40                 "Class 1 PD";
41         }
42         enum class2 {
43             value 2;
44             description
45                 "Class 2 PD";
46         }
47         enum class3 {
48             value 3;
49             description
50                 "Class 3 PD";
51         }
52         enum class4 {
53             value 4;
54             description
55                 "Class 4 PD";
56         }
57     }
58 }
59 description
60     "Enumeration for the PD class";
61 reference
62     "IEEE Std 802.3, 30.12.2.1.10";
63
64
65
```

```
1     }
2
3     typedef power-class-ext-AB-type {
4         type enumeration {
5             enum singlesig {
6                 value 0;
7                 description
8                     "Single-signature PD or 2-pair only PSE";
9             }
10            enum class1 {
11                value 1;
12                description
13                    "Class 1";
14            }
15            enum class2 {
16                value 2;
17                description
18                    "Class 2";
19            }
20            enum class3 {
21                value 3;
22                description
23                    "Class 3";
24            }
25            enum class4 {
26                value 4;
27                description
28                    "Class 4";
29            }
30            enum class5 {
31                value 5;
32                description
33                    "Class 5";
34            }
35        }
36        description
37            "Enumeration for the assigned power class ";
38        reference
39            "IEEE Std 802.3, 30.12.3.1.26";
40    }
41
42    typedef power-class-ext-type {
43        type enumeration {
44            enum dualsig {
45                value 0;
46                description
47                    "Dual-signature PD";
48            }
49            enum class1 {
50                value 1;
51                description
52                    "Class 1";
53            }
54            enum class2 {
55                value 2;
56                description
57                    "Class 2";
58            }
59            enum class3 {
```

```
1         value 3;
2         description
3           "Class 3";
4       }
5       enum class4 {
6         value 4;
7         description
8           "Class 4";
9       }
10      }
11      enum class5 {
12        value 5;
13        description
14          "Class 5";
15      }
16      enum class6 {
17        value 6;
18        description
19          "Class 6";
20      }
21      }
22      enum class7 {
23        value 7;
24        description
25          "Class 7";
26      }
27      enum class8 {
28        value 8;
29        description
30          "Class 8";
31      }
32      }
33      }
34      description
35        "Enumeration for the assigned power class ";
36      reference
37        "IEEE Std 802.3, 30.12.3.1.28";
38    }
39
40    typedef power-type {
41      type enumeration {
42        enum type4dualsigPD {
43          value 0;
44          description
45            "Type 4 dual-signature PD";
46        }
47        enum type4singlesigPD {
48          value 1;
49          description
50            "Type 4 single-signature PD";
51        }
52        enum type3dualsigPD {
53          value 2;
54          description
55            "Type 3 dual-signature PD";
56        }
57        enum type3singlesigPD {
58          value 3;
59          description
60            "Type 3 single-signature PD";
61        }
62        enum type4PSE {
```

```
1         value 4;
2         description
3             "Type 4 PSE";
4     }
5     enum type3PSE {
6         value 5;
7         description
8             "Type 3 PSE";
9     }
10 }
11 }
12 description
13     "Enumeration for the PD class";
14 reference
15     "IEEE Std 802.3, 30.12.2.1.29";
16 }
17
18 typedef power-priority-type {
19     type enumeration {
20         enum low {
21             value 0;
22             description
23                 "low priority PD";
24         }
25         enum high {
26             value 1;
27             description
28                 "high priority PD";
29         }
30         enum critical {
31             value 2;
32             description
33                 "critical priority PD";
34         }
35         enum unknown {
36             value 3;
37             description
38                 "priority unknown";
39         }
40     }
41 }
42 description
43     "Enumeration for possible priorities of a PD system";
44 reference
45     "IEEE Std 802.3, 30.12.2.1.16";
46 }
47
48 typedef power-source-type {
49     type enumeration {
50         enum pse-primary {
51             value 0;
52             description
53                 "PSE powered by a primary power source";
54         }
55         enum pse-backup {
56             value 1;
57             description
58                 "PSE powered by a backup power source";
59         }
60         enum pse-unknown {
61             value 2;
```

```
1         description
2             "PSE powered by an unknown power source";
3     }
4     enum pd-pse-and-local {
5         value 3;
6         description
7             "PD powered by a PSE and locally";
8     }
9
10    enum pd-local-only {
11        value 4;
12        description
13            "PD powered only locally";
14    }
15    enum pd-pse-only {
16        value 5;
17        description
18            "PD powered by PD only";
19    }
20    enum pd-unknown {
21        value 6;
22        description
23            "PD powered by an unknown source";
24    }
25 }
26
27 description
28     "Enumeration for the power sources of the
29     remote system. When the remote system is a PSE, it
30     indicates whether it is being powered by a primary
31     power source; a backup power source; or unknown.
32     When the remote system is a PD, it indicates whether
33     it is being powered by a PSE and locally;
34     locally only; by a PSE only; or unknown.";
35 reference
36     "IEEE Std 802.3, 30.12.2.1.15";
37 }
38
39
40 typedef powering-status-type {
41     type enumeration {
42         enum 4PdualsigPD {
43             value 0;
44             description
45                 "4-pair powering a dual-signature PD";
46         }
47         enum 4PsigPD {
48             value 1;
49             description
50                 "4-pair powering a single-signature PD";
51         }
52         enum 2P {
53             value 2;
54             description
55                 "2-pair powering";
56         }
57     }
58 }
59
60 description
61     "Enumeration for the power status of the PSE";
62 reference
63     "IEEE Std 802.3, 30.12.2.1.23";
64 }
65
```

```
1
2 typedef powered-status-type {
3     type enumeration {
4         enum 4PdualsigPD {
5             value 0;
6             description
7                 "4-pair powered dual-signature PD";
8         }
9
10        enum 2PdualsigPD {
11            value 1;
12            description
13                "2-pair powered dual-signature PD";
14        }
15        enum singlesigPD {
16            value 2;
17            description
18                "powered single-signature PD";
19        }
20    }
21    description
22        "Enumeration for the power status of the PSE";
23    reference
24        "IEEE Std 802.3, 30.12.2.1.24";
25 }
26
27
28 typedef power-pairs-type {
29     type enumeration {
30         enum altA {
31             value 0;
32             description
33                 "Alternative A";
34         }
35
36         enum altB {
37             value 1;
38             description
39                 "Alternative B";
40         }
41
42         enum both {
43             value 2;
44             description
45                 "both";
46         }
47     }
48    description
49        "Enumeration for the PSE Pinout Alternative";
50    reference
51        "IEEE Std 802.3, 30.12.2.1.25";
52 }
53
54
55 typedef measurement-source-type {
56     type enumeration {
57         enum noRequest {
58             description
59                 "No Request";
60         }
61
62         enum altA {
63             description
64                 "Pairset Alternative A / Mode A";
65         }
66     }
67 }
```

```
1      enum altB {
2          description
3              "Pairset Alternative B / Mode B";
4      }
5      enum total {
6          description
7              "Port total";
8      }
9  }
10 }
11 description
12     "PoE measurement source";
13 reference
14     "IEEE Std 802.3, 30.12.2.1.42 and 30.12.3.1.42";
15 }
16
17 augment "/lldp:lldp/lldp:port" {
18     description
19         "Augments port with 802.3 port read-only objects";
20     leaf auto-negotiation-supported {
21         type boolean;
22         config false;
23         description
24             "True if the port supports Auto-negotiation";
25         reference
26             "IEEE Std 802.3, 30.12.2.1.1";
27     }
28     leaf auto-negotiation-enabled {
29         type boolean;
30         config false;
31         description
32             "True if Auto-negotiation is enabled";
33         reference
34             "IEEE Std 802.3, 30.12.2.1.2";
35     }
36     /* Deprecated MAU objects */
37     leaf auto-negotiation-cap {
38         type binary {
39             length "2";
40         }
41         config false;
42         status deprecated;
43         description
44             "A read-only 2-octet value that contains the value (bitmap)
45             of the ifMauAutoNegCapAdvertisedBits object
46             (defined in IETF RFC 4836) which is associated with the
47             given port on the local system.
48             Superseeded by auto-negotiation-cap-bits.";
49         reference
50             "IEEE Std 802.3, 30.12.2.1.3";
51     }
52     leaf operational-mau-type {
53         type int32;
54         config false;
55         status deprecated;
56         description
57             "32-bit integer value that indicates the operational MAU
58             type of the given port.
59             Superseeded by operational-pmd-type.";
60         reference
```

```
1      "IEEE Std 802.3, 30.12.2.1.4";
2  }
3  leaf auto-negotiation-cap-bits {
4      type ieee802-phy:auto-neg-ability-bits;
5      description
6          "The capabilities advertised by the local Auto-Negotiation
7           entity.";
8      reference
9          "IEEE Std 802.3, 30.6.1.1.6,
10           aAutoNegadvertisedTechnologyAbility.";
11  }
12  leaf operational-pmd-type {
13      type identityref {
14          base ieee802-phy:pmd-type;
15      }
16      description
17          "Operational PMD type of the given port";
18      reference
19          "IEEE Std 802.3, Clause 30.5.1.1.2 aMAUType";
20  }
21  /* Clause 30 & 145 PoE related LLDP definitions */
22  leaf power-port-class {
23      type port-class-type;
24      description
25          "A read-only value that identifies the port Class of the
26           given port";
27      reference
28          "IEEE Std 802.3, 30.12.2.1.5";
29  }
30  leaf mdi-power-supported {
31      type boolean;
32      description
33          "True if MDI power is supported";
34      reference
35          "IEEE Std 802.3, 30.12.2.1.6";
36  }
37  leaf mdi-power-enabled {
38      type boolean;
39      description
40          "True if MDI power is enabled";
41      reference
42          "IEEE Std 802.3, 30.12.2.1.7";
43  }
44  leaf power-pair-controlable {
45      type boolean;
46      description
47          "True if the pair selection can be controlled";
48      reference
49          "IEEE Std 802.3, 30.12.2.1.8";
50  }
51  leaf power-pairs {
52      type pse-pinout-type;
53      description
54          "Indicates which pinout alternative is used for PD detection
55           and power";
56      reference
57          "IEEE Std 802.3, 30.12.2.1.9";
58  }
59  leaf local-power-class {
```

```
1      type pse-power-class-type;
2      description
3          "PD Power Class";
4      reference
5          "IEEE Std 802.3, 30.12.2.1.10";
6  }
7  /* Obsolete Link Aggregation objects
8     Link aggregation has not been part of 802.3 since IEEE 802.1AX-2008
9     */
10
11  leaf link-aggregation-status {
12      type bits {
13          bit aggregation-capability {
14              position 0;
15              description
16                  "IEEE Std 802.3, 79.3.3.1";
17          }
18          bit aggregation-status {
19              position 1;
20              description
21                  "IEEE Std 802.3, 79.3.3.1";
22          }
23      }
24  }
25  config false;
26  status obsolete;
27  description
28      "The bitmap value which contains the link aggregation
29      capabilities and the current aggregation
30      status of the link";
31  reference
32      "IEEE Std 802.3, 30.12.2.1.11";
33  }
34
35  leaf aggregation-port-id {
36      type int32;
37      config false;
38      status obsolete;
39      description
40          "The unique identifier allocated to this Aggregation Port
41          by the local System.";
42      reference
43          "IEEE Std 802.3, 30.12.2.1.12";
44  }
45
46  leaf local-max-frame-size {
47      type int32;
48      config false;
49      status obsolete;
50      description
51          "An integer value indicating the maximum supported frame
52          size in octets on the given port of the local system.";
53      reference
54          "IEEE Std 802.3, 30.12.2.1.13";
55  }
56
57  leaf power-type {
58      type bits {
59          bit typel-or-greater {
60              position 0;
61              description
62                  "0-typel, 1-greater than typel";
63          }
64      }
65      bit pse-or-pd {
```

```
1         position 1;
2         description
3             "0-pse, 1-pd";
4     }
5 }
6 config false;
7 description
8     "A read-only attribute that returns a bit string indicating
9     whether the local system is a PSE or a PD and whether it
10    is Type 1 or greater than Type 1. The first bit indicates
11    Type 1 or greater than Type 1. The second bit indicates
12    PSE or PD. A PSE sets this bit to indicate a PSE. A PD sets
13    this bit to indicate a PD. See also aLldpXdot3LocPowerTypeExt.
14    Superseeded by power-support";
15 reference
16     "IEEE Std 802.3, 30.12.2.1.14";
17 }
18 leaf power-support {
19     type bits {
20         bit pse-pd {
21             description
22                 "0-PD, 1-PSE";
23         }
24         bit pse-power-support {
25             description
26                 "0-not supported, 1-supported";
27         }
28         bit pse-power-state {
29             description
30                 "0-disabled 1-enabled";
31         }
32         bit pse-pairs-selection {
33             description
34                 "0-pair selection not controllable
35                 1-pair selection controllable";
36         }
37     }
38     description
39         "A read-only attribute that returns a bit string indicating
40         whether local system PoE support.";
41     reference
42         "IEEE Std 802.3, 79.3.2.1 MDI power support";
43 }
44 leaf power-source {
45     type power-source-type;
46     config false;
47     description
48         "Indicates the power sources of the local system. A PSE
49         indicates whether it is being powered by a primary power
50         source; a backup power source; or unknown. A PD indicates
51         whether it is being powered by a PSE and locally; by a PSE
52         only; or unknown.";
53     reference
54         "IEEE Std 802.3, 30.12.2.1.15";
55 }
56 leaf pd-requested-power-value {
57     type int32;
58     config false;
59     description
```

```
1         "PD requested power value. For a PD, it is the power value
2         that the PD has currently requested from the remote
3         system. For a PSE, it is the power value that the PSE
4         mirrors back to the remote system";
5     reference
6         "IEEE Std 802.3, 30.12.2.1.17";
7 }
8
9 leaf pd-requested-power-value-a {
10     type int32;
11     config false;
12     description
13         "A read-only attribute that returns the PD requested power
14         value for the Mode A pairset in units of 0.1 W. For a PD,
15         it's the power value that the PD has currently requested
16         from the remote system for the Mode A pairset. For a PSE,
17         it is the power value for the Alternative A pairset that
18         the PSE echoes back to the remote system";
19     reference
20         "IEEE Std 802.3, 30.12.2.1.18";
21 }
22
23 leaf pd-requested-power-value-b {
24     type int32;
25     config false;
26     description
27         "A read-only attribute that returns the PD requested power
28         value for the Mode B pairset in units of 0.1 W.
29         For a PD, it is the power value that the PD has currently
30         requested from the remote system for the Mode B pairset.
31         For a PSE, it is the power value for the Alternative B
32         pairset that the PSE echoes back to the remote system";
33     reference
34         "IEEE Std 802.3, 30.12.2.1.19";
35 }
36
37 leaf pse-allocated-power-value {
38     type int32;
39     config false;
40     description
41         "PSE allocated power value. For a PSE, it is the power
42         value that the PSE has currently allocated to the remote
43         system. For a PD, it is the power value that the PD mirrors
44         back to the remote system";
45     reference
46         "IEEE Std 802.3, 30.12.2.1.20";
47 }
48
49 leaf pse-allocated-power-value-a {
50     type int32;
51     config false;
52     description
53         " PSE allocated power value for the Alternative A pairset
54         in units of 0.1 W. For a PSE, it is the power value for the
55         Alternative A pairset that the PSE has currently allocated
56         to the remote system. For a PD, it is the power value for
57         the Mode A pairset that the PD echoes back to the remote
58         system.";
59     reference
60         "IEEE Std 802.3, 30.12.2.1.21";
61 }
62
63 leaf pse-allocated-power-value-b {
64     type int32;
```

```
1      config false;
2      description
3          "PSE allocated power value for the Alternative B pairset
4          in units of 0.1 W. For a PSE, it is the power value for the
5          Alternative B pairset that the PSE has currently
6          allocated to the remote system. For a PD, it is the power
7          value for the Mode B pairset that the PD echoes back to the
8          remote system.";
9
10     reference
11         "IEEE Std 802.3, 30.12.2.1.22";
12 }
13 leaf pse-powering-status {
14     type powering-status-type;
15     config false;
16     description
17         "A read only value that indicates the powering status of
18         the PSE. For a PD, the contents of this
19         attribute are undefined.";
20     reference
21         "IEEE Std 802.3, 30.12.2.1.23";
22 }
23
24 leaf pd-powered-status {
25     type powered-status-type;
26     config false;
27     description
28         "A read only value that indicates the powering status of
29         the PD. For a PSE, the contents of this attribute are
30         undefined";
31     reference
32         "IEEE Std 802.3, 30.12.2.1.24";
33 }
34
35 leaf power-pairs-ext {
36     type power-pairs-type;
37     config false;
38     description
39         "A read-only value that identifies the supported PSE
40         Pinout Alternative specified in 145.2.4. For a PSE, this
41         attribute contains the value of the aPSEPowerPairs
42         attribute (see 30.9.1.1.4). For a PD, the contents of this
43         attribute are undefined";
44     reference
45         "IEEE Std 802.3, 30.12.2.1.25";
46 }
47
48 leaf power-class-ext-A {
49     type power-class-ext-AB-type;
50     config false;
51     description
52         "For a dual-signature PD, a read-only value that indicates
53         the requested Class for Mode A during Physical Layer
54         Classification (see 145.3.6). For a single-signature PD, a
55         read-only value set to â€šinglesigâ€™. For a PSE connected to
56         a dual-signature PD, a read-only value that indicates the
57         currently assigned Class for Mode A (see 145.2.8). For a
58         PSE connected to a single-signature PD or a PSE that
59         operates only in 2-pair mode, a read-only value set to
60         â€šinglesigâ€™";
61     reference
62         "IEEE Std 802.3, 30.12.2.1.26";
63 }
64
65 }
```

```
1     leaf power-class-ext-B {
2         type power-class-ext-AB-type;
3         config false;
4         description
5             "For a dual-signature PD, a read-only value that indicates
6             the requested Class for Mode B during Physical Layer
7             Classification (see 145.3.6). For a single-signature PD,
8             a read-only value set to 'single'. For a PSE connected to
9             a dual-signature PD, a read-only value that indicates the
10            currently assigned Class for Mode B (see 145.2.8). For a
11            PSE connected to a single-signature PD or a PSE that
12            operates only in 2-pair mode, a read-only value set to
13            'single'";
14        reference
15            "IEEE Std 802.3, 30.12.2.1.27";
16    }
17    leaf power-class-ext {
18        type power-class-ext-type;
19        config false;
20        description
21            "For a single-signature PD, a read-only value that
22            indicates the requested Class during Physical
23            Layer Classification (see 145.3.6). For a dual-signature
24            PD, a read-only value set to 'dual'.
25            For a PSE connected to a single-signature PD or a PSE that
26            operates only in 2-pair mode, a read-only value that
27            indicates the currently assigned Class (see 145.2.8).
28            For a PSE connected to a dual-signature PD,
29            a read-only value set to 'dual'.";
30        reference
31            "IEEE Std 802.3, 30.12.2.1.28";
32    }
33    leaf power-type-ext {
34        type power-type;
35        config false;
36        description
37            "A read-only attribute that returns a value to indicate if
38            the local system is a Type 3 or Type 4 PSE or PD and, in
39            the case of a Type 3 or Type 4 PD, if it is a
40            single-signature PD or a dual-signature
41            PD";
42        reference
43            "IEEE Std 802.3, 30.12.2.1.29";
44    }
45    leaf pd-load {
46        type boolean;
47        config false;
48        description
49            "For a dual-signature PD, a read-only attribute that
50            returns whether the load of a dual-signature PD is
51            electrically isolated, as defined in 79.3.2.10.2. For a
52            single-signature PD or a PSE, the value of this attribute
53            is FALSE";
54        reference
55            "IEEE Std 802.3, 30.12.2.1.30";
56    }
57    leaf pd-4pid {
58        type boolean;
59        config false;
```

```
1      description
2          "A read-only Boolean attribute indicating whether the local
3          PD system supports powering of both PD Modes.";
4      reference
5          "IEEE Std 802.3, 30.12.2.1.31";
6  }
7  leaf pse-max-avail-power {
8      type int32;
9      config false;
10     description
11         "A read-only attribute that returns the local PSE maximum
12         available power value in units of 0.1 W";
13     reference
14         "IEEE Std 802.3, 30.12.2.1.32";
15 }
16 leaf pse-autoclass-support {
17     type boolean;
18     config false;
19     description
20         "Indicates whether the local PSE system supports Autoclass.";
21     reference
22         "IEEE Std 802.3, 30.12.2.1.33";
23 }
24 leaf autoclass-completed {
25     type boolean;
26     config false;
27     description
28         "Indicates whether the local PSE system has completed the
29         Autoclass measurement.";
30     reference
31         "IEEE Std 802.3, 30.12.2.1.34";
32 }
33 leaf autoclass-request {
34     type boolean;
35     config false;
36     description
37         "A read-only Boolean attribute indicating whether the local
38         PD system is requesting an Autoclass measurement.";
39     reference
40         "IEEE Std 802.3, 30.12.2.1.35";
41 }
42 leaf power-down-request {
43     type int32;
44     description
45         "A read-write attribute that indicates the local PD system
46         is requesting a power down when the value is 0x1D.";
47     reference
48         "IEEE Std 802.3, 30.12.2.1.36";
49 }
50 leaf power-down-time {
51     type int32;
52     description
53         "A read-write attribute that indicates the number of
54         seconds the PD requests to stay powered off. A value of
55         zero indicates an indefinite amount of time.";
56     reference
57         "IEEE Std 802.3, 30.12.2.1.37";
58 }
59 /* PoE measurement objects */
```

```
1     leaf meas-voltage-support {
2         type boolean;
3         description
4             "A read-only attribute that indicates the local device is
5             capable of providing a voltage measurement.;";
6         reference
7             "IEEE Std 802.3, 30.12.2.1.38";
8     }
9
10    leaf meas-current-support {
11        type boolean;
12        description
13            "A read-only attribute that indicates the local device is
14            capable of providing a current measurement.;";
15        reference
16            "IEEE Std 802.3, 30.12.2.1.39";
17    }
18
19    leaf meas-power-support {
20        type boolean;
21        description
22            "A read-only attribute that indicates the local device is
23            capable of providing a power measurement.;";
24        reference
25            "IEEE Std 802.3, 30.12.2.1.40";
26    }
27
28    leaf meas-energy-support {
29        type boolean;
30        description
31            "A read-only attribute that indicates the local device is
32            capable of providing a energy measurement.;";
33        reference
34            "IEEE Std 802.3, 30.12.2.1.41";
35    }
36    /* NOTE - What do these values mean? */
37    leaf measurement-source {
38        type bits {
39            bit bit1 {
40                position 0;
41                description
42                    "???";
43            }
44            bit bit2 {
45                position 1;
46                description
47                    "-";
48            }
49        }
50    }
51
52    status obsolete;
53    description
54        "A read-write attribute value that indicates to local
55        device on which Alternative or Mode the measurement
56        is to be taken";
57    reference
58        "IEEE Std 802.3, 30.12.2.1.42";
59
60    leaf meas-voltage-request {
61        type boolean;
62        config false;
63        description
64            "A read-only attribute that indicates the local device is
```

```
1         requesting a voltage measurement from the remote device.";
2     reference
3         "IEEE Std 802.3, 30.12.2.1.43";
4 }
5 leaf meas-current-request {
6     type boolean;
7     config false;
8     description
9         "A read-only attribute that indicates the local device is
10        requesting a current measurement from the remote device.";
11     reference
12         "IEEE Std 802.3, 30.12.2.1.44";
13 }
14 leaf meas-power-request {
15     type boolean;
16     config false;
17     description
18         "A read-only attribute that indicates the local device is
19        requesting a power measurement from the remote device.";
20     reference
21         "IEEE Std 802.3, 30.12.2.1.45";
22 }
23 leaf meas-energy-request {
24     type boolean;
25     config false;
26     description
27         "A read-only attribute that indicates the local device is
28        requesting an energy measurement from the remote device.";
29     reference
30         "IEEE Std 802.3, 30.12.2.1.46";
31 }
32 leaf meas-voltage-valid {
33     type boolean;
34     config false;
35     description
36         "A read-only attribute that indicates the local deviceâ€™s
37        voltage measurement is valid.";
38     reference
39         "IEEE Std 802.3, 30.12.2.1.47";
40 }
41 leaf meas-current-valid {
42     type boolean;
43     config false;
44     description
45         "A read-only attribute that indicates the local deviceâ€™s
46        current measurement is valid.";
47     reference
48         "IEEE Std 802.3, 30.12.2.1.48";
49 }
50 leaf meas-power-valid {
51     type boolean;
52     config false;
53     description
54         "A read-only attribute that indicates the local deviceâ€™s
55        power measurement is valid.";
56     reference
57         "IEEE Std 802.3, 30.12.2.1.49";
58 }
59 leaf meas-energy-valid {
```

```
1      type boolean;
2      config false;
3      description
4          "A read-only attribute that indicates the local deviceâ€™s
5          energy measurement is valid.";
6      reference
7          "IEEE Std 802.3, 30.12.2.1.50";
8  }
9
10 leaf meas-voltage-uncertainty {
11     type int32;
12     config false;
13     description
14         "A read-only attribute that indicates the expanded
15         uncertainty (coverage factor k = 2) for the deviceâ€™s
16         voltage measurement.";
17     reference
18         "IEEE Std 802.3, 30.12.2.1.51";
19 }
20
21 leaf meas-current-uncertainty {
22     type int32;
23     config false;
24     description
25         "A read-only attribute that indicates the expanded
26         uncertainty (coverage factor k = 2) for the deviceâ€™s
27         current measurement.";
28     reference
29         "IEEE Std 802.3, 30.12.2.1.52";
30 }
31
32 leaf meas-power-uncertainty {
33     type int32;
34     config false;
35     description
36         "A read-only attribute that indicates the expanded
37         uncertainty (coverage factor k = 2) for the deviceâ€™s
38         power measurement.";
39     reference
40         "IEEE Std 802.3, 30.12.2.1.53";
41 }
42
43 leaf meas-energy-uncertainty {
44     type int32;
45     config false;
46     description
47         "A read-only attribute that indicates the expanded
48         uncertainty (coverage factor k = 2) for the deviceâ€™s
49         energy measurement.";
50     reference
51         "IEEE Std 802.3, 30.12.2.1.54";
52 }
53
54 leaf voltage-measurement {
55     type int32;
56     config false;
57     description
58         "A read-only attribute that returns the measured device
59         voltage.";
60     reference
61         "IEEE Std 802.3, 30.12.2.1.55";
62 }
63
64 leaf current-measurement {
65     type int32;
```

```
1      config false;
2      description
3          "A read-only attribute that returns the measured device
4          current.";
5      reference
6          "IEEE Std 802.3, 30.12.2.1.56";
7  }
8
9  leaf power-measurement {
10     type int32;
11     config false;
12     description
13         "A read-only attribute that returns the measured device
14         power.";
15     reference
16         "IEEE Std 802.3, 30.12.2.1.57";
17 }
18
19 leaf energy-measurement {
20     type int32;
21     config false;
22     description
23         "A read-only attribute that returns the measured device
24         energy.";
25     reference
26         "IEEE Std 802.3, 30.12.2.1.58";
27 }
28
29 leaf pse-power-price-index {
30     type int32;
31     config false;
32     description
33         "A read-only attribute that returns an index of the price
34         of power being sourced by the PSE. For a PD, this value
35         is undefined";
36     reference
37         "IEEE Std 802.3, 30.12.2.1.59";
38 }
39
40 leaf local-response {
41     type int32;
42     config false;
43     description
44         "The maximum time required to update
45         pse-allocated-power-value";
46     reference
47         "IEEE Std 802.3, 30.12.2.1.60";
48 }
49
50 leaf local-system-ready {
51     type boolean;
52     config false;
53     description
54         "Initialization status of the Data Link Layer
55         classification engine on the local system";
56     reference
57         "IEEE Std 802.3, 30.12.2.1.61";
58 }
59
60 leaf tx-system-value {
61     type int32;
62     config false;
63     description
64         "Returns the value of Tw_sys_tx that the local system can
65         support in the transmit direction.";
```

```
1      reference
2          "IEEE Std 802.3, 30.12.2.1.62";
3  }
4  leaf tx-system-value-echo {
5      type int32;
6      config false;
7      description
8          "Returns the value of Tw_sys_tx that the emote system is
9          advertising that it can support in the transmit direction
10         and is echoed by the local system under the control of the
11         EEE DLL receiver state diagram.";
12     reference
13         "IEEE Std 802.3, 30.12.2.1.63";
14 }
15 leaf rx-system-value {
16     type int32;
17     config false;
18     description
19         "Returns the value of Tw_sys_tx that the local system is
20         requesting in the receive direction.";
21     reference
22         "IEEE Std 802.3, 30.12.2.1.64";
23 }
24 leaf rx-system-value-echo {
25     type int32;
26     config false;
27     description
28         "Returns the value of Tw_sys_tx that the remote system is
29         advertising that it is requesting in the receive direction
30         and is echoed by the local system under the control of the
31         EEE DLL transmitter state diagram.";
32     reference
33         "IEEE Std 802.3, 30.12.2.1.65";
34 }
35 leaf fallback-system-value {
36     type int32;
37     config false;
38     description
39         "Returns the value of the fallback Tw_sys_tx that the local
40         system is advertising to the remote system.";
41     reference
42         "IEEE Std 802.3, 30.12.2.1.66";
43 }
44 /* LDP EEE objects */
45 leaf tx-dll-ready {
46     type boolean;
47     config false;
48     description
49         "Returns the initialization status of the EEE transmit Data
50         Link Layer management function on the local system.";
51     reference
52         "IEEE Std 802.3, 30.12.2.1.67";
53 }
54 leaf rx-dll-ready {
55     type boolean;
56     config false;
57     description
58         "Returns the initialization status of the EEE receive Data
59         Link Layer management function on the local system.";
```

```
1      reference
2      "IEEE Std 802.3, 30.12.2.1.68";
3  }
4  leaf dll-enabled {
5      type boolean;
6      config false;
7      description
8          "Returns the status of the EEE capability negotiation on
9          the local system.";
10     reference
11         "IEEE Std 802.3, 30.12.2.1.69";
12 }
13
14 leaf tx-system-fw {
15     type boolean;
16     config false;
17     description
18         "Returns the value of LPI_FW that the local system can
19         support in the transmit direction.";
20     reference
21         "IEEE Std 802.3, 30.12.2.1.70";
22 }
23
24 leaf tx-system-fw-echo {
25     type boolean;
26     config false;
27     description
28         "Returns the value of LPI_FW that the remote system is
29         advertising that it can support in the transmit direction
30         and is echoed by the local system under the control of the
31         EEE DLL receiver state diagram.";
32     reference
33         "IEEE Std 802.3, 30.12.2.1.71";
34 }
35
36 leaf rx-system-fw {
37     type boolean;
38     config false;
39     description
40         "Returns the value of LPI_FW that the local system is
41         requesting in the receive direction.";
42     reference
43         "IEEE Std 802.3, 30.12.2.1.72";
44 }
45
46 leaf rx-system-fw-echo {
47     type boolean;
48     config false;
49     description
50         "Returns the value of LPI_FW that the remote system is
51         advertising that it is requesting in the receive direction
52         and is echoed by the local system under the control of the
53         EEE DLL transmitter state diagram.";
54     reference
55         "IEEE Std 802.3, 30.12.2.1.73";
56 }
57
58 /* LLDP preemption objects */
59 leaf preemption-supported {
60     type boolean;
61     config false;
62     description
63         "Indicates whether the given port (associated with the local
64         System) supports the preemption capability.";
```

```
1      reference
2          "IEEE Std 802.3, 30.12.2.1.74";
3  }
4  leaf preemption-enabled {
5      type boolean;
6      config false;
7      description
8          "Indicates whether the preemption capability is enabled on
9          the given port associated with the local System.";
10     reference
11         "IEEE Std 802.3, 30.12.2.1.75";
12 }
13 leaf preemption-active {
14     type boolean;
15     config false;
16     description
17         "Indicates whether the preemption capability is active on
18         the given port associated with the local System.";
19     reference
20         "IEEE Std 802.3, 30.12.2.1.76";
21 }
22 leaf additional-fragment-size {
23     type int32;
24     config false;
25     description
26         "Indicate the minimum size of non-final fragments supported
27         by the receiver on the given port associated with the local
28         System. This value is expressed in units of 64 octets of
29         additional fragment length.";
30     reference
31         "IEEE Std 802.3, 30.12.2.1.77";
32 }
33 }
34
35 augment "/lldp:lldp/lldp:port" {
36     description
37         "Augments port with 802.3 port read-write objects";
38     leaf tlvs-port-config-enable {
39         type bits {
40             bit mac-phy-config-status {
41                 position 0;
42                 description
43                     "IEEE Std 802.3, 30.12.1.1.1";
44             }
45             bit power-via-mdi {
46                 position 1;
47                 description
48                     "IEEE Std 802.3, 30.12.1.1.1";
49             }
50             bit unused {
51                 position 2;
52                 description
53                     "IEEE Std 802.3, 30.12.1.1.1";
54             }
55             bit max-frame-size {
56                 position 3;
57                 description
58                     "IEEE Std 802.3, 30.12.1.1.1";
59             }
60         }
61     }
62 }
```

```
1      bit eee-tlv {
2          position 4;
3          description
4              "IEEE Std 802.3, 30.12.1.1.1";
5      }
6      bit eee-fast-wake-tlv {
7          position 5;
8          description
9              "IEEE Std 802.3, 30.12.1.1.1";
10     }
11     bit additional-ethernet-capabilities-tlv {
12         position 6;
13         description
14             "IEEE Std 802.3, 30.12.1.1.1";
15     }
16 }
17
18 description
19     "Bitmap that corresponds to an IEEE 802.3 subtype
20      associated with a specific IEEE 802.3 port config TLV";
21 reference
22     "IEEE Std 802.3, 30.12.1.1.1";
23 }
24
25 leaf local-power-priority {
26     type power-priority-type;
27     description
28         "Priority of a PD system. For a PSE, this is the priority
29          that the PSE assigns to the PD. For a PD, this is the
30          priority that the PD requests from the PSE.";
31     reference
32         "IEEE Std 802.3, 30.12.2.1.16";
33 }
34
35 leaf local-measurement-source {
36     type measurement-source-type;
37     description
38         "A read-write attribute value that indicates to local
39          device on which Alternative or Mode the measurement
40          is to be taken.";
41     reference
42         "IEEE Std 802.3, 30.12.2.1.42";
43 }
44 }
45
46
47 /* NOTE - lldp:remote-systems-data container is "config false;" */
48
49 augment "/lldp:lldp/lldp:port/lldp:remote-systems-data" {
50     description
51         "Augments port with 802.3 port config tlvs";
52     leaf auto-negotiation-supported {
53         type boolean;
54         description
55             "True if the port supports Auto-negotiation";
56         reference
57             "IEEE Std 802.3, 30.12.3.1.1";
58     }
59     leaf auto-negotiation-enabled {
60         type boolean;
61         description
62             "True if Auto-negotiation is enabled";
63         reference
64             "IEEE Std 802.3, 30.12.3.1.1";
65     }
```

```
1      "IEEE Std 802.3, 30.12.3.1.2";
2  }
3  /* Obsolete */
4  leaf auto-negotiation-cap {
5      type binary {
6          length "2";
7      }
8      status obsolete;
9      description
10         "A read-only 2-octet value that contains the value (bitmap)
11         of the ifMauAutoNegCapAdvertisedBits object (defined in
12         IETF RFC 4836) which is associated with the given port on
13         the local system.
14         Supersceded by auto-negotiation-cap-bits.";
15     reference
16         "IEEE Std 802.3, 30.12.3.1.3";
17 }
18 leaf operational-mau-type {
19     type int32;
20     status obsolete;
21     description
22         "32-bit integer value that indicates the operational MAU
23         type of the given port.
24         Supersceded by operational-pmd-type.";
25     reference
26         "IEEE Std 802.3, 30.12.3.1.4";
27 }
28 leaf auto-negotiation-cap-bits {
29     type ieee802-phy:auto-neg-ability-bits;
30     description
31         "The capabilities advertised by the remote Auto-Negotiation
32         entity.";
33     reference
34         "IEEE Std 802.3, 30.12.3.1.3,
35         aAutoNegadvertisedTechnologyAbility.";
36 }
37 leaf operational-pmd-type {
38     type identityref {
39         base ieee802-phy:pmd-type;
40     }
41     description
42         "The operational PMD type of the given port";
43     reference
44         "IEEE Std 802.3, 30.12.3.1.4";
45 }
46 leaf power-port-class {
47     type port-class-type;
48     description
49         "A read-only value that identifies the port Class of the
50         given port";
51     reference
52         "IEEE Std 802.3, 30.12.3.1.5";
53 }
54 leaf mdi-power-supported {
55     type boolean;
56     description
57         "True if MDI power is supported";
58     reference
59         "IEEE Std 802.3, 30.12.3.1.6";
60 }
```

```
1      }
2      leaf mdi-power-enabled {
3          type boolean;
4          description
5              "True if MDI power is enabled";
6          reference
7              "IEEE Std 802.3, 30.12.3.1.7";
8      }
9
10     leaf power-pair-controlable {
11         type boolean;
12         description
13             "True if pair selection can be controlled";
14         reference
15             "IEEE Std 802.3, 30.12.3.1.8";
16     }
17     leaf power-pairs {
18         type pse-pinout-type;
19         description
20             "Indicates which pinout alternative is used for PD
21              detection and power";
22         reference
23             "IEEE Std 802.3, 30.12.3.1.9";
24     }
25
26     leaf power-class {
27         type pse-power-class-type;
28         description
29             "PD Power Class";
30         reference
31             "IEEE Std 802.3, 30.12.3.1.10";
32     }
33
34     /* Obsolete Link Aggregation objects
35        Link aggregation has not been part of 802.3 since IEEE 802.1AX-2008
36        */
37     leaf link-aggregation-status {
38         type bits {
39             bit aggregation-capability {
40                 position 0;
41                 description
42                     "IEEE Std 802.3, 79.3.3.1";
43             }
44             bit aggregation-status {
45                 position 1;
46                 description
47                     "IEEE Std 802.3, 79.3.3.1";
48             }
49             bit bit2-reserved {
50                 position 2;
51                 description
52                     "IEEE Std 802.3, 79.3.3.1";
53             }
54             bit bit3-reserved {
55                 position 3;
56                 description
57                     "IEEE Std 802.3, 79.3.3.1";
58             }
59             bit bit4-reserved {
60                 position 4;
61                 description
62                     "IEEE Std 802.3, 79.3.3.1";
63             }
64         }
65     }
```

```
1      }
2      bit bit5-reserved {
3          position 5;
4          description
5              "IEEE Std 802.3, 79.3.3.1";
6      }
7      bit bit6-reserved {
8          position 6;
9          description
10             "IEEE Std 802.3, 79.3.3.1";
11     }
12     bit bit7-reserved {
13         position 7;
14         description
15             "IEEE Std 802.3, 79.3.3.1";
16     }
17 }
18 }
19 status obsolete;
20 description
21     "The bitmap value which contains the link aggregation
22     capabilities and the current aggregation status
23     of the link";
24 reference
25     "IEEE Std 802.3, 30.12.3.1.11";
26 }
27 leaf aggregation-port-id {
28     type int32;
29     status obsolete;
30     description
31         "The unique identifier allocated to this Aggregation Port
32         by the local System.";
33     reference
34         "IEEE Std 802.3, 30.12.3.1.12";
35 }
36 leaf local-max-frame-size {
37     type int32;
38     status obsolete;
39     description
40         "An integer value indicating the maximum supported frame
41         size in octets on the given port of the local system.";
42     reference
43         "IEEE Std 802.3, 30.12.3.1.13";
44 }
45 /* POE objects
46     Consistency errors between PoE definitions in 30.9 (PoE),
47     30.12 (LLDP), 79.3.2 (PoE TLV), 79.3.8 (PoE measurement TLV).
48     Need to get these clauses reviewed against clause 145 and each other
49     */
50 leaf power-type {
51     type bits {
52         bit type1-or-greater {
53             position 0;
54             description
55                 "0-type1, 1-greater than type1";
56         }
57         bit pse-or-pd {
58             position 1;
59             description
60                 "0-pse, 1-pd";
```

```
1      }
2    }
3    status obsolete;
4    description
5      "A read-only attribute that returns a bit string indicating
6      whether the local system is a PSE or a PD and whether it
7      is Type 1 or greater than Type 1. The first bit indicates
8      Type 1 or greater than Type 1. The second bit indicates
9      PSE or PD. A PSE sets this bit to indicate a PSE. A PD
10     sets this bit to indicate a PD. See also
11     aLldpXdot3LocPowerTypeExt.";
12    reference
13      "IEEE Std 802.3, 30.12.3.1.14";
14  }
15  leaf power-source {
16    type power-source-type;
17    description
18      "Indicates the power sources of the remote system. A PSE
19      indicates whether it is being powered by a primary power
20      source; a backup power source; or unknown. A PD indicates
21      whether it is being powered by a PSE and locally; by a PSE
22      only; or unknown.";
23    reference
24      "IEEE Std 802.3, 30.12.3.1.15";
25  }
26  leaf power-priority {
27    type power-priority-type;
28    description
29      "The priority of the PD system received from the remote
30      system";
31    reference
32      "IEEE Std 802.3, 30.12.3.1.16";
33  }
34  leaf pd-requested-power-value {
35    type int32;
36    description
37      "PD requested power value that was used by the remote
38      system to compute the power value that is has currently
39      allocated to the PD.";
40    reference
41      "IEEE Std 802.3, 30.12.3.1.17";
42  }
43  leaf pd-requested-power-value-a {
44    type int32;
45    description
46      "A read-only attribute that returns the PD requested power
47      value for the Mode A pairset that was used by the remote
48      system to compute the power value that it has currently
49      allocated to the PD. For a PSE, it is the PD requested
50      power value for the Alternative A pairset received from
51      the remote system. For a PD, it is the PD requested power
52      value for the Alternative A pairset that the PSE echoes
53      back to the remote system. The definition and encoding of
54      PD requested power value for the Mode A pairset is the same
55      as described in aLldpXdot3LocPDRequestedPowerValueA";
56    reference
57      "IEEE Std 802.3, 30.12.3.1.18";
58  }
59  leaf pd-requested-power-value-b {
```

```
1      type int32;
2      description
3          "A read-only attribute that returns the PD requested power
4           value for the Mode B pairset that was used by the remote
5           system to compute the power value that it has currently
6           allocated to the PD. For a PSE, it is the PD requested
7           power value for the Alternative B pairset received from
8           the remote system. For a PD, it is the PD requested power
9           value for the Alternative B pairset that the PSE echoes
10          back to the remote system. The definition and encoding of
11          PD requested power value for the Mode B pairset is the
12          same as described in aLldpXdot3LocPDRequestedPowerValueB";
13      reference
14          "IEEE Std 802.3, 30.12.3.1.19";
15  }
16  leaf pse-allocated-power-value {
17      type int32;
18      description
19          "PSE allocated power value. For a PSE, it is the power value
20           that the PSE has currently allocated to the remote system.
21           For a PD, it is the power value that the PD mirrors back to
22           the remote system";
23      reference
24          "IEEE Std 802.3, 30.12.3.1.20";
25  }
26  leaf pse-allocated-power-value-a {
27      type int32;
28      description
29          "A read-only attribute that returns the PSE allocated power
30           value for the Alternative A pairset received from the
31           remote system. For a PSE, it is the PSE allocated power
32           value for the Alternative A pairset that was echoed back
33           by the remote PD. For a PD, it is the PSE allocated power
34           value for the Mode A pairset received from the remote
35           system. The definition and encoding of PSE allocated power
36           value for the Alternative A pairset is the same as
37           described in aLldpXdot3LocPSEAllocatedPowerValueA";
38      reference
39          "IEEE Std 802.3, 30.12.3.1.21";
40  }
41  leaf pse-allocated-power-value-b {
42      type int32;
43      description
44          "A read-only attribute that returns the PSE allocated power
45           value for the Alternative B pairset received from the
46           value for the Alternative B pairset that was echoed back
47           by the remote PD. For a PD, it is the PSE allocated power
48           value for the Mode B pairset received from the remote
49           system. The definition and encoding of PSE allocated power
50           value for the Alternative B pairset is the same as
51           described in aLldpXdot3LocPSEAllocatedPowerValueB";
52      reference
53          "IEEE Std 802.3, 30.12.3.1.22";
54  }
55  leaf pse-powering-status {
56      type powering-status-type;
57      description
58          "A read only value that indicates the powering status of
59           the remote PSE. For a PD, the contents of this attribute
```

```
1         are undefined.";
2     reference
3         "IEEE Std 802.3, 30.12.3.1.23";
4 }
5 leaf pd-powered-status {
6     type powered-status-type;
7     description
8         "A read only value that indicates the powering status of
9         the PD. For a PSE, the contents of this attribute are
10        undefined";
11    reference
12        "IEEE Std 802.3, 30.12.3.1.24";
13 }
14 leaf power-pairs-ext {
15     type power-pairs-type;
16     description
17         "A read-only value that identifies the supported PSE
18         Pinout Alternative specified in 145.2.4. For a PD, this
19         attribute contains the value of the aPSEPowerPairs
20         attribute (see 30.9.1.1.4). For a PSE, the contents of
21         this attribute are undefined";
22    reference
23        "IEEE Std 802.3, 30.12.3.1.25";
24 }
25 leaf power-class-ext-A {
26     type power-class-ext-AB-type;
27     description
28         "For a dual-signature PD, a read-only value that indicates
29         the currently assigned Class for Mode A by the remote
30         4-pair PSE. For a single-signature PD or a dual-signature
31         PD connected to a 2-pair only PSE, a read-only value set
32         to â€šinglesigâ€™ by the remote PSE. For a PSE connected to a
33         dual-signature PD, a read-only value that indicates the
34         requested Class for Mode A during Physical Layer
35         classification (see 145.2.8) by the remote PD. For a PSE
36         connected to a single-signature PD, a read-only value set
37         to â€šinglesigâ€™ by the remote PD";
38    reference
39        "IEEE Std 802.3, 30.12.3.1.26";
40 }
41 leaf power-class-ext-B {
42     type power-class-ext-AB-type;
43     description
44         "For a dual-signature PD, a read-only value that indicates
45         the currently assigned Class for Mode B by the remote
46         4-pair PSE. For a single-signature PD or a dual-signature
47         PD connected to a 2-pair only PSE, a read-only value set
48         to â€šinglesigâ€™ by the remote PSE. For a PSE connected to a
49         dual-signature PD, a read-only value that indicates the
50         requested Class for Mode B during Physical Layer
51         classification (see 145.2.8) by the remote PD. For a PSE
52         connected to a single-signature PD, a read-only value set
53         to â€šinglesigâ€™ by the remote PD";
54    reference
55        "IEEE Std 802.3, 30.12.3.1.27";
56 }
57 leaf power-class-ext {
58     type power-class-ext-type;
59     description
```

```
1      "For a single-signature PD or a dual-signature PD connected
2      to a 2-pair only PSE, a read-only value that indicates the
3      currently assigned Class by the remote PSE. For a
4      dual-signature PD connected to a 4-pair capable PSE, a
5      read-only value set to "Dualsig" by the remote PSE. For a
6      PSE connected to a single-signature PD, a read-only value
7      that indicates the requested Class during Physical Layer
8      classification (see 145.2.8) by the remote PD. For a PSE
9      connected to a dual-signature PD, a read-only value set to
10     "Dualsig" by the remote PD.";
11     reference
12         "IEEE Std 802.3, 30.12.3.1.28";
13 }
14 leaf power-type-ext {
15     type power-type;
16     description
17         "A read-only attribute that returns a value to indicate if
18         the remote system is a Type 3 or Type 4 PSE or PD and, in
19         the case of a Type 3 or Type 4 PD, if it is a
20         single-signature PD or dual-signature PD.";
21     reference
22         "IEEE Std 802.3, 30.12.3.1.29";
23 }
24 leaf pd-load {
25     type boolean;
26     description
27         "For a PSE, a read-only attribute that returns whether the
28         load of the remote dual-signature PD is electrically
29         isolated, as defined in 79.3.2.10.2. For a PD, this
30         attribute is set to FALSE.";
31     reference
32         "IEEE Std 802.3, 30.12.3.1.30";
33 }
34 leaf pd-4pid {
35     type boolean;
36     description
37         "A read-only Boolean attribute indicating whether the
38         remote PD system supports powering of both PD Modes.";
39     reference
40         "IEEE Std 802.3, 30.12.3.1.31";
41 }
42 leaf pse-max-avail-power {
43     type int32;
44     description
45         "A read-only attribute that returns the remote PSE maximum
46         available power value in units of 0.1 W";
47     reference
48         "IEEE Std 802.3, 30.12.3.1.32";
49 }
50 leaf pse-autoclass-support {
51     type boolean;
52     description
53         "Indicates whether the remote PSE system supports
54         Autoclass.";
55     reference
56         "IEEE Std 802.3, 30.12.3.1.33";
57 }
58 leaf autoclass-completed {
59     type boolean;
```

```
1      description
2          "Indicates whether the remote PSE system has completed the
3          Autoclass measurement.";
4      reference
5          "IEEE Std 802.3, 30.12.3.1.34";
6  }
7  leaf autoclass-request {
8      type boolean;
9      description
10         "A read-only Boolean attribute indicating whether the
11         remote PD system is requesting an Autoclass measurement.";
12     reference
13         "IEEE Std 802.3, 30.12.3.1.35";
14 }
15 leaf power-down-request {
16     type int32;
17     description
18         "A read-write attribute that indicates the remote PD system
19         is requesting a power down when the value is 0x1D.";
20     reference
21         "IEEE Std 802.3, 30.12.3.1.36";
22 }
23 leaf power-down-time {
24     type int32;
25     description
26         "A read-only attribute that indicates the number of seconds
27         the remote PD requests to stay powered off. A value of zero
28         indicates an indefinite amount of time";
29     reference
30         "IEEE Std 802.3, 30.12.3.1.37";
31 }
32 /* PoE measurement objects */
33 leaf meas-voltage-support {
34     type boolean;
35     description
36         "A read-only attribute that indicates the remote device is
37         capable of providing a voltage measurement.";
38     reference
39         "IEEE Std 802.3, 30.12.3.1.38";
40 }
41 leaf meas-current-support {
42     type boolean;
43     description
44         "A read-only attribute that indicates the remote device is
45         capable of providing a current measurement.";
46     reference
47         "IEEE Std 802.3, 30.12.3.1.39";
48 }
49 leaf meas-power-support {
50     type boolean;
51     description
52         "A read-only attribute that indicates the remote device is
53         capable of providing a power measurement.";
54     reference
55         "IEEE Std 802.3, 30.12.3.1.40";
56 }
57 leaf meas-energy-support {
58     type boolean;
59     description
```

```
1         "A read-only attribute that indicates the remote device is
2         capable of providing a energy measurement.>";
3     reference
4         "IEEE Std 802.3, 30.12.3.1.41 ";
5 }
6 leaf measurement-source {
7     type bits {
8         bit bit1 {
9             position 0;
10            description
11                "-";
12        }
13        bit bit2 {
14            position 1;
15            description
16                "-";
17        }
18    }
19 }
20 status obsolete;
21 description
22     "A read-write attribute value that indicates on which
23     Alternative or Mode the measurement was taken by the remote
24     device.
25     Superseded by rem-measurement-source.";
26 reference
27     "IEEE Std 802.3, 30.12.3.1.42";
28 }
29 leaf rem-measurement-source {
30     type measurement-source-type;
31     description
32         "A read-only attribute that indicates which Alternative or
33         Mode the measurement is to be taken";
34     reference
35         "IEEE Std 802.3, 30.12.3.1.42";
36 }
37 leaf meas-voltage-request {
38     type boolean;
39     description
40         "A read-only attribute that indicates the remote device is
41         requesting a voltage measurement from the local device.";
42     reference
43         "IEEE Std 802.3, 30.12.3.1.43";
44 }
45 leaf meas-current-request {
46     type boolean;
47     description
48         "A read-only attribute that indicates the remote device is
49         requesting a current measurement from the local device.";
50     reference
51         "IEEE Std 802.3, 30.12.3.1.44";
52 }
53 leaf meas-power-request {
54     type boolean;
55     description
56         "A read-only attribute that indicates the remote device is
57         requesting a power measurement from the local device.";
58     reference
59         "IEEE Std 802.3, 30.12.3.1.45";
60 }
61 }
```

```
1     leaf meas-energy-request {
2         type boolean;
3         description
4             "A read-only attribute that indicates the remote device is
5             requesting an energy measurement from the local device.";
6         reference
7             "IEEE Std 802.3, 30.12.3.1.46";
8     }
9
10    leaf meas-voltage-valid {
11        type boolean;
12        description
13            "A read-only attribute that indicates the remote deviceâ€™s
14            voltage measurement is valid.";
15        reference
16            "IEEE Std 802.3, 30.12.3.1.47";
17    }
18
19    leaf meas-current-valid {
20        type boolean;
21        description
22            "A read-only attribute that indicates the remote deviceâ€™s
23            current measurement is valid.";
24        reference
25            "IEEE Std 802.3, 30.12.3.1.48";
26    }
27
28    leaf meas-power-valid {
29        type boolean;
30        description
31            "A read-only attribute that indicates the remote deviceâ€™s
32            power measurement is valid.";
33        reference
34            "IEEE Std 802.3, 30.12.3.1.49";
35    }
36
37    leaf meas-energy-valid {
38        type boolean;
39        description
40            "A read-only attribute that indicates the remote deviceâ€™s
41            energy measurement is valid.";
42        reference
43            "IEEE Std 802.3, 30.12.3.1.50";
44    }
45
46    leaf meas-voltage-uncertainty {
47        type int32;
48        description
49            "A read-only attribute that indicates the expanded
50            uncertainty (coverage factor k = 2) for the remote deviceâ€™s
51            voltage measurement.";
52        reference
53            "IEEE Std 802.3, 30.12.3.1.51";
54    }
55
56    leaf meas-current-uncertainty {
57        type int32;
58        description
59            "A read-only attribute that indicates the expanded
60            uncertainty (coverage factor k = 2) for the remote deviceâ€™s
61            current measurement.";
62        reference
63            "IEEE Std 802.3, 30.12.3.1.52";
64    }
65
66    leaf meas-power-uncertainty {
```

```
1      type int32;
2      description
3          "A read-only attribute that indicates the expanded
4            uncertainty (coverage factor k = 2) for the remote device's
5            power measurement.";
6      reference
7          "IEEE Std 802.3, 30.12.3.1.53";
8  }
9
10     leaf meas-energy-uncertainty {
11         type int32;
12         description
13             "A read-only attribute that indicates the expanded
14               uncertainty (coverage factor k = 2) for the remote device's
15               energy measurement.";
16         reference
17             "IEEE Std 802.3, 30.12.3.1.54";
18     }
19
20     leaf voltage-measurement {
21         type int32;
22         description
23             "A read-only attribute that returns the measured remote
24               device voltage.";
25         reference
26             "IEEE Std 802.3, 30.12.3.1.55";
27     }
28
29     leaf current-measurement {
30         type int32;
31         description
32             "A read-only attribute that returns the measured remote
33               device current.";
34         reference
35             "IEEE Std 802.3, 30.12.3.1.56";
36     }
37
38     leaf power-measurement {
39         type int32;
40         description
41             "A read-only attribute that returns the measured remote
42               device power.";
43         reference
44             "IEEE Std 802.3, 30.12.3.1.57";
45     }
46
47     leaf energy-measurement {
48         type int32;
49         description
50             "A read-only attribute that returns the measured remote
51               device energy.";
52         reference
53             "IEEE Std 802.3, 30.12.3.1.58";
54     }
55
56     leaf pse-power-price-index {
57         type int32;
58         description
59             "A read-only attribute that returns an index of the price
60               of power being sourced by the remote PSE. For a PSE, this
61               value is undefined.";
62         reference
63             "IEEE Std 802.3, 30.12.3.1.59";
64     }
65
66     leaf tx-system-value {
```

```
1      type int32;
2      description
3          "Returns the value of Tw_sys_tx that the remote system can
4          support in the transmit direction.";
5      reference
6          "IEEE Std 802.3, 30.12.3.1.60";
7  }
8
9  leaf tx-system-value-echo {
10     type int32;
11     description
12         "Returns the value of Tw_sys_tx that the local system is
13         advertising that it can support in the transmit direction
14         and is echoed by the local system under the control of the
15         EEE DLL receiver state diagram.";
16     reference
17         "IEEE Std 802.3, 30.12.3.1.61";
18 }
19
20 leaf rx-system-value {
21     type int32;
22     description
23         "Returns the value of Tw_sys_tx that the remote system is
24         requesting in the receive direction.";
25     reference
26         "IEEE Std 802.3, 30.12.3.1.62";
27 }
28
29 leaf rx-system-value-echo {
30     type int32;
31     description
32         "Returns the value of Tw_sys_tx that the local system is
33         advertising that it is requesting in the receive direction
34         and is echoed by the local system under the control of the
35         EEE DLL transmitter state diagram.";
36     reference
37         "IEEE Std 802.3, 30.12.3.1.63 ";
38 }
39
40 leaf fallback-system-value {
41     type int32;
42     description
43         "Returns the value of the fallback Tw_sys_tx that the
44         remote system is advertising to the remote system.";
45     reference
46         "IEEE Std 802.3, 30.12.3.1.64";
47 }
48 /* LDP EEE objects */
49 leaf tx-system-fw {
50     type boolean;
51     description
52         "Returns the value of LPI_FW that the remote system can
53         support in the transmit direction.";
54     reference
55         "IEEE Std 802.3, 30.12.3.1.65";
56 }
57
58 leaf tx-system-fw-echo {
59     type boolean;
60     description
61         "Returns the value of LPI_FW that the local system is
62         advertising that it can support in the transmit direction
63         and is echoed by the local system under the control of the
64         EEE DLL receiver state diagram.";
```

```

1      reference
2          "IEEE Std 802.3, 30.12.3.1.66";
3      }
4      leaf rx-system-fw {
5          type boolean;
6          description
7              "Returns the value of LPI_FW that the remote system is
8              requesting in the receive direction.";
9          reference
10             "IEEE Std 802.3, 30.12.3.1.67";
11      }
12      leaf rx-system-fw-echo {
13          type boolean;
14          description
15              "Returns the value of LPI_FW that the local system is
16              advertising that it is requesting in the receive direction
17              and is echoed by the local system under the control of the
18              IEEE DDL transmitter state diagram.";
19          reference
20             "IEEE Std 802.3, 30.12.3.1.68";
21      }
22      /* LLDP preemption objects */
23      leaf preemption-supported {
24          type boolean;
25          description
26              "Indicates whether the given port (associated with the
27              remote System) supports the preemption capability.";
28          reference
29             "IEEE Std 802.3, 30.12.3.1.69";
30      }
31      leaf preemption-enabled {
32          type boolean;
33          description
34              "Indicates whether the preemption capability is enabled on
35              the given port associated with the remote System.";
36          reference
37             "IEEE Std 802.3, 30.12.3.1.70";
38      }
39      leaf preemption-active {
40          type boolean;
41          description
42              "Indicates whether the preemption capability is active on
43              the given port associated with the remote System.";
44          reference
45             "IEEE Std 802.3, 30.12.3.1.72";
46      }
47      leaf additional-fragment-size {
48          type int32;
49          description
50              "Indicate the minimum size of non-final fragments supported
51              by the receiver on the given port associated with the
52              remote System. This value is expressed in units of 64
53              octets of additional fragment length.";
54          reference
55             "IEEE Std 802.3, 30.12.3.1.72 ";
56      }
57  }
58  }
59  }
60  }
61  }
62  }
63  }
64  }
65  }

```

5.3.2.5 Ethernet PHY

Editor's Note (to be removed prior to publication):

Pretty printing of ieee802-ethernet-phy.yang file may change the appearance by adding whitespace and reformatting lines

Editor's Note (to be removed prior to publication):

IEEE Std 802.3.1 and IEEE Std 802.3.2 to be updated at the publication time

```
module ieee802-ethernet-phy {
  yang-version 1.1;
  namespace "urn:ieee:std:802.3:yang:ieee802-ethernet-phy";
  prefix ieee802-eth-phy;

  import ietf-yang-types {
    prefix yang;
    reference
      "IETF RFC 6991 YANG Types";
  }
  import iana-if-type {
    prefix ianaif;
    reference
      "IANA Interface Types";
  }
  import ietf-interfaces {
    prefix if;
    reference
      "IETF RFC 8343";
  }
  import ieee802-ethernet-phy-type {
    prefix ieee802-phy;
    reference
      "IEEE Std 802.3-2022";
  }

  organization
    "IEEE Std 802.3 Ethernet Working Group";
  contact
    "Web URL: http://www.ieee802.org/3/";
  description
    "This module contains YANG definitions for 802.3 PHYs and PMDs.";

  revision 2025-04-17 {
    description
      "Initial revision derived from IEEE8023-MAU-MIB
        using smidump.";
    reference
      "IEEE Std 802.3-2022, unless dated explicitly";
  }
}
```

```
1     }
2
3     feature fec-support {
4         description
5             "This object indicates support for one or more FEC modes.";
6     }
7
8
9     feature baset-snr-support {
10        description
11            "This object indicates support for SNR measurement for
12            BASE-T interfaces.";
13        }
14
15
16        feature time-sync-support {
17            description
18                "This object indicates support for time synchronisation.";
19        }
20
21
22        feature pcs-lane-statistics-support {
23            description
24                "This object indicates support for PCS Lane statistics.";
25        }
26
27
28        feature auto-negotiation-support {
29            description
30                "This object indicates support for Auto-Negotiation.";
31        }
32
33
34        typedef snr-value {
35            type int32 {
36                range "-127..127";
37            }
38            units "0.1dB";
39            description
40                "SNR units of 0.1 dB to an accuracy of 0.5 dB within
41                the range of -12.7 dB to 12.7 dB.";
42        }
43
44
45        typedef counter-1000-second {
46            type yang:counter64;
47            description
48                "This counter increments no more than 1000 times per second.";
49        }
50
51
52        typedef counter-5000-second {
53            type yang:counter64;
54            description
55                "This counter increments no more than 5000 times per second.";
56        }
57
58
59        typedef counter-pcs-lane {
60            type yang:counter64;
61            description
62                "This counter has a maximum per second increment rate of:
```

```

1      1.2 million for 1000 Mb/s
2      5 million for 10 Gb/s and 40 Gb/s
3      2.5 million for 100 Gb/s";
4  }
5
6
7  typedef enabled-state {
8      type enumeration {
9          enum unknown {
10             description
11                 "Initializing.";
12         }
13         enum disabled {
14             description
15                 "Disabled";
16         }
17         enum enabled {
18             description
19                 "Enabled";
20         }
21     }
22     description
23         "Enabled, disabled or unknown.";
24 }
25
26
27
28
29
30 augment "/if:interfaces/if:interface" {
31     when "derived-from-or-self(if:type,'ianaift:ethernetCsmacd') " {
32         description
33             "Applies to all Ethernet interfaces.";
34     }
35     description
36         "Augment interface model with Ethernet interface
37         information";
38     container phy-table {
39         description
40             "Table of descriptive and status information
41             about the Ethernet interface physical layer.";
42         container control {
43             description
44                 "Control information for the Ethernet interface
45                 physical layer.";
46             leaf phy-reset {
47                 type boolean;
48                 description
49                     "Setting this to 'True' resets the interface in the
50                     same manner as would a power-off, power-on cycle of
51                     at least 0.5 s duration.";
52                 reference
53                     "IEEE Std 802.3, 30.5.1.2.1 acResetMAU";
54             }
55         }
56         leaf phy-admin-control {
57             type enumeration {
58                 enum operational {
59                     description
60                         "Operational";
61                 }
62             }
63         }
64     }
65 }

```

```

1         }
2         enum standby {
3             description
4                 "Standby";
5         }
6         enum shutdown {
7             description
8                 "Shutdown";
9         }
10    }
11    }
12    }
13    description
14        "The configured state of the interface. This object
15        MAY be implemented as a read-only object by devices
16        that do not implement software control of the interface
17        state. Some devices may not support setting the value
18        of this object to some of the enumerated values.
19
20        This can be set to "standby", "operational", and
21        "shutdown" state, except that a "standby" action
22        to a multidrop interface will cause the interface
23        to enter the "shutdown" management state.";
24    reference
25        "IEEE Std 802.3, 30.5.1.2.2 acMAUAdminControl";
26    }
27    container fec-control {
28        if-feature "fec-support";
29        presence "The port supports one or more FEC modes.";
30        description
31            "FEC Control.";
32        leaf fec-mode-control {
33            type enumeration {
34                enum disabled {
35                    description
36                        "FEC disabled";
37                }
38                enum enabled {
39                    description
40                        "FEC enabled";
41                }
42                enum BASE-R-enabled {
43                    description
44                        "BASE-R FEC enabled";
45                }
46                enum RS-FEC-enabled {
47                    description
48                        "RS-FEC enabled";
49                }
50                enum RS-FEC-Int-enabled {
51                    description
52                        "RS-FEC-Int enabled";
53                }
54            }
55        }
56        description
57            "The FEC mode across the MDI of the port.

```

```
1         BASE-R-enabled, RS-FEC-enabled and
2         RS-FEC-Int-enabled are only used for PHYs which
3         support more than one type of FEC operation.
4         When Clause 73 Auto-Negotiation is enabled, this
5         leaf becomes read-only.";
6     reference
7         "IEEE Std 802.3, 30.5.1.1.16";
8 }
9
10 leaf rs-fec-bypass-control {
11     type enabled-state;
12     description
13         "Is RS-FEC bypass enabled?";
14     reference
15         "IEEE Std 802.3, 30.5.1.1.31";
16 }
17 }
18 }
19 }
20 }
21 container status {
22     config false;
23     description
24         "Status for the port.";
25     leaf phy-type {
26         type identityref {
27             base ieee802-phy:phy-type;
28         }
29         description
30             "This object identifies the PHY type.";
31         reference
32             "IEEE Std 802.3, 30.5.1.1.2, aPHYType.";
33     }
34     leaf pmd-type {
35         type identityref {
36             base ieee802-phy:pmd-type;
37         }
38         description
39             "This object identifies the PMD type.";
40         reference
41             "IEEE Std 802.3, 30.5.1.1.2, aMAUType.";
42     }
43     leaf pmd-type-list {
44         type ieee802-phy:pmd-type-list;
45         description
46             "PMD types this interface supports.";
47     }
48     leaf phy-status {
49         type enumeration {
50             enum unknown {
51                 description
52                     "Initializing, true state
53                     not yet known";
54             }
55             enum operational {
56                 description
57                     "Powered and connected";
58             }
59         }
60     }
61 }
```

```
1         }
2         enum standby {
3             description
4                 "Inactive but on";
5         }
6         enum shutdown {
7             description
8                 "Inactive and powered down";
9         }
10        }
11    }
12    description
13        "The current state of the interface.";
14    reference
15        "IEEE Std 802.3, 30.5.1.1.7, aMAUAdminState";
16    }
17    leaf phy-media-available {
18        type ieee802-phy:media-available-type;
19        description
20            "This object reports if the media is available.";
21        reference
22            "IEEE Std 802.3, 30.5.1.1.4, aMediaAvailable.";
23    }
24    /* Jabber */
25    leaf jabber-state {
26        type enumeration {
27            enum other {
28                description
29                    "undefined";
30            }
31            enum unknown {
32                description
33                    "initializing, true state
34                    not yet known";
35            }
36            enum normal {
37                description
38                    "state is true or normal";
39            }
40            enum fault {
41                description
42                    "state is false, fault, or
43                    abnormal";
44            }
45        }
46    }
47    description
48        "The JABBER state of the multidrop interface.";
49    reference
50        "IEEE Std 802.3, 30.5.1.1.6, aJabber.jabberFlag.";
51    }
52    leaf jabbering-state-enters {
53        type yang:counter64;
54        description
55            "The number of times this interface detected JABBER.";
56        reference
```

```
1         "IEEE Std 802.3,  
2         30.5.1.1.6, aJabber.jabberCounter.";  
3     }  
4     /* FEC */  
5     container fec-status {  
6         if-feature "fec-support";  
7         presence "The port supports one or more FEC modes.";  
8         description  
9             "FEC Status.";  
10        leaf fec-ability {  
11            type bits {  
12                bit other {  
13                    description  
14                        "Other";  
15                }  
16                bit base-r {  
17                    description  
18                        "BASE-R FEC supported";  
19                }  
20                bit rs-fec {  
21                    description  
22                        "RS-FEC supported";  
23                }  
24                bit rs-fec-bypass {  
25                    description  
26                        "RS-FEC bypass supported";  
27                }  
28                bit rs-fec-bypass-indication {  
29                    description  
30                        "RS-FEC error indication bypass supported";  
31                }  
32                bit pcs-fec-bypass {  
33                    description  
34                        "PCS FEC error bypass supported";  
35                }  
36                bit pcs-fec-bypass-indication {  
37                    description  
38                        "PCS FEC error indication bypass supported";  
39                }  
40            }  
41            description  
42                "The FEC modes this port supports.";  
43            reference  
44                "IEEE Std 802.3, 30.5.1.1.15, 30.5.1.1.16  
45                30.5.1.1.29, 30.5.1.1.30, 30.5.1.1.33,  
46                30.5.1.1.34";  
47        }  
48        leaf fec-mode-status {  
49            type enumeration {  
50                enum unknown {  
51                    description  
52                        "Initializing, true state not yet known.";  
53                }  
54                enum disabled {  
55
```

```

1         description
2             "FEC disabled";
3     }
4     enum enabled {
5         description
6             "FEC enabled";
7     }
8     enum BASE-R-enabled {
9         description
10            "BASE-R FEC enabled";
11    }
12    enum RS-FEC-enabled {
13        description
14            "RS-FEC enabled";
15    }
16    enum RS-FEC-Int-enabled {
17        description
18            "RS-FEC-Int enabled";
19    }
20    }
21    description
22        "The FEC mode of the PHY.
23        BASE-R-enabled, RS-FEC-enabled and
24        RS-FEC-Int-enabled are only used by PHYs which
25        support more than one type of FEC operation.";
26    reference
27        "IEEE Std 802.3, 30.5.1.1.16";
28    }
29    leaf rs-fec-bypass-enabled-status {
30        type enabled-state;
31        description
32            "Is RS-FEC bypass indication enabled?";
33        reference
34            "IEEE Std 802.3, 30.5.1.1.32";
35    }
36    }
37    /* MultiGBASE-T SNR */
38    container baset-snr {
39        if-feature "baset-snr-support";
40        presence "The port supports BASE-T SNR measurements.";
41        description
42            "BASE-T SNR Status.";
43        leaf snr-op-margin-chnl-a {
44            type snr-value;
45            description
46                "The current SNR operating margin on channel A
47                on this port for MultiGBASE-T PMAs.";
48            reference
49                "IEEE Std 802.3, 30.5.1.1.19";
50        }
51        leaf snr-op-margin-chnl-b {
52            type snr-value;
53            description
54                "The current SNR operating margin on channel B

```

```
1         on this port for MultiGBASE-T PMAs.";
2     reference
3         "IEEE Std 802.3, 30.5.1.1.20";
4 }
5 leaf snr-op-margin-chnl-c {
6     type snr-value;
7     description
8         "The current SNR operating margin on channel C
9         on this port for MultiGBASE-T PMAs.";
10    reference
11        "IEEE Std 802.3, 30.5.1.1.21";
12 }
13 leaf snr-op-margin-chnl-d {
14     type snr-value;
15     description
16         "The current SNR operating margin on channel CD
17         on this port for MultiGBASE-T PMAs.";
18    reference
19        "IEEE Std 802.3, 30.5.1.1.22";
20 }
21 }
22 /* TimeSync */
23 container time-sync {
24     if-feature "time-sync-support";
25     presence "The port supports Time Sync.";
26     description
27         "Time Sync Status.";
28     leaf time-sync-ability {
29         type bits {
30             bit transmit {
31                 description
32                     "Transmit";
33             }
34             bit receive {
35                 description
36                     "Receive";
37             }
38         }
39     }
40     description
41         "The TimeSync capabilities of this port.";
42     reference
43         "IEEE Std 802.3, 30.13.1.1, 30.13.1.2";
44 }
45 leaf time-sync-delay-tx-max {
46     type uint32;
47     units "nano-seconds";
48     description
49         "The maximum transmit data delay between
50         the xMII input to the MDI output.";
51     reference
52         "IEEE Std 802.3, 30.13.1.3";
53 }
54 leaf time-sync-delay-tx-min {
55     type uint32;
```

```
1         units "nano-seconds";
2         description
3             "The minimum transmit data delay between
4             the xMII input to the MDI output.";
5         reference
6             "IEEE Std 802.3, 30.13.1.4";
7     }
8     leaf time-sync-delay-rx-max {
9         type int32;
10        units "nano-seconds";
11        description
12            "The maximum receive data delay between
13            the MDI input to the xMII output.";
14        reference
15            "IEEE Std 802.3, 30.13.5";
16    }
17    leaf time-sync-delay-rx-min {
18        type int32;
19        units "nano-seconds";
20        description
21            "The minimum receive data delay between
22            the MDI input to the xMII output.";
23        reference
24            "IEEE Std 802.3, 30.13.5";
25    }
26 }
27
28 container statistics {
29     config false;
30     description
31         "Counters for the port.";
32     leaf phy-media-losts {
33         type yang:counter64;
34         description
35             "The number of times phy-media-available changed
36             from 'available'.";
37         reference
38             "IEEE Std 802.3, 30.5.1.1.5 aLoseMediaCounter.";
39     }
40     leaf jabbering-state-enters {
41         type yang:counter64;
42         description
43             "The number of times this interface detected JABBER.";
44         reference
45             "IEEE Std 802.3, 30.5.1.1.6, aJabber.jabberCounter.";
46     }
47     leaf false-carriers {
48         type yang:counter64;
49         description
50             "The number of false carrier
51             events during IDLE in 100BASE-X and
52             1000BASE-X links.
53
54             For all other interface types, this counter
```

```
1         will always indicate zero.
2
3         It can increment at a maximum rate of once
4         per 100 ms for 100BASE-X and once per 10us
5         for 1000BASE-X.";
6     reference
7         "IEEE Std 802.3, 30.5.1.1.10, aFalseCarriers.";
8 }
9
10 leaf pcs-coding-violations {
11     type yang:counter64;
12     description
13         "The number of PCS coding violations for 100Mb/s and
14         1000Mb/s ports.
15
16         This counter increments at a maximum rate of 25 thousand
17         per second for 100 Mb/s implementations and
18         125 thousand per second for 1000 Mb/s
19         implementations.
20
21         The contents of this
22         attribute is undefined when FEC is operating.";
23     reference
24         "IEEE Std 802.3, 30.5.1.1.14";
25 }
26 leaf fast-retrain-count-local {
27     type counter-1000-second;
28     description
29         "The number of fast retrains initiated
30         by the local device.";
31     reference
32         "IEEE Std 802.3, 30.5.1.1.24";
33 }
34 leaf fast-retrain-count-peer {
35     type counter-1000-second;
36     description
37         "The number of fast retrains initiated
38         by the link partner.";
39     reference
40         "IEEE Std 802.3, 30.5.1.1.25";
41 }
42 }
43
44 container pcs-lane-statistics {
45     if-feature "pcs-lane-statistics-support";
46     config false;
47     description
48         "Per PCS lane counters for the port.
49         These counters apply to the following:
50         1000BASE-PX,
51         10/25/40/50/100/200/400GBASE-R
52         100GBASE-P
53         10GBASE-PR
54         10/1GBASE-PRX";
55     list pcs-lane {
56         key "pcs-lane-index";
57     }
58 }
```

```
1         description
2             "Per PCS lane counters for the port.";
3         leaf pcs-lane-index {
4             type uint32 {
5                 range "0..255";
6             }
7             description
8                 "PCS lane index.";
9         }
10    }
11    leaf fec-corrected-blocks {
12        type counter-pcs-lane;
13        description
14            "The number blocks that FEC corrected.";
15        reference
16            "IEEE Std 802.3, 30.5.1.1.17";
17    }
18    leaf fec-uncorrected-blocks {
19        type counter-pcs-lane;
20        description
21            "The number blocks that FEC could not correct.";
22        reference
23            "IEEE Std 802.3, 30.5.1.1.18";
24    }
25 }
26 }
27 }
28 }
29 }
30 }
31 }
32 container auto-neg-table {
33     if-feature "auto-negotiation-support";
34     description
35         "Configuration and status objects for Auto-Negotiation
36         on Ethernet interfaces.";
37     container control {
38         description
39             "Auto-Negotiation control.";
40         leaf auto-neg-control {
41             type boolean;
42             default "true";
43             description
44                 "Enable/disable Auto-Negotiation for this interface.";
45             reference
46                 "IEEE Std 802.3, 30.6.1.1.2, acAutoNegAdminControl.";
47         }
48         leaf auto-neg-restart {
49             type boolean;
50             description
51                 "Setting this to true restarts Auto-Negotiation if enabled.";
52             reference
53                 "IEEE Std 802.3,
54                 30.6.1.2.1, acAutoNegRestartAutoConfig.";
55         }
56     }
57 }
58 }
59 container status {
60     config false;
61     description
```

```
1         "Auto-Negotiation status.";
2     leaf auto-neg-enabled {
3         type boolean;
4         description
5             "Is Auto-Negotiation enabled for this interface.";
6         reference
7             "IEEE Std 802.3,
8             30.6.1.1.2 aAutoNegAdminState.";
9     }
10
11     leaf auto-neg-remote-signaling {
12         type boolean;
13         description
14             "Is the link partner using Auto-Negotiation?";
15         reference
16             "IEEE Std 802.3, 30.6.1.1.3, aAutoNegRemoteSignaling.";
17     }
18
19     leaf auto-neg-status {
20         type enumeration {
21             enum other {
22                 description
23                     "Unknown state.";
24             }
25             enum configuring {
26                 description
27                     "Auto-Negotiation is active.";
28             }
29             enum complete {
30                 description
31                     "Auto-Negotiation is complete.";
32             }
33             enum disabled {
34                 description
35                     "Auto-Negotiation s disabled.";
36             }
37             enum parallelDetectFail {
38                 description
39                     "Auto-Negotiation failed because the
40                     link partner does not suport, or has
41                     disabled, Auto-Negotiation.";
42             }
43         }
44         description
45             "The current status of Auto-Negotiation.";
46         reference
47             "IEEE Std 802.3, 30.6.1.1.4, aAutoNegAutoConfig.";
48     }
49
50     leaf auto-neg-capability-bits {
51         type ieee802-phy:auto-neg-ability-bits;
52         description
53             "The capabilities of the local Auto-Negotiation entity.";
54         reference
55             "IEEE Std 802.3, 30.6.1.1.5,
56             aAutoNegLocalTechnologyAbility.";
57     }
58 }
```

```
1         leaf auto-neg-cap-advertised-bits {
2             type ieee802-phy:auto-neg-ability-bits;
3             description
4                 "The capabilities advertised by the local
5                 Auto-Negotiation entity.";
6             reference
7                 "IEEE Std 802.3, 30.6.1.1.6,
8                 aAutoNegadvertisedTechnologyAbility.";
9         }
10        leaf auto-neg-cap-received-bits {
11            type ieee802-phy:auto-neg-ability-bits;
12            description
13                "Auto-Negotiation capabilities received from
14                the link partner.";
15            reference
16                "IEEE Std 802.3, 30.6.1.1.7,
17                aAutoNegreceivedTechnologyAbility.";
18        }
19    }
20 }
21 }
22 }
23 }
```

5.3.2.6 Ethernet PHY Types

Editor's Note (to be removed prior to publication):

Pretty printing of ieee802-ethernet-phy-type.yang file may change the appearance by adding whitespace and reformatting lines

Editor's Note (to be removed prior to publication):

IEEE Std 802.3.1 and IEEE Std 802.3.2 to be updated at the publication time

```
43 module ieee802-ethernet-phy-type {
44     yang-version 1.1;
45     namespace "urn:ieee:std:802.3:yang:ieee802-ethernet-phy-type";
46     prefix ieee802-eth-phy-type;
47
48     organization
49         "IEEE Std 802.3 Ethernet Working Group
50         Web URL: http://www.ieee802.org/3/";
51     contact
52         "Web URL: http://www.ieee802.org/3/";
53     description
54         "This module contains YANG definitions for Physical Layer device
55         (PHY) and Physical Medium Dependent (PMD) identities and related
56         types.";
57
58     revision 2025-04-17 {
59         description
60             "Initial revision derived from IANA-MAU-MIB REVISION
61             202309070000Z using smidump.";
62     }
```

```
1      reference
2      "IEEE Std 802.3-2022, unless dated explicitly";
3  }
4
5  /* PHY type identities */
6
7
8  identity phy-type {
9      description
10         "Base type for PHY types.";
11      reference
12         "IEEE Std 802.3, Clause 30.3.2.1.2 aPhyType";
13  }
14
15
16  identity phy-type-other {
17      base phy-type;
18      description
19         "Undefined";
20  }
21
22
23  identity phy-type-unknown {
24      base phy-type;
25      description
26         "Initializing, true state or type not yet known";
27  }
28
29
30
31  identity phy-type-none {
32      base phy-type;
33      description
34         "MII present and nothing connected";
35  }
36
37
38  identity phy-type-10BASE-T {
39      base phy-type;
40      description
41         "Clause 14 10 Mb/s PAM3";
42  }
43
44
45  identity phy-type-10BASE-T1L {
46      base phy-type;
47      description
48         "Clause 146 10 Mb/s PAM3";
49  }
50
51
52  identity phy-type-10BASE-T1S {
53      base phy-type;
54      description
55         "Clause 147 10 Mb/s DME";
56  }
57
58
59
60  identity phy-type-100BASE-T1 {
61      base phy-type;
62      description
63         "Clause 96 100 Mb/s PAM3";
64  }
65  }
```

```
1
2 identity phy-type-100BASE-X {
3     base phy-type;
4     description
5         "Clause 24 or subclause 66.1 100 Mb/s 4B/5B";
6 }
7
8
9 identity phy-type-1000BASE-H {
10     base phy-type;
11     description
12         "Clause 115 1000 Mb/s PAM16-THP";
13 }
14
15
16 identity phy-type-1000BASE-T {
17     base phy-type;
18     description
19         "Clause 40 1000 Mb/s 4D-PAM5";
20 }
21
22
23 identity phy-type-1000BASE-T1 {
24     base phy-type;
25     description
26         "Clause 97 1000 Mb/s PAM3";
27 }
28
29
30 identity phy-type-1000BASE-X {
31     base phy-type;
32     description
33         "Clause 36 or subclause 66.2 1000 Mb/s 8B/10B";
34 }
35
36
37 identity phy-type-2.5GBASE-T {
38     base phy-type;
39     description
40         "Clause 126 2.5 Gb/s PAM16";
41 }
42
43
44 identity phy-type-2.5GBASE-T1 {
45     base phy-type;
46     description
47         "Clause 149 2.5 Gb/s PAM4";
48 }
49
50
51 identity phy-type-2.5GBASE-X {
52     base phy-type;
53     description
54         "Clause 127 2.5 Gb/s 8B/10B";
55 }
56
57
58 identity phy-type-5GBASE-R {
59     base phy-type;
60     description
61         "Clause 129 5 Gb/s 64/66B";
62 }
63
64
65 }
```

```
1
2 identity phy-type-5GBASE-T {
3     base phy-type;
4     description
5         "Clause 126 5 Gb/s PAM16";
6 }
7
8
9 identity phy-type-5GBASE-T1 {
10     base phy-type;
11     description
12         "Clause 149 5 Gb/s PAM4";
13 }
14
15
16 identity phy-type-10G-1GBASE-PRX {
17     base phy-type;
18     description
19         "Clause 76 10/1G-EPON 10 Gb/s 64B/66B downstream and
20         1 Gb/s 8B/10B upstream";
21 }
22
23
24 identity phy-type-10GBASE-PR {
25     base phy-type;
26     description
27         "Clause 76 10/10G-EPON 10 Gb/s 64B/66B";
28 }
29
30
31 identity phy-type-10GBASE-R {
32     base phy-type;
33     description
34         "Clause 49 10 Gb/s 64B/66B";
35 }
36
37
38 identity phy-type-10GBASE-T {
39     base phy-type;
40     description
41         "Clause 55 10 Gb/s DSQ128";
42 }
43
44
45 identity phy-type-10GBASE-T1 {
46     base phy-type;
47     description
48         "Clause 149 10 Gb/s PAM4";
49 }
50
51
52 identity phy-type-10GBASE-X {
53     base phy-type;
54     description
55         "Clause 48 10 Gb/s 4 lane 8B/10B";
56 }
57
58
59 identity phy-type-10GPASS-XR {
60     base phy-type;
61     description
62         "Clause 101 PCS up to 10 Gb/s 64B/66B OFDM downstream
63         10 Gb/s 64B/66B OFDM upstream";
64 }
65
```

```
1         and up to 1.6 Gb/s 64B/66B OFDMA upstream";
2     }
3
4     identity phy-type-10G-2p5GGBASE-SP {
5         base phy-type;
6         description
7             "Clause 164.3 10 Gb/s downstream and 2.5 Gb/s upstream
8              256B/257B";
9     }
10
11     identity phy-type-10GBASE-SP {
12         base phy-type;
13         description
14             "Clause 164.3 10 Gb/s 256B/257B";
15     }
16
17     identity phy-type-25G-10GBASE-PQ {
18         base phy-type;
19         description
20             "Clause 142 25/10G-EPON 256B/257B";
21     }
22
23     identity phy-type-25GBASE-PQ {
24         base phy-type;
25         description
26             "Clause 142 25/25G-EPON 256B/257B";
27     }
28
29     identity phy-type-25GBASE-R {
30         base phy-type;
31         description
32             "Clause 107 25 Gb/s 64B/66B";
33     }
34
35     identity phy-type-25GBASE-T {
36         base phy-type;
37         description
38             "Clause 113 25 Gb/s DSQ128";
39     }
40
41     identity phy-type-40GBASE-R {
42         base phy-type;
43         description
44             "Clause 82 40 Gb/s multi-PCS lane 64B/66B";
45     }
46
47     identity phy-type-40GBASE-T {
48         base phy-type;
49         description
50             "Clause 113 40 Gb/s DSQ128";
51     }
52
53     identity phy-type-50G-10GBASE-PQ {
54         base phy-type;
```

```
1      description
2          "Clause 142 50/10G-EPON 256B/257B";
3      }
4
5
6      identity phy-type-50G-25GBASE-PQ {
7          base phy-type;
8          description
9              "Clause 142 50/25G-EPON 256B/257B";
10         }
11
12
13         identity phy-type-50GBASE-PQ {
14             base phy-type;
15             description
16                 "Clause 142 50/50G-EPON 256B/257B";
17         }
18
19
20         identity phy-type-50GBASE-R {
21             base phy-type;
22             description
23                 "Clause 133 50 Gb/s multi-PCS lane 64B/66B";
24         }
25
26
27         identity phy-type-100GBASE-P {
28             base phy-type;
29             description
30                 "Clause 82 100 Gb/s multi-PCS lane using >2-level PAM";
31         }
32
33
34         identity phy-type-100GBASE-R {
35             base phy-type;
36             description
37                 "Clause 82 100 Gb/s multi-PCS lane using 2-level PAM";
38         }
39
40
41         identity phy-type-200GBASE-R {
42             base phy-type;
43             description
44                 "Clause 119 200 Gb/s multi-PCS lane 64B/66B";
45         }
46
47
48         identity phy-type-400GBASE-R {
49             base phy-type;
50             description
51                 "Clause 119 400 Gb/s multi-PCS lane 64B/66B";
52         }
53
54
55         identity phy-type-800GBASE-R {
56             base phy-type;
57             description
58                 "Clause 172 800 Gb/s multi-PCS lane 64B/66B";
59         }
60
61
62     /* PMD identities */
63
64
65
```

```
1  identity pmd-type {
2      description
3          "Base type for Physical Medium Dependent (PMD) sublayer types.";
4      reference
5          "IEEE Std 802.3, Clause 30.5.1.1.2 aMAUType";
6  }
7
8
9  identity pmd-type-global {
10     base pmd-type;
11     description
12         "Undefined";
13 }
14
15
16 identity pmd-type-other {
17     base pmd-type;
18     description
19         "Nonconformant PMD.";
20 }
21
22
23 identity pmd-type-unknown {
24     base pmd-type;
25     description
26         "Initializing, true state or type not yet known.";
27 }
28
29
30
31 identity pmd-type-10BASE-T {
32     base pmd-type;
33     description
34         "Twisted-pair cabling MAU as specified in Clause 14";
35 }
36
37
38 identity pmd-type-10BASE-T1L {
39     base pmd-type;
40     description
41         "Single balanced pair PHY as specified in Clause 146";
42 }
43
44
45 identity pmd-type-10BASE-T1S {
46     base pmd-type;
47     description
48         "Single balanced pair PHY as specified in Clause 147, full
49         duplex mode";
50 }
51
52
53
54 identity pmd-type-10BASE-T1SMD {
55     base pmd-type;
56     description
57         "Single balanced pair PHY as specified in Clause 147,
58         multidrop mode";
59 }
60
61
62 identity pmd-type-100BASE-BX10D {
63     base pmd-type;
64     description
65
```

```
1      "One single-mode fiber OLT PHY as specified in Clause 58";
2  }
3
4  identity pmd-type-100BASE-BX10U {
5      base pmd-type;
6      description
7          "One single-mode fiber ONU PHY as specified in Clause 58";
8  }
9
10
11
12 identity pmd-type-100BASE-FX {
13     base pmd-type;
14     description
15         "100BASE-FX PMD as specified in Clause 26";
16 }
17
18
19 identity pmd-type-100BASE-LX10 {
20     base pmd-type;
21     description
22         "Two fiber PHY as specified in Clause 58";
23 }
24
25
26 identity pmd-type-100BASE-T1 {
27     base pmd-type;
28     description
29         "Single balanced twisted-pair copper cabling PHY as specified
30         in Clause 96";
31 }
32
33
34
35 identity pmd-type-100BASE-TX {
36     base pmd-type;
37     description
38         "Two-pair Category 5 twisted-pair cabling as specified in
39         Clause 25";
40 }
41
42
43 identity pmd-type-1000BASE-BX10D {
44     base pmd-type;
45     description
46         "One single-mode fiber OLT PHY as specified in Clause 59";
47 }
48
49
50 identity pmd-type-1000BASE-BX10U {
51     base pmd-type;
52     description
53         "One single-mode fiber ONU PHY as specified in Clause 59";
54 }
55
56
57 identity pmd-type-1000BASE-KX {
58     base pmd-type;
59     description
60         "1000BASE-KX PCS/PMA over an electrical backplane PMD as
61         specified in Clause 70";
62 }
63
64
65
```

```
1  identity pmd-type-1000BASE-LX {
2      base pmd-type;
3      description
4          "1000BASE-LX fiber over long-wavelength laser PMD as specified
5          in Clause 38";
6  }
7
8
9  identity pmd-type-1000BASE-LX10 {
10     base pmd-type;
11     description
12         "Two fiber 10 km PHY as specified in Clause 59";
13 }
14
15
16 identity pmd-type-1000BASE-PX10-D {
17     base pmd-type;
18     description
19         "One single-mode fiber OMP OLT PHY, as specified in Clause 60,
20         supporting a distance of at least 10 km, and a split of at
21         least 1:16";
22 }
23
24
25
26 identity pmd-type-1000BASE-PX10-U {
27     base pmd-type;
28     description
29         "One single-mode fiber OMP ONU PHY, as specified in Clause 60,
30         supporting a distance of at least 10 km, and a split of at
31         least 1:16";
32 }
33
34
35
36 identity pmd-type-1000BASE-PX20-D {
37     base pmd-type;
38     description
39         "One single-mode fiber OMP OLT PHY, as specified in Clause 60,
40         supporting a distance of at least 20 km, and a split of at
41         least 1:16";
42 }
43
44
45
46 identity pmd-type-1000BASE-PX20-U {
47     base pmd-type;
48     description
49         "One single-mode fiber OMP ONU PHY, as specified in Clause 60,
50         supporting a distance of at least 20 km, and a split of at
51         least 1:16";
52 }
53
54
55
56 identity pmd-type-1000BASE-PX30-D {
57     base pmd-type;
58     description
59         "One single-mode fiber OMP OLT PHY, as specified in Clause 60,
60         supporting a distance of at least 20 km, and a split of at
61         least 1:32";
62 }
63
64
65 identity pmd-type-1000BASE-PX30-U {
```

```
1      base pmd-type;
2      description
3          "One single-mode fiber OMP ONU PHY, as specified in Clause 60,
4              supporting a distance of at least 20 km, and a split of at
5              least 1:32";
6      }
7
8
9      identity pmd-type-1000BASE-PX40-D {
10         base pmd-type;
11         description
12             "One single-mode fiber OMP OLT PHY as specified in Clause 60,
13                 supporting a distance of at least 20 km, and a split of at
14                 least 1:64";
15         }
16
17
18         identity pmd-type-1000BASE-PX40-U {
19             base pmd-type;
20             description
21                 "One single-mode fiber OMP ONU PHY as specified in Clause 60,
22                     supporting a distance of at least 20 km, and a split of at
23                     least 1:64";
24             }
25
26
27         identity pmd-type-1000BASE-RHA {
28             base pmd-type;
29             description
30                 "Plastic optical fiber PHY as specified in Clause 115.";
31         }
32
33
34         identity pmd-type-1000BASE-RHB {
35             base pmd-type;
36             description
37                 "Plastic optical fiber PHY as specified in Clause 115.";
38         }
39
40
41         identity pmd-type-1000BASE-RHC {
42             base pmd-type;
43             description
44                 "Plastic optical fiber PHY as specified in Clause 115.";
45         }
46
47
48         identity pmd-type-1000BASE-SX {
49             base pmd-type;
50             description
51                 "1000BASE-SX fiber over short-wavelength laser PMD as
52                 specified in Clause 38";
53         }
54
55
56         identity pmd-type-1000BASE-T {
57             base pmd-type;
58             description
59                 "Four-pair Category 5 twisted-pair cabling PHY as specified in
60                 Clause 40";
61         }
62
63
64
65
```

```
1
2 identity pmd-type-1000BASE-T1 {
3     base pmd-type;
4     description
5         "Single twisted-pair copper cable PHY as specified in Clause
6         97";
7 }
8
9
10 identity pmd-type-1000BASE-X {
11     base pmd-type;
12     description
13         "1000BASE-X PCS/PMA as specified in Clause 36 over undefined
14         PMD";
15 }
16
17
18 identity pmd-type-2.5GBASE-KX {
19     base pmd-type;
20     description
21         "2.5GBASE-X PMD as specified in Clause 128 over an electrical
22         backplane as specified in Clause 128";
23 }
24
25
26 identity pmd-type-2.5GBASE-T {
27     base pmd-type;
28     description
29         "Four-pair twisted-pair balanced copper cabling PHY as
30         specified in Clause 126";
31 }
32
33
34 identity pmd-type-2.5GBASE-T1 {
35     base pmd-type;
36     description
37         "Single balanced pair of conductors PHY as specified in Clause
38         149";
39 }
40
41
42 identity pmd-type-2.5GBASE-X {
43     base pmd-type;
44     description
45         "2.5GBASE-X PCS/PMA as specified in Clause 127 over undefined
46         PMD";
47 }
48
49
50 identity pmd-type-5GBASE-KR {
51     base pmd-type;
52     description
53         "5GBASE-KR PMD as specified in Clause 130 over an electrical
54         backplane as specified in Clause 130";
55 }
56
57
58 identity pmd-type-5GBASE-R {
59     base pmd-type;
60     description
61         "5GBASE-R PCS/PMA as specified in Clause 129 over undefined
62         PMD";
63 }
```

```
1         PMD";
2     }
3
4     identity pmd-type-5GBASE-T {
5         base pmd-type;
6         description
7             "Four-pair twisted-pair balanced copper cabling PHY as
8             specified in Clause 126";
9     }
10
11
12
13     identity pmd-type-5GBASE-T1 {
14         base pmd-type;
15         description
16             "Single balanced pair of conductors PHY as specified in Clause
17             149";
18     }
19
20
21
22     identity pmd-type-10G-1GBASE-PRX-D1 {
23         base pmd-type;
24         description
25             "One single-mode fiber 10.3125 GBd continuous downstream /
26             1.25 GBd burst mode upstream OLT PHY as specified in Clause 75";
27     }
28
29
30     identity pmd-type-10G-1GBASE-PRX-D2 {
31         base pmd-type;
32         description
33             "One single-mode fiber 10.3125 GBd continuous downstream /
34             1.25 GBd burst mode upstream OLT PHY as specified in Clause 75";
35     }
36
37
38     identity pmd-type-10G-1GBASE-PRX-D3 {
39         base pmd-type;
40         description
41             "One single-mode fiber 10.3125 GBd continuous downstream /
42             1.25 GBd burst mode upstream OLT PHY as specified in Clause 75";
43     }
44
45
46
47     identity pmd-type-10G-1GBASE-PRX-D4 {
48         base pmd-type;
49         description
50             "One single-mode fiber 10.3125 GBd continuous downstream /
51             1.25 GBd burst mode upstream OLT PHY as specified in Clause 75";
52     }
53
54
55     identity pmd-type-10G-1GBASE-PRX-U1 {
56         base pmd-type;
57         description
58             "One single-mode fiber 10.3125 GBd continuous downstream /
59             1.25 GBd burst mode upstream ONU PHY as specified in Clause 75";
60     }
61
62
63
64     identity pmd-type-10G-1GBASE-PRX-U2 {
65         base pmd-type;
```

```
1      description
2          "One single-mode fiber 10.3125 GBd continuous downstream /
3          1.25 GBd burst mode upstream ONU PHY as specified in Clause 75";
4      }
5
6
7      identity pmd-type-10G-1GBASE-PRX-U3 {
8          base pmd-type;
9          description
10             "One single-mode fiber 10.3125 GBd continuous downstream /
11             1.25 GBd burst mode upstream ONU PHY as specified in Clause 75";
12         }
13
14
15         identity pmd-type-10G-1GBASE-PRX-U4 {
16             base pmd-type;
17             description
18                 "One single-mode fiber 10.3125 GBd continuous downstream /
19                 1.25 GBd burst mode upstream ONU PHY as specified in Clause 75";
20             }
21
22
23         identity pmd-type-10GBASE-BR10-D {
24             base pmd-type;
25             description
26                 "One single-mode fiber OLT PHY supporting a distance of at
27                 least 10 km as specified in Clause 158";
28             }
29
30
31         identity pmd-type-10GBASE-BR10-U {
32             base pmd-type;
33             description
34                 "One single-mode fiber ONU PHY supporting a distance of at
35                 least 10 km as specified in Clause 158";
36             }
37
38
39         identity pmd-type-10GBASE-BR20-D {
40             base pmd-type;
41             description
42                 "One single-mode fiber OLT PHY supporting a distance of at
43                 least 20 km as specified in Clause 158";
44             }
45
46
47         identity pmd-type-10GBASE-BR20-U {
48             base pmd-type;
49             description
50                 "One single-mode fiber ONU PHY supporting a distance of at
51                 least 20 km as specified in Clause 158";
52             }
53
54
55         identity pmd-type-10GBASE-BR40-D {
56             base pmd-type;
57             description
58                 "One single-mode fiber OLT PHY supporting a distance of at
59                 least 40 km as specified in Clause 158";
60             }
61
62
63
64
65
```

```
1  identity pmd-type-10GBASE-BR40-U {
2      base pmd-type;
3      description
4          "One single-mode fiber ONU PHY supporting a distance of at
5          least 40 km as specified in Clause 158";
6  }
7
8
9  identity pmd-type-10GBASE-CX4 {
10     base pmd-type;
11     description
12         "10GBASE-CX4 copper over 8 pair 100-Ohm balanced cable as
13         specified in Clause 54";
14     }
15
16
17  identity pmd-type-10GBASE-ER {
18     base pmd-type;
19     description
20         "10GBASE-ER fiber over 1550nm optics as specified in Clause 52";
21     }
22
23
24
25  identity pmd-type-10GBASE-EW {
26     base pmd-type;
27     description
28         "10GBASE-EW fiber over 1550nm optics as specified in Clause 52";
29     }
30
31
32
33  identity pmd-type-10GBASE-KR {
34     base pmd-type;
35     description
36         "10GBASE-KR PCS/PMA over an electrical backplane PMD as
37         specified in Clause 72";
38     }
39
40
41  identity pmd-type-10GBASE-KX4 {
42     base pmd-type;
43     description
44         "10GBASE-KX4 PCS/PMA over an electrical backplane PMD as
45         specified in Clause 71";
46     }
47
48
49  identity pmd-type-10GBASE-LR {
50     base pmd-type;
51     description
52         "10GBASE-LR fiber over 1310nm optics as specified in Clause 52";
53     }
54
55
56  identity pmd-type-10GBASE-LRM {
57     base pmd-type;
58     description
59         "10GBASE-LRM fiber over 1310 nm optics as specified in Clause
60         68";
61     }
62
63
64
65  identity pmd-type-10GBASE-LW {
```

```
1      base pmd-type;
2      description
3          "10GBASE-LW fiber over 1310nm optics as specified in Clause 52";
4      }
5
6
7      identity pmd-type-10GBASE-LX4 {
8          base pmd-type;
9          description
10             "10GBASE-LX4 fiber over 4 lane 1310nm optics as specified in
11             Clause 53";
12         }
13
14
15         identity pmd-type-10GBASE-PR-D1 {
16             base pmd-type;
17             description
18                 "One single-mode fiber 10.3125 GBd continuous downstream /
19                 1.25 GBd burst mode upstream OLT PHY as specified in Clause 75";
20             }
21
22
23         identity pmd-type-10GBASE-PR-D2 {
24             base pmd-type;
25             description
26                 "One single-mode fiber 10.3125 GBd continuous downstream /
27                 1.25 GBd burst mode upstream OLT PHY as specified in Clause 75";
28             }
29
30
31         identity pmd-type-10GBASE-PR-D3 {
32             base pmd-type;
33             description
34                 "One single-mode fiber 10.3125 GBd continuous downstream /
35                 1.25 GBd burst mode upstream OLT PHY as specified in Clause 75";
36             }
37
38
39         identity pmd-type-10GBASE-PR-D4 {
40             base pmd-type;
41             description
42                 "One single-mode fiber 10.3125 GBd continuous downstream /
43                 1.25 GBd burst mode upstream OLT PHY as specified in Clause 75";
44             }
45
46
47         identity pmd-type-10GBASE-PR-U1 {
48             base pmd-type;
49             description
50                 "One single-mode fiber 10.3125 GBd continuous downstream /
51                 1.25 GBd burst mode upstream ONU PHY as specified in Clause 75";
52             }
53
54
55         identity pmd-type-10GBASE-PR-U3 {
56             base pmd-type;
57             description
58                 "One single-mode fiber 10.3125 GBd continuous downstream /
59                 1.25 GBd burst mode upstream ONU PHY as specified in Clause 75";
60             }
61
62
63
64
65
```

```
1  identity pmd-type-10GBASE-PR-U4 {
2      base pmd-type;
3      description
4          "One single-mode fiber 10.3125 GBd continuous downstream /
5          1.25 GBd burst mode upstream ONU PHY as specified in Clause 75";
6  }
7
8
9  identity pmd-type-10GBASE-R {
10     base pmd-type;
11     description
12         "10GBASE-R PCS/PMA as specified in Clause 49 over undefined
13         PMD";
14     }
15
16
17  identity pmd-type-10GBASE-SR {
18     base pmd-type;
19     description
20         "10GBASE-SR fiber over 850nm optics as specified in Clause 52";
21     }
22
23
24  identity pmd-type-10GBASE-SW {
25     base pmd-type;
26     description
27         "10GBASE-SW fiber over 850nm optics as specified in Clause 52";
28     }
29
30
31  identity pmd-type-10GBASE-T {
32     base pmd-type;
33     description
34         "Four-pair twisted-pair balanced copper cabling PHY as
35         specified in Clause 55";
36     }
37
38
39  identity pmd-type-10GBASE-T1 {
40     base pmd-type;
41     description
42         "Single balanced pair of conductors PHY as specified in Clause
43         149";
44     }
45
46
47  identity pmd-type-10GBASE-X {
48     base pmd-type;
49     description
50         "10GBASE-X PCS/PMA as specified in Clause 48 over undefined
51         PMD";
52     }
53
54
55  identity pmd-type-10GPASS-XR {
56     base pmd-type;
57     description
58         "Coax cable distribution network PHY continuous
59         downstream/burst mode upstream PHY as specified in Clause 100
60         and Clause 101";
61     }
62
63
64
65 }
```

```
1
2  identity pmd-type-10G-2p5GGBASE-SP1-Dx {
3      base pmd-type;
4      description
5          "One single-mode fiber 10.3125 GBd continuous downstream /
6            2.578125 GBd burst mode upstream OLT PHY supporting
7            channel x of FSR set 1, as specified in 164.2,";
8      }
9
10
11  identity pmd-type-10G-2p5GGBASE-SP1-Uxy {
12      base pmd-type;
13      description
14          "One single-mode fiber 10.3125 GBd continuous downstream /
15            2.578125 GBd burst mode upstream ONU PHY supporting
16            channels x to y of FSR set 1, as specified in 164.2";
17      }
18
19
20  identity pmd-type-10GBASE-SP {
21      base pmd-type;
22      description
23          "Clause 164.3 10 Gb/s 256B/257B";
24      }
25
26
27  identity pmd-type-25G-10GBASE-PQG-D2 {
28      base pmd-type;
29      description
30          "One single mode fiber, 1 Å– 25.78125 GBd continuous
31            transmission / 1 Å– 10.3125 GBd burst mode reception, medium
32            power class, as specified in Clause 141";
33      }
34
35
36  identity pmd-type-25G-10GBASE-PQG-D3 {
37      base pmd-type;
38      description
39          "One single mode fiber, 1 Å– 25.78125 GBd continuous
40            transmission / 1 Å– 10.3125 GBd burst mode reception, high
41            power class, as specified in Clause 141";
42      }
43
44
45  identity pmd-type-25G-10GBASE-PQG-U2 {
46      base pmd-type;
47      description
48          "One single mode fiber, 1 Å– 25.78125 GBd continuous reception
49            / 1 Å– 10.3125 GBd burst mode transmission, medium power class,
50            as specified in Clause 141";
51      }
52
53
54  identity pmd-type-25G-10GBASE-PQG-U3 {
55      base pmd-type;
56      description
57          "One single mode fiber, 1 Å– 25.78125 GBd continuous reception
58            / 1 Å– 10.3125 GBd burst mode transmission, high power class,
59            as specified in Clause 141";
60      }
61
62
63
64
65
```

```
1
2  identity pmd-type-25G-10GBASE-PQX-D2 {
3      base pmd-type;
4      description
5          "One single mode fiber, 1 Å– 25.78125 GBd continuous
6          transmission / 1 Å– 10.3125 GBd burst mode reception, medium
7          power class, as specified in Clause 141";
8      }
9
10
11  identity pmd-type-25G-10GBASE-PQX-D3 {
12      base pmd-type;
13      description
14          "One single mode fiber, 1 Å– 25.78125 GBd continuous
15          transmission / 1 Å– 10.3125 GBd burst mode reception, high
16          power class, as specified in Clause 141";
17      }
18
19
20  identity pmd-type-25G-10GBASE-PQX-U2 {
21      base pmd-type;
22      description
23          "One single mode fiber, 1 Å– 25.78125 GBd continuous reception
24          / 1 Å– 10.3125 GBd burst mode transmission, medium power class,
25          as specified in Clause 141";
26      }
27
28
29  identity pmd-type-25G-10GBASE-PQX-U3 {
30      base pmd-type;
31      description
32          "One single mode fiber, 1 Å– 25.78125 GBd continuous reception
33          / 1 Å– 10.3125 GBd burst mode transmission, high power class,
34          as specified in Clause 141";
35      }
36
37
38  identity pmd-type-25GBASE-BR10-D {
39      base pmd-type;
40      description
41          "One single-mode fiber OLT PHY supporting a distance of at
42          least 10 km as specified in Clause 159";
43      }
44
45
46  identity pmd-type-25GBASE-BR10-U {
47      base pmd-type;
48      description
49          "One single-mode fiber ONU PHY supporting a distance of at
50          least 10 km as specified in Clause 159";
51      }
52
53
54  identity pmd-type-25GBASE-BR20-D {
55      base pmd-type;
56      description
57          "One single-mode fiber OLT PHY supporting a distance of at
58          least 20 km as specified in Clause 159";
59      }
60
61
62
63
64
65
```

```
1  identity pmd-type-25GBASE-BR20-U {
2      base pmd-type;
3      description
4          "One single-mode fiber ONU PHY supporting a distance of at
5          least 20 km as specified in Clause 159";
6  }
7
8
9  identity pmd-type-25GBASE-BR40-D {
10     base pmd-type;
11     description
12         "One single-mode fiber OLT PHY supporting a distance of at
13         least 40 km as specified in Clause 159";
14     }
15
16
17  identity pmd-type-25GBASE-BR40-U {
18     base pmd-type;
19     description
20         "One single-mode fiber ONU PHY supporting a distance of at
21         least 40 km as specified in Clause 159";
22     }
23
24
25
26  identity pmd-type-25GBASE-CR {
27     base pmd-type;
28     description
29         "25GBASE-R PCS/PMA over shielded balanced copper cable PMD as
30         specified in Clause 110";
31     }
32
33
34  identity pmd-type-25GBASE-CR-S {
35     base pmd-type;
36     description
37         "25GBASE-R PCS/PMA over shielded balanced copper cable PMD as
38         specified in Clause 110 without support for RS-FEC ";
39     }
40
41
42
43  identity pmd-type-25GBASE-ER {
44     base pmd-type;
45     description
46         "25GBASE-R PCS/PMA over single-mode fiber PMD, with extended
47         reach, as specified in Clause 114";
48     }
49
50
51
52  identity pmd-type-25GBASE-KR {
53     base pmd-type;
54     description
55         "25GBASE-R PCS/PMA over an electrical backplane PMD as
56         specified in Clause 111";
57     }
58
59
60  identity pmd-type-25GBASE-KR-S {
61     base pmd-type;
62     description
63         "25GBASE-R PCS/PMA over an electrical backplane PMD as
64         specified in Clause 111 without support for RS-FEC";
65     }
```

```
1      }
2
3      identity pmd-type-25GBASE-LR {
4          base pmd-type;
5          description
6              "25GBASE-R PCS/PMA over single-mode fiber PMD, with long
7              reach, as specified in Clause 114";
8          }
9
10
11      identity pmd-type-25GBASE-PQG-D2 {
12          base pmd-type;
13          description
14              "One single mode fiber, 1 Å– 25.78125 GBd continuous
15              transmission / 1 Å– 25.78125 GBd burst mode reception, medium
16              power class, as specified in Clause 141";
17          }
18
19      identity pmd-type-25GBASE-PQG-D3 {
20          base pmd-type;
21          description
22              "One single mode fiber, 1 Å– 25.78125 GBd continuous
23              transmission / 1 Å– 25.78125 GBd burst mode reception, high
24              power class, as specified in Clause 141";
25          }
26
27      identity pmd-type-25GBASE-PQG-U2 {
28          base pmd-type;
29          description
30              "One single mode fiber, 1 Å– 25.78125 GBd continuous reception
31              / 1 Å– 25.78125 GBd burst mode transmission, medium power
32              class, as specified in Clause 141";
33          }
34
35      identity pmd-type-25GBASE-PQG-U3 {
36          base pmd-type;
37          description
38              "One single mode fiber, 1 Å– 25.78125 GBd continuous reception
39              / 1 Å– 25.78125 GBd burst mode transmission, high power class,
40              as specified in Clause 141";
41          }
42
43      identity pmd-type-25GBASE-PQX-D2 {
44          base pmd-type;
45          description
46              "One single mode fiber, 1 Å– 25.78125 GBd continuous
47              transmission / 1 Å– 25.78125 GBd burst mode reception, medium
48              power class, as specified in Clause 141";
49          }
50
51      identity pmd-type-25GBASE-PQX-D3 {
52          base pmd-type;
53          description
54              "One single mode fiber, 1 Å– 25.78125 GBd continuous
55              transmission / 1 Å– 25.78125 GBd burst mode reception, high
56              power class, as specified in Clause 141";
57          }
58
59      identity pmd-type-25GBASE-PQX-D3 {
60          base pmd-type;
61          description
62              "One single mode fiber, 1 Å– 25.78125 GBd continuous
63              transmission / 1 Å– 25.78125 GBd burst mode reception, high
64              power class, as specified in Clause 141";
65          }
```

```
1         power class, as specified in Clause 141";
2     }
3
4     identity pmd-type-25GBASE-PQX-U2 {
5         base pmd-type;
6         description
7             "One single mode fiber, 1 Å– 25.78125 GBd continuous reception
8             / 1 Å– 25.78125 GBd burst mode transmission, medium power
9             class, as specified in Clause 141";
10    }
11
12    identity pmd-type-25GBASE-PQX-U3 {
13        base pmd-type;
14        description
15            "One single mode fiber, 1 Å– 25.78125 GBd continuous reception
16            / 1 Å– 25.78125 GBd burst mode transmission, high power class,
17            as specified in Clause 141";
18    }
19
20    identity pmd-type-25GBASE-R {
21        base pmd-type;
22        description
23            "PCS as specified in Clause 107 with PMA as specified in
24            Clause 109 over undefined PMD";
25    }
26
27    identity pmd-type-25GBASE-SR {
28        base pmd-type;
29        description
30            "25GBASE-R PCS/PMA over multimode fiber PMD as specified in
31            Clause 112";
32    }
33
34    identity pmd-type-25GBASE-T {
35        base pmd-type;
36        description
37            "Four-pair twisted-pair balanced copper cabling PHY as
38            specified in Clause 113";
39    }
40
41    identity pmd-type-40GBASE-CR4 {
42        base pmd-type;
43        description
44            "40GBASE-R PCS/PMA over 4 lane shielded copper balanced cable
45            PMD as specified in Clause 85";
46    }
47
48    identity pmd-type-40GBASE-ER4 {
49        base pmd-type;
50        description
51            "40GBASE-R PCS/PMA over 4 WDM lane single mode fiber PMD, with
52            extended reach, as specified in Clause 87";
53    }
54
55    }
```

```
1  identity pmd-type-40GBASE-FR {
2      base pmd-type;
3      description
4          "40GBASE-R PCS/PMA over single mode fiber PMD as specified in
5          Clause 89";
6  }
7
8
9  identity pmd-type-40GBASE-KR4 {
10     base pmd-type;
11     description
12         "40GBASE-R PCS/PMA over an electrical backplane PMD as
13         specified in Clause 84";
14     }
15
16
17  identity pmd-type-40GBASE-LR4 {
18     base pmd-type;
19     description
20         "40GBASE-R PCS/PMA over 4 WDM lane single mode fiber PMD, with
21         long reach, as specified in Clause 87";
22     }
23
24
25  identity pmd-type-40GBASE-R {
26     base pmd-type;
27     description
28         "Multi-lane PCS as specified in Clause 82 over undefined
29         PMA/PMD";
30     }
31
32
33  identity pmd-type-40GBASE-SR4 {
34     base pmd-type;
35     description
36         "40GBASE-R PCS/PMA over 4 lane multimode fiber PMD as
37         specified in Clause 86";
38     }
39
40
41  identity pmd-type-40GBASE-T {
42     base pmd-type;
43     description
44         "Four-pair twisted-pair balanced copper cabling PHY as
45         specified in Clause 113";
46     }
47
48
49  identity pmd-type-50G-10GBASE-PQG-D2 {
50     base pmd-type;
51     description
52         "One single mode fiber, 2 Å– 25.78125 GBd continuous
53         transmission / 1 Å– 10.3125 GBd burst mode reception, medium
54         power class, as specified in Clause 141";
55     }
56
57
58  identity pmd-type-50G-10GBASE-PQG-D3 {
59     base pmd-type;
60     description
61         "One single mode fiber, 2 Å– 25.78125 GBd continuous
```

```
1      transmission / 1 Å– 10.3125 GBd burst mode reception, high
2      power class, as specified in Clause 141";
3  }
4
5
6  identity pmd-type-50G-10GBASE-PQG-U2 {
7      base pmd-type;
8      description
9          "One single mode fiber, 2 Å– 25.78125 GBd continuous reception
10         / 1 Å– 10.3125 GBd burst mode transmission, medium power class,
11         as specified in Clause 141";
12  }
13
14
15  identity pmd-type-50G-10GBASE-PQG-U3 {
16      base pmd-type;
17      description
18          "One single mode fiber, 2 Å– 25.78125 GBd continuous reception
19         / 1 Å– 10.3125 GBd burst mode transmission, high power class,
20         as specified in Clause 141";
21  }
22
23
24
25  identity pmd-type-50G-10GBASE-PQX-D2 {
26      base pmd-type;
27      description
28          "One single mode fiber, 2 Å– 25.78125 GBd continuous
29         transmission / 1 Å– 10.3125 GBd burst mode reception, medium
30         power class, as specified in Clause 141";
31  }
32
33
34
35  identity pmd-type-50G-10GBASE-PQX-D3 {
36      base pmd-type;
37      description
38          "One single mode fiber, 2 Å– 25.78125 GBd continuous
39         transmission / 1 Å– 10.3125 GBd burst mode reception, high
40         power class, as specified in Clause 141";
41  }
42
43
44
45  identity pmd-type-50G-10GBASE-PQX-U2 {
46      base pmd-type;
47      description
48          "One single mode fiber, 2 Å– 25.78125 GBd continuous reception
49         / 1 Å– 10.3125 GBd burst mode transmission, medium power class,
50         as specified in Clause 141";
51  }
52
53
54  identity pmd-type-50G-10GBASE-PQX-U3 {
55      base pmd-type;
56      description
57          "One single mode fiber, 2 Å– 25.78125 GBd continuous reception
58         / 1 Å– 10.3125 GBd burst mode transmission, high power class,
59         as specified in Clause 141";
60  }
61
62
63
64  identity pmd-type-50G-25GBASE-PQG-D2 {
65      base pmd-type;
```

```
1      description
2      "One single mode fiber, 2 Å– 25.78125 GBd continuous
3      transmission / 1 Å– 25.78125 GBd burst mode reception, medium
4      power class, as specified in Clause 141";
5  }
6
7
8  identity pmd-type-50G-25GBASE-PQG-D3 {
9      base pmd-type;
10     description
11     "One single mode fiber, 2 Å– 25.78125 GBd continuous
12     transmission / 1 Å– 25.78125 GBd burst mode reception, high
13     power class, as specified in Clause 141";
14 }
15
16
17 identity pmd-type-50G-25GBASE-PQG-U2 {
18     base pmd-type;
19     description
20     "One single mode fiber, 2 Å– 25.78125 GBd continuous reception
21     / 1 Å– 25.78125 GBd burst mode transmission, medium power
22     class, as specified in Clause 141";
23 }
24
25
26 identity pmd-type-50G-25GBASE-PQG-U3 {
27     base pmd-type;
28     description
29     "One single mode fiber, 2 Å– 25.78125 GBd continuous reception
30     / 1 Å– 25.78125 GBd burst mode transmission, high power class,
31     as specified in Clause 141";
32 }
33
34
35 identity pmd-type-50G-25GBASE-PQX-D2 {
36     base pmd-type;
37     description
38     "One single mode fiber, 2 Å– 25.78125 GBd continuous
39     transmission / 1 Å– 25.78125 GBd burst mode reception, medium
40     power class, as specified in Clause 141";
41 }
42
43
44 identity pmd-type-50G-25GBASE-PQX-D3 {
45     base pmd-type;
46     description
47     "One single mode fiber, 2 Å– 25.78125 GBd continuous
48     transmission / 1 Å– 25.78125 GBd burst mode reception, high
49     power class, as specified in Clause 141";
50 }
51
52
53 identity pmd-type-50G-25GBASE-PQX-U2 {
54     base pmd-type;
55     description
56     "One single mode fiber, 2 Å– 25.78125 GBd continuous reception
57     / 1 Å– 25.78125 GBd burst mode transmission, medium power
58     class, as specified in Clause 141";
59 }
60
61
62 }
```

```
1  identity pmd-type-50G-25GBASE-PQX-U3 {
2      base pmd-type;
3      description
4          "One single mode fiber, 2 Å– 25.78125 GBd continuous reception
5          / 1 Å– 25.78125 GBd burst mode transmission, high power class,
6          as specified in Clause 141";
7      }
8
9
10 identity pmd-type-50GBASE-BR10-D {
11     base pmd-type;
12     description
13         "One single-mode fiber OLT PHY supporting a distance of at
14         least 10 km as specified in Clause 160";
15     }
16
17
18 identity pmd-type-50GBASE-BR10-U {
19     base pmd-type;
20     description
21         "One single-mode fiber ONU PHY supporting a distance of at
22         least 10 km as specified in Clause 160";
23     }
24
25
26 identity pmd-type-50GBASE-BR20-D {
27     base pmd-type;
28     description
29         "One single-mode fiber OLT PHY supporting a distance of at
30         least 20 km as specified in Clause 160";
31     }
32
33
34 identity pmd-type-50GBASE-BR20-U {
35     base pmd-type;
36     description
37         "One single-mode fiber ONU PHY supporting a distance of at
38         least 20 km as specified in Clause 160";
39     }
40
41
42 identity pmd-type-50GBASE-BR40-D {
43     base pmd-type;
44     description
45         "One single-mode fiber OLT PHY supporting a distance of at
46         least 40 km as specified in Clause 160";
47     }
48
49
50 identity pmd-type-50GBASE-BR40-U {
51     base pmd-type;
52     description
53         "One single-mode fiber ONU PHY supporting a distance of at
54         least 40 km as specified in Clause 160";
55     }
56
57
58 identity pmd-type-50GBASE-CR {
59     base pmd-type;
60     description
61         "50GBASE-R PCS/PMA over shielded copper balanced cable PMD as
```

```
1         specified in Clause 136 ";
2     }
3
4     identity pmd-type-50GBASE-ER {
5         base pmd-type;
6         description
7             "50GBASE-R PCS/PMA over single-mode fiber PMD with reach up to
8             at least 40 km as specified in Clause 139";
9     }
10
11
12
13     identity pmd-type-50GBASE-FR {
14         base pmd-type;
15         description
16             "50GBASE-R PCS/PMA over single mode fiber PMD as specified in
17             Clause 139";
18     }
19
20
21
22     identity pmd-type-50GBASE-KR {
23         base pmd-type;
24         description
25             "50GBASE-R PCS/PMA over an electrical backplane PMD as
26             specified in Clause 137";
27     }
28
29
30
31     identity pmd-type-50GBASE-LR {
32         base pmd-type;
33         description
34             "50GBASE-R PCS/PMA over single mode fiber PMD as specified in
35             Clause 139";
36     }
37
38
39     identity pmd-type-50GBASE-PQG-D2 {
40         base pmd-type;
41         description
42             "One single mode fiber, 2 Å– 25.78125 GBd continuous
43             transmission / 2 Å– 25.78125 GBd burst mode reception, medium
44             power class, as specified in Clause 141";
45     }
46
47
48     identity pmd-type-50GBASE-PQG-D3 {
49         base pmd-type;
50         description
51             "One single mode fiber, 2 Å– 25.78125 GBd continuous
52             transmission / 2 Å– 25.78125 GBd burst mode reception, high
53             power class, as specified in Clause 141";
54     }
55
56
57
58     identity pmd-type-50GBASE-PQG-U2 {
59         base pmd-type;
60         description
61             "One single mode fiber, 2 Å– 25.78125 GBd continuous reception
62             / 2 Å– 25.78125 GBd burst mode transmission, medium power
63             class, as specified in Clause 141";
64     }
65 }
```

```
1
2 identity pmd-type-50GBASE-PQG-U3 {
3     base pmd-type;
4     description
5         "One single mode fiber, 2 Å– 25.78125 GBd continuous reception
6         / 2 Å– 25.78125 GBd burst mode transmission, high power class,
7         as specified in Clause 141";
8     }
9
10
11 identity pmd-type-50GBASE-PQX-D2 {
12     base pmd-type;
13     description
14         "One single mode fiber, 2 Å– 25.78125 GBd continuous
15         transmission / 2 Å– 25.78125 GBd burst mode reception, medium
16         power class, as specified in Clause 141";
17     }
18
19 identity pmd-type-50GBASE-PQX-D3 {
20     base pmd-type;
21     description
22         "One single mode fiber, 2 Å– 25.78125 GBd continuous
23         transmission / 2 Å– 25.78125 GBd burst mode reception, high
24         power class, as specified in Clause 141";
25     }
26
27 identity pmd-type-50GBASE-PQX-U2 {
28     base pmd-type;
29     description
30         "One single mode fiber, 2 Å– 25.78125 GBd continuous reception
31         / 2 Å– 25.78125 GBd burst mode transmission, medium power
32         class, as specified in Clause 141";
33     }
34
35 identity pmd-type-50GBASE-PQX-U3 {
36     base pmd-type;
37     description
38         "One single mode fiber, 2 Å– 25.78125 GBd continuous reception
39         / 2 Å– 25.78125 GBd burst mode transmission, high power class,
40         as specified in Clause 141";
41     }
42
43 identity pmd-type-50GBASE-R {
44     base pmd-type;
45     description
46         "Multi-lane PCS as specified in Clause 133 with PMA as
47         specified in Clause 135 over undefined PMD";
48     }
49
50 identity pmd-type-50GBASE-SR {
51     base pmd-type;
52     description
53         "50GBASE-R PCS/PMA over multimode fiber PMD as specified in
54         Clause 138";
55     }
56
57 }
```

```
1
2 identity pmd-type-100CR1 {
3     base pmd-type;
4     description
5         "100GBASE-CR1 as specified in Clause 162 or 100GBASE-KR1 as
6         specified in Clause 163";
7 }
8
9
10 identity pmd-type-100GBASE-CR2 {
11     base pmd-type;
12     description
13         "100GBASE-R PCS/PMA over 2 lane shielded copper balanced cable
14         PMD as specified in Clause 136";
15 }
16
17
18 identity pmd-type-100GBASE-CR4 {
19     base pmd-type;
20     description
21         "100GBASE-R PCS/PMA over 4 lane shielded copper balanced cable
22         PMD as specified in Clause 92";
23 }
24
25
26 identity pmd-type-100GBASE-CR10 {
27     base pmd-type;
28     description
29         "100GBASE-R PCS/PMA over 10 lane shielded copper balanced
30         cable PMD as specified in Clause 85";
31 }
32
33
34 identity pmd-type-100GBASE-DR {
35     base pmd-type;
36     description
37         "100GBASE-R PCS/PMA over single mode fiber PMD as specified in
38         Clause 140";
39 }
40
41
42 identity pmd-type-100GBASE-ER4 {
43     base pmd-type;
44     description
45         "100GBASE-R PCS/PMA over 4 WDM lane single mode fiber PMD,
46         with extended reach, as specified in Clause 88";
47 }
48
49
50 identity pmd-type-100GBASE-FR1 {
51     base pmd-type;
52     description
53         "100GBASE-R PCS/PMA over single-mode fiber PMD with reach up
54         to at least 2 km as specified in Clause 140";
55 }
56
57
58 identity pmd-type-100GBASE-KP4 {
59     base pmd-type;
60     description
61         "100GBASE-P PCS/PMA over an electrical backplane PMD as
```

```
1         specified in Clause 94";
2     }
3
4     identity pmd-type-100GBASE-KR2 {
5         base pmd-type;
6         description
7             "100GBASE-R PCS/PMA over an electrical backplane PMD as
8             specified in Clause 137";
9     }
10
11
12
13     identity pmd-type-100GBASE-KR4 {
14         base pmd-type;
15         description
16             "100GBASE-R PCS/PMA over an electrical backplane PMD as
17             specified in Clause 93";
18     }
19
20
21
22     identity pmd-type-100GBASE-LR1 {
23         base pmd-type;
24         description
25             "100GBASE-R PCS/PMA over single-mode fiber PMD with reach up
26             to at least 10 km as specified in Clause 140";
27     }
28
29
30
31     identity pmd-type-100GBASE-LR4 {
32         base pmd-type;
33         description
34             "100GBASE-R PCS/PMA over 4 WDM lane single mode fiber PMD,
35             with long reach, as specified in Clause 88";
36     }
37
38
39     identity pmd-type-100GBASE-R {
40         base pmd-type;
41         description
42             "Multi-lane PCS as specified in Clause 82 over undefined
43             100GBASE-R or 100GBASE-P PMA/PMD";
44     }
45
46
47     identity pmd-type-100GBASE-SR1 {
48         base pmd-type;
49         description
50             "100GBASE-R PCS/PMA over multimode fiber PMD with
51             reach up to at least 100 m as specified in Clause 167";
52     }
53
54
55
56     identity pmd-type-100GBASE-SR2 {
57         base pmd-type;
58         description
59             "100GBASE-R PCS/PMA over 2 lane multimode fiber PMD as
60             specified in Clause 138";
61     }
62
63
64     identity pmd-type-100GBASE-SR4 {
65         base pmd-type;
```

```
1      description
2          "100GBASE-R PCS/PMA over 4 lane multimode fiber PMD as
3              specified in Clause 95";
4      }
5
6
7      identity pmd-type-100GBASE-SR10 {
8          base pmd-type;
9          description
10             "100GBASE-R PCS/PMA over 10 lane multimode fiber PMD as
11                 specified in Clause 86";
12     }
13
14
15     identity pmd-type-100GBASE-VR1 {
16         base pmd-type;
17         description
18             "100GBASE-R PCS/PMA over multimode fiber PMD with
19                 reach up to at least 50 m as specified in Clause 167";
20     }
21
22
23
24     identity pmd-type-100GBASE-ZR {
25         base pmd-type;
26         description
27             "100GBASE-R PCS/100GBASE-ZR PMA over a PMD with reach up to at
28                 least 80 km as specified in Clause 154";
29     }
30
31
32
33     identity pmd-type-200GBASE-CR2 {
34         base pmd-type;
35         description
36             "200GBASE-CR2 as specified in Clause 162 or 200GBASE-KR2 as
37                 specified in Clause 163";
38     }
39
40
41
42     identity pmd-type-200GBASE-CR4 {
43         base pmd-type;
44         description
45             "200GBASE-R PCS/PMA over 4 lane shielded copper balanced cable
46                 PMD as specified in Clause 136 ";
47     }
48
49
50     identity pmd-type-200GBASE-DR4 {
51         base pmd-type;
52         description
53             "200GBASE-R PCS/PMA over 4-lane single-mode fiber PMD as
54                 specified in Clause 121";
55     }
56
57
58     identity pmd-type-200GBASE-ER4 {
59         base pmd-type;
60         description
61             "200GBASE-R PCS/PMA over 4 WDM lane single-mode fiber PMD with
62                 reach up to at least 40 km as specified in Clause 122";
63     }
64
65
```

```
1  identity pmd-type-200GBASE-FR4 {
2      base pmd-type;
3      description
4          "200GBASE-R PCS/PMA over 4 WDM lane single-mode fiber PMD with
5          reach up to at least 2 km as specified in Clause 122";
6  }
7
8
9  identity pmd-type-200GBASE-KR2 {
10     base pmd-type;
11     description
12         "200GBASE-R PCS/PMA over an electrical backplane PMD
13         as specified in Clause 163";
14     }
15
16
17  identity pmd-type-200GBASE-KR4 {
18     base pmd-type;
19     description
20         "200GBASE-R PCS/PMA over an electrical backplane PMD as
21         specified in Clause 137";
22     }
23
24
25
26  identity pmd-type-200GBASE-LR4 {
27     base pmd-type;
28     description
29         "200GBASE-R PCS/PMA over 4 WDM lane single-mode fiber PMD with
30         reach up to at least 10 km as specified in Clause 122";
31     }
32
33
34
35  identity pmd-type-200GBASE-R {
36     base pmd-type;
37     description
38         "Multi-lane PCS as specified in Clause 119 over undefined
39         PMA/PMD";
40     }
41
42
43  identity pmd-type-200GBASE-SR2 {
44     base pmd-type;
45     description
46         "200GBASE-R PCS/PMA over 2 lane multimode fiber PMD with
47         reach up to at least 100 m as specified in Clause 167";
48     }
49
50
51  identity pmd-type-200GBASE-SR4 {
52     base pmd-type;
53     description
54         "200GBASE-R PCS/PMA over 4 lane multimode fiber PMD as
55         specified in Clause 138";
56     }
57
58
59
60  identity pmd-type-200GBASE-VR2 {
61     base pmd-type;
62     description
63         "200GBASE-R PCS/PMA over 2 lane multimode fiber PMD with
64         reach up to at least 50 m as specified in Clause 167";
65     }
```

```
1      }
2
3      identity pmd-type-400GBASE-DR4 {
4          base pmd-type;
5          description
6              "400GBASE-R PCS/PMA over 4-lane single-mode fiber PMD as
7              specified in Clause 124";
8      }
9
10
11      identity pmd-type-400GBASE-DR4-2 {
12          base pmd-type;
13          description
14              "400GBASE-R PCS/PMA over 4 single-mode fibers in each
15              direction PMD with reach up to at least 2 km as specified
16              in Clause 124";
17      }
18
19
20
21      identity pmd-type-400GBASE-ER8 {
22          base pmd-type;
23          description
24              "400GBASE-R PCS/PMA over 8 WDM lane single-mode fiber PMD with
25              reach up to at least 40 km as specified in Clause 122";
26      }
27
28
29
30      identity pmd-type-400GBASE-FR4 {
31          base pmd-type;
32          description
33              "400GBASE-R PCS/PMA over 4 WDM lane single-mode fiber PMD with
34              reach up to at least 2 km as specified in Clause 151";
35      }
36
37
38
39      identity pmd-type-400GBASE-FR8 {
40          base pmd-type;
41          description
42              "400GBASE-R PCS/PMA over 8 WDM lane single-mode fiber PMD with
43              reach up to at least 2 km as specified in Clause 122";
44      }
45
46
47      identity pmd-type-400GBASE-LR4-6 {
48          base pmd-type;
49          description
50              "400GBASE-R PCS/PMA over 4 WDM lane single-mode fiber PMD with
51              reach up to at least 6 km as specified in Clause 151";
52      }
53
54
55      identity pmd-type-400GBASE-LR8 {
56          base pmd-type;
57          description
58              "400GBASE-R PCS/PMA over 8 WDM lane single-mode fiber PMD with
59              reach up to at least 10 km as specified in Clause 122";
60      }
61
62
63      identity pmd-type-400GBASE-R {
64          base pmd-type;
```

```
1      description
2          "Multi-lane PCS as specified in Clause 119 over undefined
3          PMA/PMD";
4      }
5
6
7      identity pmd-type-400GBASE-SR4 {
8          base pmd-type;
9          description
10             "400GBASE-R PCS/PMA over 4 lane multimode fiber PMD with
11             reach up to at least 100 m as specified in Clause 167";
12         }
13
14
15         identity pmd-type-400GBASE-SR4d2 {
16             base pmd-type;
17             description
18                 "400GBASE-R PCS/PMA over 8-lane multimode fiber PMD as
19                 specified in Clause 150";
20         }
21
22
23         identity pmd-type-400GBASE-SR8 {
24             base pmd-type;
25             description
26                 "400GBASE-R PCS/PMA over 8-lane multimode fiber PMD as
27                 specified in Clause 138";
28         }
29
30
31         identity pmd-type-400GBASE-SR16 {
32             base pmd-type;
33             description
34                 "400GBASE-R PCS/PMA over 16-lane multimode fiber PMD as
35                 specified in Clause 123";
36         }
37
38
39         identity pmd-type-400GBASE-VR4 {
40             base pmd-type;
41             description
42                 "400GBASE-R PCS/PMA over 4 lane multimode fiber PMD with
43                 reach up to at least 50 m as specified in Clause 167";
44         }
45
46
47         identity pmd-type-800GBASE-CR8 {
48             base pmd-type;
49             description
50                 "800GBASE-R PCS/PMA over 8-lane shielded balanced copper cable
51                 PMD as specified in Clause 162";
52         }
53
54
55         identity pmd-type-800GBASE-DR8 {
56             base pmd-type;
57             description
58                 "800GBASE-R PCS/PMA over 8 single-mode fibers in each
59                 direction PMD with reach up to at least 500 m as specified
60                 in Clause 124";
61         }
62
63
64     }
```

```
1
2 identity pmd-type-800GBASE-DR8-2 {
3     base pmd-type;
4     description
5         "800GBASE-R PCS/PMA over 8 single-mode fibers in each
6         direction PMD with reach up to at least 2 km as specified
7         in Clause 124";
8 }
9
10
11 identity pmd-type-800GBASE-KR8 {
12     base pmd-type;
13     description
14         "800GBASE-R PCS/PMA over an electrical backplane PMD as
15         specified in Clause 163";
16 }
17
18
19 identity pmd-type-800GBASE-R {
20     base pmd-type;
21     description
22         "Multi-lane PCS as specified in Clause 172 over
23         undefined PMA/PMDi€ ";
24 }
25
26
27 identity pmd-type-800GBASE-SR8 {
28     base pmd-type;
29     description
30         "800GBASE-R PCS/PMA over 8 multimode fibers in each
31         direction PMD with reach up to at least 100 m as specified
32         in Clause 167";
33 }
34
35
36 identity pmd-type-800GBASE-VR8 {
37     base pmd-type;
38     description
39         "800GBASE-R PCS/PMA over 8 multimode fibers in each
40         direction PMD with reach up to at least 50 m as specified
41         in Clause 167";
42 }
43
44
45 typedef pmd-type-list {
46     type bits {
47         /* 802.3-2022 */
48         bit bOther {
49             description
50                 "Other";
51         }
52         bit b10baseT {
53             description
54                 "10BASE-T";
55         }
56         bit b10baseFL {
57             description
58                 "10BASE-FL";
59         }
60     }
61 }
```

```
1      bit b100baseTX {
2          description
3              "100BASE-TX";
4      }
5
6      bit b100baseFX {
7          description
8              "100BASE-FX";
9      }
10
11     bit b1000baseX {
12         description
13             "1000BASE-X";
14     }
15
16     bit b1000baseLX {
17         description
18             "1000BASE-LX";
19     }
20
21     bit b1000baseSX {
22         description
23             "1000BASE-SX";
24     }
25
26     bit b1000baseCX {
27         description
28             "1000BASE-CX";
29     }
30
31     bit b1000baseT {
32         description
33             "1000BASE-T";
34     }
35
36     bit b10GbaseX {
37         description
38             "10GBASE-X";
39     }
40
41     bit b10GbaseLX4 {
42         description
43             "10GBASE-LX4";
44     }
45
46     bit b10GbaseR {
47         description
48             "10GBASE-R";
49     }
50
51     bit b10GbaseER {
52         description
53             "10GBASE-ER";
54     }
55
56     bit b10GbaseLR {
57         description
58             "10GBASE-LR";
59     }
60
61     bit b10GbaseSR {
62         description
63             "10GBASE-SR";
64     }
65
66     bit b10GbaseSW {
67         description
```

```
1         "10GBASE-SW";
2     }
3     bit b10GbaseCX4 {
4         description
5             "10GBASE-CX4";
6     }
7     bit b100BaseBX10D {
8         description
9             "100BASE-BX10D";
10    }
11    bit b100BaseBX10U {
12        description
13            "100BASE-BX10U";
14    }
15    bit b100BaseLX10 {
16        description
17            "100BASE-LX10";
18    }
19    bit b1000BaseBX10D {
20        description
21            "1000BASE-BX10D";
22    }
23    bit b1000BaseBX10U {
24        description
25            "1000BASE-BX10U";
26    }
27    bit b1000BaseLX10 {
28        description
29            "1000BASE-LX10";
30    }
31    bit b1000BasePX10D {
32        description
33            "1000BASE-PX10D";
34    }
35    bit b1000BasePX10U {
36        description
37            "1000BASE-PX10U";
38    }
39    bit b1000BasePX20D {
40        description
41            "1000BASE-PX20D";
42    }
43    bit b1000BasePX20U {
44        description
45            "1000BASE-PX20U";
46    }
47    bit b10GbaseT {
48        description
49            "10GBASE-T";
50    }
51    bit b10GbaseLRM {
52        description
53            "10GBASE-LRM";
54    }
55    }
```

```
1      bit b1000baseKX {
2          description
3              "1000BASE-KX";
4      }
5
6      bit b10GbaseKX4 {
7          description
8              "10GBASE-KX4";
9      }
10
11     bit b10GbaseKR {
12         description
13             "10GBASE-KR";
14     }
15
16     bit b10G1GbasePRXD1 {
17         description
18             "10/1GBASE-PRXD1";
19     }
20
21     bit b10G1GbasePRXD2 {
22         description
23             "10/1GBASE-PRXD2";
24     }
25
26     bit b10G1GbasePRXD3 {
27         description
28             "10/1GBASE-PRXD3";
29     }
30
31     bit b10G1GbasePRXU1 {
32         description
33             "10/1GBASE-PRXU1";
34     }
35
36     bit b10G1GbasePRXU2 {
37         description
38             "10/1GBASE-PRXU2";
39     }
40
41     bit b10G1GbasePRXU3 {
42         description
43             "10/1GBASE-PRXU3";
44     }
45
46     bit b10GbasePRD1 {
47         description
48             "10GBASE-PRD1";
49     }
50
51     bit b10GbasePRD2 {
52         description
53             "10GBASE-PRD2";
54     }
55
56     bit b10GbasePRD3 {
57         description
58             "10GBASE-PRD3";
59     }
60
61     bit b10GbasePRU1 {
62         description
63             "10GBASE-PRU1";
64     }
65
66     bit b10GbasePRU3 {
67         description
```

```
1         "10GBASE-PRU3";
2     }
3     bit b40GbaseKR4 {
4         description
5             "40GBASE-KR4";
6     }
7     bit b40GbaseCR4 {
8         description
9             "40GBASE-CR4";
10    }
11    bit b40GbaseSR4 {
12        description
13            "40GBASE-SR4";
14    }
15    bit b40GbaseFR {
16        description
17            "40GBASE-FR";
18    }
19    bit b40GbaseLR4 {
20        description
21            "40GBASE-LR4";
22    }
23    bit b100GbaseCR10 {
24        description
25            "100GBASE-CR10";
26    }
27    bit b100GbaseSR10 {
28        description
29            "100GBASE-SR10";
30    }
31    bit b100GbaseLR4 {
32        description
33            "100GBASE-LR4";
34    }
35    bit b100GbaseER4 {
36        description
37            "100GBASE-ER4";
38    }
39    bit b1000baseT1 {
40        description
41            "1000BASE-T1";
42    }
43    bit b1000basePX30D {
44        description
45            "1000BASE-PX30D";
46    }
47    bit b1000basePX30U {
48        description
49            "1000BASE-PX30U";
50    }
51    bit b1000basePX40D {
52        description
53            "1000BASE-PX40D";
54    }
55    }
```

```
1      bit b1000basePX40U {
2          description
3              "1000BASE-PX40U";
4      }
5
6      bit b10G1GbasePRXD4 {
7          description
8              "10/1GBASE-PRXD4";
9      }
10
11     bit b10G1GbasePRXU4 {
12         description
13             "10/1GBASE-PRXU4";
14     }
15
16     bit b10GbasePRD4 {
17         description
18             "10GBASE-PRD4";
19     }
20
21     bit b10GbasePRU4 {
22         description
23             "10GBASE-PRU4";
24     }
25
26     bit b25GbaseCR {
27         description
28             "25GBASE-CR";
29     }
30
31     bit b25GbaseCRS {
32         description
33             "25GBASE-CRS";
34     }
35
36     bit b25GbaseKR {
37         description
38             "25GBASE-KR";
39     }
40
41     bit b25GbaseKRS {
42         description
43             "25GBASE-KRS";
44     }
45
46     bit b25GbaseR {
47         description
48             "25GBASE-R";
49     }
50
51     bit b25GbaseSR {
52         description
53             "25GBASE-SR";
54     }
55
56     bit b25GbaseT {
57         description
58             "25GBASE-T";
59     }
60
61     bit b40GbaseER4 {
62         description
63             "40GBASE-ER4";
64     }
65
66     bit b40GbaseR {
67         description
```

```
1         "40GBASE-R";
2     }
3     bit b40GbaseT {
4         description
5             "40GBASE-T";
6     }
7     bit b100GbaseCR4 {
8         description
9             "100GBASE-CR4";
10    }
11    bit b100GbaseKR4 {
12        description
13            "100GBASE-KR4";
14    }
15    bit b100GbaseKP4 {
16        description
17            "100GBASE-KP4";
18    }
19    bit b100GbaseR {
20        description
21            "100GBASE-R";
22    }
23    bit b100GbaseSR4 {
24        description
25            "100GBASE-SR4";
26    }
27    bit b2p5GbaseT {
28        description
29            "2.5GBASE-T";
30    }
31    bit b5GbaseT {
32        description
33            "5GBASE-T";
34    }
35    bit b100baseT1 {
36        description
37            "100BASE-T1";
38    }
39    bit b1000baseRHA {
40        description
41            "1000BASE-RHA";
42    }
43    bit b1000baseRHB {
44        description
45            "1000BASE-RHB";
46    }
47    bit b1000baseRHC {
48        description
49            "1000BASE-RHC";
50    }
51    bit b2p5GbaseKX {
52        description
53            "2.5GBASE-KX";
54    }
55 }
```

```
1      bit b2p5GbaseX {
2          description
3              "2.5GBASE-X";
4      }
5
6      bit b5GbaseKR {
7          description
8              "5GBASE-KR";
9      }
10
11     bit b5GbaseR {
12         description
13             "5GBASE-R";
14     }
15
16     bit b10GpassXR {
17         description
18             "10GPASS-XR";
19     }
20
21     bit b25GbaseLR {
22         description
23             "25GBASE-LR";
24     }
25
26     bit b25GbaseER {
27         description
28             "25GBASE-ER";
29     }
30
31     bit b50GbaseR {
32         description
33             "50GBASE-R";
34     }
35
36     bit b50GbaseCR {
37         description
38             "50GBASE-CR";
39     }
40
41     bit b50GbaseKR {
42         description
43             "50GBASE-KR";
44     }
45
46     bit b50GbaseSR {
47         description
48             "50GBASE-SR";
49     }
50
51     bit b50GbaseFR {
52         description
53             "50GBASE-FR";
54     }
55
56     bit b50GbaseLR {
57         description
58             "50GBASE-LR";
59     }
60
61     bit b50GbaseER {
62         description
63             "50GBASE-ER";
64     }
65
66     bit b100GbaseCR2 {
67         description
```

```
1         "100GBASE-CR2";
2     }
3     bit b100GbaseKR2 {
4         description
5             "100GBASE-KR2";
6     }
7     bit b100GbaseSR2 {
8         description
9             "100GBASE-SR2";
10    }
11    bit b100GbaseDR {
12        description
13            "100GBASE-DR";
14    }
15    bit b200GbaseR {
16        description
17            "200GBASE-R";
18    }
19    bit b200GbaseDR4 {
20        description
21            "200GBASE-DR4";
22    }
23    bit b200GbaseFR4 {
24        description
25            "200GBASE-FR4";
26    }
27    bit b200GbaseLR4 {
28        description
29            "200GBASE-LR4";
30    }
31    bit b200GbaseCR4 {
32        description
33            "200GBASE-CR4";
34    }
35    bit b200GbaseKR4 {
36        description
37            "200GBASE-KR4";
38    }
39    bit b200GbaseSR4 {
40        description
41            "200GBASE-SR4";
42    }
43    bit b200GbaseER4 {
44        description
45            "200GBASE-ER4";
46    }
47    bit b400GbaseR {
48        description
49            "400GBASE-R";
50    }
51    bit b400GbaseSR16 {
52        description
53            "400GBASE-SR16";
54    }
55    }
```

```
1      bit b400GbaseDR4 {
2          description
3              "400GBASE-DR4";
4      }
5      bit b400GbaseFR8 {
6          description
7              "400GBASE-FR8";
8      }
9      bit b400GbaseLR8 {
10         description
11             "400GBASE-LR8";
12     }
13     bit b400GbaseER8 {
14         description
15             "400GBASE-ER8";
16     }
17     bit b10baseT1L {
18         description
19             "10BASE-T1L";
20     }
21     bit b10baseT1S {
22         description
23             "10BASE-T1S";
24     }
25     bit b10baseT1SMD {
26         description
27             "10BASE-T1SMD";
28     }
29     bit b100GbaseFR1 {
30         description
31             "100GBASE-FR1";
32     }
33     bit b100GbaseLR1 {
34         description
35             "100GBASE-LR1";
36     }
37     bit b100GBASE-CR1 {
38         description
39             "100GBASE-LR1";
40     }
41     bit b400GbaseFR4 {
42         description
43             "400GBASE-FR4";
44     }
45     bit b400GbaseLR46 {
46         description
47             "400GBASE-LR46";
48     }
49     bit b400GbaseSR8 {
50         description
51             "400GBASE-SR8";
52     }
53     bit b400GbaseSR4p2 {
54         description
```

```
1         "400GBASE-SR4p2";
2     }
3     bit b2p5GbaseT1 {
4         description
5             "2.5GBASE-T1";
6     }
7     bit b5GbaseT1 {
8         description
9             "5GBASE-T1";
10    }
11    bit b10GbaseT1 {
12        description
13            "10GBASE-T1";
14    }
15    bit b25G10GbasePQGD2 {
16        description
17            "25/10GBASE-PQGD2";
18    }
19    bit b25G10GbasePQGD3 {
20        description
21            "25/10GBASE-PQGD3";
22    }
23    bit b25G10GbasePQGU2 {
24        description
25            "25/10GBASE-PQGU2";
26    }
27    bit b25G10GbasePQGU3 {
28        description
29            "25/10GBASE-PQGU3";
30    }
31    bit b25G10GbasePQXD2 {
32        description
33            "25/10GBASE-PQXD2";
34    }
35    bit b25G10GbasePQXD3 {
36        description
37            "25/10GBASE-PQXD3";
38    }
39    bit b25G10GbasePQXU2 {
40        description
41            "25/10GBASE-PQXU2";
42    }
43    bit b25G10GbasePQXU3 {
44        description
45            "25/10GBASE-PQXU3";
46    }
47    bit b25GbasePQGD2 {
48        description
49            "25GBASE-PQGD2";
50    }
51    bit b25GbasePQGD3 {
52        description
53            "25GBASE-PQGD3";
54    }
55    }
56    }
57    }
58    }
59    }
60    }
61    }
62    }
63    }
64    }
65    }
```

```
1      bit b25GbasePQGU2 {
2          description
3              "25GBASE-PQGU2";
4      }
5
6      bit b25GbasePQGU3 {
7          description
8              "25GBASE-PQGU3";
9      }
10
11     bit b25GbasePQXD2 {
12         description
13             "25GBASE-PQXD2";
14     }
15
16     bit b25GbasePQXD3 {
17         description
18             "25GBASE-PQXD3";
19     }
20
21     bit b25GbasePQXU2 {
22         description
23             "25GBASE-PQXU2";
24     }
25
26     bit b25GbasePQXU3 {
27         description
28             "25GBASE-PQXU3";
29     }
30
31     bit b50G10GbasePQGD2 {
32         description
33             "50/10GBASE-PQGD2";
34     }
35
36     bit b50G10GbasePQGD3 {
37         description
38             "50/10GBASE-PQGD3";
39     }
40
41     bit b50G10GbasePQGU2 {
42         description
43             "50/10GBASE-PQGU2";
44     }
45
46     bit b50G10GbasePQGU3 {
47         description
48             "50/10GBASE-PQGU3";
49     }
50
51     bit b50G10GbasePQXD2 {
52         description
53             "50/10GBASE-PQXD2";
54     }
55
56     bit b50G10GbasePQXD3 {
57         description
58             "50/10GBASE-PQXD3";
59     }
60
61     bit b50G10GbasePQXU2 {
62         description
63             "50/10GBASE-PQXU2";
64     }
65
66     bit b50G10GbasePQXU3 {
67         description
```

```
1         "50/10GBASE-PQXU3";
2     }
3     bit b50G25GbasePQGD2 {
4         description
5             "50/25GBASE-PQGD2";
6     }
7     bit b50G25GbasePQGD3 {
8         description
9             "50/25GBASE-PQGD3";
10    }
11    bit b50G25GbasePQGU2 {
12        description
13            "50/25GBASE-PQGU2";
14    }
15    bit b50G25GbasePQGU3 {
16        description
17            "50/25GBASE-PQGU3";
18    }
19    bit b50G25GbasePQXD2 {
20        description
21            "50/25GBASE-PQXD2";
22    }
23    bit b50G25GbasePQXD3 {
24        description
25            "50/25GBASE-PQXD3";
26    }
27    bit b50G25GbasePQXU2 {
28        description
29            "50/25GBASE-PQXU2";
30    }
31    bit b50G25GbasePQXU3 {
32        description
33            "50/25GBASE-PQXU3";
34    }
35    bit b50GbasePQGD2 {
36        description
37            "50GBASE-PQGD2";
38    }
39    bit b50GbasePQGD3 {
40        description
41            "50GBASE-PQGD3";
42    }
43    bit b50GbasePQGU2 {
44        description
45            "50GBASE-PQGU2";
46    }
47    bit b50GbasePQGU3 {
48        description
49            "50GBASE-PQGU3";
50    }
51    bit b50GbasePQXD2 {
52        description
53            "50GBASE-PQXD2";
54    }
55    }
```

```
1      bit b50GbasePQXD3 {
2          description
3              "50GBASE-PQXD3";
4      }
5      bit b50GbasePQXU2 {
6          description
7              "50GBASE-PQXU2";
8      }
9      bit b50GbasePQXU3 {
10         description
11             "50GBASE-PQXU3";
12     }
13     bit b100GbaseZR {
14         description
15             "100GBASE-ZR";
16     }
17     bit b10GbaseBR10D {
18         description
19             "10GBASE-BR10D";
20     }
21     bit b10GbaseBR10U {
22         description
23             "10GBASE-BR10U";
24     }
25     bit b10GbaseBR20D {
26         description
27             "10GBASE-BR20D";
28     }
29     bit b10GbaseBR20U {
30         description
31             "10GBASE-BR20U";
32     }
33     bit b10GbaseBR40D {
34         description
35             "10GBASE-BR40D";
36     }
37     bit b10GbaseBR40U {
38         description
39             "10GBASE-BR40U";
40     }
41     bit b25GbaseBR10D {
42         description
43             "25GBASE-BR10D";
44     }
45     bit b25GbaseBR10U {
46         description
47             "25GBASE-BR10U";
48     }
49     bit b25GbaseBR20D {
50         description
51             "25GBASE-BR20D";
52     }
53     bit b25GbaseBR20U {
54         description
```

```
1         "25GBASE-BR20U";
2     }
3     bit b25GbaseBR40D {
4         description
5             "25GBASE-BR40D";
6     }
7     bit b25GbaseBR40U {
8         description
9             "25GBASE-BR40U";
10    }
11    bit b50GbaseBR10D {
12        description
13            "50GBASE-BR10D";
14    }
15    bit b50GbaseBR10U {
16        description
17            "50GBASE-BR10U";
18    }
19    bit b50GbaseBR20D {
20        description
21            "50GBASE-BR20D";
22    }
23    bit b50GbaseBR20U {
24        description
25            "50GBASE-BR20U";
26    }
27    bit b50GbaseBR40D {
28        description
29            "50GBASE-BR40D";
30    }
31    bit b50GbaseBR40U {
32        description
33            "50GBASE-BR40U";
34    }
35    /* 802.3ck-2022 */
36    bit b100GBaseCR1 {
37        description
38            "100GBASE-CR1";
39    }
40    bit b100GBaseKR1 {
41        description
42            "100GBASE-KR1";
43    }
44    bit b200GBaseCR2 {
45        description
46            "200GBASE-CR2";
47    }
48    bit b200GBaseKR2 {
49        description
50            "200GBASE-KR2";
51    }
52    bit b400GBaseCR4 {
53        description
54            "400GBASE-CR4";
```

```
1      }
2      bit b400GBaseKR4 {
3          description
4              "400GBASE-KR4";
5      }
6      /* 802.3cs-2022 */
7      bit b10G2p5GBaseSP1Dx {
8          description
9              "10/2.5GBASE-SP1-Dx";
10     }
11     bit b10G2p5GBaseSP1Uxy {
12         description
13             "10/2.5GBASE-SP1-Uxy";
14     }
15     bit b10GBaseSP1Dx {
16         description
17             "10GBASE-SP1-Dx";
18     }
19     bit b10GBaseSP1Uxy {
20         description
21             "10GBASE-SP1-Uxy";
22     }
23     /* 802.3db-2022 */
24     bit b100GBaseSR1 {
25         description
26             "100GBASE-SR1";
27     }
28     bit b100GBaseVR1 {
29         description
30             "100GBASE-VR1";
31     }
32     bit bit200GBaseSR2 {
33         description
34             "200GBASE-SR2";
35     }
36     bit bit200GBaseVR2 {
37         description
38             "200GBASE-VR2";
39     }
40     bit bit400GBaseSR4 {
41         description
42             "400GBASE-SR4";
43     }
44     bit bit400GBaseVR4 {
45         description
46             "400GBASE-VR4";
47     }
48     /* 802.3cz-2023 */
49     bit b2p5GbaseAU {
50         description
51             "2.5GBASE-AU";
52     }
53     bit b5GbaseAU {
54         description
```

```
1         "5GBASE-AU";
2     }
3     bit b10GbaseAU {
4         description
5             "10GBASE-AU";
6     }
7     bit b25GbaseAU {
8         description
9             "25GBASE-AU";
10    }
11    bit b50GbaseAU {
12        description
13            "50GBASE-AU";
14    }
15    /* 802.3cy-2023 */
16    bit b25GbaseT1 {
17        description
18            "25GBASE-T1";
19    }
20    /* 802.3ca-2023 */
21    bit bit400GBaseDR4d2 {
22        description
23            "400GBASE-DR4-2";
24    }
25    bit bit800GBaseCR8 {
26        description
27            "800GBASE-CR8";
28    }
29    bit bit800GBaseDR8 {
30        description
31            "800GBASE-DR8";
32    }
33    bit bit800GBaseDR8d2 {
34        description
35            "800GBASE-DR8-2";
36    }
37    bit bit800GBaseKR8 {
38        description
39            "800GBASE-KR8";
40    }
41    bit bit800GBaseR {
42        description
43            "800GBASE-R";
44    }
45    bit bit800GBaseSR8 {
46        description
47            "800GBASE-SR8";
48    }
49    bit bit800GBaseVR8 {
50        description
51            "800GBASE-VR8";
52    }
53    }
54    }
55    description
```

```
1      "This data type is used to report the set of pmd-type values
2      an interface can implement.";
3  reference
4      "IEEE Std 802.3, Clause 30.5.1.1.3 aMAUTypeList";
5  }
6
7
8  typedef media-available-type {
9      type enumeration {
10         /* 802.3-2022 */
11         enum other {
12             description
13                 "Undefined";
14         }
15         enum unknown {
16             description
17                 "initializing, true state not yet known";
18         }
19         enum available {
20             description
21                 "link or light normal, loopback normal";
22         }
23         enum notAvailable {
24             description
25                 "link loss or low light, no loopback";
26         }
27         enum remoteFault {
28             description
29                 "remote fault with no detail";
30         }
31         enum remoteJabber {
32             description
33                 "remote fault, reason known to be jabber";
34         }
35         enum remoteLinkLoss {
36             description
37                 "remote fault, reason known to be far-end link loss";
38         }
39         enum remoteTest {
40             description
41                 "remote fault, reason known to be test";
42         }
43         enum offline {
44             description
45                 "offline, applies only to Clause 37 Auto-Negotiation";
46         }
47         enum autoNegError {
48             description
49                 "Auto-Negotiation Error, applies only to Clause 37
50                 auto-Negotiation";
51         }
52         enum pmdLinkFault {
53             description
54                 "PMD/PMA receive link fault";
55         }
56     }
57 }
```

```
1      enum pcsLinkFault {
2          description
3              "PCS receive link fault";
4      }
5      enum excessiveBER {
6          description
7              "PCS Bit Error Ratio monitor reporting excessive error
8              ratio";
9      }
10     enum dxsLinkFault {
11         description
12             "DTE XGXS receive link fault, applies only to XAUI ";
13     }
14     enum pxsLinkFault {
15         description
16             "PHY XGXS transmit link fault, applies only to XAUI";
17     }
18 }
19 description
20     "This data type is used to report media availability.";
21 reference
22     "IEEE Std 802.3, Clause 30.5.1.1.4 aMediaAvailable";
23 }
24
25 typedef auto-neg-ability-bits {
26     type bits {
27         /* Not PHY specific */
28         bit bOther {
29             description
30                 "Other abilities beyond the scope of this module.";
31         }
32         bit bHalfDuplex {
33             description
34                 "Half duplex";
35         }
36         bit bFullDuplex {
37             description
38                 "Full duplex";
39         }
40         bit bPause {
41             description
42                 "PAUSE operation as defined in Annex 31B";
43         }
44         bit bAPause {
45             description
46                 "Asymmetric PAUSE operation as defined in
47                 Clause 37, Annex 28B, and Annex 31B";
48         }
49         bit bSPause {
50             description
51                 "Symmetric PAUSE operation as defined in
52                 Clause 37, Annex 28B, and Annex 31B";
53         }
54         bit bBPause {
```

```
1      description
2          "Asymmetric and Symmetric PAUSE operation
3          as defined in Clause 37, Annex 28B, and Annex 31B";
4      }
5  bit bRemFault {
6      description
7          "Remote fault bit (RF) as specified in Clause 73
8          and Clause 98";
9  }
10
11  bit bRemFault1 {
12      description
13          "Remote fault bit 1 (RF1) as specified in Clause 37";
14  }
15
16  bit bRemFault2 {
17      description
18          "Remote fault bit 2 (RF2) as specified in Clause 37";
19  }
20
21  bit bFECCapable {
22      description
23          "FEC ability as specified in Clause 73 (see 73.6.5)
24          and Clause 74";
25  }
26
27  bit bFECRequested {
28      description
29          "FEC requested as specified in Clause 73 (see 73.6.5)
30          and Clause 74";
31  }
32
33  bit bRSFEC25Greq {
34      description
35          "25G RS-FEC requested as specified in Clause 73 (see 73.6.5)
36          and Clause 108";
37  }
38
39  bit bBaseFEC25Greq {
40      description
41          "25Gb/s BASE-R FEC ";
42  }
43
44  bit bForceMS {
45      description
46          "Force MASTER-SLAVE as specified in Clause 98 (see
47          98.2.1.2.5)";
48  }
49
50  /* PHY/PMD */
51  bit b10baseT {
52      description
53          "10BASE-T as defined in Clause 14";
54  }
55
56  bit b10baseT1L {
57      description
58          "10BASE-T1L as specified in Clause 146";
59  }
60
61  bit b10baseT1S {
62      description
63          "10BASE-T1S as specified in Clause 147";
64  }
65  }
```

```
1      bit b100baseTX {
2          description
3              "100BASE-TX as defined in Clause 25";
4      }
5
6      bit b1000baseX {
7          description
8              "1000BASE-X half duplex as specified in Clause 36";
9      }
10
11     bit b1000baseT {
12         description
13             "1000BASE-T PHY as specified in Clause 40";
14     }
15
16     bit b1000baseT1 {
17         description
18             "1000BASE-T1 as specified in Clause 97";
19     }
20
21     bit b2p5GbaseT {
22         description
23             "2.5GBASE-T PHY as specified in Clause 126";
24     }
25
26     bit b2p5GbaseT1 {
27         description
28             "2.5GBASE-T1 as specified in Clause 149";
29     }
30
31     bit b2p5GbaseKX {
32         description
33             "2.5GBASE-KX as specified in Clause 128";
34     }
35
36     bit b5GbaseT {
37         description
38             "5GBASE-T PHY as specified in Clause 126";
39     }
40
41     bit b5GbaseT1 {
42         description
43             "5GBASE-T1 as specified in Clause 149";
44     }
45
46     bit b5GbaseKR {
47         description
48             "5GBASE-KR as specified in Clause 130";
49     }
50
51     bit b10GbaseT {
52         description
53             "10GBASE-T PHY as specified in Clause 55";
54     }
55
56     bit b1000baseKX {
57         description
58             "1000BASE-KX as specified in Clause 70";
59     }
60
61     bit b10GbaseKX4 {
62         description
63             "10GBASE-KX4 as specified in Clause 71";
64     }
65
66     bit b10GbaseKR {
67         description
```

```
1         "10GBASE-KR as specified in Clause 72";
2     }
3     bit b25GbaseT {
4         description
5             "25GBASE-T as specified in Clause 113";
6     }
7     bit b25GbaseRS {
8         description
9             "25GBASE-CR-S as specified in Clause 110 or
10             25GBASE-KR-S as specified in Clause 111";
11     }
12     bit b25GbaseR {
13         description
14             "25GBASE-CR as specified in Clause 110 or
15             25GBASE-KR as specified in Clause 111";
16     }
17     bit b25GbaseT1 {
18         description
19             "25GBASE-T1 as specified in Clause 165";
20     }
21     bit b40GbaseKR4 {
22         description
23             "40GBASE-KR4 as specified in Clause 84";
24     }
25     bit b40GbaseCR4 {
26         description
27             "40GBASE-CR4 as specified in Clause 85";
28     }
29     bit b40GbaseT {
30         description
31             "40GBASE-T as specified in Clause 113";
32     }
33     bit b50GbaseR {
34         description
35             "50GBASE-CR as specified in Clause 136 or
36             50GBASE-KR as specified in Clause 137";
37     }
38     bit b100GbaseCR10 {
39         description
40             "100GBASE-CR10 as specified in Clause 85";
41     }
42     bit b100GbaseCR4 {
43         description
44             "100GBASE-CR4 as specified in Clause 92";
45     }
46     bit b100GbaseKR4 {
47         description
48             "100GBASE-KR4 as specified in Clause 93";
49     }
50     bit b100GbaseKP4 {
51         description
52             "100GBASE-KP4 as specified in Clause 94";
53     }
54     bit b100GbaseR2 {
```

```
1         description
2             "100GBASE-CR2 as specified in Clause 136 or
3             100GBASE-KR2 as specified in Clause 137";
4         }
5     bit b200GbaseR4 {
6         description
7             "200GBASE-CR4 as specified in Clause 136
8             or 200GBASE-KR4 as specified in Clause 137";
9     }
10 }
11 }
12 description
13     "This data type is used as the syntax of the
14     ieee802-eth-phy-auto-neg-capability-bits,
15     ieee802-eth-phy-auto-neg-cap-advertised-bits,
16     and ieee802-eth-phy-auto-neg-cap-received-bits
17     objects in ieee802-eth-phy-auto-neg-table.";
18 reference
19     "IEEE Std 802.3, Clause 30.6.1.1.5";
20 }
21 }
22 }
23 }
24 }
```

6. YANG module for Ethernet data terminal equipment (DTE) power via medium dependent interface (MDI) and Power over Data Lines (PoDL)

6.1 Introduction

This clause defines a YANG module to manage power via MDI Power Sourcing Equipment (PSE) and Power over Data Line (PoDL) PSE.

IEEE Std 802.3 defines the hardware registers that allow management interfaces to be built for a DTE Power via MDI and Power over Data Line device. The YANG module defined in this clause extends the Ethernet-interface YANG data modules defined in Clause 5 with the management objects required for the management of PoE and PoDL devices and ports.

6.2 YANG module structure

The deprecated *ieee802-ethernet-pse* superseded by *ieee802-ethernet-pse-2* YANG module of this clause is focused on the configuration and monitoring of the Power over Ethernet (PoE) function defined in IEEE Std 802.3, including power via MDI, as well as Power over Data Line which can also be considered as the single pair PoE. The module augments the *ieee802-ethernet-interface* YANG module with attributes for the PoE function. The module is partitioned into two major containers.

The PoE PSE container describes a multi-pair PSE, while the PoDL PSE describes a single-pair PSE.

6.3 Security considerations for Ethernet data terminal equipment (DTE) power via medium dependent interface (MDI) and Power over Data Line Module

There are a number of data nodes defined in this YANG module that are configurable as read-write. Such data nodes can be considered sensitive or vulnerable in some network environments. The support for configuration operations in a non-secure environment without proper protection can have a negative effect on network operations.

Setting the following data nodes to incorrect values can result in improper operation of the PSE, including the possibility that the Powered Device (PD) does not receive power from the PSE port:

- `pse-enable`
- `powering-pairs`

Some of the readable operational states in this module can be considered sensitive or vulnerable in some network environments. These are as follows:

- `pairs-control-ability`
- `classifications`
- `pd-power-class`
- `pse-type`
- `detected-pd-type`

It is thus important to control GET access to these data nodes and to possibly encrypt their values when sending them over the network.

6.4 Mapping of IEEE Std 802.3, Clause 30 managed objects

This subclause contains the mapping between YANG data nodes included in *ieee802-ethernet-pse* (see Table 6-1) YANG module, managed objects, and attributes defined in IEEE Std 802.3, Clause 30.

Table 6–1—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-ethernet-pse* YANG data nodes

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-pse</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
N/A	N/A		interfaces/interface/ethernet/pse	supported-pse-type	R
oPSE	aPSEAdminState	30.9.1	interfaces/interface/ethernet/pse/multi-pair	pse-enable	R
	aPSEPowerPairs			powering-pairs	R/W
	aPSEPowerPairsControlAbility			pairs-control-ability	R
	aPSEPowerDetectionStatus			detection-status	R
	aPSEPowerClassification			classifications	R
	aPSEActualPower			actual-power	R
	aPSEPowerAccuracy			power-accuracy	R
	aPSEInvalidSignatureCounter		interfaces/interface/ethernet/pse/multi-pair/statistics	invalid-signature	R
	aPSEPowerDeniedCounter			power-denied	R
	aPSEOverLoadCounter			overload	R
	aPSEShortCounter			short	R
	aPSEMPSAbsentCounter			mps-absent	R
	aPSECumulativeEnergy			cumulative-energy	R

Table 6–1—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-ethernet-pse* YANG data nodes (continued)

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-pse</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
oPoDLPSE	aPoDLPSEAdminState	30.15	interfaces/interface/ethernet/pse/single-pair	pse-enable	R
	aPoDLPSEPowerDetectionStatus			detection-status	R
	aPoDLPSEType			podl-type	R
	aPoDLPSEDetectedPDType			detected-pd-type	R
	aPoDLPSEDetectedPDPowerClass			pd-power-class	R
	aPoDLPSEActualPower			actual-power	R
	aPoDLPSEPowerAccuracy			power-accuracy	R
	aPoDLPSEInvalidSignatureCounter		interfaces/interface/ethernet/pse/single-pair/statistics	invalid-signature	R
	aPoDLPSEInvalidClassCounter			invalid-class	R
	aPoDLPSEPowerDeniedCounter			power-denied	R
	aPoDLPSEOverLoadCounter			overload	R
	aPoDLPSEMaintainFullVoltageSignatureAbsentCounter			fvs-absent	R
	aPoDLPSECumulativeEnergy			cumulative-energy	R

6.5 YANG module definitionⁱ

The YANG module tree hierarchy uses terms defined in IETF RFC 8407.

6.5.1 Tree hierarchy

6.5.1.1 ieee802-ethernet-pse

```

module: ieee802-ethernet-pse
  augment /if:interfaces/if:interface/ieee802-eth-if:ethernet:
    +--rw pse
      +--ro supported-pse-type?   identityref
      x--rw multi-pair!
        | +--rw pse-enable?       boolean
        | +--rw powering-pairs?   identityref
        | +--ro pairs-control-ability? boolean
        | x--ro detection-status?  multi-pair-detection-state
        | x--ro classifications?   power-class
        | x--ro statistics
        | | +--ro power-denied?    yang:counter64
        | | +--ro invalid-signature? yang:counter64
        | | +--ro mps-absent?      yang:counter64
        | | x--ro overload?        yang:counter64
        | | x--ro short?           yang:counter64
        | | +--ro cumulative-energy? yang:counter64
        | +--ro actual-power?      decimal64
        | +--ro power-accuracy?    int64
      x--rw single-pair!
        x--rw pse-enable?         boolean
        x--ro detection-status?    single-pair-detection-state
        +--ro pod1-type?          enumeration
        +--ro detected-pd-type?   enumeration
        x--ro pd-power-class?     power-class
        x--ro statistics
        | +--ro power-denied?      yang:counter64

```

ⁱ Copyright release for YANG modules: Users of this standard may freely reproduce the YANG module contained in this subclause so that it can be used for its intended purpose.

```

1      |  +--ro invalid-signature?   yang:counter64
2      |  +--ro invalid-class?     yang:counter64
3      |  +--ro overload?          yang:counter64
4      |  +--ro fvs-absence?       yang:counter64
5      |  +--ro cumulative-energy?   yang:counter64
6      |  +--ro actual-power?       decimal64
7      |  +--ro power-accuracy?     int64
8
9

```

6.5.1.2 ieee802-ethernet-pse-2

```

15 module: ieee802-ethernet-pse-2
16
17 augment /if:interfaces/if:interface/ieee802-eth-if:ethernet:
18   +--rw pse-2
19     +--ro supported-pse-type?   pse-support
20     +--rw multi-pair! {multi-pair-pse}?
21       |  +--rw pse-enable?      boolean
22       |  +--ro pse-state?       boolean
23       |  +--rw multi-pair-powering-pairs? multi-pair-powering-pairs
24       |  +--ro pairs-control-ability? boolean
25       |  +--ro detection-status? multi-pair-detection-state
26       |  +--ro classifications? multi-pair-power-class
27       |  +--ro statistics
28       |  |  +--ro power-denied?   yang:counter64
29       |  |  +--ro invalid-signature? yang:counter64
30       |  |  +--ro mps-absent?     yang:counter64
31       |  |  +--ro overload?      yang:counter64
32       |  |  +--ro short?         yang:counter64
33       |  |  +--ro cumulative-energy? yang:counter64
34       |  +--ro actual-power?     uint32
35       |  +--ro power-accuracy?    int64
36     +--rw single-pair! {single-pair-pse}?
37       +--rw pse-enable?          boolean
38       +--ro detection-status?    single-pair-detection-state
39       +--ro single-pair-type?    single-pair-pse-type
40       +--ro detected-pd-type?    single-pair-pd-type
41       +--ro pd-power-class?      single-pair-power-class
42
43

```

```
1  +---ro statistics
2  |   +---ro power-denied?      yang:counter64
3  |   +---ro invalid-signature? yang:counter64
4  |   +---ro invalid-class?    yang:counter64
5  |   +---ro overload?         yang:counter64
6  |   +---ro fvs-absence?      yang:counter64
7  |   +---ro cumulative-energy?  yang:counter64
8  +---ro actual-power?         uint32
9  +---ro power-accuracy?       int64
```

6.5.2 YANG module

In the YANG module definition, should any discrepancy between the text of the description for individual YANG nodes in the YANG modules of this clause and the corresponding definition in 6.2 through 6.5 of this clause occur, the definitions in the YANG modules attached to this standard shall take precedence.

An ASCII text version of the YANG module can be found at the following URL:^j

Editor's Note (to be removed prior to publication):

Yang files contained in <https://github.com/YangModels/yang/tree/main/standard/ieee/published/802.3> are IEEE 802.3.2-2019 version and will be updated at the publication time.

<https://github.com/YangModels/yang/tree/master/standard/ieee/published/802.3>

6.5.2.1 Ethernet PSE Module

Editor's Note (to be removed prior to publication):

Pretty printing of `ieee802-ethernet-pse.yang` file may change the appearance by adding whitespace and reformatting lines

Editor's Note (to be removed prior to publication):

IEEE Std 802.3.1 and IEEE Std 802.3.2 to be updated at the publication time

```
module ieee802-ethernet-pse {
  yang-version 1.1;
  namespace "urn:ieee:std:802.3:yang:ieee802-ethernet-pse";
  prefix ieee802-pse;

  import ietf-interfaces {
    prefix if;
    reference
      "IETF RFC 8343";
  }
  import ietf-yang-types {
    prefix yang;
    reference
      "IETF RFC 6991";
  }
  import ieee802-ethernet-interface {
    prefix ieee802-eth-if;
  }

  organization
    "IEEE 802.3 Ethernet Working Group
    Web URL: http://www.ieee802.org/3/";
  contact
    "Web URL: http://www.ieee802.org/3/";
  description
    "This module is deprecated and superceded by
    ieee802-ethernet-pse-2."
```

^jCopyright release for YANG modules: Users of this standard may freely reproduce the YANG module contained in this subclause so that it can be used for its intended purpose.

```
1      This module contains YANG definitions for configuring and
2      managing ports with Power Over Ethernet feature defined by
3      IEEE 802.3. It provides functionality roughly equivalent to
4      that of the POWER-ETHERNET-MIB defined in IETF RFC 3621.";
5
6      revision 2025-04-17 {
7          description
8              "Deprecated in IEEE Std 802.3.2-202x, Draft D3.3, superceeded by
9              ieee802-ethernet-pse-2";
10         reference
11             "IEEE Std IEEE Std 802.3.2-202x, unless dated explicitly";
12     }
13
14     identity pse-type {
15         description
16             "Base type for PSE.";
17     }
18
19     identity all {
20         base powering-pairs;
21         description
22             "All pairs are in use.";
23     }
24
25     identity four-pair {
26         base pse-type;
27         description
28             "PSE support IEEE Std 802.3, Clause 145.";
29     }
30
31     identity two-pair {
32         base pse-type;
33         description
34             "PSE supports IEEE Std 802.3, Clause 33.";
35     }
36
37     identity single-pair {
38         base pse-type;
39         description
40             "PSE support IEEE Std 802.3, Clause 104.";
41     }
42
43     identity powering-pairs {
44         description
45             "Base type for powering pairs.";
46     }
47
48     identity signal {
49         base powering-pairs;
50         description
51             "The signal pairs are in use.";
52     }
53
54     identity spare {
55         base powering-pairs;
56         description
57             "The spare pairs are in use.";
58     }
59
60 }
```

```
1  typedef multi-pair-detection-state {
2      type enumeration {
3          enum disabled {
4              value 1;
5              description
6                  "PSE disabled.";
7          }
8          enum searching {
9              value 2;
10             description
11                 "PSE is searching.";
12             }
13             enum deliveringPower {
14                 value 3;
15                 description
16                     "PSE is delivering power.";
17             }
18             enum fault {
19                 value 4;
20                 description
21                     "PSE fault detected.";
22             }
23             enum test {
24                 value 5;
25                 description
26                     "PSE test mode.";
27             }
28             enum otherFault {
29                 value 6;
30                 description
31                     "PSE implementation specific fault detected.";
32             }
33         }
34         status deprecated;
35         description
36             "Deprecated in IEEE Std 802.3.2-202x, Draft D3.1, superceeded by
37             ieee802-ethernet-pse-2
38             Detection state of a multi-pair PSE.";
39         reference
40             "IEEE Std 802.3, 30.9.1.1.5";
41     }
42
43     typedef single-pair-detection-state {
44         type enumeration {
45             enum unknown {
46                 value 1;
47                 description
48                     "True detection state unknown.";
49             }
50             enum disabled {
51                 value 2;
52                 description
53                     "PoDL PSE is disabled.";
54             }
55             enum searching {
56                 value 3;
57                 description
58                     "PoDL PSE is searching.";
59             }
60         }
61     }
```

```
1      enum deliveringPower {
2          value 4;
3          description
4              "PoDL PSE is delivering power.";
5      }
6      enum sleep {
7          value 5;
8          description
9              "PoDL PSE is in sleep state.";
10     }
11     enum idle {
12         value 6;
13         description
14             "PoDL PSE is idle.";
15     }
16     enum error {
17         value 7;
18         description
19             "PoDL PSE error.";
20     }
21 }
22
23
24 status deprecated;
25 description
26     "Deprecated in IEEE Std 802.3.2-202x, Draft D3.1, superceeded by
27     ieee802-ethernet-pse-2
28     Detection state of a PoDL PSE.";
29 reference
30     "IEEE Std 802.3, 30.15.1.1.3";
31 }
32
33
34 typedef power-class {
35     type enumeration {
36         enum class0 {
37             value 1;
38             description
39                 "Class 0";
40         }
41         enum class1 {
42             value 2;
43             description
44                 "Class 1";
45         }
46         enum class2 {
47             value 3;
48             description
49                 "Class 2";
50         }
51         enum class3 {
52             value 4;
53             description
54                 "Class 3";
55         }
56         enum class4 {
57             value 5;
58             description
59                 "Class 4";
60         }
61         enum class5 {
62             value 6;
```

```

1      description
2          "Class 5 (for PoDL-only)";
3      }
4      enum class6 {
5          value 7;
6          description
7              "Class 6 (for PoDL-only)";
8      }
9
10     enum class7 {
11         value 8;
12         description
13             "Class 7 (for PoDL-only)";
14     }
15     enum class8 {
16         value 9;
17         description
18             "Class 8 (for PoDL-only)";
19     }
20
21     enum class9 {
22         value 10;
23         description
24             "Class 9 (for PoDL-only)";
25     }
26     enum unknown {
27         value 11;
28         description
29             "Initializing, true Power Class not yet known
30             (only for PoDL PSE).";
31     }
32 }
33
34 status deprecated;
35 description
36     "Deprecated in IEEE Std 802.3.2-202x, Draft D3.1, superceeded by
37     ieee802-ethernet-pse-2
38     The power class.";
39
40 reference
41     "IEEE Std 802.3, 30.9.1.1.6 aPSEPowerClassification and
42     IEEE Std 802.3, 30.15.1.1.6 aPoDLPSEDetectedPDPowerClass.";
43 }
44
45 augment "/if:interfaces/if:interface/ieee802-eth-if:ethernet" {
46     status deprecated;
47     description
48         "Deprecated in IEEE Std 802.3.2-202x, Draft D3.1, superceeded by
49         ieee802-ethernet-pse-2
50         Augments ethernet interface configuration model with
51         nodes specific to DTE Power via MDI devices and ports";
52     container pse {
53         description
54             "DTE Power via MDI port configuration";
55         reference
56             "IEEE Std 802.3, 30.9.1 PoE PSE & IEEE Std 802.3, 30.15.1
57             PoDL PSE";
58         leaf supported-pse-type {
59             type identityref {
60                 base ieee802-pse:pse-type;
61             }
62             config false;
63             description
64

```

```
1         "PSE supports one or more of IEEE Std 802.3 Clause 33,  
2         Clause 104, or Clause 145.";
3     }
4     container multi-pair {
5         presence "PSE port supports IEEE Std 802.3, Clause 33.";
6         status deprecated;
7         description
8             "Deprecated in IEEE Std 802.3.2-202x, Draft D3.1, superceded by  
9             ieee802-ethernet-pse-2PSE port configuration in IEEE Std 802.3,  
10            30.9.1.";
11    }
12    leaf pse-enable {
13        type boolean;
14        default "false";
15        description
16            "When true enables the PSE function on the interface,  
17            when false disables the PSE function on the  
18            interface.";
19        reference
20            "IEEE Std 802.3, 30.9.1.1.2 aPSEAdminState";
21    }
22    leaf powering-pairs {
23        type identityref {
24            base powering-pairs;
25        }
26        description
27            "Describes or controls the PSE pairs in use. If the  
28            value of pairs-control-ability is true, this object  
29            is writeable.";
30        reference
31            "IEEE Std 802.3, 30.9.1.1.4 aPSEPowerPairs";
32    }
33    leaf pairs-control-ability {
34        type boolean;
35        default "true";
36        config false;
37        description
38            "Describes the ability to control switching the  
39            power sourcing pins of the PSE.";
40        reference
41            "IEEE Std 802.3, 30.9.1.1.3  
42            aPSEPowerPairsControlAbility";
43    }
44    leaf detection-status {
45        type multi-pair-detection-state;
46        config false;
47        status deprecated;
48        description
49            "Deprecated in IEEE Std 802.3.2-202x, Draft D3.1, superceded by  
50            ieee802-ethernet-pse-2  
51            Describes the operational status of the port  
52            PD detection.";
53        reference
54            "IEEE Std 802.3, 30.9.1.1.5 aPSEPowerDetectionStatus";
55    }
56    leaf classifications {
57        when "../detection-status = 'deliveringPower'" {
58            description
59                "Deprecated in IEEE Std 802.3.2-202x, Draft D3.1, superceded by  
60                ieee802-ethernet-pse-2
```

```

1         This node only applies when the detection status is
2         delivering power.";
3     }
4     type power-class;
5     config false;
6     status deprecated;
7     description
8         "The power class of the PSE port.";
9     reference
10        "IEEE Std 802.3, 30.9.1.1.6 aPSEPowerClassification";
11 }
12 container statistics {
13     config false;
14     status deprecated;
15     description
16         "Deprecated in IEEE Std 802.3.2-202x, Draft D3.1, superceeded by
17         ieee802-ethernet-pse-2statistics information of the multi-pair
18         port.";
19     leaf power-denied {
20         type yang:counter64;
21         description
22             "This counter is incremented when the PSE state
23             diagram enters the POWER_DENIED state, per
24             IEEE Std 802.3, Figure 33-9.";
25         reference
26             "IEEE Std 802.3, 30.9.1.1.14";
27     }
28     leaf invalid-signature {
29         type yang:counter64;
30         description
31             "This counter is incremented when the PSE state
32             diagram enters the SIGNATURE_INVALID state per
33             IEEE Std 802.3, Figure 33-9.";
34         reference
35             "IEEE Std 802.3, 30.9.1.1.11";
36     }
37     leaf mps-absent {
38         type yang:counter64;
39         description
40             "This counter is incremented when the PSE
41             transitions directly from the POWER_ON state to the
42             IDLE state due to tmpdo_timer_done being asserted,
43             per IEEE Std 802.3, Figure 33-9.";
44         reference
45             "IEEE Std 802.3, 30.9.1.1.20";
46     }
47     leaf overload {
48         type yang:counter64;
49         status deprecated;
50         description
51             "This counter is incremented when the PSE state
52             diagram enters the ERROR_DELAY state due to the
53             ovld_detected variable being TRUE, per
54             IEEE Std 802.3, Figure 33-9.";
55         reference
56             "IEEE Std 802.3, 30.9.1.1.17";
57     }
58     leaf short {
59         type yang:counter64;
60     }
61 }
62 
```

```
1         status deprecated;
2         description
3             "This Yang object is deprecated as its not defined in
4             base standard.
5             This counter is incremented when the PSE state
6             diagram enters the ERROR_DELAY state due to the
7             short_detected variable being TRUE, per
8             IEEE Std 802.3, Figure 33-9.";
9         reference
10            "IEEE Std 802.3, 30.9.1.1.10 aPSEShortCounter";
11    }
12    leaf cumulative-energy {
13        type yang:counter64;
14        units "millijoules";
15        description
16            "The cumulative energy supplied by the PSE as
17            measured at the MDI in millijoules.";
18        reference
19            "IEEE Std 802.3, 30.9.1.1.25";
20    }
21    }
22    leaf actual-power {
23        type decimal64 {
24            fraction-digits 4;
25        }
26        units "milliwatts";
27        config false;
28        description
29            "The actual power drawn by a PD over the port.";
30        reference
31            "IEEE Std 802.3, 30.9.1.1.23";
32    }
33    leaf power-accuracy {
34        type int64;
35        units "milliwatts";
36        config false;
37        description
38            "An integer value indicating the accuracy
39            associated with power-accuracy in +/- milliwatts.";
40        reference
41            "IEEE Std 802.3, 30.9.1.1.24";
42    }
43    }
44    container single-pair {
45        presence "PSE port working in PoDL.";
46        status deprecated;
47        description
48            "Deprecated in IEEE Std 802.3.2-202x, Draft D3.1, superceded by
49            ieee802-ethernet-pse-2
50            PoDL PSE configuration as defined in
51            IEEE Std 802.3, 30.15.1.";
52        leaf pse-enable {
53            type boolean;
54            default "false";
55            status deprecated;
56            description
57                "Deprecated in IEEE Std 802.3.2-202x, Draft D3.1, superceded by
58                ieee802-ethernet-pse-2
59                When true enables the PSE function on the interface,
```

```
1         when false disables the PSE function on the
2         interface.";
3     reference
4         "IEEE Std 802.3, 30.15.1.1.2 aPoDLPSEAdminState";
5 }
6 leaf detection-status {
7     type single-pair-detection-state;
8     config false;
9     status deprecated;
10    description
11        "Deprecated in IEEE Std 802.3.2-202x, Draft D3.1, superceeded by
12        ieee802-ethernet-pse-2
13        Indicates the current status of the PoDL PSE.";
14    reference
15        "IEEE Std 802.3, 30.15.1.1.3
16        aPoDLPSEPowerDetectionStatus";
17 }
18 leaf podl-type {
19     type enumeration {
20         enum unknown {
21             description
22                 "Unknown PSE type.";
23         }
24         enum typeA {
25             description
26                 "TypeA PSE";
27         }
28         enum typeB {
29             description
30                 "TypeB PSE";
31         }
32         enum typeC {
33             description
34                 "Type PSEC";
35         }
36         enum typeD {
37             description
38                 "TypeD PSE";
39         }
40         enum typeE {
41             description
42                 "TypeE PSE";
43         }
44         enum typeF {
45             description
46                 "TypeF PSE";
47         }
48     }
49     config false;
50     description
51         "PSE type specified in and
52         IEEE Std 802.3, 30.15.1.1.4.";
53 }
54 leaf detected-pd-type {
55     when "../detection-status = 'deliveringPower'" {
56         description
57             "This node only applies when the detection status is
58             delivering power.";
59     }
60 }
```

```
1      type enumeration {
2          enum unknown {
3              description
4                  "Unknown PD type";
5          }
6          enum typeA {
7              description
8                  "TypeA PD";
9          }
10         enum typeB {
11             description
12                 "TypeB PD";
13         }
14         enum typeC {
15             description
16                 "TypeC PD";
17         }
18         enum typeD {
19             description
20                 "TypeD PD";
21         }
22         enum typeE {
23             description
24                 "TypeE PD";
25         }
26         enum typeF {
27             description
28                 "TypeF PD";
29         }
30     }
31     config false;
32     description
33         "Indicates the Type of the detected PoDL PD as
34         specified in IEEE Std 802.3, 104.5.1.";
35     reference
36         "IEEE Std 802.3, 30.15.1.1.5 aPoDLPSEDetectedPDType";
37 }
38 leaf pd-power-class {
39     when "../detection-status = 'deliveringPower'" {
40         description
41             "This node only applies when the detection status is
42             delivering power.";
43     }
44     type power-class;
45     config false;
46     status deprecated;
47     description
48         "Deprecated in IEEE Std 802.3.2-202x, Draft D3.1, superceeded by
49         ieee802-ethernet-pse-2
50         Power class of the PD detected on the PSE port.";
51     reference
52         "IEEE Std 802.3, 30.15.1.1.6
53         aPoDLPSEDetectedPDPowerClass";
54 }
55 container statistics {
56     config false;
57     status deprecated;
58     description
59         "Deprecated in IEEE Std 802.3.2-202x, Draft D3.1, superceeded by
```

```
1         ieee802-ethernet-pse-2
2         PSE Discontinuities in the values of counters in this
3         container can occur at re-initialization of the
4         management system, and at other times as indicated by the
5         value of the 'discontinuity-time' leaf defined in the
6         ietf-interfaces YANG module (IETF RFC 8343).";
7     leaf power-denied {
8         type yang:counter64;
9         description
10            "This counter is incremented when the PoDL PSE state
11            diagram variable power_available transitions from
12            true to false (see IEEE Std 802.3, 104.4.3.3).";
13        reference
14            "IEEE Std 802.3, 30.15.1.1.9
15            aPoDLPSEPowerDeniedCounter";
16    }
17    leaf invalid-signature {
18        type yang:counter64;
19        description
20            "This counter is incremented when the PSE state
21            diagram enters the SIGNATURE_INVALID state per
22            IEEE Std 802.3, Figure 33-9.";
23        reference
24            "IEEE Std 802.3, 30.15.1.1.7
25            aPoDLPSEInvalidSignatureCounter";
26    }
27    leaf invalid-class {
28        type yang:counter64;
29        description
30            "This counter is incremented when the PoDL PSE state
31            diagram variable tclass_timer_done transitions from
32            false to true or when the valid_class variable
33            transitions from true to false
34            (see IEEE Std 802.3, 104.4.3.3).";
35        reference
36            "IEEE Std 802.3, 30.15.1.1.8
37            aPoDLPSEInvalidClassCounter";
38    }
39    leaf overload {
40        type yang:counter64;
41        description
42            "This counter is incremented when the PSE state
43            diagram variable overload_held transitions from
44            false to true (see IEEE Std 802.3, 104.4.3.3).";
45        reference
46            "IEEE Std 802.3, 30.15.1.1.10
47            aPoDLPSEOverLoadCounter";
48    }
49    leaf fvs-absence {
50        type yang:counter64;
51        description
52            "Maintain Full Voltage Signature absent counter.
53            This counter is incremented when the PoDL PSE state
54            diagram variable mfvs_timeout transitions from false
55            to true (see IEEE Std 802.3, 104.4.3.3).";
56        reference
57            "IEEE Std 802.3, 30.15.1.1.11
58            aPoDLPSEMaintainFullVoltageSignatureAbsentCounter";
59    }
60 }
```

```
1         leaf cumulative-energy {
2             type yang:counter64;
3             units "millijoules";
4             description
5                 "A count of the cumulative energy supplied by the
6                 PoDL PSE, measured at the MDI, and expressed in
7                 units of millijoules.";
8             reference
9                 "IEEE Std 802.3, 30.15.1.1.14
10                aPoDLPSECumulativeEnergy";
11        }
12    }
13    leaf actual-power {
14        type decimal64 {
15            fraction-digits 4;
16        }
17        units "milliwatts";
18        config false;
19        description
20            "An integer value indicating present (actual) power
21            being supplied by the PoDL PSE as measured at the MDI
22            in milliwatts.";
23        reference
24            "IEEE Std 802.3, 30.15.1.1.12 aPoDLPSEActualPower";
25    }
26    leaf power-accuracy {
27        type int64;
28        units "milliwatts";
29        config false;
30        description
31            "A signed integer value indicating the accuracy
32            associated
33            with power-accuracy in milliwatts.";
34        reference
35            "IEEE Std 802.3, 30.15.1.1.13 aPoDLPSEPowerAccuracy";
36    }
37 }
38 }
39 }
40 }
41 }
42 }
43 }
44 }
```

59.5.2.2 Ethernet PSE-2 Module

Editor's Note (to be removed prior to publication):

Pretty printing of ieee802-ethernet-interface.yang file may change the appearance by adding whitespace and reformatting lines

Editor's Note (to be removed prior to publication):

IEEE Std 802.3.1 and IEEE Std 802.3.2 to be updated at the publication time

```
61 module ieee802-ethernet-pse-2 {
62     yang-version 1.1;
63     namespace "urn:ieee:std:802.3:yang:ieee802-ethernet-pse-2";
64     prefix ieee802-pse-2024;
```

```
1
2   import ietf-interfaces {
3     prefix if;
4     reference
5       "IETF RFC 8343";
6   }
7   import ietf-yang-types {
8     prefix yang;
9     reference
10      "IETF RFC 6991";
11   }
12   import ieee802-ethernet-interface {
13     prefix ieee802-eth-if;
14   }
15
16   organization
17     "IEEE 802.3 Ethernet Working Group
18     Web URL: http://www.ieee802.org/3/";
19   contact
20     "Web URL: http://www.ieee802.org/3/";
21   description
22     "This module contains YANG definitions for configuring and
23     managing ports with Power Over Ethernet features defined by
24     IEEE Std 802.3-2022.
25     This module is based on, and supersedes ieee802-ethernet-pse.yang
26     from IEEE Std 802.3.2-2019.";
27
28   revision 2025-04-17 {
29     description
30       "Updates under IEEE Std 802.3.2-202x, Draft D3.3";
31     reference
32       "IEEE Std IEEE Std 802.3.2-202x, unless dated explicitly";
33   }
34
35   feature multi-pair-pse {
36     description
37       "This object indicates the support for IEEE Std 802.3
38       Clause 33 and/or 145.";
39   }
40
41   feature single-pair-pse {
42     description
43       "This object indicates the support for IEEE Std 802.3
44       Clause 104.";
45   }
46
47   typedef multi-pair-detection-state {
48     type enumeration {
49       enum disabled {
50         description
51           "PSE disabled.";
52       }
53       enum searching {
54         description
55           "PSE is searching.";
56       }
57       enum deliveringPower {
58         description
59           "PSE is delivering power.";
60       }
61     }
62   }
```

```
1      }
2      enum fault {
3          description
4              "PSE fault detected.";
5      }
6      enum test {
7          description
8              "PSE test mode.";
9      }
10     }
11     enum otherFault {
12         description
13             "PSE implementation specific fault detected.";
14     }
15 }
16 description
17     "Detection state of a multi-pair PSE.";
18 reference
19     "IEEE Std 802.3, 30.9.1.1.5";
20 }
21
22
23 typedef single-pair-detection-state {
24     type enumeration {
25         enum unknown {
26             description
27                 "True detection state unknown.";
28         }
29         enum disabled {
30             description
31                 "PSE is disabled.";
32         }
33         enum searching {
34             description
35                 "PSE is searching.";
36         }
37         enum deliveringPower {
38             description
39                 "PSE is delivering power.";
40         }
41         enum sleep {
42             description
43                 "PSE is in sleep state.";
44         }
45         enum idle {
46             description
47                 "PSE is idle.";
48         }
49         enum error {
50             description
51                 "PSE error.";
52         }
53     }
54 }
55
56 description
57     "Detection state of a single-pair PSE.";
58 reference
59     "IEEE Std 802.3, 30.15.1.1.3";
60 }
61
62
63 typedef multi-pair-power-class {
64     type enumeration {
```

```
1      enum class0 {
2          description
3              "Class 0";
4      }
5      enum class1 {
6          description
7              "Class 1";
8      }
9
10     enum class2 {
11         description
12             "Class 2";
13     }
14     enum class3 {
15         description
16             "Class 3";
17     }
18     enum class4 {
19         description
20             "Class 4";
21     }
22
23     enum class5 {
24         description
25             "Class 5";
26     }
27     enum class6 {
28         description
29             "Class 6";
30     }
31
32     enum class7 {
33         description
34             "Class 7";
35     }
36     enum class8 {
37         description
38             "Class 8";
39     }
40 }
41
42 description
43     "Multi-pair PoE power class.";
44 reference
45     "IEEE Std 802.3, 30.9.1.1.6 aPSEPowerClassification ";
46 }
47
48 typedef pse-support {
49     type enumeration {
50         enum two-pair {
51             description
52                 "PSE port supports IEEE Std 802.3, Clause 33.";
53         }
54         enum four-pair {
55             description
56                 "PSE port supports IEEE Std 802.3, Clause 33 and 145.";
57         }
58         enum single-pair {
59             description
60                 "PSE port supports IEEE Std 802.3, Clause 104.";
61         }
62     }
63 }
64
65 description
```

```
1      "PSE support.";
2  }
3
4  typedef multi-pair-powering-pairs {
5      type enumeration {
6          enum signal {
7              description
8                  "Using signal pairs.";
9          }
10         enum spare {
11             description
12                 "Using spare pairs.";
13         }
14         enum both {
15             description
16                 "Using signal and spare pairs.";
17         }
18     }
19 }
20
21 description
22     "Powering pairs.";
23 }
24
25 typedef single-pair-pd-type {
26     type enumeration {
27         enum unknown {
28             description
29                 "Unknown PD type";
30         }
31         enum typeA {
32             description
33                 "TypeA PD";
34         }
35         enum typeB {
36             description
37                 "TypeB PD";
38         }
39         enum typeC {
40             description
41                 "TypeC PD";
42         }
43         enum typeD {
44             description
45                 "TypeD PD";
46         }
47         enum typeE {
48             description
49                 "TypeE PD";
50         }
51         enum typeF {
52             description
53                 "TypeF PD";
54         }
55     }
56 }
57
58 description
59     "Single-pair PoE PD type.";
60
61 reference
62     "IEEE Std 802.3, 30.15.1.1.5 aPoDLPSEDetectedPDType";
63
64 }
65
```

```
1  typedef single-pair-pse-type {
2      type enumeration {
3          enum unknown {
4              description
5                  "Unknown PSE type.";
6          }
7          enum typeA {
8              description
9                  "TypeA PSE";
10         }
11         enum typeB {
12             description
13                 "TypeB PSE";
14         }
15         enum typeC {
16             description
17                 "TypeC PSE";
18         }
19         enum typeD {
20             description
21                 "TypeD PSE";
22         }
23         enum typeE {
24             description
25                 "TypeE PSE";
26         }
27         enum typeF {
28             description
29                 "TypeF PSE";
30         }
31     }
32     description
33         "Single-pair PoE PSE type.";
34     reference
35         "IEEE Std 802.3, 30.15.1.1.5 aPoDLPSEDetectedPDType";
36 }
37
38 typedef single-pair-power-class {
39     type enumeration {
40         enum class0 {
41             description
42                 "Class 0";
43         }
44         enum class1 {
45             description
46                 "Class 1";
47         }
48         enum class2 {
49             description
50                 "Class 2";
51         }
52         enum class3 {
53             description
54                 "Class 3";
55         }
56         enum class4 {
57             description
58                 "Class 4";
59         }
60     }
61 }
```

```
1      enum class5 {
2          description
3              "Class 5";
4      }
5      enum class6 {
6          description
7              "Class 6";
8      }
9
10     enum class7 {
11         description
12             "Class 7";
13     }
14     enum class8 {
15         description
16             "Class 8";
17     }
18     enum class9 {
19         description
20             "Class 9";
21     }
22
23     enum class10 {
24         description
25             "Class 10";
26     }
27     enum class11 {
28         description
29             "Class 11";
30     }
31     enum class12 {
32         description
33             "Class 12";
34     }
35
36     enum class13 {
37         description
38             "Class 13";
39     }
40
41     enum class14 {
42         description
43             "Class 14";
44     }
45     enum class15 {
46         description
47             "Class 15";
48     }
49     enum unknown {
50         description
51             "Initializing, true Power Class not yet known.";
52     }
53 }
54
55 description
56     "Single-pair PoE power class.";
57 reference
58     "IEEE Std 802.3, 30.15.1.1.6 aPoDLPSEDetectedPDPowerClass";
59
60
61 augment "/if:interfaces/if:interface/ieee802-eth-if:ethernet" {
62     description
63         "Augments ethernet interface configuration model with
64         nodes specific to Multi-pair and Single-pair PoE.";
65 }
```

```
1     container pse-2 {
2         description
3             "Multi-pair and Single-pair PoE port configuration.
4
5             Discontinuities in the values of counters in this container
6             can occur at re-initialization of the management system,
7             and at other times as indicated by the value of the
8             'discontinuity-time' leaf defined in the ietf-interfaces
9             YANG module (IETF RFC 8343).";
10        reference
11            "IEEE Std 802.3, 30.9.1 multi-pair PoE PSE and
12              IEEE Std 802.3, 30.15.1 single-pair PoE PSE";
13        leaf supported-pse-type {
14            type pse-support;
15            config false;
16            description
17                "The PSE types are supported.";
18        }
19        /* Multi-pair objects */
20        container multi-pair {
21            if-feature "multi-pair-pse";
22            presence "PSE port supports IEEE Std 802.3 Clause 33
23                      and/or Clause 145.";
24            description
25                "PSE port configuration in IEEE Std 802.3, 30.9.1.";
26            leaf pse-enable {
27                type boolean;
28                default "false";
29                description
30                    "When true enables the PSE function on the port,
31                     when false disables the PSE function on the port.";
32                reference
33                    "IEEE Std 802.3, 30.9.1.2.1 acPSEAdminControl";
34            }
35            leaf pse-state {
36                type boolean;
37                config false;
38                description
39                    "The PSE enabled state on the port.";
40                reference
41                    "IEEE Std 802.3, 30.9.1.1.2 aPSEAdminState";
42            }
43            leaf multi-pair-powering-pairs {
44                type multi-pair-powering-pairs;
45                description
46                    "The PSE pairs in use. If the value of
47                     pairs-control-ability is true, this object is
48                     writeable.";
49                reference
50                    "IEEE Std 802.3, 30.9.1.1.4 aPSEPowerPairs";
51            }
52            leaf pairs-control-ability {
53                type boolean;
54                default "true";
55                config false;
56                description
57                    "Can the PSE Port control switching the
58                     power sourcing pins.";
59                reference
```

```
1      "IEEE Std 802.3, 30.9.1.1.3 aPSEPowerPairsControlAbility";
2  }
3  leaf detection-status {
4      type multi-pair-detection-state;
5      config false;
6      description
7          "The operational status of the port PD detection.";
8      reference
9          "IEEE Std 802.3, 30.9.1.1.5 aPSEPowerDetectionStatus";
10 }
11 leaf classifications {
12     when "../detection-status = 'deliveringPower'" {
13         description
14             "This node only applies when the detection status is
15              deliveringPower.";
16     }
17     type multi-pair-power-class;
18     config false;
19     description
20         "The power class of the detected PD.";
21     reference
22         "IEEE Std 802.3, 30.9.1.1.6 aPSEPowerClassification";
23 }
24 /* Multi-pair statistics */
25 container statistics {
26     config false;
27     description
28         "Multi-pair PoE Statistics information for the port.";
29     leaf power-denied {
30         type yang:counter64;
31         description
32             "The number of times the PSE denied power.";
33         reference
34             "IEEE Std 802.3, 30.9.1.1.14";
35     }
36     leaf invalid-signature {
37         type yang:counter64;
38         description
39             "The number of times the PSE detected an
40              invalid signature.";
41         reference
42             "IEEE Std 802.3, 30.9.1.1.11";
43     }
44     leaf mps-absent {
45         type yang:counter64;
46         description
47             "The number of times the PSE
48              lost the 'Maintain Power Signature' (MPS).";
49         reference
50             "IEEE Std 802.3, 30.9.1.1.20";
51     }
52     leaf overload {
53         type yang:counter64;
54         status deprecated;
55         description
56             "The number of times the PSE detected
57              an output current overload.";
58         reference
59             "IEEE Std 802.3, 30.9.1.1.17";
60     }
61 }
```

```
1      }
2      leaf short {
3          type yang:counter64;
4          status deprecated;
5          description
6              "This object is deprecated as it's not defined in base
7              standard.";
8      }
9
10     leaf cumulative-energy {
11         type yang:counter64;
12         units "millijoules";
13         description
14             "The cumulative energy supplied by the PSE.";
15         reference
16             "IEEE Std 802.3, 30.9.1.1.25";
17     }
18 }
19
20 /* Multi-pair measurement */
21 leaf actual-power {
22     type uint32;
23     units "milliwatts";
24     config false;
25     description
26         "The actual power drawn by a PD over the port.";
27     reference
28         "IEEE Std 802.3, 30.9.1.1.23";
29 }
30
31 leaf power-accuracy {
32     type int64;
33     units "milliwatts";
34     config false;
35     description
36         "The power accuracy of the port.";
37     reference
38         "IEEE Std 802.3, 30.9.1.1.24";
39 }
40
41 }
42 /* Single-pair objects */
43 container single-pair {
44     if-feature "single-pair-pse";
45     presence "PSE port supports IEEE Std 802.3 Clause 104.";
46     description
47         "Single-pair PSE configuration as defined in
48         IEEE Std 802.3, 30.15.1.";
49     leaf pse-enable {
50         type boolean;
51         default "false";
52         description
53             "When true enables the PSE function on the port,
54             when false disables the PSE function on the port.";
55         reference
56             "IEEE Std 802.3, 30.15.1.1.2 aPoDLPSEAdminState";
57     }
58 }
59 leaf detection-status {
60     type single-pair-detection-state;
61     config false;
62     description
63         "Indicates the current status of the Single-pair PSE.";
64     reference
65 
```

```
1         "IEEE Std 802.3,  
2         30.15.1.1.3 aPoDLPSEPowerDetectionStatus";  
3     }  
4     leaf single-pair-type {  
5         type single-pair-pse-type;  
6         config false;  
7         description  
8             "Single-pair PSE type.";  
9         reference  
10            "IEEE Std 802.3, 30.15.1.1.4 aPoDLPSEType";  
11    }  
12    leaf detected-pd-type {  
13        when "../detection-status = 'deliveringPower'" {  
14            description  
15                "This node only applies when the detection status is  
16                delivering power.";  
17            }  
18            type single-pair-pd-type;  
19            config false;  
20            description  
21                "Indicates the detected type of the single-pair PD.";  
22            reference  
23                "IEEE Std 802.3, 30.15.1.1.5 aPoDLPSEDetectedPDType";  
24        }  
25    leaf pd-power-class {  
26        when "../detection-status = 'deliveringPower'" {  
27            description  
28                "This node only applies when the detection status is  
29                delivering power.";  
30            }  
31            type single-pair-power-class;  
32            config false;  
33            description  
34                "The power class of the detected PD.";  
35            reference  
36                "IEEE Std 802.3,  
37                30.15.1.1.6 aPoDLPSEDetectedPDPowerClass";  
38        }  
39    }  
40    /* Single-pair statistics */  
41    container statistics {  
42        config false;  
43        description  
44            "Single-pair PoE Statistics information for the port.";  
45        leaf power-denied {  
46            type yang:counter64;  
47            description  
48                "The number of times the PSE denied power.";  
49            reference  
50                "IEEE Std 802.3,  
51                30.15.1.1.9 aPoDLPSEPowerDeniedCounter";  
52        }  
53        leaf invalid-signature {  
54            type yang:counter64;  
55            description  
56                "The number of times the PSE detected an  
57                invalid signature.";  
58            reference  
59                "IEEE Std 802.3, 30.15.1.1.7  
60                aPoDLPSEInvalidSignatureCounter";  
61        }  
62    }  
63    }  
64    }  
65    }
```

```

1      }
2      leaf invalid-class {
3          type yang:counter64;
4          description
5              "The number of times PD classification timed out
6              or did not get valid class information from SCCP.";
7          reference
8              "IEEE Std 802.3,
9              30.15.1.1.8 aPoDLPSEInvalidClassCounter";
10     }
11     leaf overload {
12         type yang:counter64;
13         description
14             "The number of times the PSE detected
15             an output current overload.";
16         reference
17             "IEEE Std 802.3, 30.15.1.1.10 aPoDLPSEOverLoadCounter";
18     }
19     leaf fvs-absence {
20         type yang:counter64;
21         description
22             "The number of times the PSE
23             lost the 'Maintain Full Voltage Signature' (MFVS).";
24         reference
25             "IEEE Std 802.3, 30.15.1.1.11
26             aPoDLPSEMaintainFullVoltageSignatureAbsentCounter";
27     }
28     leaf cumulative-energy {
29         type yang:counter64;
30         units "millijoules";
31         description
32             "The cumulative energy supplied by the PSE as
33             measured at the MDI.";
34         reference
35             "IEEE Std 802.3, 30.15.1.1.14 aPoDLPSECumulativeEnergy";
36     }
37 }
38 /* Single-pair measurement */
39 leaf actual-power {
40     type uint32;
41     units "milliwatts";
42     config false;
43     description
44         "The actual power drawn by a PD over the port.";
45     reference
46         "IEEE Std 802.3, 30.15.1.1.12 aPoDLPSEActualPower";
47 }
48 leaf power-accuracy {
49     type int64;
50     units "milliwatts";
51     config false;
52     description
53         "The power accuracy of the port.";
54     reference
55         "IEEE Std 802.3, 30.15.1.1.13 aPoDLPSEPowerAccuracy";
56 }
57 }
58 }
59 }
60 }
61 }
62 }
63 }
64 }
65 }

```

1 }
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7. YANG module for Ethernet Passive Optical Network (EPON)

7.1 Introduction

This clause defines a YANG module to manage Ethernet Passive Optical Network (EPON).

EPON is defined in IEEE Std 802.3, covering Physical Layers and Media Access Control sublayers. The Passive Optical Network (PON) is comprised of sections of single-mode fiber connected with passive optical splitter/coupler devices, forming a passive optical tree, as shown in Figure 7–1. Individual branches of the PON are terminated with the Optical Line Terminal (OLT) in the Central Office or at remote optical nodes, and Optical Network Units (ONUs) near the subscribers. ONUs can be located either in some remote location (e.g., basement in a multi-dwelling unit) or directly at the subscriber premises. Various types of Customer Premises Equipment (CPE) can be connected to ONUs or even integrated with such devices.

7.2 YANG module structure

The *ieee802-ethernet-pon* YANG module of this clause is focused on the configuration and monitoring of EPON.

7.3 Mapping of IEEE Std 802.3, Clause 30 managed objects

This sub-clause contains the mapping between YANG data nodes included in *ieee802-ethernet-pon* (see Table 7–1) YANG module, managed objects, and attributes defined in IEEE Std 802.3.1, IEEE8023-DOT3-EPON-MIB.

**Table 7–1—Mapping between IEEE Std 802.3.1, IEEE8023-DOT3-EPON-MIB managed objects
and *ieee802-ethernet-pon* YANG data nodes**

IEEE Std 802.3.1, IEEE8023-DOT3-EPON-MIB		Corresponding <i>ieee802-ethernet-pon</i> YANG data nodes		
Managed object(s)	Attribute(s)	Container(s)	Data node(s)	R/W
dot3EponFecTable	dot3EponFecMode		fec-mode	R/W
	dot3EponFecPCSCodingViolation	statistics-pon-fec	fec-code-group-violations	R
	dot3EponFecAbility		fec-capability	R
	dot3EponFecCorrectedBlocks	statistics-pon-fec	fec-code-word-corrected-errors	R
	dot3EponFecUncorrectableBlocks	statistics-pon-fec	fec-code-word-uncorrected-errors	R
	dot3EponFecBufferHeadCodingViolation	statistics-pon-fec	fec-buffer-head-coding-violation	R
dot3MpcpControl Table	dot3MpcpAdminState		mpcp-admin-state	R/W
	dot3MpcpMode		mpcp-mode	R
	dot3MpcpLinkID		mpcp-logical-link-id	R
	dot3MpcpRemoteMACAddress		mpcp-remote-mac-address	R
	dot3MpcpRegistrationState		mpcp-logical-link-state	R
	dot3MpcpSyncTime		mpcp-sync-time	R
	dot3MpcpTransmitElapsed		mpcp-elapsed-time-out	R
	dot3MpcpReceiveElapsed		mpcp-elapsed-time-in	R
	dot3MpcpRoundTripTime		mpcp-round-trip-time	R
	dot3MpcpMaximumPendingGrants		mpcp-maximum-grant-count	R

Table 7–1—Mapping between IEEE Std 802.3.1, IEEE8023-DOT3-EPON-MIB managed objects and *ieee802-ethernet-pon* YANG data nodes (continued)

IEEE Std 802.3.1, IEEE8023-DOT3-EPON-MIB		Corresponding <i>ieee802-ethernet-pon</i> YANG data nodes		
Managed object(s)	Attribute(s)	Container(s)	Data node(s)	R/W
dot3ExtPkgQueueSets Table	dot3QueueSetIndex	mpcp-queue-thresholds	mpcp-queue-set-index	R/W
	dot3ExtPkgObjectReportThreshold		mpcp-queue-set-threshold	R/W
	dot3QueueIndex	mpcp-queues	mpcp-queue-index	R/W
	dot3ExtPkgObjectReportNumThreshold		mpcp-queue-threshold-count	R/W
	dot3ExtPkgObjectReportMaximumNumThreshold		mpcp-queue-threshold-count-max	R
	dot3ExtPkgStatTxFramesQueue		in-mpcp-queue-frames	R
	dot3ExtPkgStatRxFramesQueue		out-mpcp-queue-frames	R
	dot3ExtPkgStatDroppedFramesQueue		mpcp-queue-frames-drop	R
dot3ExtPkgControl Table	dot3ExtPkgObjectReset dot3MpcpOperStatus		mpcp-logical-link-admin-state	R/W
	dot3ExtPkgObjectNumberOfLLIDs		mpcp-logical-link-count	R
	dot3ExtPkgObjectReportMaximumNumQueues		mpcp-maximum-queue-count-per-report	R
dot3RecognizedMulticast-IDs Table	dot3RecognizedMulticastID	multicast-IDs	multicast-ID	R/W

**Table 7–1—Mapping between IEEE Std 802.3.1, IEEE8023-DOT3-EPON-MIB managed objects
and *ieee802-ethernet-pon* YANG data nodes (continued)**

IEEE Std 802.3.1, IEEE8023-DOT3-EPON-MIB		Corresponding <i>ieee802-ethernet-pon</i> YANG data nodes		
Managed object(s)	Attribute(s)	Container(s)	Data node(s)	R/W
dot3OmpEmulation Table	dot3OmpEmulationType	statistics-ompe	ompe-mode	R
	dot3OmpEmulationSLDErrors		in-ompe-frames-errored-sld	R
	dot3OmpEmulationCRC8Errors		in-ompe-frames-errored-crc8	R
	dot3OmpEmulationBadLLID		in-ompe-frames-with-bad-llid	R
	dot3OmpEmulationGoodLLID		in-ompe-frames-with-good-llid	R
	dot3OmpEmulationBroadcastBitNotOnuLlid		in-ompe-frames-not-match-onu-llid-broadcast	R
	dot3OmpEmulationOnuLLIDNotBroadcast		in-ompe-frames-match-onu-llid-not-broadcast	R
	dot3OmpEmulationBroadcastBitPlusOnuLlid		in-ompe-frames-match-onu-llid-broadcast	R
	dot3OmpEmulationNotBroadcastBitNotOnuLlid		in-ompe-frames-not-match-onu-llid-not-broadcast	R
			in-ompe-frames	R
			ompe-onu-frames-with-good-llid-good-crc8	
			ompe-olt-frames-with-good-llid-good-crc8	

**Table 7–1—Mapping between IEEE Std 802.3.1, IEEE8023-DOT3-EPON-MIB managed objects
and *ieee802-ethernet-pon* YANG data nodes (continued)**

IEEE Std 802.3.1, IEEE8023-DOT3-EPON-MIB		Corresponding <i>ieee802-ethernet-pon</i> YANG data nodes		
Managed object(s)	Attribute(s)	Container(s)	Data node(s)	R/W
dot3MpcpStat Table	dot3MpcpMACCtrlFramesTransmitted	statistics-mpcp	out-mpcp-mac-ctrl-frames	R
	dot3MpcpMACCtrlFramesReceived		in-mpcp-mac-ctrl-frames	R
	dot3MpcpDiscoveryWindowsSent		mpcp-discovery-window-count	R
	dot3MpcpDiscoveryTimeout		mpcp-discovery-timeout-count	R
	dot3MpcpTxRegRequest		out-mpcp-register-req	R
	dot3MpcpRxRegRequest		in-mpcp-register-req	R
	dot3MpcpTxRegAck		out-mpcp-register-ack	R
	dot3MpcpRxRegAck		in-mpcp-register-ack	R
	dot3MpcpTxReport		out-mpcp-report	R
	dot3MpcpRxReport		in-mpcp-report	R
	dot3MpcpTxGate		out-mpcp-gate	R
	dot3MpcpRxGate		in-mpcp-gate	R
	dot3MpcpTxRegister		out-mpcp-register	R
	dot3MpcpRxRegister		in-mpcp-register	R

Table 7–1—Mapping between IEEE Std 802.3.1, IEEE8023-DOT3-EPON-MIB managed objects and *ieee802-ethernet-pon* YANG data nodes (continued)

IEEE Std 802.3.1, IEEE8023-DOT3-EPON-MIB		Corresponding <i>ieee802-ethernet-pon</i> YANG data nodes		
Managed object(s)	Attribute(s)	Container(s)	Data node(s)	R/W
dot3ExtPkgOptIf Table	dot3ExtPkgOptIfLowerInputPowerThreshold	thresholds-trx	in-trx-power-low-threshold	R/W
	dot3ExtPkgOptIfUpperInputPowerThreshold		in-trx-power-high-threshold	R/W
	dot3ExtPkgOptIfLowerOutputPowerThreshold		out-trx-power-low-threshold	R/W
	dot3ExtPkgOptIfUpperOutputPowerThreshold		out-trx-power-high-threshold	R/W
	dot3ExtPkgOptIfSignalDetect		in-trx-power-signal-detect	R
	dot3ExtPkgOptIfInputPower		in-trx-power	R
	dot3ExtPkgOptIfLowInputPower		in-trx-power-low-15-minutes-bin	R
	dot3ExtPkgOptIfHighInputPower		in-trx-power-high-15-minutes-bin	R
	dot3ExtPkgOptIfTransmitEnable		out-trx-power-signal-detect	R/W
	dot3ExtPkgOptIfOutputPower		out-trx-power	R
	dot3ExtPkgOptIfLowOutputPower		out-trx-power-low-15-minutes-bin	R
	dot3ExtPkgOptIfHighOutputPower		out-trx-power-high-15-minutes-bin	R
	dot3ExtPkgOptIfSuspectedFlag		trx-data-reliable	R

7.4 YANG module definition^k

The YANG module tree hierarchy uses terms defined in IETF RFC 8407.

7.4.1 Tree hierarchy

7.4.1.1 ieee802-ethernet-pon

```

module: ieee802-ethernet-pon
  augment /if:interfaces/if:interface/ieee802-eth-if:ethernet:
    +--rw fec-mode?                fec-mode {fec-supported}?
    +--rw mpcp-admin-state?        mpcp-admin-state
    +--ro mpcp-logical-link-admin-state? mpcp-logical-link-admin-state
    +--rw trx-transmit-admin-state? trx-admin-state {trx-power-level-reporting-supported}?
    +--ro capabilities
      | +--ro mpcp-supported?      mpcp-supported
      +--ro statistics-mpcp
        | +--ro out-mpcp-mac-ctrl-frames? yang:counter64
        | +--ro in-mpcp-mac-ctrl-frames?  yang:counter64
        | +--ro mpcp-discovery-window-count? yang:counter64
        | +--ro mpcp-discovery-timeout-count? yang:counter64
        | +--ro out-mpcp-register-req?     yang:counter64
        | +--ro in-mpcp-register-req?      yang:counter64
        | +--ro out-mpcp-register-ack?     yang:counter64
        | +--ro in-mpcp-register-ack?      yang:counter64
        | +--ro out-mpcp-report?           yang:counter64
        | +--ro in-mpcp-report?           yang:counter64
        | +--ro out-mpcp-gate?            yang:counter64
        | +--ro in-mpcp-gate?            yang:counter64
        | +--ro out-mpcp-register?        yang:counter64
        | +--ro in-mpcp-register?        yang:counter64
      +--rw statistics-ompe
        | +--ro in-ompe-frames-errored-sld? yang:counter64
        | +--ro in-ompe-frames-errored-crc8? yang:counter64
        | +--ro ompe-onu-frames-with-good-llid-good-crc8? yang:counter64

```

^kCopyright release for YANG modules: Users of this standard may freely reproduce the YANG module contained in this subclause so that it can be used for its intended purpose.

```

1 | +---ro ompe-olt-frames-with-good-llid-good-crc8?          yang:counter64
2 | +---ro in-ompe-frames-with-bad-llid?                     yang:counter64
3 | +---ro in-ompe-frames-with-good-llid?                    yang:counter64
4 | +---ro in-ompe-frames?                                    yang:counter64
5 | +---ro in-ompe-frames-not-match-onu-llid-broadcast?      yang:counter64
6 | +---ro in-ompe-frames-match-onu-llid-not-broadcast?      yang:counter64
7 | +---ro in-ompe-frames-match-onu-llid-broadcast?          yang:counter64
8 | +---ro in-ompe-frames-not-match-onu-llid-not-broadcast?  yang:counter64
9 | +---ro in-ompe-frames-not-match-onu-llid-not-broadcast?  yang:counter64
10 +---rw thresholds-trx {trx-power-level-reporting-supported}?
11 | +---rw in-trx-power-low-threshold?      power-level {trx-power-level-reporting-supported}?
12 | +---rw in-trx-power-high-threshold?     power-level {trx-power-level-reporting-supported}?
13 | +---rw out-trx-power-low-threshold?     power-level {trx-power-level-reporting-supported}?
14 | +---rw out-trx-power-high-threshold?    power-level {trx-power-level-reporting-supported}?
15 x---rw statistics-trx {trx-power-level-reporting-supported}?
16 | +---ro in-trx-power-signal-detect?      boolean
17 | +---ro in-trx-power?                    power-level
18 | +---ro in-trx-power-low-15-minutes-bin? power-level
19 | +---ro in-trx-power-high-15-minutes-bin? power-level
20 | +---ro out-trx-power-signal-detect?      boolean
21 | +---ro out-trx-power?                    power-level
22 | +---ro out-trx-power-low-15-minutes-bin? power-level
23 | +---ro out-trx-power-high-15-minutes-bin? power-level
24 | +---ro out-trx-power-high-15-minutes-bin? power-level
25 | +---ro trx-data-reliable?                boolean {trx-power-level-reporting-supported}?
26 +---rw monitoring-trx {trx-power-level-reporting-supported}?
27 | +---ro in-trx-power-signal-detect?      boolean
28 | +---ro in-trx-power?                    power-level
29 | +---ro in-trx-power-low-15-minutes-bin? power-level
30 | +---ro in-trx-power-high-15-minutes-bin? power-level
31 | +---ro out-trx-power-signal-detect?      boolean
32 | +---ro out-trx-power?                    power-level
33 | +---ro out-trx-power-low-15-minutes-bin? power-level
34 | +---ro out-trx-power-high-15-minutes-bin? power-level
35 | +---ro out-trx-power-high-15-minutes-bin? power-level
36 | +---ro trx-data-reliable?                boolean {trx-power-level-reporting-supported}?
37 +---ro statistics-pon-fec {fec-supported}?
38 | +---ro fec-code-group-violations?        yang:counter64
39 | +---ro fec-buffer-head-coding-violations? yang:counter64
40 | +---ro fec-code-word-corrected-errors?   yang:counter64
41 | +---ro fec-code-word-uncorrected-errors? yang:counter64
42 | +---ro fec-code-word-uncorrected-errors? yang:counter64
43 +---rw mpcp-logical-link-admin-actions

```

```

1 | +---x state-change-action-type
2 | | +---w input
3 | | +---w state-change-action-type? identityref
4 | +---x reset-action-type
5 | | +---w input
6 | | +---w reset-action-type? identityref
7 | +---x register-type
8 | | +---w input
9 | | +---w register-type? identityref
10 |
11 +--rw mpcp-queues* [mpcp-queue-index]
12 | +--rw mpcp-queue-index uint8
13 | +--rw mpcp-queue-threshold-count? uint8
14 | +--ro mpcp-queue-threshold-count-max? uint8
15 | +--rw mpcp-queue-thresholds* [mpcp-queue-set-index]
16 | | +--rw mpcp-queue-set-index uint8
17 | | +--rw mpcp-queue-set-threshold? uint64
18 | +--ro in-mpcp-queue-frames? yang:counter64
19 | +--ro out-mpcp-queue-frames? yang:counter64
20 | +--ro mpcp-queue-frames-drop? yang:counter64
21 |
22 +--rw multicast-IDs* [multicast-ID]
23 | +--rw multicast-ID uint32
24 |
25 +--ro fec-capability? fec-capability
26 +--ro mpcp-mode? mpcp-mode
27 +--ro mpcp-sync-time? uint64
28 +--ro mpcp-logical-link-id? mpcp-supported
29 +--ro mpcp-remote-mac-address? ieee:mac-address
30 +--ro mpcp-logical-link-state? mpcp-logical-link-state
31 +--ro mpcp-elapsed-time-out? uint64
32 +--ro mpcp-elapsed-time-in? uint64
33 +--ro mpcp-round-trip-time? uint16
34 +--ro mpcp-maximum-grant-count? uint8
35 +--ro mpcp-logical-link-count? mpcp-llid-count
36 +--ro mpcp-maximum-queue-count-per-report? mpcp-maximum-queue-count-per-report
37 +--ro ompe-mode? ompe-mode
38
39
40
41
42
43

```

7.4.2 YANG module

In the YANG module definition, should any discrepancy between the text of the description for individual YANG nodes in the YANG modules of this clause and the corresponding definition in 7.2 through 7.4 of this clause occur, the definitions in the YANG modules attached to this standard shall take precedence.

An ASCII text version of the YANG module can be found at the following URL:¹

Editor's Note (to be removed prior to publication):

Yang files contained in <https://github.com/YangModels/yang/tree/main/standard/ieee/published/802.3> are IEEE 802.3.2-2019 version and will be updated at the publication time.

<https://github.com/YangModels/yang/tree/master/standard/ieee/published/802.3>

Editor's Note (to be removed prior to publication):

Pretty printing of `ieee802-ethernet-pon.yang` file may change the appearance by adding whitespace and reformatting lines

Editor's Note (to be removed prior to publication):

IEEE Std 802.3.1 and IEEE Std 802.3.2 to be updated at the publication time

```
module ieee802-ethernet-pon {
  yang-version 1.1;
  namespace "urn:ieee:std:802.3:yang:ieee802-ethernet-pon";
  prefix ieee802-eth-pon;

  import ieee802-types {
    prefix ieee;
    reference
      "IEEE 802 types";
  }
  import ietf-yang-types {
    prefix yang;
    reference
      "IETF RFC 6991";
  }
  import ietf-interfaces {
    prefix if;
    reference
      "IETF RFC 8343";
  }
  import ieee802-ethernet-interface {
    prefix ieee802-eth-if;
  }

  organization
    "IEEE 802.3 Ethernet Working Group"
    Web URL: http://www.ieee802.org/3/";
  contact
    "Web URL: http://www.ieee802.org/3/";
  description
```

¹Copyright release for YANG modules: Users of this standard may freely reproduce the YANG module contained in this subclause so that it can be used for its intended purpose.

```
1      "This module contains a collection of YANG definitions for
2      managing the Multi Point Control Protocol for
3      Ethernet PON (EPON), as defined in IEEE Std 802.3, Clause 64
4      and Clause 77.
5
6      This YANG module augments the 'ethernet' module.";
7
8      revision 2025-04-17 {
9          description
10             "Updates under IEEE Std 802.3.2-202x, Draft D3.3";
11          reference
12             "IEEE Std 802.3-2022, unless dated explicitly";
13      }
14
15      feature trx-power-level-reporting-supported {
16          description
17             "This object indicates the support for optical transceiver
18             power level monitoring and reporting capability.
19             When 'true', the given interface supports the optical power
20             level monitoring and reporting function. Otherwise,
21             the value is 'false'.";
22      }
23
24      feature fec-supported {
25          description
26             "This object indicates the support of operation of the
27             optional FEC sublayer of the 1G-EPON PHY specified in
28             IEEE Std 802.3, 65.2. The value of 'unknown' is reported in
29             the initialization, for non FEC support state or type not
30             yet known. The value of 'not supported' is reported when the
31             sublayer is not supported. The value of 'supported' is
32             reported when the sublayer is supported. This object is
33             applicable for an OLT, with the same value for all logical
34             links, and for an ONU.";
35          reference
36             "IEEE Std 802.3, 30.5.1.1.15";
37      }
38
39      identity state-change-action-type {
40          description
41             "Type of interface state change requested.";
42      }
43
44      identity power-down {
45          base state-change-action-type;
46          description
47             "Power down the EPON logical interface.
48             Power-down actions are applicable for the OLT and ONU. A
49             power down of a specific logical interface affects only
50             the logical interface (and not the physical interface).
51             the logical interface will be unavailable while the
52             power-down occurs and data may be lost. Other logical
53             interface are unaffected by power-down.
54
55             This action is relevant when the admin state is active.";
56      }
57
58      identity power-up {
59          base state-change-action-type;
```

```
1      description
2          "Exit EPON logical interface power-down state.";
3      }
4
5      identity reset-action-type {
6          description
7              "Type of reset action requested.";
8      }
9
10
11     identity reset-interface {
12         base reset-action-type;
13         description
14             "Reset the EPON logical interface. Resetting an interface
15             can lead an interruption of service for the users connected
16             to the respective EPON interface.
17
18             This object is applicable for an OLT and an ONU. At the
19             OLT, it has a distinct value for each logical interface.
20             A reset for a specific logical interface resets only
21             this logical interface and not the physical interface.
22
23             Thus, a logical link that is malfunctioning can be
24             reset without affecting the operation of other logical
25             interfaces.
26
27             The reset can cause Discontinuities in the values of the
28             counters of the interface, similar to re-initialization
29             of the management system.";
30     }
31
32
33     identity register-type {
34         description
35             "Type of registration requested.";
36     }
37
38
39     identity register {
40         base register-type;
41         description
42             "Register indicates a request to register an LLID.
43             This action applies to an OLT or ONU logical interface.";
44     }
45
46
47     identity reregister {
48         base register-type;
49         description
50             "Re-register indicates an request to re-register an LLID.
51             This action applies to an OLT or ONU logical interface.";
52     }
53
54
55     identity deregister {
56         base register-type;
57         description
58             "De-register indicates an request to de-register an LLID.
59             This action applies to an OLT or ONU logical interface.
60             Deregister may result in an interruption of service to
61             users connected to the respective EPON interface.";
62     }
63
64
65     typedef mpcp-supported {
```

```
1     type boolean;
2     description
3         "This object indicates that the given interface supports
4         MPCP, i.e., it is an Ethernet PON (EPON) interface.";
5 }
6
7 typedef mpcp-llid {
8     type uint64 {
9         range "0 .. 32767";
10    }
11    description
12        "Logical Link Identifiers (LLIDs) are used to identify a
13        single MAC from a number of MACs which may be present in the
14        EPON OLT or ONU. LLIDs between the value of 0x07FFE
15        and 0x7FFF are assigned for ONU discovery and registration.
16        Other LLIDs are dynamically assigned by the OLT during the
17        registration process. For a complete description of how the
18        LLID is used in an EPON device, see IEEE Std 802.3,
19        Clause 65 for 1G-EPON and Clause 76 for 10G-EPON.";
20    reference
21        "IEEE Std 802.3, 65.1.3.3 for 1G-EPON and
22        76.2.6.1.3 for 10G-EPON";
23 }
24
25 typedef mpcp-maximum-queue-count-per-report {
26     type uint8 {
27         range "0..7";
28     }
29     default "0";
30     description
31         "Defines the maximum number of queues (0-7) in the REPORT
32         MPCPDU as defined in IEEE Std 802.3, Clause 64 and
33         Clause 77.";
34 }
35
36 typedef mpcp-llid-count {
37     type uint64 {
38         range "0 .. 32767";
39     }
40     description
41         "Indicates the number of registered LLIDs. The initialization
42         value is 0. This is applicable for an OLT with the same
43         value for all logical interfaces and for an ONU.";
44     reference
45         "IEEE Std 802.3, 65.1.3.3 for 1G-EPON and
46         76.2.6.1.3 for 10G-EPON";
47 }
48
49 typedef mpcp-admin-state {
50     type enumeration {
51         enum enabled {
52             description
53                 "When selecting the value of 'enabled', the MultiPoint
54                 Control Protocol sublayer on the OLT / ONU is enabled.";
55         }
56         enum disabled {
57             description
58                 "When selecting the value of 'disabled', the MultiPoint
59                 Control Protocol sublayer on the OLT / ONU is
```

```
1         disabled.";
2     }
3 }
4 description
5     "Enumeration of valid administrative states for a
6     MultiPoint MAC Control sublayer on the OLT or ONU.";
7 reference
8     "IEEE Std 802.3, 30.3.5.2.1";
9 }
10
11
12 typedef mpcp-mode {
13     type enumeration {
14         enum olt {
15             description
16                 "MPCP mode: olt";
17         }
18         enum onu {
19             description
20                 "MPCP mode: onu";
21         }
22     }
23 }
24 description
25     "Enumeration of valid MPCP modes for EPON interfaces.";
26 reference
27     "IEEE Std 802.3, 30.3.5.1.3";
28 }
29
30 typedef mpcp-logical-link-state {
31     type enumeration {
32         enum unregistered {
33             description
34                 "MPCP registration state: logical link is
35                 NOT registered.";
36         }
37         enum registering {
38             description
39                 "MPCP registration state: logical link is currently in
40                 the process of registering.";
41         }
42         enum registered {
43             description
44                 "MPCP registration state: logical link is currently
45                 registered.";
46         }
47     }
48 }
49 description
50     "Enumeration of valid MPCP registration states for EPON
51     interfaces.";
52 reference
53     "IEEE Std 802.3, 30.3.5.1.6";
54 }
55
56
57
58 typedef mpcp-logical-link-admin-state {
59     type enumeration {
60         enum reset {
61             description
62                 "When read, the value of 'reset' indicates that the given
63                 logical link on the OLT / ONU has been reset.";
64         }
65     }
```

```
1      enum operate {
2          description
3              "When read, the value of 'operate' indicates that the
4              given logical link on the OLT / ONU has moved into
5              operating mode.";
6      }
7      enum unknown {
8          description
9              "When read, the value of 'unknown' indicates that the
10             status of the given logical link on the OLT / ONU is
11             currently not known.";
12      }
13      enum registered {
14          description
15              "When read, the value of 'registered' indicates that the
16              given logical link on the OLT / ONU has been
17              registered.";
18      }
19      enum deregistered {
20          description
21              "When read, the value of 'deregistered' indicates that the
22              given logical link on the OLT / ONU has been
23              deregistered.";
24      }
25      enum reregistered {
26          description
27              "When read, the value of 'reregistered' indicates that the
28              given logical link on the OLT / ONU has been
29              reregistered.";
30      }
31      }
32      description
33          "Enumeration of valid administrative states for a logical
34          link on the OLT or ONU.";
35      }
36
37      typedef ompe-mode {
38          type enumeration {
39              enum unknown {
40                  description
41                      "omp-emulation mode: unknown = system is initializing";
42              }
43              enum olt {
44                  description
45                      "omp-emulation mode: olt";
46              }
47              enum onu {
48                  description
49                      "omp-emulation mode: onu";
50              }
51          }
52          description
53              "Enumeration of valid OMP-Emulation modes for EPON
54              interfaces.";
55          reference
56              "IEEE Std 802.3, 30.3.7.1.2";
57      }
58
59      typedef fec-capability {
```

```
1      type enumeration {
2          enum unknown {
3              description
4                  "FEC capability: unknown = system is initializing.";
5          }
6          enum supported {
7              description
8                  "FEC capability: supported.";
9          }
10         enum NotSupported {
11             description
12                 "FEC capability: not supported.";
13         }
14     }
15 }
16 description
17     "Enumeration of valid FEC capability values for EPON
18     interfaces with enabled MPCP.";
19 reference
20     "IEEE Std 802.3, 30.5.1.1.15";
21 }
22
23 typedef fec-mode {
24     type enumeration {
25         enum unknown {
26             description
27                 "FEC mode: unknown = system is initializing.";
28         }
29         enum disabled {
30             description
31                 "FEC mode: disabled = FEC is disabled for the given
32                 logical link (both Tx and Rx directions).";
33         }
34         enum enabled-Tx-Rx {
35             description
36                 "FEC mode: enabled-Tx-Rx = FEC is enabled for the given
37                 logical link in both Tx and Rx directions.";
38         }
39         enum enabled-Tx-only {
40             description
41                 "FEC mode: enabled-Tx-only = FEC is enabled for
42                 the given logical link but only in Tx direction.";
43         }
44         enum enabled-Rx-only {
45             description
46                 "FEC mode: enabled-Rx-only = FEC is enabled for
47                 the given logical link but only in Rx direction.";
48         }
49     }
50 }
51 description
52     "Enumeration of valid FEC modes for EPON interfaces.";
53 reference
54     "IEEE Std 802.3, 30.5.1.1.16";
55 }
56
57 typedef power-level {
58     type int32;
59     units "0.1 dBm";
60     description
61         "Power level reflects the value of power, as measured at the
```

```

1      optical transceiver, expressed in units of 0.1 dBm.";
2  }
3
4  typedef trx-admin-state {
5      type enumeration {
6          enum enabled {
7              description
8                  "When read as 'enabled', the transmitter is enabled and
9                  operating under the control of the logical control
10                 protocol. When set to 'enabled', the transmitter is
11                 enabled to operate under the control of the logical
12                 control protocol.";
13             }
14         }
15         enum disabled {
16             description
17                 "When read as 'disabled', the transmitter is currently
18                 disabled (not transmitting). When set to 'disabled',
19                 the transmitter is expected to be disabled
20                 (to stop transmitting).";
21         }
22     }
23 }
24 description
25     "Enumeration of valid administrative states for an optical
26     transceiver.";
27 reference
28     "IEEE Std 802.3.1, dot3ExtPkgOptIfTransmitEnable";
29 }
30
31 augment "/if:interfaces/if:interface/ieee802-eth-if:ethernet" {
32     description
33         "Augments the definition of Ethernet interface (
34         /if:interfaces/if:interface/ieee802-eth-if:ethernet)
35         module with nodes specific to Ethernet PON (EPON).";
36     leaf fec-mode {
37         if-feature "fec-supported";
38         type fec-mode;
39         description
40             "This object reflects the current administrative state of
41             the FEC function for the given logical link on an ONU or
42             OLT.
43
44             When reading the value of 'disabled', the FEC function on
45             the given logical link is disabled.
46
47             When reading the value of 'enabled-Tx-Rx', the FEC function
48             on the given logical link is enabled in both Tx and Rx
49             directions.
50
51             When reading the value of 'enabled-Tx-only', the FEC
52             function on the given logical link is enabled in Tx
53             direction only.
54
55             When reading the value of 'enabled-Rx-only', the FEC
56             function on the given logical link is enabled in Rx
57             direction only.
58
59             When reading the value of 'unknown', the state of the FEC
60             function on the given logical link is unknown or the FEC
61             function is currently initializing.
62
63             When reading the value of 'unknown', the state of the FEC
64             function on the given logical link is unknown or the FEC
65             function is currently initializing.

```

```
1
2     This object is applicable for an OLT and an ONU. This
3     object has the same value for each logical link.";
4     reference
5         "IEEE Std 802.3, 30.5.1.1.16";
6
7 }
8 leaf mpcp-admin-state {
9     type mpcp-admin-state;
10    description
11        "This object reflects the current administrative state of
12        the MultiPoint MAC Control sublayer, as defined in
13        IEEE Std 802.3, Clause 64 and Clause 77, for the
14        OLT / ONU.
15
16        When reading the value of 'enabled', the MultiPoint
17        Control Protocol on the OLT / ONU is enabled.
18
19        When reading the value of 'disabled', the MultiPoint
20        Control Protocol on the OLT / ONU is disabled.
21
22        This object is applicable for an OLT and an ONU. It has
23        the same value for all logical links.";
24    reference
25        "IEEE Std 802.3, 30.3.5.1.2";
26
27 }
28 leaf mpcp-logical-link-admin-state {
29     type mpcp-logical-link-admin-state;
30     config false;
31     description
32         "This object reflects the current administrative state of a
33         logical link on an ONU or OLT.
34
35         When reading the value of 'reset', the given logical link
36         is undergoing a reset.
37
38         When reading the value of 'unknown', the current status of
39         the given logical link is unknown and the link might be
40         undergoing initialization.
41
42         When reading the value of 'operate', the given logical
43         link is operating normally.
44
45         When reading the value of 'registered', the given logical
46         link was requested to perform registration.
47
48         When reading the value of 'deregistered', the given
49         logical link was requested to perform deregistration.
50
51         When reading the value of 'reregistered', the given
52         logical link was requested to perform reregistration.
53
54         This object is applicable for an OLT and an ONU. It has a
55         distinct value for each logical link.";
56     reference
57         "IEEE Std 802.3.1, dot3ExtPkgObjectRegisterAction";
58
59 }
60 leaf trx-transmit-admin-state {
61     when "../..//ieee802-eth-if:ethernet/
62         ieee802-eth-pon:mpcp-admin-state = 'enabled'";
63
64 }
```

```
1      if-feature "trx-power-level-reporting-supported";
2      type trx-admin-state;
3      description
4          "This object reflects the current status of the transmitter
5           in the optical transceiver.
6
7           When read as 'enabled', the optical transmitter is enabled
8           and operating under the control of the logical control
9           protocol.
10
11          When read as 'disabled', the optical transmitter is
12          disabled.
13
14          This object is applicable for an OLT and an ONU.
15          At the OLT, this object has a distinct value for each
16          logical link.
17
18          The value of this object is only reliable when
19          /if:interfaces-state/if:interface/ieee802-eth-if:ethernet/
20          'mpcp-admin-state' is equal to 'enabled'.";
21      reference
22          "IEEE Std 802.3.1, dot3ExtPkgOptIfTransmitEnable";
23  }
24  container capabilities {
25      config false;
26      description
27          "This container includes all EPON interface-specific
28           capabilities.";
29      leaf mpcp-supported {
30          type mpcp-supported;
31          default "true";
32          description
33              "This object indicates that the given interface supports
34               MPCP, i.e., it is an Ethernet PON (EPON) interface.";
35      }
36  }
37  container statistics-mpcp {
38      config false;
39      description
40          "This container defines a set of MPCP-related statistics
41           counters of an EPON interface, as defined in
42           IEEE Std 802.3, Clause 64 and Clause 77.
43
44           Discontinuities in the values of counters in this
45           container can occur at re-initialization of the management
46           system, and at other times as indicated by the value of
47           the 'discontinuity-time' leaf defined in the ietf-interfaces
48           YANG module (IETF RFC 8343).";
49      leaf out-mpcp-mac-ctrl-frames {
50          type yang:counter64;
51          units "frames";
52          config false;
53          description
54              "A count of MPCP frames passed to the MAC sublayer for
55               transmission.
56
57               This counter is incremented when a MA_CONTROL.request
58               service primitive is generated within the MAC control
59               sublayer with an opcode indicating an MPCP frame.
```

```
1
2     This object is applicable for an OLT and an ONU. It has
3     a distinct value for each logical link.";
4 reference
5     "IEEE Std 802.3, 30.3.5.1.7";
6 }
7 leaf in-mpcp-mac-ctrl-frames {
8     type yang:counter64;
9     units "frames";
10    config false;
11    description
12        "A count of MPCP frames passed by the MAC sublayer to the
13        MAC Control sublayer.
14
15        This counter is incremented when a frame is received at
16        the interface which is an MPCP frame or has a
17        Length/Type Ethernet header field value equal to the
18        Type assigned for 802.3_MAC_Control as specified in
19        IEEE Std 802.3, 31.4.1.3.
20
21        This object is applicable for an OLT and an ONU. It has
22        a distinct value for each logical link.";
23 reference
24     "IEEE Std 802.3, 30.3.5.1.8";
25 }
26 leaf mpcp-discovery-window-count {
27     when "../ompe-mode = 'olt'";
28     type yang:counter64;
29     units "discovery windows";
30     config false;
31     description
32         "A count of discovery windows generated by the OLT.
33
34         The counter is incremented by one for each generated
35         discovery window.
36
37         This object is applicable for an OLT and has the same
38         value for each logical link.";
39 reference
40     "IEEE Std 802.3, 30.3.5.1.22";
41 }
42 leaf mpcp-discovery-timeout-count {
43     when "../ompe-mode = 'olt'";
44     type yang:counter64;
45     units "discovery timeouts";
46     config false;
47     description
48         "A count of the number of times a discovery timeout
49         occurs.
50
51         This counter is incremented by one for each discovery
52         processing state-machine reset resulting from timeout
53         waiting for message arrival.
54
55         This object is applicable for an OLT and has the same
56         value for each logical link.";
57 reference
58     "IEEE Std 802.3, 30.3.5.1.23";
59 }
60
61
62
63
64
65
```

```
1     leaf out-mpcp-register-req {
2         when "../ompe-mode = 'onu'";
3         type yang:counter64;
4         units "frames";
5         config false;
6         description
7             "A count of the number of times a REGISTER_REQ MPCP frame
8             transmission occurs.
9
10            This counter is incremented by one for each
11            REGISTER_REQ MPCP frame transmitted as defined in
12            IEEE Std 802.3, Clause 64 and Clause 77.
13
14            This object is applicable for an ONU and has the same
15            value for each logical link.";
16         reference
17             "IEEE Std 802.3, 30.3.5.1.12";
18     }
19     leaf in-mpcp-register-req {
20         when "../ompe-mode = 'olt'";
21         type yang:counter64;
22         units "frames";
23         config false;
24         description
25             "A count of the number of times a REGISTER_REQ MPCP frame
26             reception occurs.
27
28            This counter is incremented by one for each
29            REGISTER_REQ MPCP frame received as defined in
30            IEEE Std 802.3, Clause 64 and Clause 77.
31
32            This object is applicable for an OLT and has the same
33            value for each logical link.";
34         reference
35             "IEEE Std 802.3, 30.3.5.1.17";
36     }
37     leaf out-mpcp-register-ack {
38         when "../ompe-mode = 'onu'";
39         type yang:counter64;
40         units "frames";
41         config false;
42         description
43             "A count of the number of times a REGISTER_ACK MPCP frame
44             transmission occurs.
45
46            This counter is incremented by one for each
47            REGISTER_ACK MPCP frame transmitted as defined in
48            IEEE Std 802.3, Clause 64 and Clause 77.
49
50            This object is applicable for an ONU and has a distinct
51            value for each logical link.";
52         reference
53             "IEEE Std 802.3, 30.3.5.1.10";
54     }
55     leaf in-mpcp-register-ack {
56         when "../ompe-mode = 'olt'";
57         type yang:counter64;
58         units "frames";
59         config false;
```

```
1      description
2          "A count of the number of times a REGISTER_ACK MPCP frame
3            reception occurs.
4
5          This counter is incremented by one for each
6          REGISTER_ACK MPCP frame received as defined in
7          IEEE Std 802.3, Clause 64 and Clause 77.
8
9          This object is applicable for an OLT and has a distinct
10         value for each logical link.";
11
12     reference
13         "IEEE Std 802.3, 30.3.5.1.15";
14 }
15 leaf out-mpcp-report {
16     when "../..ompe-mode = 'onu'";
17     type yang:counter64;
18     units "frames";
19     config false;
20     description
21         "A count of the number of times a REPORT MPCP frame
22           transmission occurs.
23
24         This counter is incremented by one for each REPORT MPCP
25         frame transmitted as defined in IEEE Std 802.3,
26         Clause 64 and Clause 77.
27
28         This object is applicable for an ONU and has a distinct
29         value for each logical link.";
30
31     reference
32         "IEEE Std 802.3, 30.3.5.1.13";
33 }
34 leaf in-mpcp-report {
35     when "../..ompe-mode = 'olt'";
36     type yang:counter64;
37     units "frames";
38     config false;
39     description
40         "A count of the number of times a REPORT MPCP frame
41           reception occurs.
42
43         This counter is incremented by one for each REPORT MPCP
44         frame received as defined in IEEE Std 802.3,
45         Clause 64 and Clause 77.
46
47         This object is applicable for an OLT and has a distinct
48         value for each logical link.";
49
50     reference
51         "IEEE Std 802.3, 30.3.5.1.18";
52 }
53 leaf out-mpcp-gate {
54     when "../..ompe-mode = 'olt'";
55     type yang:counter64;
56     units "frames";
57     config false;
58     description
59         "A count of the number of times a GATE MPCP frame
60           transmission occurs.
61
62         This counter is incremented by one for each GATE MPCP
```

```
1         frame transmitted as defined in IEEE Std 802.3,  
2         Clause 64 and Clause 77.  
3  
4         This object is applicable for an OLT and has a distinct  
5         value for each logical link.";  
6     reference  
7         "IEEE Std 802.3, 30.3.5.1.9";  
8  
9     }  
10    leaf in-mpcp-gate {  
11        when "../..ompe-mode = 'onu'";  
12        type yang:counter64;  
13        units "frames";  
14        config false;  
15        description  
16            "A count of the number of times a GATE MPCP frame  
17            reception occurs.  
18  
19            This counter is incremented by one for each GATE MPCP  
20            frame received as defined in IEEE Std 802.3,  
21            Clause 64 and Clause 77.  
22  
23            This object is applicable for an ONU and has a distinct  
24            value for each logical link.";  
25        reference  
26            "IEEE Std 802.3, 30.3.5.1.14";  
27    }  
28    leaf out-mpcp-register {  
29        when "../..ompe-mode = 'olt'";  
30        type yang:counter64;  
31        units "frames";  
32        config false;  
33        description  
34            "A count of the number of times a REGISTER MPCP frame  
35            transmission occurs.  
36  
37            This counter is incremented by one for each  
38            REGISTER MPCP frame transmitted as defined in  
39            IEEE Std 802.3, Clause 64 and Clause 77.  
40  
41            This object is applicable for an OLT and has a distinct  
42            value for each logical link.";  
43        reference  
44            "IEEE Std 802.3, 30.3.5.1.11";  
45    }  
46    leaf in-mpcp-register {  
47        when "../..ompe-mode = 'onu'";  
48        type yang:counter64;  
49        units "frames";  
50        config false;  
51        description  
52            "A count of the number of times a REGISTER MPCP frame  
53            reception occurs.  
54  
55            This counter is incremented by one for each  
56            REGISTER MPCP frame received as defined in  
57            IEEE Std 802.3, Clause 64 and Clause 77.  
58  
59            This object is applicable for an ONU and has a distinct  
60            value for each logical link.";  
61  
62  
63  
64  
65
```

```
1         reference
2             "IEEE Std 802.3, 30.3.5.1.16";
3     }
4 }
5 container statistics-ompe {
6     description
7         "This container defines a set of OMP-Emulation-related
8         statistics counters of an EPON interface, as defined in
9         IEEE Std 802.3, Clause 65 and Clause 76.
10
11         Discontinuities in the values of counters in this
12         container can occur at re-initialization of the management
13         system, and at other times as indicated by the value of
14         the 'discontinuity-time' leaf defined in the
15         ietf-interfaces YANG module (IETF RFC 8343).";
16     reference
17         "IEEE Std 802.3.1, dot3OmpEmulationStatEntry";
18     leaf in-ompe-frames-errored-sld {
19         type yang:counter64;
20         units "frames";
21         config false;
22         description
23             "A count of frames received that do not contain a valid
24             SLD field as defined in IEEE Std 802.3, 65.1.3.3.1 or
25             76.2.6.1.3.1, as appropriate.
26
27             This object is applicable for an OLT and an ONU.
28             It has a distinct value for each logical link.";
29         reference
30             "IEEE Std 802.3, 30.3.7.1.3";
31     }
32     leaf in-ompe-frames-errored-crc8 {
33         type yang:counter64;
34         units "frames";
35         config false;
36         description
37             "A count of frames received that contain a valid SLD
38             field, as defined in IEEE Std 802.3, 65.1.3.3.1 or
39             76.2.6.1.3.1 as appropriate, but do not pass the CRC-8
40             check as defined in IEEE Std 802.3, 65.1.3.3.3 or
41             76.2.6.1.3.3 as appropriate.
42
43             This object is applicable for an OLT and an ONU.
44             It has a distinct value for each logical link.";
45         reference
46             "IEEE Std 802.3, 30.3.7.1.4";
47     }
48     leaf ompe-onu-frames-with-good-llid-good-crc8 {
49         when "../ompe-mode = 'onu'";
50         type yang:counter64;
51         units "frames";
52         config false;
53         description
54             "A count of frames received that 1) contain a valid SLD
55             field in an ONU, 2) meet the rule for frame acceptance,
56             and 3) pass the CRC-8 check.
57
58             The SLD is defined in IEEE Std 802.3, 65.1.3.3.1 or
59             76.2.6.1.3.1, as appropriate.
```

```
1
2     The rules for LLID acceptance are defined in
3     IEEE Std 802.3, 65.1.3.3.2 or 76.2.6.1.3.2,
4     as appropriate.
5
6     The CRC-8 check is defined in IEEE Std 802.3,
7     65.1.3.3.3 or 76.2.6.1.3.3, as appropriate.
8
9
10    This object is applicable for an ONU and has a distinct
11    value for each logical link.";
12    reference
13    "IEEE Std 802.3, 30.3.7.1.6";
14  }
15  leaf ompe-olt-frames-with-good-llid-good-crc8 {
16    when "../.. /ompe-mode = 'olt'";
17    type yang:counter64;
18    units "frames";
19    config false;
20    description
21    "A count of frames received that 1) contain a valid SLD
22    field in an OLT, and 2) pass the CRC-8 check.
23
24    The SLD is defined in IEEE Std 802.3, 65.1.3.3.1 or
25    76.2.6.1.3.1, as appropriate.
26
27    The frame acceptance are defined in IEEE Std 802.3,
28    65.1.3.3.2 or 76.2.6.1.3.2, as appropriate.
29
30    The CRC-8 check is defined in IEEE Std 802.3,
31    65.1.3.3.3 or 76.2.6.1.3.3, as appropriate.
32
33    This object is applicable for an OLT and has a distinct
34    value for each logical link.";
35    reference
36    "IEEE Std 802.3, 30.3.7.1.6";
37  }
38  leaf in-ompe-frames-with-bad-llid {
39    when "../.. /ompe-mode = 'olt'";
40    type yang:counter64;
41    units "frames";
42    config false;
43    description
44    "A count of frames received that contain a valid SLD
45    field, as defined in IEEE Std 802.3, 65.1.3.3.1 or
46    76.2.6.1.3.1, as appropriate, and pass the CRC-8 check
47    as defined in IEEE Std 802.3, 65.1.3.3.3 or
48    76.2.6.1.3.3, as appropriate, but are discarded due to
49    the LLID check.
50
51    This object is applicable for an OLT and has a distinct
52    value for each logical link.";
53    reference
54    "IEEE Std 802.3, 30.3.7.1.8";
55  }
56  leaf in-ompe-frames-with-good-llid {
57    type yang:counter64;
58    units "frames";
59    config false;
60    description
```

```
1      "A count of frames received that contain a valid SLD
2      field, as defined in IEEE Std 802.3, 65.1.3.3.1 or
3      76.2.6.1.3.1 as appropriate, but do not pass the CRC-8
4      check as defined in IEEE Std 802.3, 65.1.3.3.3 or
5      76.2.6.1.3.3 as appropriate.
6
7      This object is applicable for an OLT and an ONU. It has
8      a distinct value for each logical link.";
9
10     reference
11     "IEEE Std 802.3, 30.3.7.1.4";
12 }
13 leaf in-ompe-frames {
14     type yang:counter64;
15     units "frames";
16     config false;
17     description
18     "A count of frames received that contain a valid SLD
19     field, as defined in IEEE Std 802.3, 65.1.3.3.1 or
20     76.2.6.1.3.1, as appropriate, and pass the CRC-8
21     check as defined in IEEE Std 802.3, 65.1.3.3.3 or
22     76.2.6.1.3.3, as appropriate.
23
24     This object is applicable for an OLT and an ONU. It has
25     a distinct value for each logical link.";
26
27     reference
28     "IEEE Std 802.3, 30.3.7.1.6 (ONU) and 30.3.7.1.7 (OLT)";
29 }
30 leaf in-ompe-frames-not-match-onu-llid-broadcast {
31     when "../ompe-mode = 'onu'";
32     type yang:counter64;
33     units "frames";
34     config false;
35     description
36     "A count of frames received that contain a valid SLD
37     field, as defined in IEEE Std 802.3, 65.1.3.3.1 or
38     76.2.6.1.3.1, as appropriate, pass the CRC-8 check, as
39     defined in IEEE Std 802.3, 65.1.3.3.3 or 76.2.6.1.3.3,
40     as appropriate, and contain the broadcast bit in the
41     LLID and not the ONU's LLID (frame accepted) as defined
42     in IEEE Std 802.3, Clause 65 and Clause 76,
43     as appropriate.
44
45     This object is applicable for an ONU only.";
46
47     reference
48     "IEEE Std 802.3.1,
49     dot3OmpEmulationBroadcastBitNotOnuLlid";
50 }
51 leaf in-ompe-frames-match-onu-llid-not-broadcast {
52     when "../ompe-mode = 'onu'";
53     type yang:counter64;
54     units "frames";
55     config false;
56     description
57     "A count of frames received that contain a valid SLD
58     field, as defined in IEEE Std 802.3, 65.1.3.3.1 or
59     76.2.6.1.3.1, as appropriate, pass the CRC-8 check, as
60     defined in IEEE Std 802.3, 65.1.3.3.3 or 76.2.6.1.3.3,
61     as appropriate, and contain the ONU's LLID
62     (frame accepted) as defined in IEEE Std 802.3, Clause 65
```

```
1         and Clause 76, as appropriate.
2
3         This object is applicable for an ONU only.";
4     reference
5         "IEEE Std 802.3.1, dot3OmpEmulationOnuLLIDNotBroadcast";
6 }
7
8 leaf in-ompe-frames-match-onu-llid-broadcast {
9     when "../ompe-mode = 'onu'";
10    type yang:counter64;
11    units "frames";
12    config false;
13    description
14        "A count of frames received that contain a valid SLD
15        field, as defined in IEEE Std 802.3, 65.1.3.3.1 or
16        76.2.6.1.3.1, as appropriate, pass the CRC-8 check, as
17        defined in IEEE Std 802.3, 65.1.3.3.3 or 76.2.6.1.3.3,
18        as appropriate, and contain the broadcast bit in the
19        LLID and the ONU's LLID (frame accepted) as defined in
20        IEEE Std 802.3, Clause 65 and Clause 76, as appropriate.
21
22        This object is applicable for an ONU only.";
23    reference
24        "IEEE Std 802.3.1,
25        dot3OmpEmulationBroadcastBitPlusOnuLlid";
26 }
27
28 leaf in-ompe-frames-not-match-onu-llid-not-broadcast {
29     when "../ompe-mode = 'onu'";
30    type yang:counter64;
31    units "frames";
32    config false;
33    description
34        "A count of frames received that contain a valid SLD
35        field, as defined in IEEE Std 802.3, 65.1.3.3.1 or
36        76.2.6.1.3.1, as appropriate, pass the CRC-8 check,
37        as defined in IEEE Std 802.3, 65.1.3.3.3 or
38        76.2.6.1.3.3, as appropriate, do not contain the
39        broadcast bit in the LLID and do not contain the ONU's
40        LLID (frame is NOT accepted) as defined in
41        IEEE Std 802.3, Clause 65 and Clause 76, as appropriate.
42
43        This object is applicable for an ONU only.";
44    reference
45        "IEEE Std 802.3.1,
46        dot3OmpEmulationNotBroadcastBitNotOnuLlid";
47 }
48 }
49
50 }
51
52 container thresholds-trx {
53     if-feature "trx-power-level-reporting-supported";
54     description
55         "This container defines a set of optical transceiver
56         thresholds of an EPON interface as defined in
57         IEEE Std 802.3, Clause 60 and Clause 75.";
58     reference
59         "IEEE Std 802.3.1, dot3ExtPkgOptIfEntry";
60     leaf in-trx-power-low-threshold {
61         if-feature "trx-power-level-reporting-supported";
62         type power-level;
63         description
64             "This object reflects the current setting of low alarm
65
```

```
1         threshold for the input power into the optical receiver.
2         If the value reported in 'in-trx-power' object drops
3         below the value set in 'in-trx-power-low-threshold', a
4         'in-trx-power-low-threshold-crossing' event is
5         generated.
6
7         This object is applicable for an OLT and an ONU. It has
8         a distinct value for each logical link.";
9
10        reference
11        "IEEE Std 802.3.1,
12        dot3ExtPkgOptIfLowerInputPowerThreshold";
13    }
14    leaf in-trx-power-high-threshold {
15        if-feature "trx-power-level-reporting-supported";
16        type power-level;
17        description
18        "This object reflects the current setting of high alarm
19        threshold for the input power into the optical receiver.
20        If the value reported in 'in-trx-power' object exceeds
21        the value set in 'in-trx-power-high-threshold', a
22        'in-trx-power-high-threshold-crossing' event is
23        generated.
24
25        This object is applicable for an OLT and an ONU. It has
26        a distinct value for each logical link.";
27
28        reference
29        "IEEE Std 802.3.1,
30        dot3ExtPkgOptIfUpperInputPowerThreshold";
31    }
32
33    leaf out-trx-power-low-threshold {
34        if-feature "trx-power-level-reporting-supported";
35        type power-level;
36        description
37        "This object reflects the current setting of low alarm
38        threshold for the output power out of the optical
39        transmitter. If the value reported in 'out-trx-power'
40        object drops below the value set in
41        'out-trx-power-low-threshold', a
42        'out-trx-power-low-threshold-crossing' event is
43        generated.
44
45        This object is applicable for an OLT and an ONU. It has
46        a distinct value for each logical link.";
47
48        reference
49        "IEEE Std 802.3.1,
50        dot3ExtPkgOptIfLowerOutputPowerThreshold";
51    }
52
53    leaf out-trx-power-high-threshold {
54        if-feature "trx-power-level-reporting-supported";
55        type power-level;
56        description
57        "This object reflects the current setting of high alarm
58        threshold for the output power out of the optical
59        transmitter. If the value reported in 'out-trx-power'
60        object exceeds the value set in
61        'out-trx-power-high-threshold', a
62        'out-trx-power-high-threshold-crossing' event is
63        generated.
64
65
```

```
1         This object is applicable for an OLT and an ONU.
2         It has a distinct value for each logical link.";
3     reference
4         "IEEE Std 802.3.1,
5         dot3ExtPkgOptIfUpperOutputPowerThreshold";
6     }
7 }
8
9 container statistics-trx {
10     if-feature "trx-power-level-reporting-supported";
11     status deprecated;
12     description
13         "This container defines a set of optical transceiver
14         statistics counters of an EPON interface as defined in
15         IEEE Std 802.3, Clause 60 and Clause 75.";
16     reference
17         "IEEE Std 802.3.1, dot3ExtPkgOptIfEntry";
18     leaf in-trx-power-signal-detect {
19         type boolean;
20         config false;
21         description
22             "This object indicates whether a valid optical signal was
23             detected (when read as 'true') or not
24             (when read as 'false') at the input to the
25             optical transceiver.
26
27             This object is applicable for an OLT and an ONU. It has
28             a distinct value for each logical link.";
29         reference
30             "IEEE Std 802.3.1, dot3ExtPkgOptIfSignalDetect";
31     }
32     leaf in-trx-power {
33         type power-level;
34         config false;
35         description
36             "This object reflects the value of the input power, as
37             measured at the optical transceiver, expressed in units
38             of 0.1 dBm.
39
40             At the ONU, the measurement is performed in a continuous
41             manner.
42
43             At the OLT, the measurement is performed in a burst-mode
44             manner, for each incoming data burst.
45
46             This object is applicable for an OLT and an ONU. It has
47             a distinct value for each logical link.";
48         reference
49             "IEEE Std 802.3.1, dot3ExtPkgOptIfInputPower";
50     }
51     leaf in-trx-power-low-15-minutes-bin {
52         type power-level;
53         config false;
54         description
55             "This object reflects the lowest value of the input power
56             during the period of the last 15 minutes, as measured at
57             the optical transceiver, and expressed in units of
58             0.1 dBm.
59
60             At the ONU, the measurement is performed in a continuous
```

```
1      manner and stored in a rolling 15-minutes' long
2      observation bin.
3
4      At the OLT, the measurement is the average power for
5      each incoming data burst, and stored in a rolling
6      15-minutes' long observation bin.
7
8      This object is applicable for an OLT and an ONU. It has
9      a distinct value for each logical link.";
10     reference
11       "IEEE Std 802.3.1, dot3ExtPkgOptIfLowInputPower";
12   }
13   leaf in-trx-power-high-15-minutes-bin {
14     type power-level;
15     config false;
16     description
17       "This object reflects the highest value of the input
18       power during the period of the last 15 minutes, as
19       measured at the optical transceiver, and expressed in
20       units of 0.1 dBm.
21
22       At the ONU, the measurement is performed in a continuous
23       manner and stored in a rolling 15-minutes' long
24       observation bin.
25
26       At the OLT, the measurement is the average power for
27       each incoming data burst, and stored in a rolling
28       15-minutes' long observation bin.
29
30       This object is applicable for an OLT and an ONU. It has
31       a distinct value for each logical link.";
32     reference
33       "IEEE Std 802.3.1, dot3ExtPkgOptIfHighInputPower";
34   }
35   leaf out-trx-power-signal-detect {
36     type boolean;
37     config false;
38     description
39       "This object indicates whether a valid optical signal was
40       detected (when read as 'true') or not
41       (when read as 'false') at the output from the
42       optical transceiver.
43
44       This object is applicable for an OLT and an ONU. It has
45       a distinct value for each logical link.";
46     reference
47       "IEEE Std 802.3.1, dot3ExtPkgOptIfTransmitAlarm";
48   }
49   leaf out-trx-power {
50     type power-level;
51     config false;
52     description
53       "This object reflects the value of the output power, as
54       measured at the optical transceiver, expressed in units
55       of 0.1 dBm.
56
57       At the ONU, the measurement is performed in a burst-mode
58       manner for each outgoing data burst.
```

```
1         At the OLT, the measurement is performed in a continuous
2         manner.
3
4         This object is applicable for an OLT and an ONU. It has
5         a distinct value for each logical link.";
6     reference
7         "IEEE Std 802.3.1, dot3ExtPkgOptIfOutputPower";
8
9 }
10 leaf out-trx-power-low-15-minutes-bin {
11     type power-level;
12     config false;
13     description
14         "This object reflects the lowest value of the output
15         power during the period of the last 15 minutes, as
16         measured at the optical transceiver, and expressed in
17         units of 0.1 dBm.
18
19         At the ONU, the measurement is performed in a burst-mode
20         manner and stored in a rolling 15-minutes' long
21         observation bin.
22
23         At the OLT, the measurement is the average power for
24         each incoming data burst, and stored in a rolling
25         15-minutes' long observation bin.
26
27         This object is applicable for an OLT and an ONU. It has
28         a distinct value for each logical link.";
29     reference
30         "IEEE Std 802.3.1, dot3ExtPkgOptIfLowOutputPower";
31
32 }
33 leaf out-trx-power-high-15-minutes-bin {
34     type power-level;
35     config false;
36     description
37         "This object reflects the highest value of the output
38         power during the period of the last 15 minutes, as
39         measured at the optical transceiver, and expressed in
40         units of 0.1 dBm.
41
42         At the ONU, the measurement is performed in a burst-mode
43         manner and stored in a rolling 15-minutes' long
44         observation bin.
45
46         At the OLT, the measurement is the average power for
47         each incoming data burst, and stored in a rolling
48         15-minutes' long observation bin.
49
50         This object is applicable for an OLT and an ONU. It has
51         a distinct value for each logical link.";
52     reference
53         "IEEE Std 802.3.1, dot3ExtPkgOptIfHighOutputPower";
54
55 }
56 leaf trx-data-reliable {
57     if-feature "trx-power-level-reporting-supported";
58     type boolean;
59     config false;
60     description
61         "This object indicates whether data contained in
62         individual counters in 'statistics-trx' container are
```

```
1         reliable (when read as 'true') or not
2         (when read as 'false').
3
4         This object is applicable for an OLT and an ONU. It has
5         a distinct value for each logical link.";
6     reference
7         "IEEE Std 802.3.1, dot3ExtPkgOptIfSuspectedFlag";
8 }
9
10 }
11 container monitoring-trx {
12     if-feature "trx-power-level-reporting-supported";
13     description
14         "This container defines a set of optical transceiver
15         statistics counters of an EPON interface as defined in
16         IEEE Std 802.3, Clause 60 and Clause 75.";
17     reference
18         "IEEE Std 802.3.1, dot3ExtPkgOptIfEntry";
19     leaf in-trx-power-signal-detect {
20         type boolean;
21         config false;
22         description
23             "This object indicates whether a valid optical signal was
24             detected (when read as 'true') or not
25             (when read as 'false') at the input to the
26             optical transceiver.
27
28             This object is applicable for an OLT and an ONU. It has
29             a distinct value for each logical link.";
30         reference
31             "IEEE Std 802.3.1, dot3ExtPkgOptIfSignalDetect";
32     }
33     leaf in-trx-power {
34         type power-level;
35         config false;
36         description
37             "This object reflects the value of the input power, as
38             measured at the optical transceiver, expressed in units
39             of 0.1 dBm.
40
41             At the ONU, the measurement is performed in a continuous
42             manner.
43
44             At the OLT, the measurement is performed in a burst-mode
45             manner, for each incoming data burst.
46
47             This object is applicable for an OLT and an ONU. It has
48             a distinct value for each logical link.";
49         reference
50             "IEEE Std 802.3.1, dot3ExtPkgOptIfInputPower";
51     }
52     leaf in-trx-power-low-15-minutes-bin {
53         type power-level;
54         config false;
55         description
56             "This object reflects the lowest value of the input power
57             during the period of the last 15 minutes, as measured at
58             the optical transceiver, and expressed in units of
59             0.1 dBm.
60
61             This object is applicable for an OLT and an ONU. It has
62             a distinct value for each logical link.";
63         reference
64             "IEEE Std 802.3.1, dot3ExtPkgOptIfInputPower";
65     }
```

```
1         At the ONU, the measurement is performed in a continuous
2         manner and stored in a rolling 15-minutes' long
3         observation bin.
4
5         At the OLT, the measurement is the average power for
6         each incoming data burst, and stored in a rolling
7         15-minutes' long observation bin.
8
9         This object is applicable for an OLT and an ONU.
10        It has a distinct value for each logical link.";
11    reference
12        "IEEE Std 802.3.1, dot3ExtPkgOptIfLowInputPower";
13    }
14    leaf in-trx-power-high-15-minutes-bin {
15        type power-level;
16        config false;
17        description
18            "This object reflects the highest value of the input
19            power during the period of the last 15 minutes,
20            as measured at the optical transceiver,
21            and expressed in units of 0.1 dBm.
22
23            At the ONU, the measurement is performed in a continuous
24            manner and stored in a rolling 15-minutes' long
25            observation bin.
26
27            At the OLT, the measurement is the average power for
28            each incoming data burst, and stored in a rolling
29            15-minutes' long observation bin.
30
31            This object is applicable for an OLT and an ONU. It has
32            a distinct value for each logical link.";
33        reference
34            "IEEE Std 802.3.1, dot3ExtPkgOptIfHighInputPower";
35    }
36    leaf out-trx-power-signal-detect {
37        type boolean;
38        config false;
39        description
40            "This object indicates whether a valid optical signal was
41            detected (when read as 'true') or not
42            (when read as 'false') at the output from the
43            optical transceiver.
44
45            This object is applicable for an OLT and an ONU. It has
46            a distinct value for each logical link.";
47        reference
48            "IEEE Std 802.3.1, dot3ExtPkgOptIfTransmitAlarm";
49    }
50    leaf out-trx-power {
51        type power-level;
52        config false;
53        description
54            "This object reflects the value of the output power, as
55            measured at the optical transceiver, expressed in units
56            of 0.1 dBm.
57
58            At the ONU, the measurement is performed in a burst-mode
59            manner for each outgoing data burst.
```

```
1
2       At the OLT, the measurement is performed in a continuous
3       manner.
4
5       This object is applicable for an OLT and an ONU. It has
6       a distinct value for each logical link.";
7   reference
8       "IEEE Std 802.3.1, dot3ExtPkgOptIfOutputPower";
9   }
10  leaf out-trx-power-low-15-minutes-bin {
11      type power-level;
12      config false;
13      description
14          "This object reflects the lowest value of the output
15          power during the period of the last 15 minutes, as
16          measured at the optical transceiver, and expressed in
17          units of 0.1 dBm.
18
19          At the ONU, the measurement is performed in a burst-mode
20          manner and stored in a rolling 15-minutes' long
21          observation bin.
22
23          At the OLT, the measurement is the average power for
24          each incoming data burst, and stored in a rolling
25          15-minutes' long observation bin.
26
27          This object is applicable for an OLT and an ONU. It has
28          a distinct value for each logical link.";
29      reference
30          "IEEE Std 802.3.1, dot3ExtPkgOptIfLowOutputPower";
31  }
32  leaf out-trx-power-high-15-minutes-bin {
33      type power-level;
34      config false;
35      description
36          "This object reflects the highest value of the output
37          power during the period of the last 15 minutes, as
38          measured at the optical transceiver, and expressed in
39          units of 0.1 dBm.
40
41          At the ONU, the measurement is performed in a burst-mode
42          manner and stored in a rolling 15-minutes' long
43          observation bin.
44
45          At the OLT, the measurement is the average power for
46          each incoming data burst, and stored in a rolling
47          15-minutes' long observation bin.
48
49          This object is applicable for an OLT and an ONU. It has
50          a distinct value for each logical link.";
51      reference
52          "IEEE Std 802.3.1, dot3ExtPkgOptIfHighOutputPower";
53  }
54  leaf trx-data-reliable {
55      if-feature "trx-power-level-reporting-supported";
56      type boolean;
57      config false;
58      description
59          "This object indicates whether data contained in
```

```
1         individual counters in 'statistics-trx' container are
2         reliable (when read as 'true') or not
3         (when read as 'false').
4
5         This object is applicable for an OLT and an ONU. It has
6         a distinct value for each logical link.";
7     reference
8         "IEEE Std 802.3.1, dot3ExtPkgOptIfSuspectedFlag";
9 }
10 }
11 }
12 container statistics-pon-fec {
13     when "(../fec-capability = 'supported') and
14         (../fec-mode = 'enabled-Tx-Rx')";
15     if-feature "fec-supported";
16     config false;
17     description
18         "This container defines a set of FEC-related statistics
19         counters of an EPON interface, as defined in
20         IEEE Std 802.3, Clause 65 and Clause 76.
21
22         Discontinuities in the value of this counter can occur at
23         re-initialization of the management system, and at other
24         times as indicated by the value of the
25         'discontinuity-time' leaf defined in the ietf-interfaces
26         YANG module (IETF RFC 8343).";
27     reference
28         "IEEE Std 802.3.1, dot3OmpEmulationStatEntry";
29     leaf fec-code-group-violations {
30         type yang:counter64;
31         units "code-group";
32         config false;
33         description
34             "For 1G-EPON this is a count of the number of events that
35             cause the PHY to indicate â€™Data reception errorâ€™ or
36             â€™Carrier Extend Errorâ€™ on the GMII (see IEEE Std 802.3,
37             Table 35-1). The contents of this counter is undefined
38             when FEC is operating. For 10G-EPON this object is not
39             applicable.
40
41             This object is applicable for an OLT and an ONU. At the
42             OLT, it has a distinct value for each logical link.";
43         reference
44             "IEEE Std 802.3, 30.5.1.1.14";
45     }
46     leaf fec-buffer-head-coding-violations {
47         type yang:counter64;
48         units "code-group";
49         config false;
50         description
51             "For 1G-EPON PHY, this object represents the count of the
52             number of invalid code-groups received directly from the
53             link when FEC is enabled. When FEC is disabled this
54             counter stops counting.
55
56             For 10G-EPON PHYs, this object is set to zero.
57
58             This object is applicable for an OLT and an ONU. It has
59             a distinct value for each logical link.";
60         reference
```

```
1      "IEEE Std 802.3.1, dot3EponFecBufferHeadCodingViolation";
2    }
3    leaf fec-code-word-corrected-errors {
4      type yang:counter64;
5      units "code-group";
6      config false;
7      description
8        "For 1G-EPON or 10G-EPON PHYs, this object represents a
9        count of corrected FEC blocks.
10
11        This counter increments by one for each received
12        FEC block that contained detected errors and was
13        corrected by the FEC function in the PHY.
14
15        This object is applicable for an OLT and an ONU. It has
16        a distinct value for each logical link.";
17      reference
18        "IEEE Std 802.3, 30.5.1.1.17";
19    }
20  leaf fec-code-word-uncorrected-errors {
21    type yang:counter64;
22    units "code-group";
23    config false;
24    description
25      "For 1G-EPON or 10G-EPON PHYs, this object represents a
26      count of uncorrectable FEC blocks.
27
28      This counter increments by one for each received FEC
29      block that contained detected errors and was not
30      corrected by the FEC function in the PHY.
31
32      This object is applicable for an OLT and an ONU. It has
33      a distinct value for each logical link.";
34    reference
35      "IEEE Std 802.3, 30.5.1.1.18";
36  }
37 }
38
39 container mpcp-logical-link-admin-actions {
40   description
41     "Container of actions.";
42   action state-change-action-type {
43     description
44       "Request a state change on the interface.";
45     input {
46       leaf state-change-action-type {
47         type identityref {
48           base state-change-action-type;
49         }
50         description
51           "Type of interface state change requested.";
52       }
53     }
54   }
55 }
56
57 action reset-action-type {
58   description
59     "Request a reset-action of the interface.";
60   input {
61     leaf reset-action-type {
62       type identityref {
```

```

1         base reset-action-type;
2     }
3     description
4         "Type of reset action requested of the interface.";
5     }
6 }
7
8 }
9 action register-type {
10     description
11         "Request a registration action.";
12     input {
13         leaf register-type {
14             type identityref {
15                 base register-type;
16             }
17             description
18                 "Type of registration action requested of the
19                 interface.";
20         }
21     }
22 }
23 }
24 }
25 list mpcp-queues {
26     key "mpcp-queue-index";
27     description
28         "An instance of this object for each value of
29         'mpcp-queue-index' is created when a new logical link is
30         registered and deleted when the logical link is
31         deregistered.
32
33         All instances of this object in the ONU associated with
34         the given logical link are then mapped to a REPORT MPCPDU,
35         when generated.
36
37         +-----+
38         |          Destination Address          |
39         +-----+
40         |          Source Address              |
41         +-----+
42         |          Length/Type                 |
43         +-----+
44         |          OpCode                      |
45         +-----+
46         |          TimeStamp                   |
47         +-----+
48         |          Number of Queue Sets        |
49         +-----+
50         |          Report bitmap               |
51         +-----+
52         |          Queue 0 report              |
53         +-----+
54         |          Queue 1 report              |
55         +-----+
56         |          Queue 2 report              |
57         +-----+
58         |          Queue 3 report              |
59         +-----+
60         |          Queue 4 report              |
61         +-----+
62
63         -|
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```

1      |          Queue 5 report          |   |
2      +-----+
3      |          Queue 6 report          |   |
4      +-----+
5      |          Queue 7 report          |   |
6      +-----+
7      |          Pad/reserved            |   -|
8      +-----+
9      |          FCS                     |   |
10     +-----+
11
12

```

The 'Queue N report' field reports the current occupancy of each upstream transmission queue associated with the given logical link.

The 'Number of Queue Sets' field defines the number of reported 'Queue N report' sets.

For each Queue Set, the 'Report bitmap' field defines which upstream transmission queues are present in the REPORT MPCPDU.

Although the REPORT MPCPDU can report current occupation for up to 8 upstream transmission queues in a single REPORT MPCPDU, the actual number is flexible.

The 'mpcp-queue-group' grouping has a variable size that is limited by value of 'mpcp-maximum-queue-count-per-report' object, allowing ONUs report the occupancy of fewer upstream transmission queues, as needed.

This object is applicable for an OLT and an ONU.
At the OLT, this object has a distinct value for each logical link and every queue. At the ONU, it has a distinct value for every queue."

```

38 reference
39   "IEEE Std 802.3.1, dot3ExtPkgQueueEntry";
40 leaf mpcp-queue-index {
41   type uint8 {
42     range "0 .. 7" {
43       description
44         "This object indicates the identity (index) of a
45          queue in the ONU. It can have a value between
46          0 and 7, limited by the value stored in the
47          'mpcp-maximum-queue-count-per-report' object.";
48       reference
49         "See 'mpcp-maximum-queue-count-per-report' object";
50     }
51   }
52 }
53
54 description
55   "An object represents the index of an upstream
56    transmission queue storing subscriber packets.
57    The size (occupancy) of the upstream transmission queue
58    identified by this object is then reported within
59    REPORT MPCPDU, defined in IEEE Std 802.3, Clause 64
60    and Clause 77.
61
62    This object indicates the identity (index) of a queue in
63    the ONU. It can have a value between 0 and 7, limited by
64    the value stored in the
65

```

```
1      'mpcp-maximum-queue-count-per-report' object.  
2  
3      This object is applicable for an OLT and an ONU.  
4      It has a distinct value for each logical link and each  
5      queue.  
6      At the ONU, it has a distinct value for each queue.";  
7  reference  
8      "IEEE Std 802.3.1, dot3QueueIndex";  
9  }  
10 leaf mpcp-queue-threshold-count {  
11     type uint8 {  
12         range "0 .. 7" {  
13             description  
14                 "This object indicates the identity (index) of a  
15                 queue in the ONU. It can have a value between  
16                 0 and 7, limited by the value stored in the  
17                 'mpcp-maximum-queue-count-per-report' object.";  
18             reference  
19                 "See 'mpcp-queue-threshold-count-max' object";  
20         }  
21     }  
22     description  
23         "This object reflects the number of reporting thresholds  
24         for the specific upstream transmission queue, reflected  
25         in the REPORT MPCPDU, as defined in IEEE Std 802.3,  
26         Clause 64 and Clause 77.  
27  
28         Each 'Queue set' provides information for the specific  
29         upstream transmission queue occupancy of frames below  
30         the matching reporting threshold.  
31  
32         A read of this object reflects the number of reporting  
33         thresholds for the specific upstream transmission queue.  
34  
35         This object is applicable for an OLT and an ONU. It has  
36         a distinct value for each logical link and each queue.  
37         At the ONU, it has a distinct value for each queue.";  
38     reference  
39         "IEEE Std 802.3.1, dot3ExtPkgObjectReportNumThreshold";  
40 }  
41 leaf mpcp-queue-threshold-count-max {  
42     type uint8 {  
43         range "0 .. 7" {  
44             description  
45                 "This object can have a value between 0 and 7.";  
46         }  
47     }  
48     config false;  
49     description  
50         "This object reflects the maximum number of reporting  
51         thresholds for the specific upstream transmission queue,  
52         reflected in the REPORT MPCPDU, as defined in  
53         IEEE Std 802.3, Clause 64 and Clause 77.  
54  
55         A read of this object reflects the maximum number of  
56         reporting thresholds for the specific upstream  
57         transmission queue.  
58  
59         This object is applicable for an OLT and an ONU. It has
```

```

    a distinct value for each logical link and each queue.
    At the ONU, it has a distinct value for each queue.";
reference
    "IEEE Std 802.3.1,
    dot3ExtPkgObjectReportMaximumNumThreshold";
}
list mpcp-queue-thresholds {
    when '../mpcp-queue-threshold-count > 0';
    key "mpcp-queue-set-index";
    max-elements 7;
    description
        "An instance of this object for each value of
        'mpcp-queue-index' is created when a new logical link is
        registered and deleted when the logical link is
        deregistered.

        All instances of this object in the ONU associated with
        the given logical link are then mapped to a
        REPORT MPCPDU, when generated.

```

Destination Address	
Source Address	
Length/Type	
OpCode	
Time Stamp	
Number of Queue Sets	
Report bitmap	
Queue 0 report	
Queue 1 report	
Queue 2 report	
Queue 3 report	
Queue 4 report	
Queue 5 report	
Queue 6 report	
Queue 7 report	
Pad/reserved	
FCS	

The 'Queue N report' field reports the current occupancy of each upstream transmission queue associated with the given logical link.

The 'Number of Queue Sets' field defines the number of reported 'Queue N report' sets.

For each Queue Set, the 'Report bitmap' field defines which upstream transmission queues are present in the REPORT MPCPDU. Although the REPORT MPCPDU can report current occupation for up to 8 upstream transmission queues in a single REPORT MPCPDU, the actual number is flexible.

The 'mpcp-queue-group' grouping has a variable size that is limited by value of 'mpcp-maximum-queue-count-per-report' object, allowing ONUs to report the occupancy of fewer upstream transmission queues, as needed.

This object is applicable for an OLT and an ONU. It has a distinct value for each logical link and every queue. At the ONU, it has a distinct value for every queue."

```
reference
  "IEEE Std 802.3.1, dot3ExtPkgQueueSetsEntry";
leaf mpcp-queue-set-index {
  type uint8 {
    range "0 .. 7" {
      description
        "This object indicates the identity (index) of a
        queue in the ONU. It can have a value between
        0 and 7, limited by the value stored in the
        'mpcp-maximum-queue-count-per-report' object.";
      reference
        "See 'mpcp-maximum-queue-count-per-report' object";
    }
  }
  description
    "This object represents the index of the Queue Set for
    the 'mpcp-queue-set-group' grouping. The size
    (occupancy) of the upstream transmission queues
    belonging to the given Queue Set is then reported
    within REPORT MPCPDU, defined in IEEE Std 802.3,
    Clause 64 and Clause 77.

    This object can have a value between 0 and 7, limited
    by the value stored in the
    'mpcp-queue-threshold-count-max' object.";
  reference
    "IEEE Std 802.3.1, dot3QueueSetIndex";
}
leaf mpcp-queue-set-threshold {
  type uint64;
  units "TQ";
  default "0";
  description
    "This object defines the value of a reporting threshold
    for each Queue Set stored in REPORT MPCPDU defined in
    IEEE Std 802.3, Clause 64 and Clause 77.

    The number of Queue Sets for each upstream
    transmission queue is defined in the
```

```
1         'mpcp-queue-threshold-count' object.  
2  
3         Within REPORT MPCPDU, each Queue Set provides  
4         information on the current upstream transmission queue  
5         occupancy for frames below the matching threshold.  
6  
7         The value stored in this object is expressed in the  
8         units of Time quanta (TQ), where 1 TQ = 16 ns.  
9  
10        A read of this object provides the current threshold  
11        value for the specific upstream transmission queue.  
12  
13        This object is applicable for an OLT and an ONU. At  
14        the OLT, it has a distinct value for each logical link,  
15        each queue, and each Queue Set.  
16  
17        At the ONU, it has a distinct value for each queue and  
18        each Queue Set.";  
19  
20        reference  
21        "IEEE Std 802.3.1, dot3ExtPkgObjectReportThreshold";  
22    }  
23 }  
24  
25 leaf in-mpcp-queue-frames {  
26     type yang:counter64;  
27     config false;  
28     description  
29         "A count of the number of times a frame reception event  
30         results in a frame being queued in (for ONUs) or  
31         received from (for OLTs) the corresponding queue. This  
32         object is incremented by one for each frame written to  
33         (in the case of the ONU) or received for (in case of the  
34         OLT) the associated queue.  
35  
36         The queue index matches the queue number in  
37         REPORT MPCPDU, as defined in IEEE Std 802.3, Clause 64  
38         and Clause 77.  
39  
40         This object is applicable for an OLT and an ONU.  
41         At the OLT, it has a distinct value for each logical link  
42         and each queue. At the ONU, it has a distinct value for  
43         each queue.";  
44  
45     reference  
46     "IEEE Std 802.3.1, dot3ExtPkgStatRxFramesQueue";  
47 }  
48  
49 leaf out-mpcp-queue-frames {  
50     when "../../mpcp-mode = 'onu'";  
51     type yang:counter64;  
52     config false;  
53     description  
54         "This object reflects the number of frame transmission  
55         events from the corresponding upstream transmission  
56         queue. This object is incremented by one for each frame  
57         transmitted, when it is output from the associated  
58         queue.  
59  
60         The queue index matches the queue number in  
61         REPORT MPCPDU, as defined in IEEE Std 802.3, Clause 64  
62         and Clause 77.  
63  
64  
65
```

```

1         This object is applicable for an ONU only. At the ONU,
2         it has a distinct value for each queue.";
3     reference
4         "IEEE Std 802.3.1, dot3ExtPkgStatTxFramesQueue";
5 }
6 leaf mpcp-queue-frames-drop {
7     when "../mpcp-mode = 'onu'";
8     type yang:counter64;
9     config false;
10    description
11        "This object reflects the number of frame drop events
12        from the corresponding upstream transmission queue.
13        This object is incremented by one for each frame dropped
14        in the associated queue.
15
16        The queue index matches the queue number in
17        REPORT MPCPDU, as defined in IEEE Std 802.3, Clause 64
18        and Clause 77.
19
20        This object is applicable for an ONU only. At the ONU,
21        it has a distinct value for each queue.";
22    reference
23        "IEEE Std 802.3.1, dot3ExtPkgStatDroppedFramesQueue";
24 }
25 }
26 list multicast-IDs {
27     key "multicast-ID";
28     description
29         "Multicast-IDs list of multicast IDs
30         to be recognized by the device.";
31     leaf multicast-ID {
32         type uint32;
33         description
34             "Multicast-IDs to be recognized by the device.";
35         reference
36             "IEEE Std 802.3, 30.3.5.1.25";
37     }
38 }
39 leaf fec-capability {
40     type fec-capability;
41     config false;
42     description
43         "This object is used to identify whether the given
44         interface is capable of supporting FEC or not.";
45 }
46 leaf mpcp-mode {
47     type mpcp-mode;
48     config false;
49     description
50         "This object is used to identify the operational state of
51         the MultiPoint MAC Control sublayer as defined in
52         IEEE Std 802.3, Clause 64 and Clause 77.
53
54         Reading 'olt' for an OLT (controller) mode and 'onu' for
55         an ONU (client) mode.
56
57         This object is used to identify the operational mode for
58         the MPCP objects.
59
60
61
62
63
64
65
```

```
1         This object is applicable for an OLT, with the same value
2         for all logical links, and for an ONU.";
3     reference
4         "IEEE Std 802.3, 30.3.5.1.3";
5 }
6 leaf mpcp-sync-time {
7     type uint64;
8     units "TQ (16ns)";
9     config false;
10    description
11        "This object reports the 'sync lock time' of the OLT
12        receiver in units of Time Quanta (TQ; 1 TQ = 16 ns; see
13        IEEE Std 802.3, Clause 64 and Clause 77).
14
15        The value returned is equal to [sync lock time ns]/16,
16        rounded up to the nearest TQ. If this value exceeds
17        4,294,967,295 TQ, the value 4,294,967,295 TQ is returned.
18
19        This object is applicable for an OLT, with distinct values
20        for all logical links, and for an ONU.";
21    reference
22        "IEEE Std 802.3.1, dot3MpcpSyncTime";
23 }
24 leaf mpcp-logical-link-id {
25     type mpcp-supported;
26     config false;
27     description
28         "This object is used to identify the operational state of
29         the MultiPoint MAC Control sublayer as defined in
30         IEEE Std 802.3, Clause 64 and Clause 77.
31
32         Reading 'olt' for an OLT (controller) mode and 'onu' for
33         an ONU (client) mode.
34
35         This object is used to identify the operational mode for
36         the MPCP objects.
37
38         This object is applicable for an OLT, with the same value
39         for all logical links, and for an ONU.";
40     reference
41         "IEEE Std 802.3, 30.3.5.1.3";
42 }
43 leaf mpcp-remote-mac-address {
44     type ieee:mac-address;
45     config false;
46     description
47         "This object identifies the source_address parameter of the
48         last MPCPDUs passed to the MAC Control. This value is
49         updated on reception of a valid frame with:
50
51         1) a destination Field equal to the multicast address
52         assigned for MAC Control as specified in
53         IEEE Std 802.3, Annex 31A;
54
55         2) the lengthOrType field value equal to the Type assigned
56         for MAC Control as specified in IEEE Std 802.3, Annex 31A;
57
58         3) an MPCP Control opcode value equal to the subtype
59         assigned for MPCP as specified in IEEE Std 802.3,
```

```
1         Annex 31A.
2
3         This object is applicable for an OLT and an ONU. It has a
4         distinct value for each logical link.
5
6         The value reflects the MAC address of the remote entity
7         and therefore the OLT holds a value for each LLID, which
8         is the MAC address of the ONU.
9
10        The ONU has a single value that is the OLT MAC address.";
11    reference
12        "IEEE Std 802.3, 30.3.5.1.5";
13    }
14    leaf mpcp-logical-link-state {
15        type mpcp-logical-link-state;
16        config false;
17        description
18            "This object identifies the registration state of the
19            MultiPoint MAC Control sublayer as defined in
20            IEEE Std 802.3, Clause 64 and Clause 77.
21
22            When this object has the enumeration 'unregistered', the
23            interface is unregistered and may be used for registering
24            a link partner.
25
26            When this object has the enumeration 'registering',
27            the interface is in the process of registering a
28            link-partner.
29
30            When this object has the enumeration 'registered', the
31            interface has an established link-partner.
32
33            This object is applicable for an OLT and an ONU. It has a
34            distinct value for each logical link.";
35        reference
36            "IEEE Std 802.3, 30.3.5.1.6";
37    }
38    leaf mpcp-elapsed-time-out {
39        type uint64;
40        units "TQ (16ns)";
41        config false;
42        description
43            "This object reports the interval from the last MPCP frame
44            transmission in increments of Time Quanta
45            (TQ; 1 TQ = 16 ns;
46            see IEEE Std 802.3, Clause 64 and Clause 77).
47
48            The value returned is equal to [interval from last MPCP
49            frame transmission on this EPON interface, expressed
50            in ns]/16. If this value exceeds 4,294,967,295 TQ, the
51            value 4,294,967,295 TQ is returned.
52
53            This object is applicable for an OLT and an ONU. It has a
54            distinct value for each logical link.";
55        reference
56            "IEEE Std 802.3, 30.3.5.1.19";
57    }
58    leaf mpcp-elapsed-time-in {
59        type uint64;
```

```
1      units "TQ (16ns)";
2      config false;
3      description
4          "This object reports the interval from the last MPCP frame
5          reception in increments of Time Quanta (TQ; 1 TQ = 16 ns;
6          see IEEE Std 802.3, Clause 64 and Clause 77).
7
8          The value returned is equal to [interval from last MPCP
9          frame reception on this EPON interface, expressed in
10         ns]/16. If this value exceeds 4,294,967,295 TQ, the value
11         4,294,967,295 TQ is returned.
12
13         This object is applicable for an OLT and an ONU. It has a
14         distinct value for each logical link.";
15     reference
16         "IEEE Std 802.3, 30.3.5.1.20";
17 }
18 leaf mpcp-round-trip-time {
19     when "../ompe-mode = 'olt'";
20     type uint16;
21     units "TQ (16ns)";
22     config false;
23     description
24         "This object reports the MPCP round trip time in increments
25         of Time Quanta (TQ; 1 TQ = 16 ns; see IEEE Std 802.3,
26         Clause 64 and Clause 77).
27
28         The value returned is equal to [round trip time in ns]/16.
29         If this value exceeds 65,535 TQ, the value 65,535 TQ is
30         returned.
31
32         This object is applicable for an OLT. It has a distinct
33         value for each logical link.";
34     reference
35         "IEEE Std 802.3, 30.3.5.1.21";
36 }
37 leaf mpcp-maximum-grant-count {
38     when "../ompe-mode = 'onu'";
39     type uint8;
40     config false;
41     description
42         "This object reports the maximum number of grants that an
43         ONU can store for handling. The maximum number of grants
44         that an ONU can store for handling has a range of
45         0 to 255.
46
47         This object is applicable for an ONU and has a distinct
48         value for each logical link.";
49     reference
50         "IEEE Std 802.3, 30.3.5.1.24";
51 }
52 leaf mpcp-logical-link-count {
53     type mpcp-llid-count;
54     units "LLID";
55     config false;
56     description
57         "This object reflects the number of logical links
58         registered on the OLT / ONU. The LLID field, as defined in
59         the IEEE Std 802.3, Clause 65 and Clause 76, is a 2-byte
```

```
1         register (15-bit field and a broadcast bit) limiting the
2         number of logical links to 32,768.
3
4         This object is initialized to the value of 0 when the
5         OLT / ONU is powered up.
6
7         This object is applicable for an OLT and an ONU. It has
8         the same value for all logical links.";
9
10        reference
11        "IEEE Std 802.3.1, dot3ExtPkgObjectNumberOfLLIDs";
12    }
13    leaf mpcp-maximum-queue-count-per-report {
14        when "../ompe-mode = 'olt'";
15        type mpcp-maximum-queue-count-per-report;
16        config false;
17        description
18        "This object reflects the maximum number of queues (0-7)
19        that can be accepted by the OLT in a single REPORT MPCPDU,
20        as defined in IEEE Std 802.3, Clause 64 and Clause 77.
21
22        This object is applicable for an OLT and has a distinct
23        value for each logical link.";
24        reference
25        "IEEE Std 802.3.1, dot3ExtPkgObjectReportMaximumNumQueues";
26    }
27    leaf ompe-mode {
28        type ompe-mode;
29        config false;
30        description
31        "This object indicates the mode of operation of the
32        Reconciliation Sublayer for Point-to-Point Emulation (see
33        IEEE Std 802.3, 65.1 or 76.2 as appropriate).
34
35        The value of 'unknown' is assigned in initialization; true
36        state or type is not yet known.
37
38        The value of 'olt' is assigned when the sublayer is
39        operating in OLT mode.
40
41        The value of 'onu' is assigned when the sublayer is
42        operating in ONU mode.
43
44        This object is applicable for an OLT and an ONU. It has
45        the same value for each logical link.";
46        reference
47        "IEEE Std 802.3, 30.3.7.1.2";
48    }
49 }
50 }
51 }
52 }
53 }
54 }
55 }
56 }
57 }
58 }
59 }
60 }
61 }
62 }
63 }
64 }
65 }
```

8. YANG module for Ethernet Link OAM

8.1 Introduction

IEEE Std 802.3, Clause 57 includes management capabilities for Ethernet-like interfaces to provide some basic operations, administration and maintenance (OAM) functions. The defined functionality includes discovery, error signaling, loopback, and link monitoring. This clause defines a portion of the YANG module for use with NETCONF or RESTCONF to manage these Ethernet-like interface capabilities.

8.2 Overview

Ethernet OAM is composed of a core set of functions and a set of optional functional groups as described in Clause 57 of IEEE Std 802.3. The core functions include discovery operations (determining if the other end of the link is OAM capable and what OAM functions it supports), state machine implementation, and some critical event flows.

Ethernet OAM provides single-hop functionality in that it works only between two directly connected Ethernet stations. Ethernet OAM has three functional objectives, which are detailed in 8.2.1 through 8.2.3. The definition of a basic Ethernet OAM protocol data unit is given in 8.2.4.

8.2.1 Remote fault indication

Remote fault indication provides a mechanism for one end of an Ethernet link to signal the other end that the receive path is non-operational. Some Ethernet Physical Layers offer mechanisms to signal this condition at the Physical Layer. Ethernet OAM added a mechanism so that some Ethernet Physical Layers can operate in unidirectional mode, allowing frames to be transmitted in one direction even when the other direction is non-operational. Traditionally, Ethernet PHYs do not allow frame transmission in one direction if the other direction is not operational. Using this mode, Ethernet OAM allows frame-based signaling of remote fault conditions while still not allowing higher layer applications to be aware of the unidirectional capability. This clause includes mechanisms for capturing that fault information and reflecting such information in data nodes and notifications within the NETCONF management framework.

8.2.2 Link monitoring

Ethernet OAM includes event signaling capability so that one end of an Ethernet link can indicate the occurrence of certain important events to the other end of the link. This happens via layer 2 protocols. This clause defines methods for incorporating the occurrence of these events, at both the local end and the far end of the link, into the YANG-based management framework.

Ethernet OAM also includes mechanisms for one Ethernet station to query another directly connected Ethernet station about the status of its Ethernet interface variables and status. This clause does not include mechanisms for controlling how one Ethernet endpoint may use this functionality to query the status or statistics of a peer Ethernet entity.

8.2.3 Remote loopback

Remote loopback is a link state where the peer Ethernet entity echoes every received packet (without modifications) back onto the link. Remote loopback is intrusive in that the other end of the link is not forwarding traffic from higher layers out over the link. This clause defines data nodes controlling loopback operation and reading the status of the loopback state.

8.2.4 Ethernet OAM protocol data units

An Ethernet OAM protocol data unit (OAMPDU) is a valid Ethernet frame with a destination Media Access Control (MAC) address equal to the MAC address assigned for Slow Protocols (see IEEE Std 802.3, Annex 57A), a lengthOrType field equal to the Type assigned for Slow Protocols, and a Slow Protocols subtype equal to that of the subtype assigned for Ethernet OAM.

OAMPDU is used throughout this clause as an abbreviation for Ethernet OAM protocol data unit. OAMPDUs are the mechanism by which two directly connected Ethernet interfaces exchange OAM information.

8.3 Security considerations for Ethernet operations, administration, and maintenance (OAM) module

The readable data nodes in this module can provide information about network traffic, and therefore, they may be considered sensitive. In particular, OAM provides mechanisms for reading the Clause 30 IEEE 802.3 management attributes from a link partner via a layer 3 protocol. IEEE Std 802.3 OAM does not include encryption or authentication mechanisms. It should not be used in environments where this interface information is considered sensitive, and where the facility terminations are unprotected. By default, OAM is disabled on Ethernet-like interfaces and is therefore not a risk.

IEEE Std 802.3 OAM is designed to support deployment in access and enterprise networks. In access networks, one end of a link is the CO-side, and the other is the CPE-side, and the facilities are often protected in wiring cages or closets. In such deployments, it is often the case that the CO-side is protected from access from the CPE-side. Within IEEE Std 802.3 OAM, this protection from remote access is accomplished by configuring the CPE-side in passive mode using the mode leaf. This prevents the CPE from accessing functions and information at the CO-side of the connection. In enterprise networks, read-only interface information is often considered non-sensitive.

The frequency of OAM PDUs on an Ethernet interface does not adversely affect data traffic, as OAM is a slow protocol with very limited bandwidth potential, and it is not required for normal link operation. Although there are a number of objects in this module with read-write or read-create MAX-ACCESS, they have limited effects on user data.

The loopback capability of OAM can have potentially disruptive effects; when remote loopback is enabled, the remote station automatically transmits all received traffic back to the local station except for OAM traffic. This completely disrupts all higher layer protocols such as bridging, IP, and NETCONF/RESTCONF.

The administrative state and mode are also configuration nodes. Disabling OAM can interrupt management activities between peer devices, potentially causing serious problems. Setting the mode node to an undesired value can allow access to Ethernet monitoring, events, and functions that may not be acceptable in a particular deployment scenario. In addition to loopback functionality, Ethernet interface statistics and events can be accessed via the OAM protocol, which may not be desired in some circumstances.

OAM event configuration also contains configuration nodes. These nodes control whether events are sent, and at what thresholds. Note that the frequency of event communication is limited by the frequency limits of Slow Protocols on Ethernet interfaces. Also, the information available via OAM events is also available via OAM Variable Requests. Access to this information via either OAM events or Variable Requests is controlled by the admin and mode nodes. As mentioned previously, inadequate protection of these variables can result in access to link information and functions.

8.4 Mapping of IEEE 802.3 managed objects

This subclause contains the mapping between the YANG data nodes defined in this clause and the attributes defined in IEEE Std 802.3, Clause 30. Table 8–1 provides the mapping between the *ieee802-ethernet-link-oam* module data nodes and the OAM attributes of IEEE Std 802.3, Clause 30.

**Table 8–1—Mapping between IEEE Std 802.3, Clause 30 managed objects
and *ieee802-ethernet-link-oam* YANG data nodes**

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-link-oam</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
oOAM	aOAMAdminState	30.3.6.1.2	interfaces/interface/ethernet/link-oam	admin	R/W
	dot3OamOperStatus	IEEE Std 802.3.1	interfaces/interface/ethernet/link-oam/ discovery-info/local	operational-status	R
	aOAMDiscoveryState	30.3.6.1.4			
	aOAMLocalFlagsField	30.3.6.1.10		loopback-mode	R
	aOAMRemoteFlagsField	30.3.6.1.11			
	aOAMLocalState	30.3.6.1.14			
	aOAMMode	30.3.6.1.3		mode	RW
	aOAMLocalRevision	30.3.6.1.12		revision	R
	aOAMLocalPDUConfiguration	30.3.6.1.8		oammtu	R
	aOAMRemoteConfiguration	30.3.6.1.7	interfaces/interface/ethernet/link-oam/ discovery-info/remote/functions-sup- ported	uni-directional-link-fault	R/W
				loopback	R/W
				mib-retrieval	R/W
	aOAMLocalConfiguration	30.3.6.1.6	interfaces/interface/ethernet/link-oam/ discovery-info/local/functions-sup- ported/link-monitor	link-monitoring	R/W
	aOAMLocalErrSymPeriodConfig	30.3.6.1.34	interfaces/interface/ethernet/link-oam/ link-monitor/event-type	threshold-type window threshold	R/W
	aOAMLocalErrFrameConfig	30.3.6.1.36			R/W
	aOAMLocalErrFramePeriodConfig	30.3.6.1.38			R/W
	aOAMLocalErrFrameSecsSummaryConfig	30.3.6.1.40			
	aOAMLocalErrSymPeriodConfig	30.3.6.1.42			
	aOAMRemoteMACAddress	30.3.6.1.5	interfaces/interface/ethernet/link-oam/ discovery-info/remote	mac-address	R
	aOAMRemoteVendorOUI	30.3.6.1.16		vendor-oui	R

Table 8–1—Mapping between IEEE Std 802.3, Clause 30 managed objects
and *ieee802-ethernet-link-oam* YANG data nodes (continued)

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-link-oam</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
	aOAMRemoteVendorSpecificInfo	30.3.6.1.17		vendor-info	R
	aOAMRemoteState	30.3.6.1.15		loopback-mode	R
	aOAMMode	30.3.6.1.3		mode	R
	aOAMRemoteRevision	30.3.6.1.13		revision	R
	aOAMRemotePDUConfiguration	30.3.6.1.9		oammtu	R

**Table 8–1—Mapping between IEEE Std 802.3, Clause 30 managed objects
and *ieee802-ethernet-link-oam* YANG data nodes (continued)**

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-link-oam</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
OAM	aOAMLocalConfiguration	30.3.6.1.6	interfaces/interface/ethernet/link-oam/ discovery-info/local/functions-sup- ported	uni-directional-link-fault	R
				loopback	R
				link-monitoring	R
				mib-retrieval	R
	dot3OamEventLogEntry	IEEE Std 802.3.1	interfaces-state/ interface/ethernet/ link-oam/event-log/event-log-entry	index	R
				oui	R
				timestamp	R
				location	R
				event-type	R
				running-total	R
				event-total	R
	aOAMLocalErrSymPeriodEvent aOAMLocalErrFrameEvent aOAMLocalErrFramePeriodConfig aOAMLocalErrFrameSecsSummaryEvent aOAMRemoteErrSymPeriodEvent aOAMRemoteErrFrameEvent aOAMRemoteErrFramePeriodEvent aOAMRemoteErrFrameSecsSummaryEvent	30.3.6.1.35 30.3.6.1.37 30.3.6.1.38 30.3.6.1.41 30.3.6.1.42 30.3.6.1.43 30.3.6.1.44 30.3.6.1.45	interfaces/interface/ethernet/link-oam/ event-log/event-log-entry/threshold	threshold-event-type window threshold value	R R R R
	Dot3OamStatsEntry	RFC-4878		out-information	R
	aOAMInformationTx	30.3.6.1.20			

**Table 8–1—Mapping between IEEE Std 802.3, Clause 30 managed objects
and *ieee802-ethernet-link-oam* YANG data nodes (continued)**

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-link-oam</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
	aOAMInformationRx	30.3.6.1.21		in-information	R
	aOAMUniqueEventNotificationTx	30.3.6.1.22		out-unique-event-notification	R
	aOAMUniqueEventNotificationRx	30.3.6.1.24		in-unique-event-notification	R
	aOAMDuplicateEventNotificationTx	30.3.6.1.23		out-duplicate-event-notification	R
	aOAMDuplicateEventNotificationRx	30.3.6.1.25		in-duplicate-event-notification	R
	aOAMLoopbackControlTx	30.3.6.1.26		out-loopback-control	R
	aOAMLoopbackControlRx	30.3.6.1.27		in-loopback-control	R
	aOAMVariableRequestTx	30.3.6.1.28		out-variable-request	R

**Table 8–1—Mapping between IEEE Std 802.3, Clause 30 managed objects
and *ieee802-ethernet-link-oam* YANG data nodes (continued)**

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-link-oam</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
OAM	aOAMVariableRequestRx	30.3.6.1.29	interfaces/interface/ethernet/link-oam/ statistics	variable-requeste-rx	R
	aOAMVariableResponseTx	30.3.6.1.30		out-variable-response	R
	aOAMVariableResponseRx	30.3.6.1.31		in-variable-response	R
	aOAMOrganizationSpecificTx	30.3.6.1.32		out-org-specific	R
	aOAMOrganizationSpecificRx	30.3.6.1.33		in-org-specific	R
	aOAMUnsupportedCodesTx	30.3.6.1.18		unsupported-condes-tx	R
	aOAMUnsupportedCodesRx	30.3.6.1.19		in-unsupported-codes	R
	aFramesLostDueToOAMError	30.3.6.1.46		frames-lost-due-to-oam	R
	aOAMLocalErrSymPeriodEvent, Errored Symbols	30.3.6.1.35		local-error-symbol-period-log-entries	R
	aOAMLocalErrFrameEvent, Errored Frames	30.3.6.1.37		local-error-frame-log-entries	R
	aOAMLocalErrFramePeriodEvent, Errored Frames	30.3.6.1.39		local-error-frame-period-log-entries	R
	aOAMLocalErrFrameSecsSummaryEvent, Errored Frame Seconds Summary	30.3.6.1.41		local-error-frame-second-log-entries	R
	aOAMRemoteErrSymPeriodEvent, Errored Symbols	30.3.6.1.42		remote-error-symbol-period-log-entries	R
	aOAMRemoteErrFrameEven, Errored Frames	30.3.6.1.43		remote-error-frame-log-entries	R
	aOAMRemoteErrFramePeriodEvent, Errored Frames	30.3.6.1.44		remote-error-frame-period-log-entries	R
	aOAMRemoteErrFrameSecsSummaryEvent, Errored Frame Seconds Summary	30.3.6.1.45		remote-error-frame-second-log-entries	R

8.5 YANG module definition^m

The YANG module tree hierarchy uses terms defined in IETF RFC 8407.

8.5.1 Tree hierarchy

8.5.1.1 ieee802-ethernet-link-oam

```

module: ieee802-ethernet-link-oam
  augment /if:interfaces/if:interface:
    +--rw link-oam!
      +---x remote-loopback {remote-loopback-initiate}?
        | +---w input
        | | +---w enable      boolean
        | +---ro output
        |   +---ro success      boolean
        |   +---ro error-message? string
      +---x reset-stats
        | +---ro output
        |   +---ro success      boolean
        |   +---ro error-message? string
      +---n non-threshold-event
        | +-- oui              vendor-oui
        | +-- timestamp         uint64
        | +-- location          event-location
        | +-- event-type        identityref
        | +-- running-total     yang:counter64
        | +-- event-total       yang:counter64
      +---n threshold-event {link-monitoring-local or
link-monitoring-remote}?
        | +-- oui              vendor-oui
        | +-- timestamp         uint64
        | +-- location          event-location
        | +-- event-type        identityref
        | +-- running-total     yang:counter64

```

^mCopyright release for YANG modules: Users of this standard may freely reproduce the YANG module contained in this subclause so that it can be used for its intended purpose.

```

1      |   +-- event-total          yang:counter64
2      |   +-- threshold {link-monitoring-local or
3 link-monitoring-remote}?
4      |       +-- threshold-event-type      threshold-event-enum
5      |       +-- window                  uint64
6      |       +-- threshold                uint64
7      |       +-- value                    uint64
8      +--rw admin?                        admin-state
9      +--rw discovery-info
10     |   +--rw local
11     |   |   +--ro operational-status      operational-state
12     |   |   +--ro loopback-mode          loopback-status {remote-loopback-initiate or
13 remote-loopback-respond}?
14     |   |   +--rw mode?                  mode
15     |   |   +--rw functions-supported
16     |   |   |   +--rw uni-directional-link-fault?  boolean {uni-directional-link-fault}?
17     |   |   |   +--rw loopback?              boolean {remote-loopback-initiate}?
18     |   |   |   +--rw link-monitor {link-monitoring-remote or
19 link-monitoring-local}?
20     |   |   |   +--rw link-monitoring?  boolean
21     |   |   |   +--rw event-type* [threshold-type] {link-monitoring-local}?
22     |   |   |   +--rw threshold-type    threshold-event-enum
23     |   |   |   +--rw window?           uint64
24     |   |   |   +--rw threshold?        uint64
25     |   |   |   x--rw mib-retrieval?     boolean {remote-mib-retrieval-initiate or
26 remote-mib-retrieval-respond}?
27     |   |   |   +--rw data-retrieval?    boolean {remote-data-retrieval-initiate or
28 remote-mib-retrieval-respond}?
29     |   |   +--ro revision?              uint64
30     |   |   +--ro oammtu?                uint16
31     |   +--ro remote
32     |   |   +--ro mac-address?            ieee:mac-address
33     |   |   +--ro vendor-oui?            vendor-oui
34     |   |   +--ro vendor-info?           uint64
35     |   |   +--ro loopback-mode          loopback-status
36     |   |   +--ro mode?                  mode
37     |   |   +--ro functions-supported
38     |   |   |   +--ro uni-directional-link-fault?  boolean
39     |   |   |   +--ro loopback?              boolean
40
41
42
43

```

```

1      |      | +--ro link-monitoring?                boolean
2      |      | +--ro mib-retrieval?              boolean
3      |      +--ro revision?                    uint64
4      |      +--ro oammtu?                      uint16
5      +--ro event-log
6      |   +--ro event-log-entry* [index]
7      |   |   +--ro index                      uint64
8      |   |   +--ro oui                       vendor-oui
9      |   |   +--ro timestamp                  uint64
10     |   |   +--ro location                   event-location
11     |   |   +--ro event-type                 identityref
12     |   |   +--ro running-total              yang:counter64
13     |   |   +--ro event-total                yang:counter64
14     |   |   +--ro threshold {link-monitoring-local or
15     |   |   link-monitoring-remote}?
16     |   |   +--ro threshold-event-type      threshold-event-enum
17     |   |   +--ro window                    uint64
18     |   |   +--ro threshold                 uint64
19     |   |   +--ro value                     uint64
20     +--ro statistics
21     |   +--ro out-information                 yang:counter64
22     |   +--ro in-information                 yang:counter64
23     |   +--ro out-unique-event-notification yang:counter64 {link-monitoring-local}?
24     |   +--ro in-unique-event-notification yang:counter64 {link-monitoring-remote}?
25     |   +--ro out-duplicate-event-notification yang:counter64 {link-monitoring-local}?
26     |   +--ro in-duplicate-event-notification yang:counter64 {link-monitoring-remote}?
27     |   +--ro out-loopback-control          yang:counter64 {remote-loopback-initiate}?
28     |   +--ro in-loopback-control          yang:counter64 {remote-loopback-respond}?
29     |   +--ro out-variable-request          yang:counter64 {remote-data-retrieval-initiate}?
30     |   +--ro in-variable-request          yang:counter64 {remote-mib-retrieval-respond}?
31     |   +--ro out-variable-response        yang:counter64 {remote-mib-retrieval-respond}?
32     |   +--ro in-variable-response        yang:counter64 {remote-data-retrieval-initiate}?
33     |   +--ro out-org-specific              yang:counter64
34     |   +--ro in-org-specific               yang:counter64
35     |   +--ro out-unsupported-codes         yang:counter64
36     |   +--ro in-unsupported-codes         yang:counter64
37     |   +--ro frames-lost-due-to-oam       yang:counter64
38     |   +--ro local-error-symbol-period-log-entries yang:counter64
39     |   +--ro local-error-frame-log-entries yang:counter64

```

```
1  +---ro local-error-frame-period-log-entries      yang:counter64
2  +---ro local-error-frame-second-log-entries      yang:counter64
3  +---ro remote-error-symbol-period-log-entries    yang:counter64 {link-monitoring-remote}?
4  +---ro remote-error-frame-log-entries            yang:counter64 {link-monitoring-remote}?
5  +---ro remote-error-frame-period-log-entries     yang:counter64 {link-monitoring-remote}?
6  +---ro remote-error-frame-second-log-entries     yang:counter64 {link-monitoring-remote}?
```

8.5.2 YANG module

In the YANG module definition, should any discrepancy between the text of the description for individual YANG nodes in the YANG modules of this clause and the corresponding definition in 8.2 through 8.5 of this clause occur, the definitions in the YANG modules attached to this standard shall take precedence.

An ASCII text version of the YANG module can be found at the following URL:ⁿ

Editor's Note (to be removed prior to publication):

Yang files contained in <https://github.com/YangModels/yang/tree/main/standard/ieee/published/802.3> are IEEE 802.3.2-2019 version and will be updated at the publication time.

<https://github.com/YangModels/yang/tree/master/standard/ieee/published/802.3>

Editor's Note (to be removed prior to publication):

Pretty printing of `ieee802-ethernet-link-oam.yang` file may change the appearance by adding whitespace and reformatting lines

Editor's Note (to be removed prior to publication):

IEEE Std 802.3.1 and IEEE Std 802.3.2 to be updated at the publication time

```
module ieee802-ethernet-link-oam {
  yang-version 1.1;
  namespace "urn:ieee:std:802.3:yang:ieee802-ethernet-link-oam";
  prefix ieee802-link-oam;

  import ieee802-types {
    prefix ieee;
    reference
      "IEEE 802 types";
  }
  import ietf-yang-types {
    prefix yang;
    reference
      "IETF RFC 6991";
  }
  import iana-if-type {
    prefix ianaift;
    reference
      "http://www.iana.org/assignments/yang-parameters/
      iana-if-type@2023-01-26.yang";
  }
  import ietf-interfaces {
    prefix if;
    reference
      "IETF RFC 8343";
  }

  organization
    "IEEE 802.3 Ethernet Working Group
    Web URL: http://www.ieee802.org/3/";
```

ⁿCopyright release for YANG modules: Users of this standard may freely reproduce the YANG module contained in this subclause so that it can be used for its intended purpose.

```
1    contact
2      "Web URL: http://www.ieee802.org/3/";
3    description
4      "This module contains a collection of YANG definitions
5      for managing the Ethernet Link OAM feature defined by IEEE
6      802.3. It provides functionality roughly equivalent to that of
7      the DOT3-OAM-MIB defined in IETF RFC 4878.";
8
9
10   revision 2025-04-17 {
11     description
12       "Updates under IEEE Std 802.3.2-202x, Draft D3.3";
13     reference
14       "IEEE Std 802.3-2022, unless dated explicitly";
15   }
16
17   feature uni-directional-link-fault {
18     description
19       "This feature means the device supports Uni Directional Link
20       Fault detection.";
21     reference
22       "IEEE Std 802.3, 57.1.2:a, 30.3.6.1.6 aOAMLocalConfiguration
23       and 30.3.6.1.7 aOAMRemoteConfiguration";
24   }
25
26
27   feature remote-loopback-initiate {
28     description
29       "This feature means the device supports being the initiator
30       of remote loopback.";
31     reference
32       "IEEE Std 802.3, 57.1.2:b, 30.3.6.1.6
33       aOAMLocalConfiguration";
34   }
35
36
37   feature remote-loopback-respond {
38     description
39       "This feature means the device supports responding to remote
40       loopback control OAMPDUs received from the peer";
41     reference
42       "IEEE Std 802.3, 57.1.2:b, 30.3.6.1.7
43       aOAMRemoteConfiguration";
44   }
45
46
47   feature link-monitoring-local {
48     description
49       "This feature means the device monitors the link at the local
50       side and can generate Link Event OAMPDUs to the peer
51       device.";
52     reference
53       "IEEE Std 802.3, 57.1.2:c:1, 30.3.6.1.6
54       aOAMLocalConfiguration, and 30.3.6.1.7
55       aOAMRemoteConfiguration";
56   }
57
58
59   feature link-monitoring-remote {
60     description
61       "This feature means the device can process Link Event OAMPDUs
62       received from the peer device and report itself about this
63       event on its own management interface.";
64     reference
```

```
1      "IEEE Std 802.3, 57.1.2:c:1,  
2      30.3.6.1.6 aOAMLocalConfiguration,  
3      and 30.3.6.1.7 aOAMRemoteConfiguration";  
4  }  
5  
6  feature remote-mib-retrieval-initiate {  
7      status deprecated;  
8      description  
9          "remote-mib-retrieval-initiate is deprecated and changed name  
10         to feature remote-data-retrieval-initiate. This feature  
11         means the device supports data retrieval from the peer  
12         device. I.e. the device can send Variable Requests OAMPDUs  
13         to the peer side and process the received Variable Response  
14         OAMPDUs.";  
15      reference  
16          "IEEE Std 802.3, 57.1.2:c:2,  
17          30.3.6.1.6 aOAMLocalConfiguration,  
18          and 30.3.6.1.7 aOAMRemoteConfiguration";  
19      }  
20  
21  
22  
23  feature remote-data-retrieval-initiate {  
24      description  
25          "This feature means the device supports data retrieval from  
26          the peer device. I.e. the device can send Variable Requests  
27          OAMPDUs to the peer side and process the received Variable  
28          Response OAMPDUs.";  
29      reference  
30          "IEEE Std 802.3, 57.1.2:c:2,  
31          30.3.6.1.6 aOAMLocalConfiguration,  
32          and 30.3.6.1.7 aOAMRemoteConfiguration";  
33      }  
34  
35  
36  feature remote-mib-retrieval-respond {  
37      description  
38          "This feature means the device allows the peer device to  
39          retrieve data from the managed device. I.e. the device can  
40          process received Variable Requests OAMPDUs and respond with  
41          Variable Response OAMPDUs.";  
42      reference  
43          "IEEE Std 802.3, 57.1.2:c:2,  
44          30.3.6.1.6 aOAMLocalConfiguration,  
45          and 30.3.6.1.7 aOAMRemoteConfiguration";  
46      }  
47  
48  
49  identity event-type {  
50      description  
51          "Base identity for all Link OAM event types.";  
52  }  
53  
54  
55  identity threshold-event-type {  
56      base event-type;  
57      description  
58          "Event type for a Link Monitoring threshold event.";  
59  }  
60  
61  identity link-fault-event {  
62      if-feature "uni-directional-link-fault";  
63      base event-type;  
64      description  
65  }
```

```
1      "Event type for a uni-directional link fault event.";
2      reference
3      "IEEE Std 802.3, 57.2.10.1";
4  }
5
6  identity dying-gasp-event {
7      base event-type;
8      description
9      "Event type for a dying gasp event.";
10     reference
11     "IEEE Std 802.3, 57.2.10.1";
12 }
13
14 identity critical-event {
15     base event-type;
16     description
17     "Event type for a critical event.";
18     reference
19     "IEEE Std 802.3, 57.2.10.1";
20 }
21
22 typedef threshold-event-enum {
23     type enumeration {
24         enum symbol-period-event {
25             value 1;
26             description
27             "Errored symbol period event.";
28         }
29         enum frame-period-event {
30             value 2;
31             description
32             "Errored frame period event.";
33         }
34         enum frame-event {
35             value 3;
36             description
37             "Errored frame event";
38         }
39         enum frame-seconds-event {
40             value 4;
41             description
42             "Errored frame seconds event.";
43         }
44     }
45     description
46     "Enumeration of the valid threshold event types.";
47     reference
48     "IEEE Std 802.3, 57.5.3";
49 }
50
51 typedef mode {
52     type enumeration {
53         enum passive {
54             value 0;
55             description
56             "Ethernet Link OAM Passive mode.";
57         }
58         enum active {
59             value 1;
```

```
1         description
2             "Ethernet Link OAM Active mode.";
3     }
4 }
5 description
6     "Enumeration of the valid modes in which Link OAM may run.";
7 reference
8     "IEEE Std 802.3, 57.2.9 and 30.3.6.1.3.";
9 }
10
11
12 typedef event-location {
13     type enumeration {
14         enum event-location-local {
15             value 1;
16             description
17                 "A local event.";
18         }
19         enum event-location-remote {
20             value 2;
21             description
22                 "A remote event.";
23         }
24     }
25 }
26 description
27     "The location of the event that caused a log entry.";
28 }
29
30
31 typedef loopback-status {
32     type enumeration {
33         enum none {
34             value 1;
35             description
36                 "Loopback is not being performed.";
37         }
38         enum initiating {
39             value 2;
40             description
41                 "Initiating master loopback.";
42         }
43         enum master-loopback {
44             value 3;
45             description
46                 "In master loopback mode.";
47         }
48         enum terminating {
49             value 4;
50             description
51                 "Terminating master loopback mode.";
52         }
53         enum local-loopback {
54             value 5;
55             description
56                 "In slave loopback mode.";
57         }
58         enum unknown {
59             value 6;
60             description
61                 "Parser and multiplexer combination unexpected.";
62         }
63     }
64 }
65
```

```
1      }
2      description
3      "The loopback mode of an OAM interface.";
4      reference
5      "IEEE Std 802.3, 57.2.11";
6  }
7
8
9  typedef operational-state {
10     type enumeration {
11         enum disabled {
12             value 1;
13             description
14             "IEEE Std 802.3 OAM is disabled.";
15         }
16         enum link-fault {
17             value 2;
18             description
19             "IEEE Std 802.3 OAM has encountered a link fault.";
20         }
21         enum passive-wait {
22             value 3;
23             description
24             "Passive OAM entity waiting to see if peer is
25             OAM capable.";
26         }
27         enum active-send-local {
28             value 4;
29             description
30             "Active OAM entity trying to determine if peer
31             is OAM capable.";
32         }
33         enum send-local-and-remote {
34             value 5;
35             description
36             "OAM discovered peer but still to accept or
37             reject peer configuration.";
38         }
39         enum send-local-and-remote-ok {
40             value 6;
41             description
42             "OAM peering is allowed by local device.";
43         }
44         enum peering-locally-rejected {
45             value 7;
46             description
47             "OAM peering rejected by local device.";
48         }
49         enum peering-remotely-rejected {
50             value 8;
51             description
52             "OAM peering rejected by remote device.";
53         }
54         enum operational {
55             value 9;
56             description
57             "IEEE Std 802.3 OAM is operational.";
58         }
59         enum operational-half-duplex {
60             value 10;
61         }
62     }
63 }
64
65
```

```
1         description
2             "IEEE Std 802.3 OAM is operating in half-duplex mode.";
3     }
4 }
5 description
6     "Operational state of an interface.";
7 reference
8     "IEEE Std 802.3, 30.3.6.1.4,
9     30.3.6.1.10, and 30.3.6.1.11";
10 }
11
12
13 typedef vendor-oui {
14     type string {
15         length "6";
16     }
17     description
18         "Organizationally Unique Identifier (OUI) as assigned to
19         vendor by the IEEE Registration Authority";
20     reference
21         "IEEE Std 802, subclause 8.2.2";
22 }
23
24
25 typedef admin-state {
26     type enumeration {
27         enum enabled {
28             value 1;
29             description
30                 "IEEE Std 802.3, Clause 57 OAM is in the
31                 enabled admin state.";
32         }
33         enum disabled {
34             value 2;
35             description
36                 "IEEE Std 802.3, Clause 57 OAM is in the
37                 disabled admin state.";
38         }
39     }
40 }
41 description
42     "Admin state of the OAM function on an interface.";
43 reference
44     "IEEE Std 802.3, 30.3.6.1.2 and 30.3.6.2";
45 }
46
47
48 grouping event-details {
49     description
50         "Nodes describing an event, used in the event log and in
51         notifications.";
52     reference
53         "IETF RFC 4878, Dot3OamEventLogEntry";
54     leaf oui {
55         type vendor-oui;
56         mandatory true;
57         description
58             "Organizationally Unique Identifier for the device that
59             generated the event.";
60     }
61     leaf timestamp {
62         type uint64;
63         units "milliseconds";
64     }
65 }
```

```
1      mandatory true;
2      description
3          "Timestamp in milliseconds since Unix epoch for when the
4              event occurred.";
5  }
6  leaf location {
7      type event-location;
8      mandatory true;
9      description
10         "Where the event occurred (local or remote).";
11  }
12  leaf event-type {
13      type identityref {
14          base event-type;
15      }
16      mandatory true;
17      description
18          "Type of event that occurred.";
19      reference
20          "IEEE Std 802.3, 30.3.6.1.10 and 30.3.6.11";
21  }
22  leaf running-total {
23      type yang:counter64;
24      mandatory true;
25      description
26          "The running total number of errors seen since OAM was
27              enabled on the interface. For threshold events, this is
28              the total number of times that particular type of error
29              (e.g. symbol error) has occurred, which may be greater
30              than the number of threshold-crossing event notifications
31              of that type generated during that time (which is conveyed
32              by the event-total leaf).";
33  }
34  leaf event-total {
35      type yang:counter64;
36      mandatory true;
37      description
38          "Total number of times this event has occurred since OAM
39              was enabled on the interface. For threshold events this is
40              the number of events generated of this type (as opposed to
41              the total number of errors of that type, which may be
42              greater, and is conveyed by the running-total leaf.";
43  }
44  }
45  }
46  grouping threshold-event-details {
47      description
48          "Nodes describing a threshold event, used in the event
49              log and in notifications";
50      reference
51          "IETF RFC 4878, Dot3OamEventLogEntry";
52      container threshold {
53          when
54              "../event-type = 'ieee802-link-oam:threshold-event-type'" {
55              description
56                  "These nodes only apply to threshold event types";
57          }
58      }
59      if-feature
60          "link-monitoring-local or
```

```
1      link-monitoring-remote";
2      description
3      "Nodes specific to threshold (link monitoring) events";
4      leaf threshold-event-type {
5          type threshold-event-enum;
6          mandatory true;
7          description
8              "The type of threshold event";
9          reference
10             "IEEE Std 802.3, 57.5.3";
11      }
12      leaf window {
13          type uint64;
14          mandatory true;
15          description
16              "Size of the window in which the event was generated.
17              Units are dependent on the threshold event type.";
18      }
19      leaf threshold {
20          type uint64;
21          mandatory true;
22          description
23              "Size of the threshold that was breached during the
24              window. Units are dependent on the threshold
25              event type.";
26      }
27      leaf value {
28          type uint64;
29          mandatory true;
30          description
31              "Breaching value. Units are dependent on the threshold
32              event type, and match that of the threshold.";
33      }
34  }
35  }
36  }
37  }
38  }
39  }
40  grouping statistics-common {
41      description
42          "Collection of Link OAM event/packet counters.";
43      reference
44          "IETF RFC 4878, Dot3OamStatsEntry";
45      leaf out-information {
46          type yang:counter64;
47          mandatory true;
48          description
49              "Number of information OAMPDUs transmitted.";
50          reference
51              "IEEE Std 802.3, 30.3.6.1.20";
52      }
53      leaf in-information {
54          type yang:counter64;
55          mandatory true;
56          description
57              "Number of information OAMPDUs received.";
58          reference
59              "IEEE Std 802.3, 30.3.6.1.21";
60      }
61      leaf out-unique-event-notification {
62          if-feature "link-monitoring-local";
63      }
64  }
65  }
```

```
1      type yang:counter64;
2      mandatory true;
3      description
4          "Number of unique event notification OAMPDUs transmitted.";
5      reference
6          "IEEE Std 802.3, 30.3.6.1.22";
7  }
8
9  leaf in-unique-event-notification {
10     if-feature "link-monitoring-remote";
11     type yang:counter64;
12     mandatory true;
13     description
14         "Number of unique event notification OAMPDUs received.";
15     reference
16         "IEEE Std 802.3, 30.3.6.1.24";
17 }
18
19 leaf out-duplicate-event-notification {
20     if-feature "link-monitoring-local";
21     type yang:counter64;
22     mandatory true;
23     description
24         "Number of duplicate event notification OAMPDUs
25             transmitted.";
26     reference
27         "IEEE Std 802.3, 30.3.6.1.23";
28 }
29
30 leaf in-duplicate-event-notification {
31     if-feature "link-monitoring-remote";
32     type yang:counter64;
33     mandatory true;
34     description
35         "Number of duplicate event notification OAMPDUs
36             received.";
37     reference
38         "IEEE Std 802.3, 30.3.6.1.25";
39 }
40
41 leaf out-loopback-control {
42     if-feature "remote-loopback-initiate";
43     type yang:counter64;
44     mandatory true;
45     description
46         "Number of loopback control OAMPDUs transmitted.";
47     reference
48         "IEEE Std 802.3, 30.3.6.1.26";
49 }
50
51 leaf in-loopback-control {
52     if-feature "remote-loopback-respond";
53     type yang:counter64;
54     mandatory true;
55     description
56         "Number of loopback control OAMPDUs received.";
57     reference
58         "IEEE Std 802.3, 30.3.6.1.27";
59 }
60
61 leaf out-variable-request {
62     if-feature "remote-data-retrieval-initiate";
63     type yang:counter64;
64     mandatory true;
65     description
```

```
1         "Number of variable request OAMPDUs transmitted.";
2     reference
3         "IEEE Std 802.3, 30.3.6.1.28";
4 }
5 leaf in-variable-request {
6     if-feature "remote-mib-retrieval-respond";
7     type yang:counter64;
8     mandatory true;
9     description
10        "Number of variable request OAMPDUs received.";
11    reference
12        "IEEE Std 802.3, 30.3.6.1.29";
13 }
14 leaf out-variable-response {
15     if-feature "remote-mib-retrieval-respond";
16     type yang:counter64;
17     mandatory true;
18     description
19        "Number of variable response OAMPDUs transmitted.";
20    reference
21        "IEEE Std 802.3, 30.3.6.1.30";
22 }
23 leaf in-variable-response {
24     if-feature "remote-data-retrieval-initiate";
25     type yang:counter64;
26     mandatory true;
27     description
28        "Number of variable response OAMPDUs received.";
29    reference
30        "IEEE Std 802.3, 30.3.6.1.31";
31 }
32 leaf out-org-specific {
33     type yang:counter64;
34     mandatory true;
35     description
36        "Number of organization specific OAMPDUs transmitted.";
37    reference
38        "IEEE Std 802.3, 30.3.6.1.32";
39 }
40 leaf in-org-specific {
41     type yang:counter64;
42     mandatory true;
43     description
44        "Number of organization specific OAMPDUs received.";
45    reference
46        "IEEE Std 802.3, 30.3.6.1.33";
47 }
48 leaf out-unsupported-codes {
49     type yang:counter64;
50     mandatory true;
51     description
52        "Number of OAMPDUs with unsupported codes transmitted.";
53    reference
54        "IEEE Std 802.3, 30.3.6.1.18";
55 }
56 leaf in-unsupported-codes {
57     type yang:counter64;
58     mandatory true;
59     description
```

```
1         "Number of OAMPDUs with unsupported codes received.";
2     reference
3         "IEEE Std 802.3, 30.3.6.1.19";
4 }
5 leaf frames-lost-due-to-oam {
6     type yang:counter64;
7     mandatory true;
8     description
9         "A count of the number of frames that were dropped by the
10         OAM multiplexer. Since the OAM multiplexer has multiple
11         inputs and a single output, there may be cases where
12         frames are dropped due to transmit resource contention.
13         This counter is incremented whenever a frame is dropped by
14         the OAM layer.";
15     reference
16         "IEEE Std 802.3, 30.3.6.1.46";
17 }
18 }
19 }
20 }
21
22 grouping discovery-remote {
23     description
24         "Nodes describing the discovery process remote end of a
25         link.";
26     leaf mode {
27         type mode;
28         description
29             "Mode (passive/active).";
30         reference
31             "IEEE Std 802.3, 30.3.6.1.3";
32     }
33 }
34 container functions-supported {
35     description
36         "The Link OAM functions supported by this interface.";
37     reference
38         "IEEE Std 802.3, 30.3.6.1.7";
39     leaf uni-directional-link-fault {
40         type boolean;
41         description
42             "Unidirectional link fault support.";
43     }
44     leaf loopback {
45         type boolean;
46         description
47             "Remote Loopback support.";
48     }
49     leaf link-monitoring {
50         type boolean;
51         description
52             "Link monitoring support.";
53     }
54     leaf mib-retrieval {
55         type boolean;
56         description
57             "MIB variable retrieval support.";
58     }
59 }
60 }
61 }
62 }
63 leaf revision {
64     type uint64;
65     config false;
```

```
1      description
2          "Configuration revision.";
3      reference
4          "IEEE Std 802.3, 30.3.6.1.12 and 30.3.6.1.13";
5  }
6  leaf oamtu {
7      type uint16;
8      units "octets";
9      config false;
10     description
11         "The maximum OAMPDU size for the remote node.
12         The peer OAM entities exchange the maximum size they can
13         support and negotiate to use the smaller of the two maximum
14         OAMPDU sizes.";
15     reference
16         "IEEE Std 802.3, 30.3.6.1.8 and 30.3.6.1.9";
17 }
18 }
19 }
20 }
21
22 grouping discovery-local {
23     description
24         "Nodes describing the local end discovery process of a
25         link.";
26     leaf mode {
27         type mode;
28         description
29             "This object configures the mode of OAM operation as active
30             or passive. Active mode provides capabilities to initiate
31             monitoring activities with the remote OAM peer entity,
32             while passive mode waits for the peer to initiate actions
33             with it. Changing this value results in incrementing the
34             revision field of locally generated OAMPDUs
35             (see IEEE Std 802.3, 30.3.6.1.12) and triggers the
36             OAM discovery process if the operational state was
37             already 'operational'. The default value is
38             implementation-dependent.";
39         reference
40             "IEEE Std 802.3, 30.3.6.1.3";
41     }
42 }
43
44 container functions-supported {
45     description
46         "The Link OAM functions supported by this interface.";
47     reference
48         "IEEE Std 802.3, 30.3.6.1.7";
49     leaf uni-directional-link-fault {
50         if-feature "uni-directional-link-fault";
51         type boolean;
52         description
53             "Unidirectional link fault support.
54             This affects the setting of the 'Unidirectional Support'
55             bit in the OAM configuration field put in the
56             Information OAMPDU.
57             This bit indicates to the peer device that it can send
58             OAM PDUs on links that are operating in unidirectional
59             mode (traffic flowing in one direction only).";
60     }
61 }
62
63 leaf loopback {
64     if-feature "remote-loopback-initiate";
65     type boolean;
```

```
1      default "true";
2      description
3          "Remote Loopback support.";
4  }
5  container link-monitor {
6      if-feature
7          "link-monitoring-remote or
8           link-monitoring-local";
9      description
10         "Configure link monitor parameters.";
11     reference
12         "IEEE Std 802.3, 57.1.2:c";
13     leaf link-monitoring {
14         type boolean;
15         default "true";
16         description
17             "Enable or disable monitoring.
18             This affects the setting of the 'Link Events' bit in
19             the OAM configuration field put in the Information
20             OAMPDU. This bit indicates to the peer device that the
21             OAM entity can send and receive Event Notification
22             OAMPDUs.";
23     }
24     list event-type {
25         if-feature "link-monitoring-local";
26         key "threshold-type";
27         description
28             "A list containing at most one entry for each of the
29             threshold event types. If there is no entry for a
30             particular event type, the default values are used for
31             both window size and threshold.";
32         leaf threshold-type {
33             type threshold-event-enum;
34             description
35                 "The type of threshold event for which this list
36                 entry is specifying the configuration.";
37             reference
38                 "IEEE Std 802.3, 57.5.3";
39         }
40         leaf window {
41             type uint64;
42             description
43                 "The size of the window to use when monitoring for
44                 this threshold event. The units, default and upper
45                 and lower bounds depend on the threshold type as
46                 follows:
47
48                 Symbol Period:
49                     Units:    number of symbols
50                     Default:  number of symbols in one second for the
51                             underlying physical layer
52                     Min:      number of symbols in one second for the
53                             underlying physical layer
54                     Max:      number of symbols in one minute for the
55                             underlying physical layer
56
57                 Frame:
58                     Units:    deciseconds
59                     Default:  1 second
60
61                 
```

```

1           Min:      1 second
2           Max:      1 minute
3
4           Frame Period:
5             Units:   number of frames
6             Default: number of minFrameSize frames in one
7                     second for the underlying physical layer
8           Min:      number of minFrameSize frames in one
9                     second for the underlying physical layer
10          Max:      number of minFrameSize frames in one
11                  minute for the underlying physical layer
12
13
14          Frame Seconds:
15            Units:   deciseconds
16            Default: 60 seconds
17            Min:     10 seconds
18            Max:     900 seconds";
19
20          reference
21            "IEEE Std 802.3, 30.3.6.1.34, 30.3.6.1.36, 30.3.6.1.38,
22            and 30.3.6.1.40";
23        }
24    leaf threshold {
25        type uint64 {
26            range "1..max";
27        }
28        default "1";
29        description
30          "The threshold value to use when determining whether
31          to generate an event given the number of errors that
32          occurred in a given window. The units depend on the
33          threshold type as follows:
34
35          Symbol Period: number of errored symbols
36          Frame:         number of errored frames
37          Frame Period:  number of errored frames
38          Frame Seconds: number of seconds containing at least
39                        1 frame error";
40        reference
41          "IEEE Std 802.3, 30.3.6.1.34, 30.3.6.1.36, 30.3.6.1.38,
42          and 30.3.6.1.40";
43    }
44  }
45
46  }
47
48  leaf mib-retrieval {
49      if-feature
50        "remote-mib-retrieval-initiate or
51        remote-mib-retrieval-respond";
52      type boolean;
53      status deprecated;
54      description
55        "leaf mib-retrieval is deprecated and changed name to
56        data-retrieval.
57        MIB variable retrieval support.
58        This affects the setting of the 'Variable Retrieval' bit
59        in the OAM configuration field put in the Information
60        OAMPDU. This bit indicates to the peer device that the
61        OAM entity can send and receive Variable Request and
62        Response OAMPDUs.";
63  }
64
65  }

```

```
1      leaf data-retrieval {
2          if-feature
3              "remote-data-retrieval-initiate or
4                  remote-mib-retrieval-respond";
5          type boolean;
6          description
7              "Variable retrieval support.
8              This affects the setting of the 'Variable Retrieval' bit
9              in the OAM configuration field put in the Information
10             OAMPDU. This bit indicates to the peer device that the
11             OAM entity can send and receive Variable Request and
12             Response OAMPDUs.";
13         }
14     }
15 }
16 leaf revision {
17     type uint64;
18     config false;
19     description
20         "Configuration revision.";
21     reference
22         "IEEE Std 802.3, 30.3.6.1.12 and 30.3.6.1.13";
23 }
24 leaf oammtu {
25     type uint16;
26     units "octets";
27     config false;
28     description
29         "The maximum OAMPDU size for the local node. The peer OAM
30         entities exchange the maximum size they can support and
31         negotiate to use the smaller of the two maximum OAMPDU
32         sizes.";
33     reference
34         "IEEE Std 802.3, 30.3.6.1.8 and 30.3.6.1.9";
35 }
36 }
37
38
39
40 grouping discovery-info {
41     description
42         "Information relating to the discovery process.";
43     container local {
44         description
45             "Properties of the local device.";
46         leaf operational-status {
47             type operational-state;
48             config false;
49             mandatory true;
50             description
51                 "Operational status.";
52             reference
53                 "IETF RFC 4878, dot3OamOperStatus; IEEE Std 802.3,
54                 30.3.6.1.4, 30.3.6.1.10, and 30.3.6.1.11";
55         }
56     }
57     leaf loopback-mode {
58         if-feature
59             "remote-loopback-initiate or
60                 remote-loopback-respond";
61         type loopback-status;
62         config false;
63         mandatory true;
64     }
65 }
```

```

1      description
2      "The loopback mode the interface is in.";
3      reference
4      "IEEE Std 802.3, 30.3.6.1.14";
5  }
6  uses discovery-local;
7  }
8  container remote {
9      config false;
10     description
11     "Properties of the remote (peer) device.";
12     leaf mac-address {
13         type ieee:mac-address;
14         description
15         "Remote MAC address.";
16         reference
17         "IEEE Std 802.3, 30.3.6.1.5";
18     }
19     leaf vendor-oui {
20         type vendor-oui;
21         description
22         "Remote vendor OUI.";
23         reference
24         "IEEE Std 802.3, 30.3.6.1.16";
25     }
26     leaf vendor-info {
27         type uint64;
28         description
29         "Remote vendor info. The semantics of this value are
30         proprietary and specific to the vendor.";
31         reference
32         "IEEE Std 802.3, 30.3.6.1.17";
33     }
34     leaf loopback-mode {
35         type loopback-status;
36         mandatory true;
37         description
38         "The loopback mode the interface is in.";
39         reference
40         "IEEE Std 802.3, 30.3.6.1.15";
41     }
42     uses discovery-remote;
43 }
44 }
45
46 augment "/if:interfaces/if:interface" {
47     when "derived-from-or-self(if:type, 'ianaift:ethernetCsmacd') or
48     derived-from-or-self(if:type, 'ianaift:ptm') " {
49         description
50         "Augments the interface model with nodes
51         specific to Ethernet Link OAM.";
52     }
53     description
54     "Augments Ethernet interface model with nodes
55     specific to Ethernet Link OAM.";
56     container link-oam {
57         presence "Implies Link OAM is configured on the interface.";
58         description
59         "Interface operational state for Ethernet Link OAM.";
60     }
61 }

```

```
1      action remote-loopback {
2          if-feature "remote-loopback-initiate";
3          description
4              "Start/stop remote loopback on the specified interface.";
5          reference
6              "IEEE Std 802.3, 57.1.2:b";
7          input {
8              leaf enable {
9                  type boolean;
10                 mandatory true;
11                 description
12                     "Whether to enable or disable remote loopback.";
13             }
14         }
15     }
16     output {
17         leaf success {
18             type boolean;
19             mandatory true;
20             description
21                 "True if the operation was successful,
22                  false otherwise.";
23         }
24         leaf error-message {
25             type string;
26             description
27                 "If the operation failed, optionally used to
28                  provide extra details.";
29         }
30     }
31 }
32
33 action reset-stats {
34     description
35         "Reset Ethernet Link OAM statistics on this interface.";
36     output {
37         leaf success {
38             type boolean;
39             mandatory true;
40             description
41                 "True if the operation was successful,
42                  false otherwise.";
43         }
44         leaf error-message {
45             type string;
46             description
47                 "If the operation failed, optionally used to provide
48                  extra details.";
49         }
50     }
51 }
52
53 }
54
55 notification non-threshold-event {
56     description
57         "This notification is sent when a local or remote
58         non-threshold crossing event is detected.";
59     uses event-details {
60         refine "event-type" {
61             must ". != 'ieee802-link-oam:threshold-event-type'" {
62                 description
63                     "This leaf is not set to
64                     'threshold-event-type'.";
65             }
66         }
67     }
68 }
```

```

1      }
2    }
3  }
4 }
5 notification threshold-event {
6   if-feature
7     "link-monitoring-local or
8     link-monitoring-remote";
9   description
10    "This notification is sent when a local or remote
11    threshold crossing event is detected.";
12  uses event-details {
13    refine "event-type" {
14      must ". = 'ieee802-link-oam:threshold-event-type'" {
15        description
16          "This leaf is set to 'threshold-event-type.'";
17      }
18    }
19  }
20 }
21 }
22 uses threshold-event-details;
23 }
24 leaf admin {
25   type admin-state;
26   default "disabled";
27   description
28     "This object is used to provision the default
29     administrative OAM mode for this interface. This object
30     represents the desired state of OAM for this interface.
31     It starts in the disabled state until an explicit
32     management action or configuration information retained
33     by the system causes a transition to the enabled(1)
34     state. When enabled(1), Ethernet OAM will attempt to
35     operate over this interface. The default value is
36     implementation-dependent.";
37 }
38 }
39 container discovery-info {
40   description
41     "Information relating to the discovery process.";
42   uses discovery-info;
43 }
44 container event-log {
45   config false;
46   description
47     "List of Ethernet Link OAM event log entries on the
48     interface.";
49   list event-log-entry {
50     key "index";
51     description
52       "Ethernet Link OAM event log entry.";
53     leaf index {
54       type uint64;
55       description
56         "Index of this event in the event log.";
57     }
58   }
59   uses event-details;
60   uses threshold-event-details;
61 }
62 }
63 }
64 }
65 container statistics {

```

```
1      config false;
2      description
3          "Statistics for an 802.3 OAM interface.";
4      uses statistics-common;
5      leaf local-error-symbol-period-log-entries {
6          type yang:counter64;
7          mandatory true;
8          description
9              "Number of local error symbol period log entries.";
10     }
11     leaf local-error-frame-log-entries {
12         type yang:counter64;
13         mandatory true;
14         description
15             "Number of local error frame log entries.";
16     }
17     leaf local-error-frame-period-log-entries {
18         type yang:counter64;
19         mandatory true;
20         description
21             "Number of local error frame period log entries.";
22     }
23     leaf local-error-frame-second-log-entries {
24         type yang:counter64;
25         mandatory true;
26         description
27             "Number of local error frame second log entries.";
28     }
29     leaf remote-error-symbol-period-log-entries {
30         if-feature "link-monitoring-remote";
31         type yang:counter64;
32         mandatory true;
33         description
34             "Number of remote error symbol period log entries.";
35     }
36     leaf remote-error-frame-log-entries {
37         if-feature "link-monitoring-remote";
38         type yang:counter64;
39         mandatory true;
40         description
41             "Number of remote error frame log entries.";
42     }
43     leaf remote-error-frame-period-log-entries {
44         if-feature "link-monitoring-remote";
45         type yang:counter64;
46         mandatory true;
47         description
48             "Number of remote error frame period log entries.";
49     }
50     leaf remote-error-frame-second-log-entries {
51         if-feature "link-monitoring-remote";
52         type yang:counter64;
53         mandatory true;
54         description
55             "Number of remote error frame second log entries.";
56     }
57 }
58 }
59 }
```

1 }

2
3
4 NOTE—IEEE Std 802.3, Annex K defines optional alternative terminology for “master” and “slave”.
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Annex A

(informative)

Bibliography

The following referenced documents are informative and not needed in the implementation of this standard, rather left here for the benefit of the reader. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments or corrigenda) applies.

[B1] IEEE Std 802®-2014, IEEE Standard for Local and Metropolitan Area Networks: Overview and Architecture.^{o, p}

[B2] IEEE Std 802d™-2017, IEEE Standard for Local and Metropolitan Area Networks: Overview and Architecture Amendment 1: Allocation of Uniform Resource Name (URN) Values in IEEE 802 Standards.

[B3] IEEE Std 802.1Q™-2014, IEEE Standard for Local and metropolitan area networks—Bridges and Bridged Networks.

[B4] IETF RFC 2819, Remote Network Monitoring Management Information Base, S. Waldbusser, May 2000.^q

[B5] IETF RFC 3621, *Power Ethernet MIB*, A. Berger, December 2003

[B6] IETF RFC 3635, *Definitions of Managed Objects for the Ethernet-like Interface Types*, J. Flick, September 2003.

[B7] IETF RFC 6991, *Common YANG Data Types*, Schoenwaelder J., July 2013.

[B8] IETF RFC 8342, *Network Management Datastore Architecture (NMDA)*, M. Bjorklund, J. Schoenwaelder, P. Shafer, K. Watsen, and R. Wilton, March 2018.

[B9] IETF RFC 8343, *A YANG Data Model for Interface Management*, Bjorklund, M., March 2018.

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^qInternet Requests for Comments (RFCs) are available on the World Wide Web at the following ftp site: venera.isi.edu; logon: anonymous; password: user's e-mail address; directory: in-inotes.